SEN00161-03

Shop Manual

107E-1 SERIES

ENGINE

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SEN00161-03

Shop Manual

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ENGINE

KOMATSU®

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SHOP MANUAL

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107E-1
SERIES

ENGINE

SEN00161-03

Instructions for Binder Inserts

This sheet includes two styles of binder insert:

Insert for earlier vinyl clear window binder.
Cut binder insert on lines indicated and insert in spine of binder. Discard the front cover binder insert.

Insert for later poly binder with half window.
Insert clear insert protection sheet in label holder. Cut binder insert on lines indicated and insert protection sheet in spine of binder.

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Organization list of the shop manual

The contents of this shop manual are shown together with Form No. in a list.

Note 1: Always keep the latest version of this manual in accordance with this list and utilize accordingly.
The marks shown to the right of Form No. denote the following:
○: New issue (to be filed additionally) ●: Revision (to be replaced for each Form No.)

Note 2: This shop manual can be supplied for each Form No.

Note 3: To file this shop manual in the special binder for management, handle it as follows:
- Place a divider on the top of each section in the file after matching the Tub No. with No. indicated next to each Section Name shown in the table below:
- File overview and other materials in sections in the order shown below and utilize them accordingly.

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Foreword and general information

(Rev. 2005/11)

Safety notice

Important safety notice

Proper service and repair are extremely important for safe machine operation. The service and repair techniques recommended by Komatsu and described in this manual are both effective and safe. Some of these techniques require the use of tools specially designed by Komatsu for the specific purpose.

To prevent injury to workers, the symbol ⚠ is used to mark safety precautions in this manual. The cautions accompanying these symbols should always be followed carefully. If any dangerous situation arises or may possibly arise, first consider safety, and take the necessary actions to deal with the situation.

1. General precautions

⚠ **Mistakes in operation are extremely dangerous. Read the Operation and Maintenance Manual carefully before operating the machine.**

1) Before carrying out any greasing or repairs, read all the safety plates stuck to the machine. For the locations of the safety plates and detailed explanation of precautions, see the Operation and Maintenance Manual.
2) Decide a place in the repair workshop to keep tools and removed parts. Always keep the tools and parts in their correct places. Always keep the work area clean and make sure that there is no dirt, water, or oil on the floor. Smoke only in the areas provided for smoking. Never smoke while working.
3) When carrying out any operation, always wear safety shoes and helmet. Do not wear loose work clothes, or clothes with buttons missing.
   - Always wear safety glasses when hitting parts with a hammer.
   - Always wear safety glasses when grinding parts with a grinder, etc.
4) When carrying out any operation with 2 or more workers, always agree on the operating procedure before starting. Always inform your fellow workers before starting any step of the operation. Before starting work, hang UNDER REPAIR signs in the operator's compartment.
5) Only qualified workers must carry out work and operation which require license or qualification.
6) Keep all tools in good condition, learn the correct way to use them, and use the proper ones of them. Before starting work, thoroughly check the tools, machine, forklift, service car, etc.
7) If welding repairs are needed, always have a trained and experienced welder carry out the work. When carrying out welding work, always wear welding gloves, apron, shielding goggles, cap and other clothes suited for welding work.
8) Before starting work, warm up your body thoroughly to start work under good condition.

Safety points

1	Good arrangement
2	Correct work clothes
3	Following work standard
4	Making and checking signs
5	Prohibition of operation and handling by unlicensed workers
6	Safety check before starting work
7	Wearing protective goggles (for cleaning or grinding work)
8	Wearing shielding goggles and protectors (for welding work)
9	Good physical condition and preparation
10	Precautions against work which you are not used to or you are used to too much

2. Preparations for work

1) Before adding oil or making any repairs, park the machine on hard and level ground, and apply the parking brake and block the wheels or tracks to prevent the machine from moving.
2) Before starting work, lower the work equipment (blade, ripper, bucket, etc.) to the ground. If this is not possible, insert the lock pin or use blocks to prevent the work equipment from falling. In addition, be sure to lock all the control levers and hang warning signs on them.

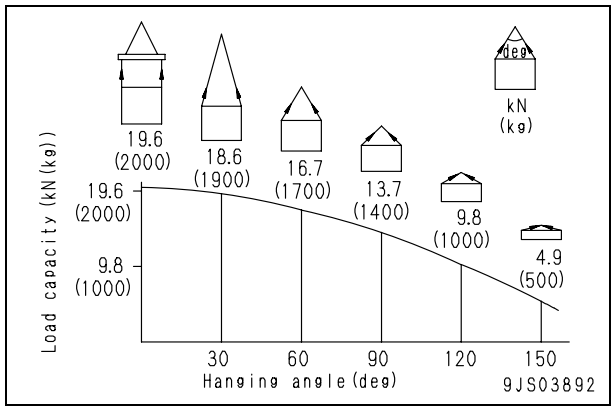
3) When disassembling or assembling, support the machine with blocks, jacks, or stands before starting work.
4) Remove all mud and oil from the steps or other places used to get on and off the machine. Always use the handrails, ladders or steps when getting on or off the machine. Never jump on or off the machine. If it is impossible to use the handrails, ladders or steps, use a stand to provide safe footing.

3. Precautions during work

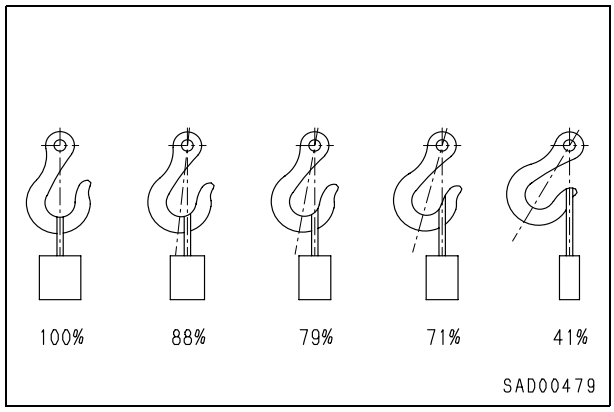
1) Before disconnecting or removing components of the oil, water, or air circuits, first release the pressure completely from the circuit. When removing the oil filler cap, a drain plug, or an oil pressure pickup plug, loosen it slowly to prevent the oil from spurting out.
2) The coolant and oil in the circuits are hot when the engine is stopped, so be careful not to get scalded. Wait for the oil and coolant to cool before carrying out any work on the oil or water circuits.
3) Before starting work, stop the engine. When working on or around a rotating part, in particular, stop the engine. When checking the machine without stopping the engine (measuring oil pressure, revolving speed, temperature, etc.), take extreme care not to get rolled or caught in rotating parts or moving parts.
4) Before starting work, remove the leads from the battery. Always remove the lead from the negative (–) terminal first.
5) When raising a heavy component (heavier than 25 kg), use a hoist or crane. Before starting work, check that the slings (wire ropes, chains, and hooks) are free from damage. Always use slings which have ample capacity and install them to proper places. Operate the hoist or crane slowly to prevent the component from hitting any other part. Do not work with any part still raised by the hoist or crane.
6) When removing a cover which is under internal pressure or under pressure from a spring, always leave 2 bolts in diagonal positions. Loosen those bolts gradually and alternately to release the pressure, and then remove the cover.
7) When removing components, be careful not to break or damage the electrical wiring. Damaged wiring may cause electrical fires.
8) When removing piping, stop the fuel or oil from spilling out. If any fuel or oil drips onto the floor, wipe it up immediately. Fuel or oil on the floor can cause you to slip and can even start fires.
9) As a general rule, do not use gasoline to wash parts. Do not use it to clean electrical parts, in particular.
10) Be sure to assemble all parts again in their original places. Replace any damaged parts and parts which must not be reused with new parts. When installing hoses and wires, be sure that they will not be damaged by contact with other parts when the machine is operated.
11) When installing high pressure hoses, make sure that they are not twisted. Damaged tubes are dangerous, so be extremely careful when installing tubes for high pressure circuits. In addition, check that connecting parts are correctly installed.
12) When assembling or installing parts, always tighten them to the specified torques. When installing protective parts such as guards, or parts which vibrate violently or rotate at high speed, be particularly careful to check that they are installed correctly.
13) When aligning 2 holes, never insert your fingers or hand. Be careful not to get your fingers caught in a hole.
14) When measuring hydraulic pressure, check that the measuring tools are correctly assembled.
15) Take care when removing or installing the tracks of track-type machines. When removing the track, the track separates suddenly, so never let anyone stand at either end of the track.
16) If the engine is operated for a long time in a place which is not ventilated well, you may suffer from gas poisoning. Accordingly, open the windows and doors to ventilate well.

4. Precautions for sling work and making signs

1) Only one appointed worker must make signs and co-workers must communicate with each other frequently. The appointed sign maker must make specified signs clearly at a place where he is seen well from the operator's seat and where he can see the working condition easily. The sign maker must always stand in front of the load and guide the operator safely.
   - Do not stand under the load.
   - Do not step on the load.
2) Check the slings before starting sling work.
3) Keep putting on gloves during sling work. (Put on leather gloves, if available.)
4) Measure the weight of the load by the eye and check its center of gravity.
5) Use proper sling according to the weight of the load and method of slinging. If too thick wire ropes are used to sling a light load, the load may slip and fall.
6) Do not sling a load with 1 wire rope alone. If it is slung so, it may rotate and may slip out of the rope. Install 2 or more wire ropes symmetrically.
   ⚠ Slinging with one rope may cause turning of the load during hoisting, untwisting of the rope, or slipping of the rope from its original winding position on the load, which can result in a dangerous accident.
7) Limit the hanging angle to 60°, as a rule. Do not sling a heavy load with ropes forming a wide hanging angle from the hook. When hoisting a load with 2 or more ropes, the force subjected to each rope will increase with the hanging angle. The table below shows the variation of allowable load in kN {kg} when hoisting is made with 2 ropes, each of which is allowed to sling up to 9.8 kN {1,000 kg} vertically, at various hanging angles. When the 2 ropes sling a load vertically, up to 19.6 kN {2,000 kg} of total weight can be suspended. This weight is reduced to 9.8 kN {1,000 kg} when the 2 ropes make a hanging angle of 120°. If the 2 ropes sling a 19.6 kN {2,000 kg} load at a lifting angle of 150°, each of them is subjected to a force as large as 39.2 kN {4,000 kg}.



8) When installing wire ropes to an angular load, apply pads to protect the wire ropes. If the load is slippery, apply proper material to prevent the wire rope from slipping.
9) Use the specified eyebolts and fix wire ropes, chains, etc. to them with shackles, etc.
10) Apply wire ropes to the middle portion of the hook.
    - Slinging near the tip of the hook may cause the rope to slip off the hook during hoisting. The hook has the maximum strength at the middle portion.



11) Do not use twisted or kinked wire ropes.
12) When lifting up a load, observe the following.
    - Wind in the crane slowly until wire ropes are stretched. When settling the wire ropes with the hand, do not grasp them but press them from above. If you grasp them, your fingers may be caught.
    - After the wire ropes are stretched, stop the crane and check the condition of the slung load, wire ropes, and pads.

- If the load is unstable or the wire rope or chains are twisted, lower the load and lift it up again.
- Do not lift up the load slantingly.

13) When lifting down a load, observe the following.
- When lifting down a load, stop it temporarily at 30 cm above the floor, and then lower it slowly.
- Check that the load is stable, and then remove the sling.
- Remove kinks and dirt from the wire ropes and chains used for the sling work, and put them in the specified place.

5. Precautions for using mobile crane

★ Read the Operation and Maintenance Manual of the crane carefully in advance and operate the crane safely.

6. Precautions for using overhead hoist crane

⚠ **When raising a heavy part (heavier than 25 kg), use a hoist, etc. In Disassembly and assembly, the weight of a part heavier than 25 kg is indicated after the mark of .**

1) Before starting work, inspect the wire ropes, brake, clutch, controller, rails, over wind stop device, electric shock prevention earth leakage breaker, crane collision prevention device, and power application warning lamp, and check safety.
2) Observe the signs for sling work.
3) Operate the hoist at a safe place.
4) Check the direction indicator plates (east, west, south, and north) and the directions of the control buttons without fail.
5) Do not sling a load slantingly. Do not move the crane while the slung load is swinging.
6) Do not raise or lower a load while the crane is moving longitudinally or laterally.
7) Do not drag a sling.
8) When lifting up a load, stop it just after it leaves the ground and check safety, and then lift it up.
9) Consider the travel route in advance and lift up a load to a safe height.
10) Place the control switch on a position where it will not be an obstacle to work and passage.
11) After operating the hoist, do not swing the control switch.
12) Remember the position of the main switch so that you can turn off the power immediately in an emergency.
13) If the hoist stops because of a power failure, turn the power switch OFF. When turning on a switch which was turned OFF by the electric shock prevention earth leakage breaker, check that the devices related to that switch are not in operation state.
14) If you find an obstacle around the hoist, stop the operation.
15) After finishing the work, stop the hoist at the specified position and raise the hook to at least 2 m above the floor. Do not leave the sling installed to the hook.

7. Selecting wire ropes

1) Select adequate ropes depending on the weight of parts to be hoisted, referring to the table below.

Wire ropes
(Standard “Z” twist ropes without galvanizing)
(JIS G3525, No. 6, Type 6X37-A)

Nominal diameter of rope	Allowable load	
mm	kN	ton
10	8.8	0.9
12	12.7	1.3
14	17.3	1.7
16	22.6	2.3
18	28.6	2.9
20	35.3	3.6
25	55.3	5.6
30	79.6	8.1
40	141.6	14.4
50	221.6	22.6
60	318.3	32.4

★ The allowable load is one-sixth of the breaking strength of the rope used (Safety coefficient: 6).

How to read the shop manual

- Some attachments and optional parts in this shop manual may not be delivered to certain areas. If one of them is required, consult KOMATSU distributors.
- Materials and specifications are subject to change without notice.
- Shop manuals are divided into the “Chassis volume” and “Engine volume”. For the engine unit, see the engine volume of the engine model mounted on the machine.

1. Composition of shop manual

This shop manual contains the necessary technical information for services performed in a workshop. For ease of understanding, the manual is divided into the following sections.

00. Index and foreword

This section explains the shop manuals list, table of contents, safety, and basic information.

01. Specification

This section explains the specifications of the machine.

10. Structure, function and maintenance standard

This section explains the structure, function, and maintenance standard values of each component. The structure and function sub-section explains the structure and function of each component. It serves not only to give an understanding of the structure, but also serves as reference material for troubleshooting. The maintenance standard sub-section explains the criteria and remedies for disassembly and service.

20. Standard value table

This section explains the standard values for new machine and judgement criteria for testing, adjusting, and troubleshooting. This standard value table is used to check the standard values in testing and adjusting and to judge parts in troubleshooting.

30. Testing and adjusting

This section explains measuring instruments and measuring methods for testing and adjusting, and method of adjusting each part. The standard values and judgement criteria for testing and adjusting are explained in Testing and adjusting.

40. Troubleshooting

This section explains how to find out failed parts and how to repair them. The troubleshooting is divided by failure modes. The “S mode” of the troubleshooting related to the engine may be also explained in the Chassis volume and Engine volume. In this case, see the Chassis volume.

50. Disassembly and assembly

This section explains the special tools and procedures for removing, installing, disassembling, and assembling each component, as well as precautions for them. In addition, tightening torque and quantity and weight of coating material, oil, grease, and coolant necessary for the work are also explained.

90. Diagrams and drawings (chassis volume)/Repair and replacement of parts (engine volume)

- Chassis volume
  This section gives hydraulic circuit diagrams and electrical circuit diagrams.
- Engine volume
  This section explains the method of reproducing, repairing, and replacing parts.

2. Revision and distribution

Any additions, revisions, or other change of notices will be sent to KOMATSU distributors. Get the most up-to-date information before you start any work.

3. Filing method

File by the brochures in the correct order of the form number printed in the shop manual composition table.

- **Revised edition mark**
  When a manual is revised, the ones and tens digits of the form number of each brochure is increased by 1. (Example: 00, 01, 02 …)

- **Revisions**
  Revised brochures are shown in the shop manual composition table.

4. Symbols

Important safety and quality portions are marked with the following symbols so that the shop manual will be used practically.

Symbol	Item	Remarks
	Safety	Special safety precautions are necessary when performing work.
★	Caution	Special technical precautions or other precautions for preserving standards are necessary when performing work.
	Weight	Weight of parts of component or parts. Caution necessary when selecting hoisting wire, or when working posture is important, etc.
	Tightening torque	Places that require special attention for tightening torque during assembly.
	Coat	Places to be coated with adhesives, etc. during assembly.
	Oil, coolant	Places where oil, etc. must be added, and capacity.
	Drain	Places where oil, etc. must be drained, and quantity to be drained.

5. Units

In this shop manual, the units are indicated with International System of units (SI). For reference, conventionally used Gravitational System of units is indicated in parentheses { }.

Explanation of terms for maintenance standard

The maintenance standard values necessary for judgment of products and parts are described by the following terms.

1. Standard size and tolerance

- To be accurate, the finishing size of parts is a little different from one to another.
- To specify a finishing size of a part, a temporary standard size is set and an allowable difference from that size is indicated.
- The above size set temporarily is called the “standard size” and the range of difference from the standard size is called the “tolerance”.
- The tolerance with the symbols of + or – is indicated on the right side of the standard size.

Example:

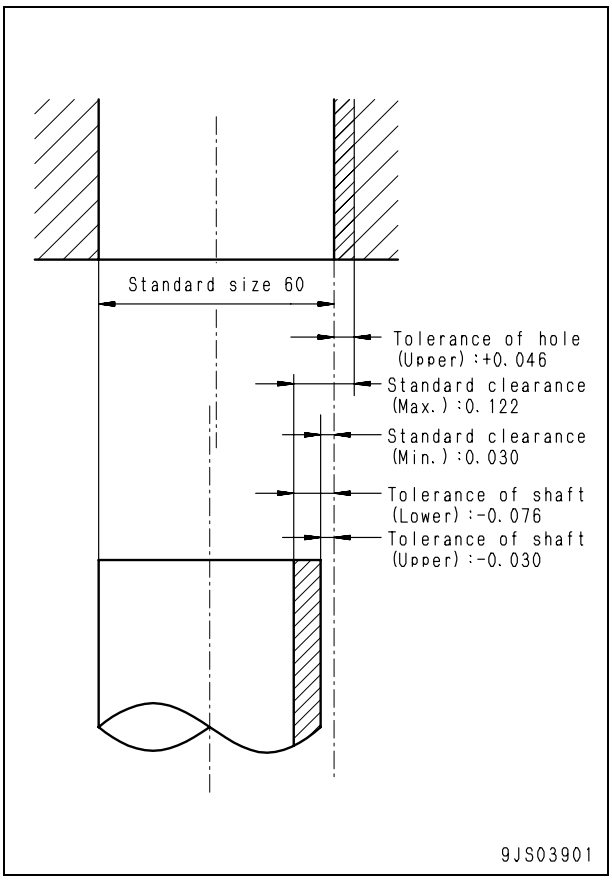
Standard size	Tolerance
120	–0.022 –0.126

★ The tolerance may be indicated in the text and a table as [standard size (upper limit of tolerance/lower limit of tolerance)].
Example) 120 (–0.022/–0.126)

- Usually, the size of a hole and the size of the shaft to be fitted to that hole are indicated by the same standard size and different tolerances of the hole and shaft. The tightness of fit is decided by the tolerance.
- Indication of size of rotating shaft and hole and relationship drawing of them

Example:

Standard size	Tolerance	
	Shaft	Hole
60	–0.030 –0.076	+0.046 0



2. Standard clearance and standard value

- The clearance made when new parts are assembled is called the “standard clearance“, which is indicated by the range from the minimum clearance to the maximum clearance.
- When some parts are repaired, the clearance is generally adjusted to the standard clearance.
- A value of performance and function of new products or equivalent is called the “standard value“, which is indicated by a range or a target value.
- When some parts are repaired, the value of performance/function is set to the standard value.

3. Standard interference

- When the size of a hole is smaller than the size of a shaft because of the standard size and tolerance, the difference between these sizes is called the “interference”.
- The range (A – B) from the difference (A) between the minimum size of the shaft and the maximum size of the hole to the difference (B) between the maximum size of the shaft and the minimum size of the hole is the “standard interference”.
- After repairing or replacing some parts, measure the size of their hole and shaft and check that the interference is in the standard range.

4. Repair limit and allowable value

- The size of a part changes because of wear and deformation while it is used. The limit of changed size is called the “repair limit”.
- If a part is worn to the repair limit must be replaced or repaired.
- The performance and function of a product lowers while it is used. A value below which the product can be used without causing a problem is called the “allowable value”.
- If a product is worn to the allowable value, it must be checked or repaired. Since the permissible value is estimated from various tests or experiences in most cases, however, it must be judged after considering the operating condition and customer's requirement.

5. Clearance limit

- Parts can be used until the clearance between them is increased to a certain limit. The limit at which those parts cannot be used is called the “clearance limit”.
- If the clearance between the parts exceeds the clearance limit, they must be replaced or repaired.

6. Interference limit

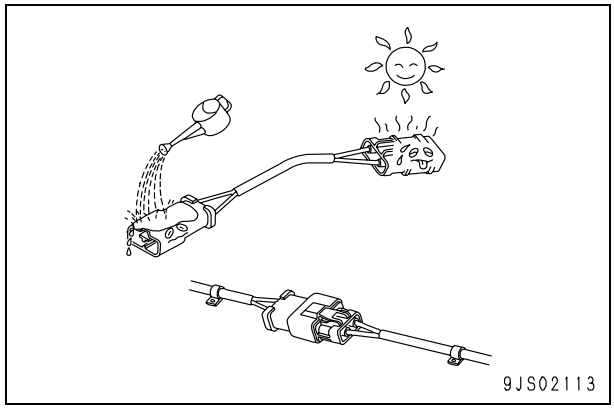
- The allowable maximum interference between the hole of a part and the shaft of another part to be assembled is called the “interference limit”.
- The interference limit shows the repair limit of the part of smaller tolerance.
- If the interference between the parts exceeds the interference limit, they must be replaced or repaired.

Handling electric equipment and hydraulic component

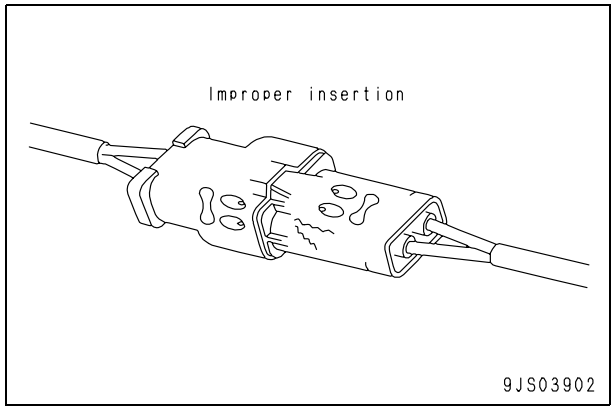
To maintain the performance of the machine over a long period, and to prevent failures or other troubles before they occur, correct “operation“, “maintenance and inspection“, “troubleshooting“, and “repairs” must be carried out. This section deals particularly with correct repair procedures for mechatronics and is aimed at improving the quality of repairs. For this purpose, it gives sections on “Handling electric equipment” and “Handling hydraulic equipment” (particularly gear oil and hydraulic oil).

Points to remember when handling electric equipment

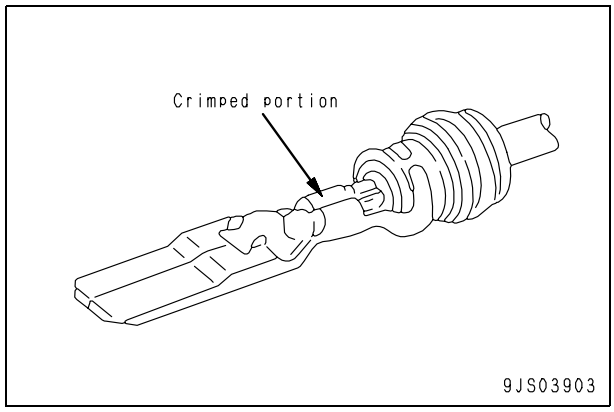
1. **Handling wiring harnesses and connectors**
   Wiring harnesses consist of wiring connecting one component to another component, connectors used for connecting and disconnecting one wire from another wire, and protectors or tubes used for protecting the wiring.
   Compared with other electrical components fitted in boxes or cases, wiring harnesses are more likely to be affected by the direct effects of rain, water, heat, or vibration. Furthermore, during inspection and repair operations, they are frequently removed and installed again, so they are likely to suffer deformation or damage. For this reason, it is necessary to be extremely careful when handling wiring harnesses.



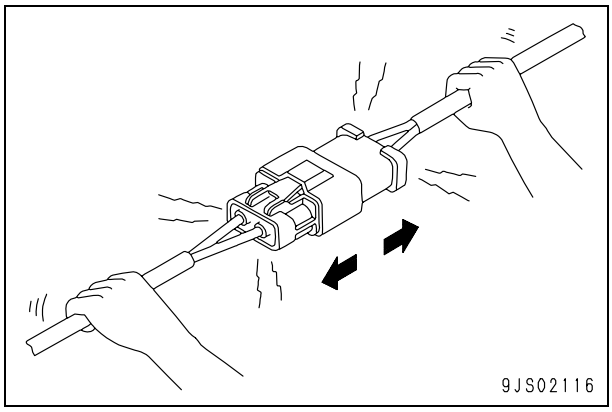
2. **Main failures occurring in wiring harness**
   1) Defective contact of connectors (defective contact between male and female)
      Problems with defective contact are likely to occur because the male connector is not properly inserted into the female connector, or because one or both of the connectors is deformed or the position is not correctly aligned, or because there is corrosion or oxidization of the contact surfaces.



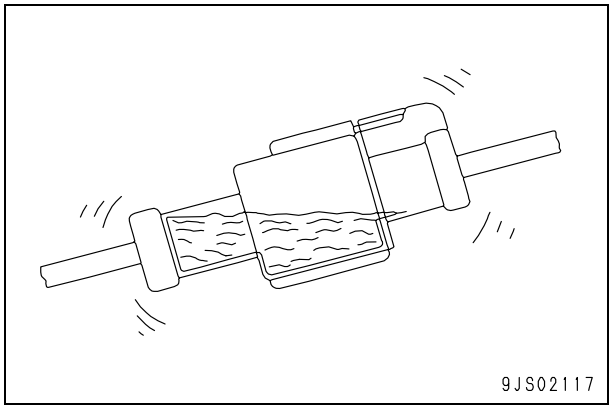
   2) Defective crimping or soldering of connectors
      The pins of the male and female connectors are in contact at the crimped terminal or soldered portion, but if there is excessive force brought to bear on the wiring, the plating at the joint will peel and cause improper connection or breakage.



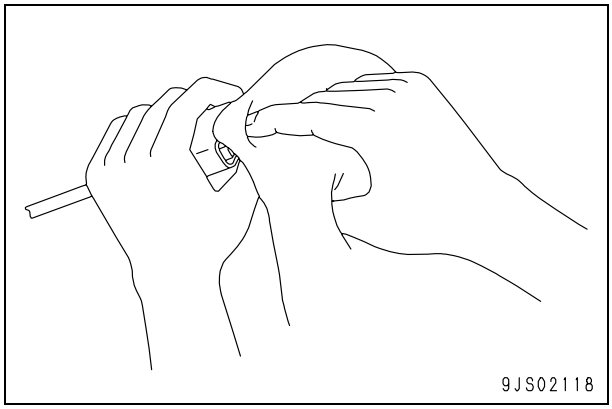
3) Disconnections in wiring
If the wiring is held and the connectors are pulled apart, or components are lifted with a crane with the wiring still connected, or a heavy object hits the wiring, the crimping of the connector may separate, or the soldering may be damaged, or the wiring may be broken.



4) High-pressure water entering connector
The connector is designed to make it difficult for water to enter (drip-proof structure), but if high-pressure water is sprayed directly on the connector, water may enter the connector, depending on the direction of the water jet. Accordingly, take care not splash water over the connector. The connector is designed to prevent water from entering, but at the same time, if water does enter, it is difficult for it to be drained. Therefore, if water should get into the connector, the pins will be short-circuited by the water, so if any water gets in, immediately dry the connector or take other appropriate action before passing electricity through it.



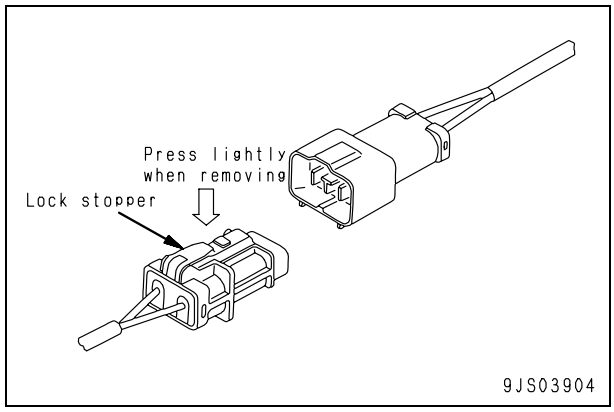
5) Oil or dirt stuck to connector
If oil or grease are stuck to the connector and an oil film is formed on the mating surface between the male and female pins, the oil will not let the electricity pass, so there will be defective contact. If there is oil or grease stuck to the connector, wipe it off with a dry cloth or blow it dry with compressed air and spray it with a contact restorer.
★ When wiping the mating portion of the connector, be careful not to use excessive force or deform the pins.
★ If there is oil or water in the compressed air, the contacts will become even dirtier, so remove the oil and water from the compressed air completely before cleaning with compressed air.



3. Removing, installing, and drying connectors and wiring harnesses

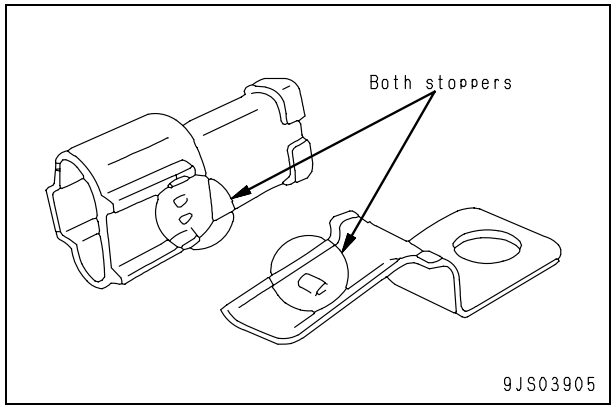
1) Disconnecting connectors

1] Hold the connectors when disconnecting.
When disconnecting the connectors, hold the connectors. For connectors held by a screw, loosen the screw fully, then hold the male and female connectors in each hand and pull apart. For connectors which have a lock stopper, press down the stopper with your thumb and pull the connectors apart.
★ Never pull with one hand.

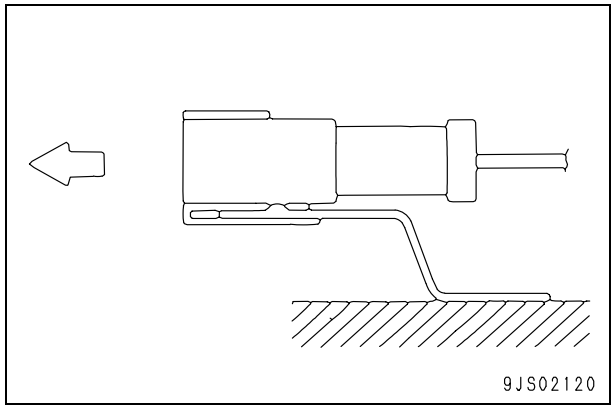


2] When removing from clips

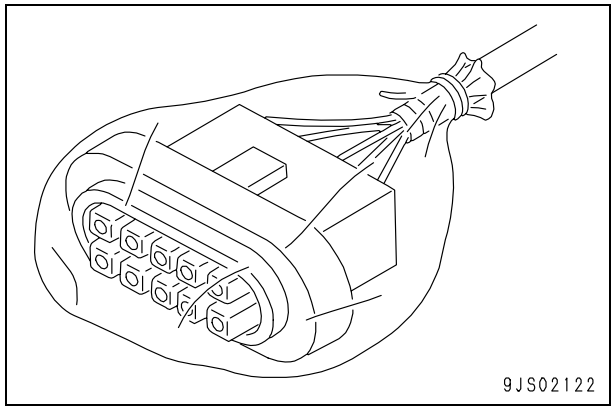
- Both of the connector and clip have stoppers, which are engaged with each other when the connector is installed.



- When removing a connector from a clip, pull the connector in a parallel direction to the clip for removing stoppers.
★ If the connector is twisted up and down or to the left or right, the housing may break.



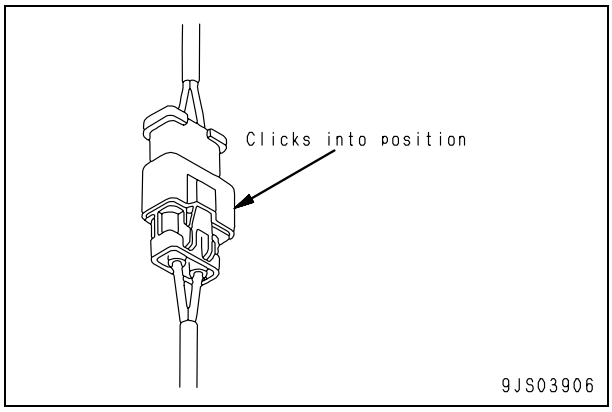
3] Action to take after removing connectors
After removing any connector, cover it with a vinyl bag to prevent any dust, dirt, oil, or water from getting in the connector portion.
★ If the machine is left disassembled for a long time, it is particularly easy for improper contact to occur, so always cover the connector.

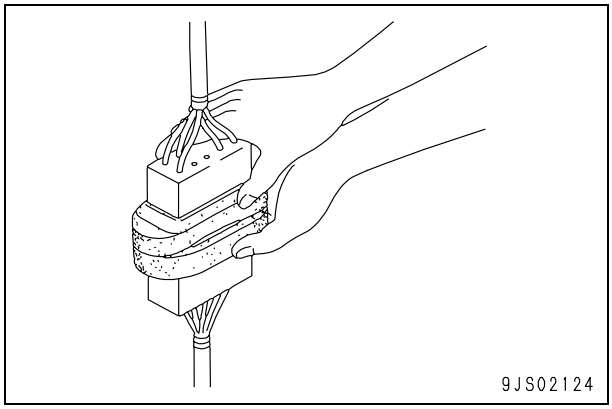


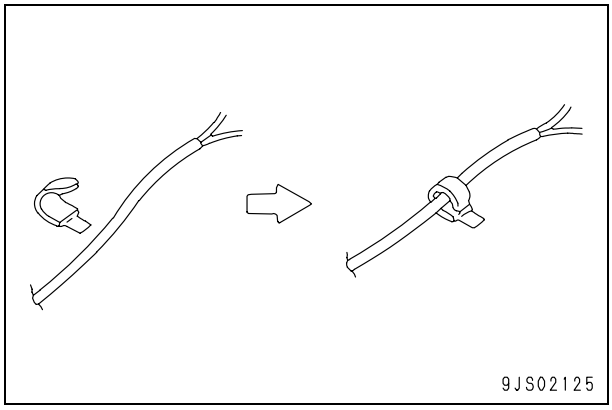
2) Connecting connectors
   1] Check the connector visually.
   Check that there is no oil, dirt, or water stuck to the connector pins (mating portion).
   Check that there is no deformation, defective contact, corrosion, or damage to the connector pins.
   Check that there is no damage or breakage to the outside of the connector.
   ★ If there is any oil, water, or dirt stuck to the connector, wipe it off with a dry cloth. If any water has got inside the connector, warm the inside of the wiring with a dryer, but be careful not to make it too hot as this will cause short circuits.
   ★ If there is any damage or breakage, replace the connector.
   2] Fix the connector securely.
   Align the position of the connector correctly, and then insert it securely. For connectors with lock stopper, push in the connector until the stopper clicks into position.
   3] Correct any protrusion of the boot and any misalignment of the wiring harness.
   For connectors fitted with boots, correct any protrusion of the boot. In addition, if the wiring harness is misaligned, or the clamp is out of position, adjust it to its correct position.
   ★ If the connector cannot be corrected easily, remove the clamp and adjust the position.
   ● If the connector clamp has been removed, be sure to return it to its original position. Check also that there are no loose clamps.

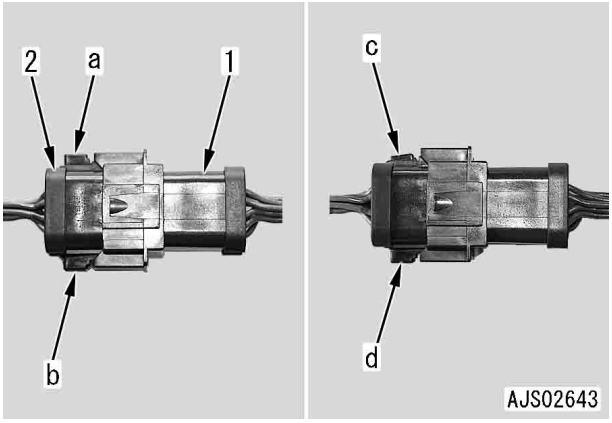
3) Connecting DT connectors
   Since the DT 8-pin and 12-pin heavy duty wire connectors have 2 latches respectively, push them in until they click 2 times.

   1. Male connector
   2. Female connector
   ● Normal locking state (Horizontal): a, b, d
   ● Incomplete locking state (Diagonal): c

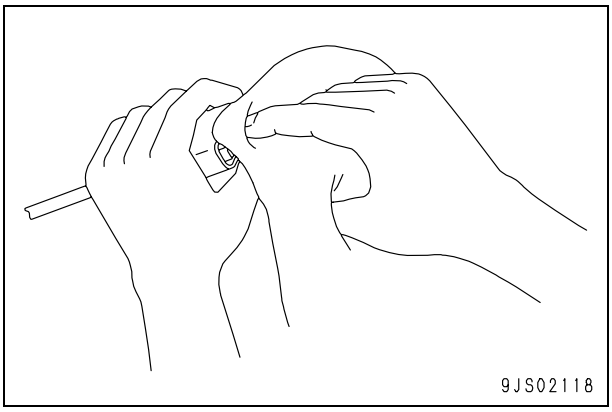


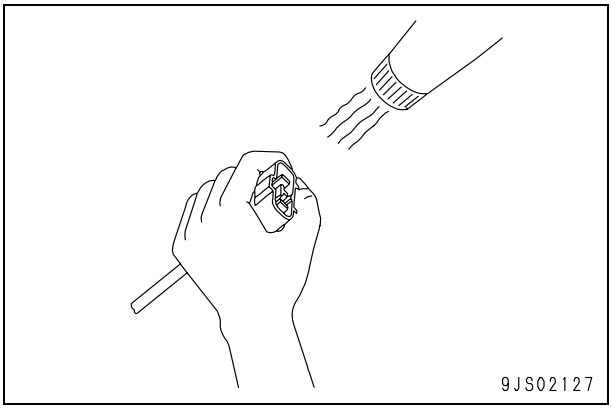


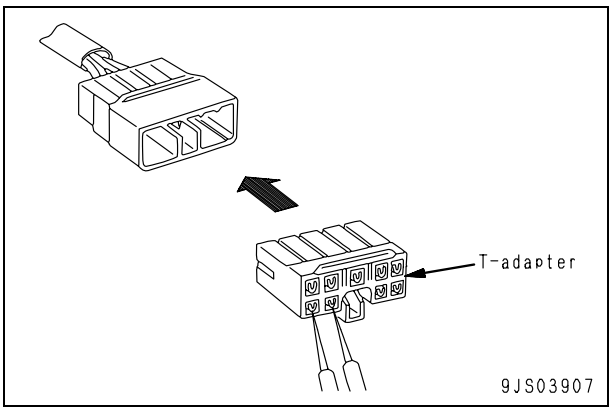




4) Drying wiring harness
If there is any oil or dirt on the wiring harness, wipe it off with a dry cloth. Avoid washing it in water or using steam. If the connector must be washed in water, do not use high-pressure water or steam directly on the wiring harness. If water gets directly on the connector, do as follows.
   1] Disconnect the connector and wipe off the water with a dry cloth.
      ★ If the connector is blown dry with compressed air, there is the risk that oil in the air may cause defective contact, so remove all oil and water from the compressed air before blowing with air.
   2] Dry the inside of the connector with a dryer.
      If water gets inside the connector, use a dryer to dry the connector.
      ★ Hot air from the dryer can be used, but regulate the time that the hot air is used in order not to make the connector or related parts too hot, as this will cause deformation or damage to the connector.
   3] Carry out a continuity test on the connector.
      After drying, leave the wiring harness disconnected and carry out a continuity test to check for any short circuits between pins caused by water.
      ★ After completely drying the connector, blow it with contact restorer and reassemble.

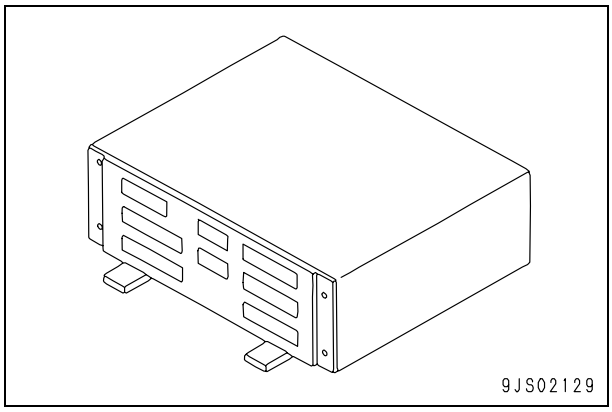


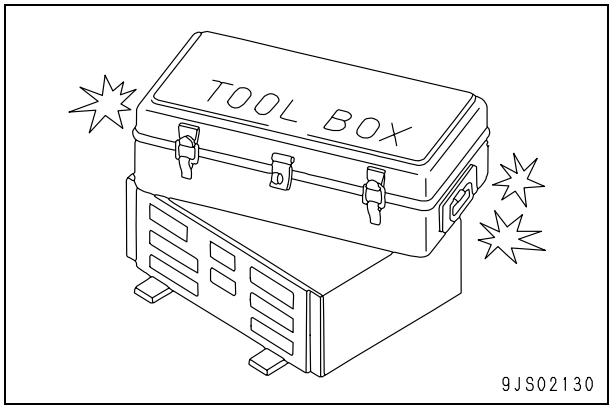




4. Handling controller

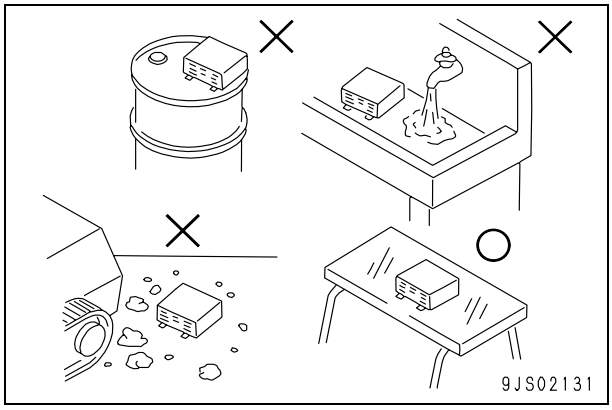
1) The controller contains a microcomputer and electronic control circuits. These control all of the electronic circuits on the machine, so be extremely careful when handling the controller.
2) Do not place objects on top of the controller.
3) Cover the control connectors with tape or a vinyl bag. Never touch the connector contacts with your hand.
4) During rainy weather, do not leave the controller in a place where it is exposed to rain.
5) Do not place the controller on oil, water, or soil, or in any hot place, even for a short time. (Place it on a suitable dry stand).
6) Precautions when carrying out arc welding
When carrying out arc welding on the body, disconnect all wiring harness connectors connected to the controller. Fit an arc welding ground close to the welding point.





5. Points to remember when troubleshooting electric circuits

1) Always turn the power OFF before disconnecting or connecting connectors.
2) Before carrying out troubleshooting, check that all the related connectors are properly inserted.
   ★ Disconnect and connect the related connectors several times to check.
3) Always connect any disconnected connectors before going on to the next step.
   ★ If the power is turned ON with the connectors still disconnected, unnecessary abnormality displays will be generated.
4) When carrying out troubleshooting of circuits (measuring the voltage, resistance, continuity, or current), move the related wiring and connectors several times and check that there is no change in the reading of the tester.
   ★ If there is any change, there is probably defective contact in that circuit.



Points to remember when handling hydraulic equipment

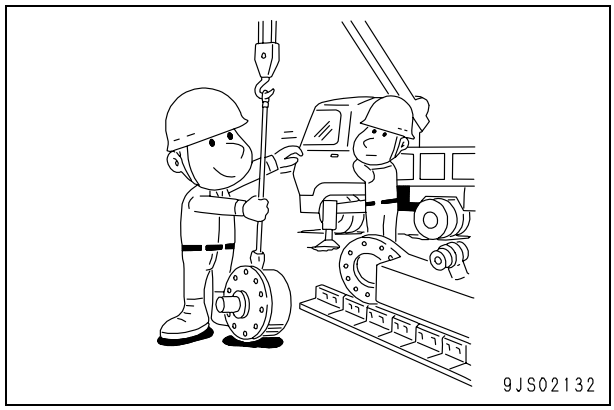
With the increase in pressure and precision of hydraulic equipment, the most common cause of failure is dirt (foreign material) in the hydraulic circuit. When adding hydraulic oil, or when disassembling or assembling hydraulic equipment, it is necessary to be particularly careful.

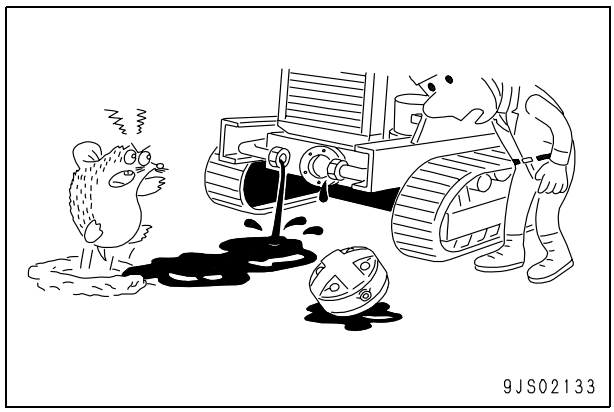
1. **Be careful of the operating environment.**
   Avoid adding hydraulic oil, replacing filters, or repairing the machine in rain or high winds, or places where there is a lot of dust.

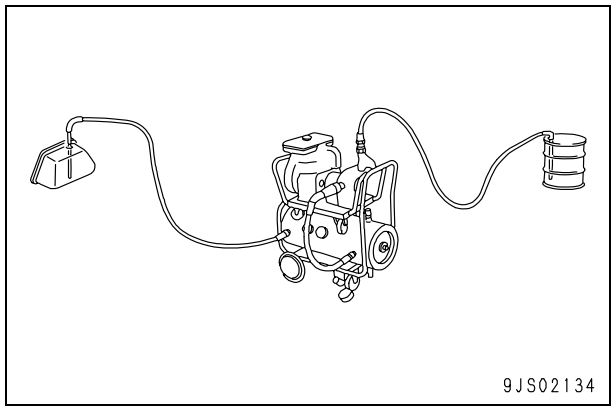
2. **Disassembly and maintenance work in the field**
   If disassembly or maintenance work is carried out on hydraulic equipment in the field, there is danger of dust entering the equipment. It is also difficult to check the performance after repairs, so it is desirable to use unit exchange. Disassembly and maintenance of hydraulic equipment should be carried out in a specially prepared dustproof workshop, and the performance should be checked with special test equipment.

3. **Sealing openings**
   After any piping or equipment is removed, the openings should be sealed with caps, tapes, or vinyl bags to prevent any dirt or dust from entering. If the opening is left open or is blocked with a rag, there is danger of dirt entering or of the surrounding area being made dirty by leaking oil so never do this. Do not simply drain oil out onto the ground, but collect it and ask the customer to dispose of it, or take it back with you for disposal.

4. **Do not let any dirt or dust get in during refilling operations**
   Be careful not to let any dirt or dust get in when refilling with hydraulic oil. Always keep the oil filler and the area around it clean, and also use clean pumps and oil containers. If an oil cleaning device is used, it is possible to filter out the dirt that has collected during storage, so this is an even more effective method.





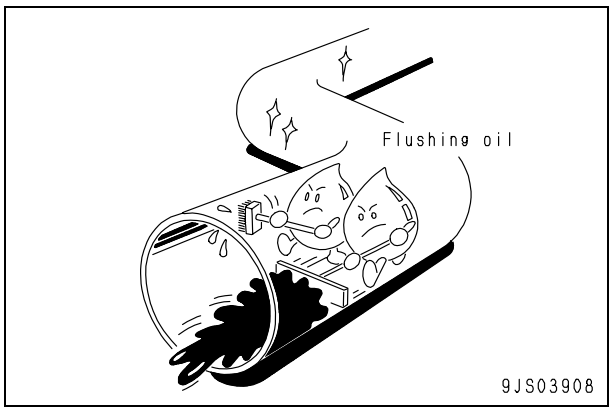


5. Change hydraulic oil when the temperature is high

When hydraulic oil or other oil is warm, it flows easily. In addition, the sludge can also be drained out easily from the circuit together with the oil, so it is best to change the oil when it is still warm. When changing the oil, as much as possible of the old hydraulic oil must be drained out. (Drain the oil from the hydraulic tank; also drain the oil from the filter and from the drain plug in the circuit.) If any old oil is left, the contaminants and sludge in it will mix with the new oil and will shorten the life of the hydraulic oil.

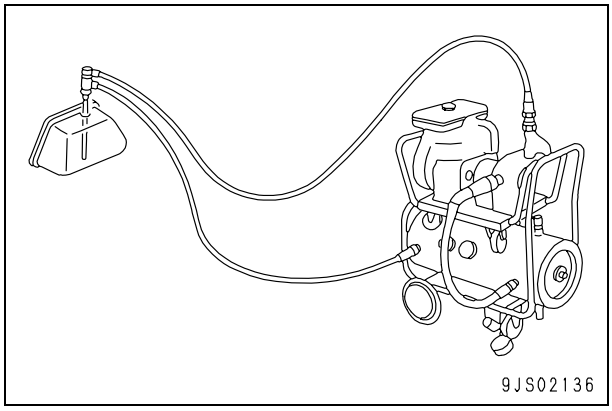
6. Flushing operations

After disassembling and assembling the equipment, or changing the oil, use flushing oil to remove the contaminants, sludge, and old oil from the hydraulic circuit. Normally, flushing is carried out twice: primary flushing is carried out with flushing oil, and secondary flushing is carried out with the specified hydraulic oil.



7. Cleaning operations

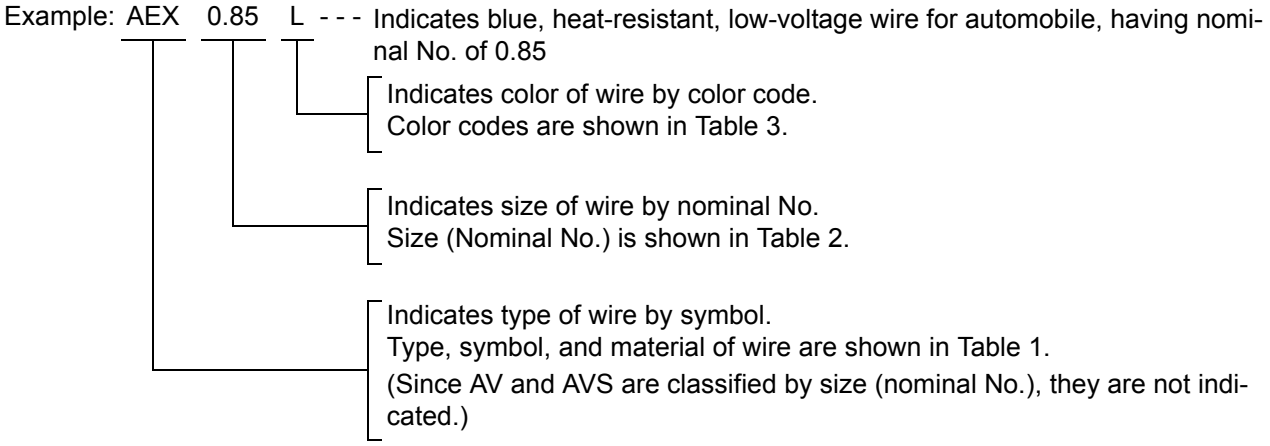
After repairing the hydraulic equipment (pump, control valve, etc.) or when running the machine, carry out oil cleaning to remove the sludge or contaminants in the hydraulic oil circuit. The oil cleaning equipment is used to remove the ultra fine (about 3 μ) particles that the filter built in the hydraulic equipment cannot remove, so it is an extremely effective device.



How to read electric wire code

★ The information about the wires unique to each machine model is described in Troubleshooting section, Relational information of troubleshooting.

In the electric circuit diagram, the material, thickness, and color of each electric wire are indicated by symbols. The electric wire code is helpful in understanding the electric circuit diagram.



1. Type, symbol, and material

AV and AVS are different in only thickness and outside diameter of the cover. AEX is similar to AV in thickness and outside diameter of AEX and different from AV and AVS in material of the cover.

(Table 1)

Type	Symbol	Material		Using temperature range (°C)	Example of use
Low-voltage wire for automobile	AV	Conductor	Annealed copper for electric appliance	–30 to +60	General wiring (Nominal No. 5 and above)
		Insulator	Soft polyvinyl chloride		
Thin-cover low-voltage wire for automobile	AVS	Conductor	Annealed copper for electric appliance		General wiring (Nominal No. 3 and below)
		Insulator	Soft polyvinyl chloride		
Heat-resistant low-voltage wire for automobile	AEX	Conductor	Annealed copper for electric appliance	–50 to +110	General wiring in extremely cold district, wiring at high-temperature place
		Insulator	Heat-resistant crosslinked polyethylene		

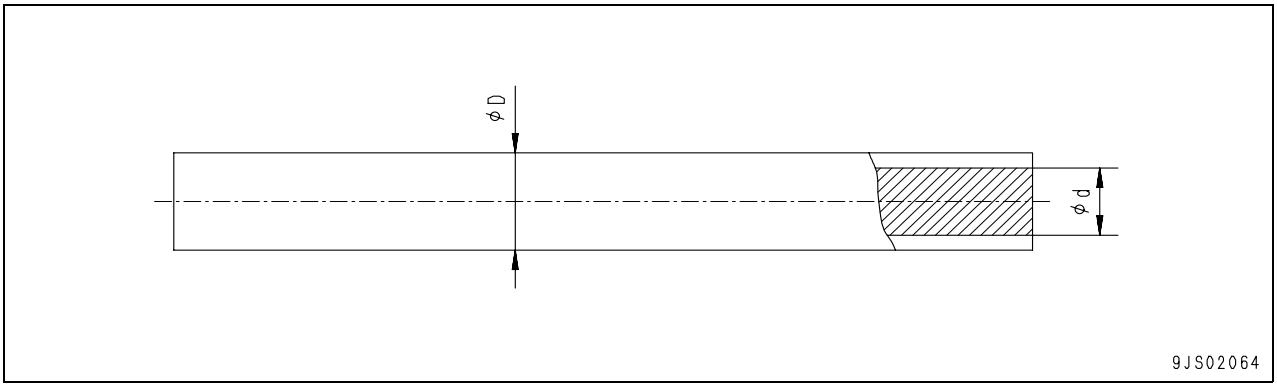
2. Dimensions

(Table 2)

Nominal No.			0.5f	(0.5)	0.75f	(0.85)	1.25f	(1.25)	2f	2	3f	3	5
Conductor		Number of strands/Diameter of strand	20/0.18	7/0.32	30/0.18	11/0.32	50/0.18	16/0.32	37/0.26	26/0.32	58/0.26	41/0.32	65/0.32
		Sectional area (mm^2)	0.51	0.56	0.76	0.88	1.27	1.29	1.96	2.09	3.08	3.30	5.23
		d (approx.)	1.0		1.2		1.5		1.9	1.9	2.3	2.4	3.0
Cover D	AVS	Standard	2.0		2.2		2.5		2.9	2.9	3.5	3.6	–
	AV	Standard	–		–		–		–	–	–	–	4.6
	AEX	Standard	2.0		2.2		2.7		3.0	3.1	–	3.8	4.6

Nominal No.			8	15	20	30	40	50	60	85	100
Conductor		Number of strands/Diameter of strand	50/0.45	84/0.45	41/0.80	70/0.80	85/0.80	108/0.80	127/0.80	169/0.80	217/0.80
		Sectional area (mm^2)	7.95	13.36	20.61	35.19	42.73	54.29	63.84	84.96	109.1
		d (approx.)	3.7	4.8	6.0	8.0	8.6	9.8	10.4	12.0	13.6
Cover D	AVS	Standard	–	–	–	–	–	–	–	–	–
	AV	Standard	5.5	7.0	8.2	10.8	11.4	13.0	13.6	16.0	17.6
	AEX	Standard	5.3	7.0	8.2	10.8	11.4	13.0	13.6	16.0	17.6

"f" of nominal No. denotes flexible".



3. Color codes table

(Table 3)

Color Code	Color of wire
B	Black
Br	Brown
BrB	Brown & Black
BrR	Brown & Red
BrW	Brown & White
BrY	Brown & Yellow
Ch	Charcoal
Dg	Dark green
G	Green
GB	Green & Black
GL	Green & Blue
Gr	Gray
GR	Green & Red
GW	Green & White
GY	Green & Yellow
L	Blue
LB	Blue & Black
Lg	Light green
LgB	Light green & Black
LgR	Light green & Red

Color Code	Color of wire
LgW	Light green & White
LgY	Light green & Yellow
LR	Blue & Red
LW	Blue & White
LY	Blue & Yellow
O	Orange
P	Pink
R	Red
RB	Red & Black
RG	Red & Green
RL	Red & Blue
RW	Red & White
RY	Red & Yellow
Sb	Sky Blue
Y	Yellow
YB	Yellow & Black
YG	Yellow &Green
YL	Yellow & Blue
YR	Yellow & Red
YW	Yellow & White

Remarks: In a color code consisting of 2 colors, the first color is the color of the background and the second color is the color of the marking.
Example: “GW” means that the background is Green and marking is White.

4. Types of circuits and color codes

(Table 4)

Type of wire		AVS or AV						AEX	
Type of circuit	Charge	R	WG	–	–	–	–	R	–
	Ground	B	–	–	–	–	–	B	–
	Start	R	–	–	–	–	–	R	–
	Light	RW	RB	RY	RG	RL	–	D	–
	Instrument	Y	YR	YB	YG	YL	YW	Y	Gr
	Signal	G	GW	GR	GY	GB	GL	G	Br
	Others	L	LW	LR	LY	LB	–	L	–
		Br	BrW	BrR	BrY	BrB	–	–	–
		Lg	LgR	LgY	LgB	LgW	–	–	–
		O	–	–	–	–	–	–	–
		Gr	–	–	–	–	–	–	–
		P	–	–	–	–	–	–	–
		Sb	–	–	–	–	–	–	–
		Dg	–	–	–	–	–	–	–
		Ch	–	–	–	–	–	–	–

Method of disassembling and connecting push-pull type coupler

⚠ Before carrying out the following work, loosen the oil filler cap of the hydraulic tank gradually to release the residual pressure from the hydraulic tank.

⚠ Even if the residual pressure is released from the hydraulic tank, some hydraulic oil flows out when the hose is disconnected. Accordingly, prepare an oil receiving container.

Type 1

1. Disconnection

1) Hold adapter (1) and push hose joint (2) into mating adapter (3). (Fig. 1)
   - ★ The adapter can be pushed in about 3.5 mm.
   - ★ Do not hold rubber cap portion (4).
2) After hose joint (2) is pushed into adapter (3), press rubber cap portion (4) against adapter (3) until it clicks. (Fig. 2)
3) Hold hose adapter (1) or hose (5) and pull it out. (Fig. 3)
   - ★ Since some hydraulic oil flows out, prepare an oil receiving container.

Fig. 1

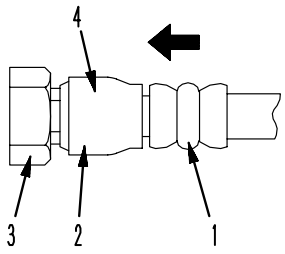


Fig. 2

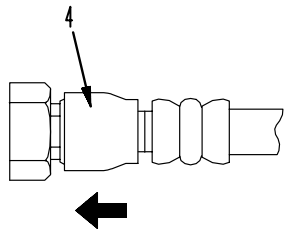
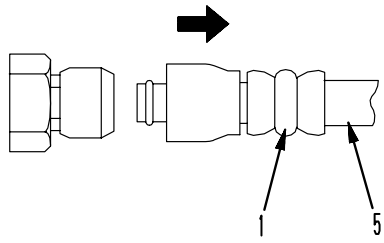


Fig. 3



9JS02109

2. Connection

1) Hold hose adapter (1) or hose (5) and insert it in mating adapter (3), aligning them with each other. (Fig. 4)
   - ★ Do not hold rubber cap portion (4).
2) After inserting the hose in the mating adapter perfectly, pull it back to check its connecting condition. (Fig. 5)
   - ★ When the hose is pulled back, the rubber cap portion moves toward the hose about 3.5 mm. This does not indicate abnormality, however.

Fig. 4

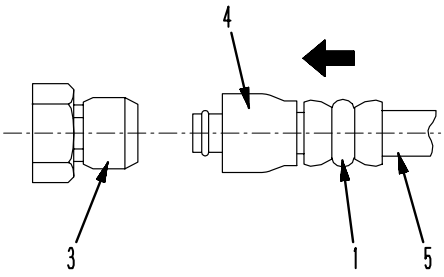
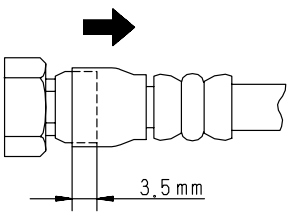


Fig. 5

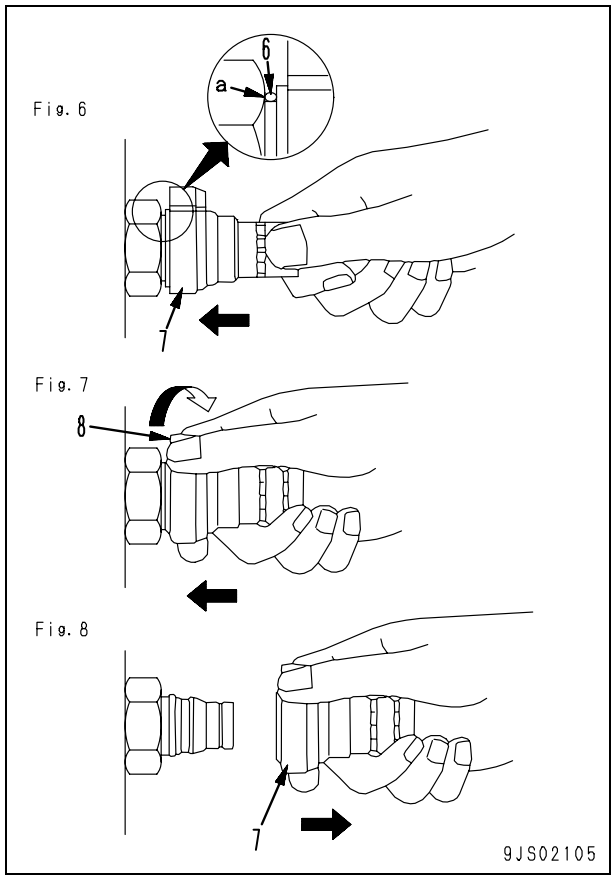


9JS02110

Type 2

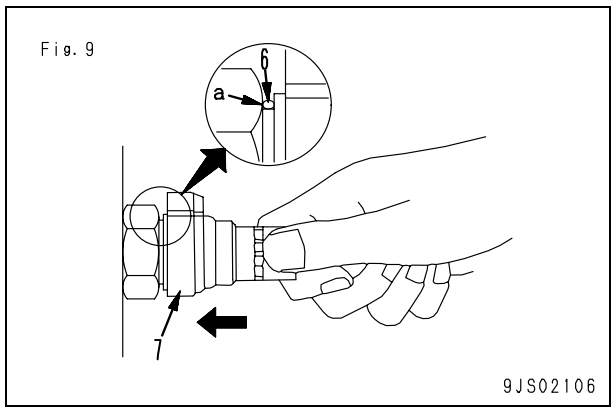
1. Disconnection

1) Hold the tightening portion and push body (7) straight until sliding prevention ring (6) contacts contact surface (a) of the hexagonal portion at the male end. (Fig. 6)
2) While holding the condition of Step 1), turn lever (8) to the right (clockwise). (Fig. 7)
3) While holding the condition of Steps 1) and 2), pull out whole body (7) to disconnect it. (Fig. 8)



2. Connection

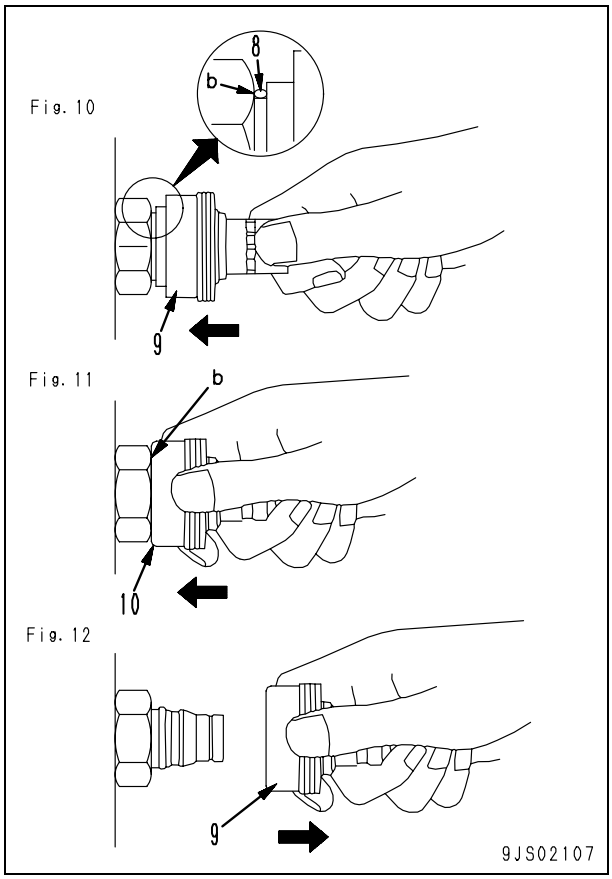
- Hold the tightening portion and push body (7) straight until sliding prevention ring (6) contacts contact surface (a) of the hexagonal portion at the male end. (Fig. 9)



Type 3

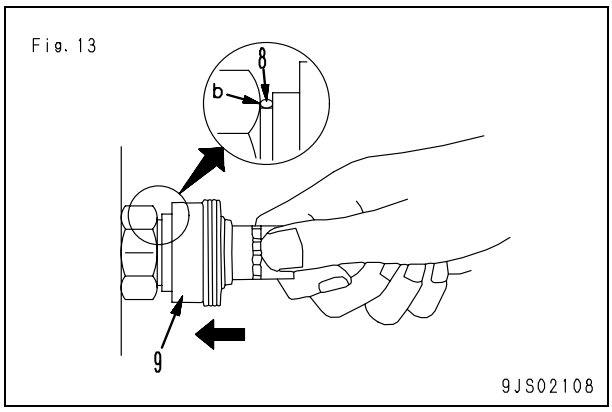
1. Disconnection

1) Hold the tightening portion and push body (9) straight until sliding prevention ring (8) contacts contact surface (b) of the hexagonal portion at the male end. (Fig. 10)
2) While holding the condition of Step 1), push cover (10) straight until it contacts contact surface (b) of the hexagonal portion at the male end. (Fig. 11)
3) While holding the condition of Steps 1) and 2), pull out whole body (9) to disconnect it. (Fig. 12)



2. Connection

- Hold the tightening portion and push body (9) straight until the sliding prevention ring contacts contact surface (b) of the hexagonal portion at the male end. (Fig. 13)



Standard tightening torque table

1. Table of tightening torques for bolts and nuts
   ★ Unless there are special instructions, tighten metric nuts and bolts to the torque below. (When using torque wrench)

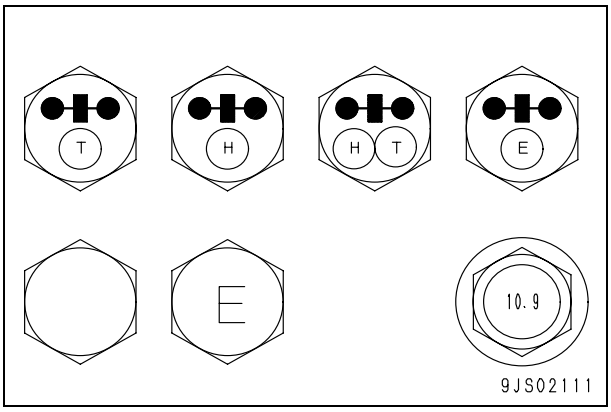
   ★ The following table corresponds to the bolts in Fig. A.

Thread diameter of bolt	Width across flats	Tightening torque	
mm	mm	Nm	kgm
6	10	11.8 – 14.7	1.2 – 1.5
8	13	27 – 34	2.8 – 3.5
10	17	59 – 74	6 – 7.5
12	19	98 – 123	10 – 12.5
14	22	153 – 190	15.5 – 19.5
16	24	235 – 285	23.5 – 29.5
18	27	320 – 400	33 – 41
20	30	455 – 565	46.5 – 58
22	32	610 – 765	62.5 – 78
24	36	785 – 980	80 – 100
27	41	1,150 – 1,440	118 – 147
30	46	1,520 – 1,910	155 – 195
33	50	1,960 – 2,450	200 – 250
36	55	2,450 – 3,040	250 – 310
39	60	2,890 – 3,630	295 – 370

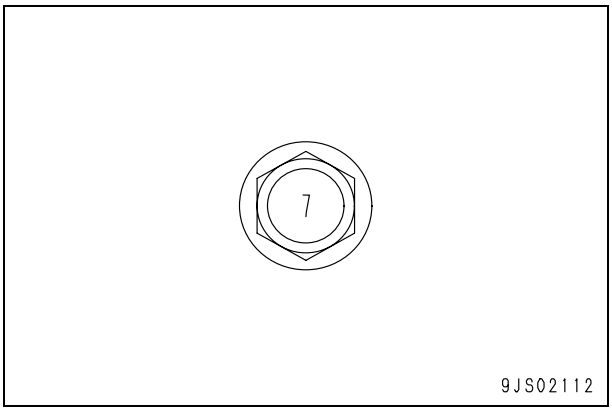
   ★ The following table corresponds to the bolts in Fig. B.

Thread diameter of bolt	Width across flats	Tightening torque	
mm	mm	Nm	kgm
6	10	5.9 – 9.8	0.6 – 1.0
8	13	13.7 – 23.5	1.4 – 2.4
10	14	34.3 – 46.1	3.5 – 4.7
12	27	74.5 – 90.2	7.6 – 9.2

★ Fig. A



★ Fig. B



2. Table of tightening torques for split flange bolts

★ Unless there are special instructions, tighten split flange bolts to the torque below.

Thread diameter of bolt	Width across flats	Tightening torque	
mm	mm	Nm	kgm
10	14	59 – 74	6.0 – 7.5
12	17	98 – 123	10.0 – 12.5
16	22	235 – 285	23.5 – 29.5

3. Table of tightening torques for O-ring boss piping joints

★ Unless there are special instructions, tighten O-ring boss piping joints to the torque below.

Nominal No.	Thread diameter	Width across flats	Tightening torque Nm {kgm}	
	mm	mm	Range	Target
02	14	Varies depending on type of connector.	35 – 63 { 3.5 – 6.5}	44 { 4.5}
03,04	20		84 – 132 { 8.5 – 13.5}	103 {10.5}
05,06	24		128 – 186 {13.0 – 19.0}	157 {16.0}
10,12	33		363 – 480 {37.0 – 49.0}	422 {43.0}
14	42		746 – 1,010 {76.0 – 103}	883 {90.0}

4. Table of tightening torques for O-ring boss plugs

★ Unless there are special instructions, tighten O-ring boss plugs to the torque below.

Nominal No.	Thread diameter	Width across flats	Tightening torque Nm {kgm}	
	mm	mm	Range	Target
08	8	14	5.88 – 8.82 {0.6 – 0.9}	7.35 {0.75}
10	10	17	9.81 – 12.74 {1.0 – 1.3}	11.27 {1.15}
12	12	19	14.7 – 19.6 {1.5 – 2.0}	17.64 {1.8}
14	14	22	19.6 – 24.5 {2.0 – 2.5}	22.54 {2.3}
16	16	24	24.5 – 34.3 {2.5 – 3.5}	29.4 {3.0}
18	18	27	34.3 – 44.1 {3.5 – 4.5}	39.2 {4.0}
20	20	30	44.1 – 53.9 {4.5 – 5.5}	49.0 {5.0}
24	24	32	58.8 – 78.4 {6.0 – 8.0}	68.6 {7.0}
30	30	32	93.1 – 122.5 { 9.5 – 12.5}	107.8 {11.0}
33	33	–	107.8 – 147.0 {11.0 – 15.0}	127.4 {13.0}
36	36	36	127.4 – 176.4 {13.0 – 18.0}	151.9 {15.5}
42	42	–	181.3 – 240.1 {18.5 – 24.5}	210.7 {21.5}
52	52	–	274.4 – 367.5 {28.0 – 37.5}	323.4 {33.0}

5. Table of tightening torques for hoses (taper seal type and face seal type)

★ Unless there are special instructions, tighten the hoses (taper seal type and face seal type) to the torque below.

★ Apply the following torque when the threads are coated (wet) with engine oil.

Nominal No. of hose	Width across flats	Tightening torque Nm {kgm}		Taper seal	Face seal	
		Range	Target	Thread size (mm)	Nominal No. - Number of threads, type of thread	Thread diameter (mm) (Reference)
02	19	34 – 54 { 3.5 – 5.5}	44 { 4.5}	–	9/16-18UN	14.3
		34 – 63 { 3.5 – 6.5}		14	–	–
03	22	54 – 93 { 5.5 – 9.5}	74 { 7.5}	–	11/16-16UN	17.5
	24	59 – 98 { 6.0 – 10.0}	78 { 8.0}	18	–	–
04	27	84 – 132 { 8.5 – 13.5}	103 {10.5}	22	13/16-16UN	20.6
05	32	128 – 186 {13.0 – 19.0}	157 {16.0}	24	1-14UNS	25.4
06	36	177 – 245 {18.0 – 25.0}	216 {22.0}	30	1-3/16-12UN	30.2
(10)	41	177 – 245 {18.0 – 25.0}	216 {22.0}	33	–	–
(12)	46	197 – 294 {20.0 – 30.0}	245 {25.0}	36	–	–
(14)	55	246 – 343 {25.0 – 35.0}	294 {30.0}	42	–	–

6. Table of tightening torques for 102, 107 and 114 engine series (Bolts and nuts)

★ Unless there are special instructions, tighten the metric bolts and nuts of the 102, 107 and 114 engine series to the torque below.

Thread size	Tightening torque			
	Bolt		BANJO bolt	
mm	Nm	kgm	Nm	kgm
6	10 ± 2	1.02 ± 0.20	8 ± 2	0.81 ± 0.20
8	24 ± 4	2.45 ± 0.41	10 ± 2	1.02 ± 0.20
10	43 ± 6	4.38 ± 0.61	12 ± 2	1.22 ± 0.20
12	77 ± 12	7.85 ± 1.22	24 ± 4	2.45 ± 0.41
14	–	–	36 ± 5	3.67 ± 0.51

7. Table of tightening torques for 102, 107 and 114 engine series (Eye joints)

★ Unless there are special instructions, tighten the metric eye joints of the 102, 107 and 114 engine series to the torque below.

Thread size	Tightening torque	
mm	Nm	kgm
6	8 ± 2	0.81 ± 0.20
8	10 ± 2	1.02 ± 0.20
10	12 ± 2	1.22 ± 0.20
12	24 ± 4	2.45 ± 0.41
14	36 ± 5	3.67 ± 0.51

8. Table of tightening torques for 102, 107 and 114 engine series (Taper screws)

★ Unless there are special instructions, tighten the taper screws (unit: inch) of the 102, 107 and 114 engine series to the torque below.

Thread size	Tightening torque	
inch	Nm	kgm
1/16	3 ± 1	0.31 ± 0.10
1/8	8 ± 2	0.81 ± 0.20
1/4	12 ± 2	1.22 ± 0.20
3/8	15 ± 2	1.53 ± 0.20
1/2	24 ± 4	2.45 ± 0.41
3/4	36 ± 5	3.67 ± 0.51
1	60 ± 9	6.12 ± 0.92

Conversion table

Method of using the conversion table

The conversion table in this section is provided to enable simple conversion of figures. For details of the method of using the conversion table, see the example given below.

Example: Method of using the conversion table to convert from millimeters to inches

1. Convert 55 mm into inches.

1) Locate the number 50 in the vertical column at the left side, take this as (A), and then draw a horizontal line from (A).
2) Locate the number 5 in the row across the top, take this as (B), then draw a perpendicular line down from (B).
3) Take the point where the two lines cross as (C). This point (C) gives the value when converting from millimeters to inches. Therefore, 55 mm = 2.165 inches.

2. Convert 550 mm into inches.

1) The number 550 does not appear in the table, so divide it by 10 (move the decimal point one place to the left) to convert it to 55 mm.
2) Carry out the same procedure as above to convert 55 mm to 2.165 inches.
3) The original value (550 mm) was divided by 10, so multiply 2.165 inches by 10 (move the decimal point one place to the right) to return to the original value. This gives 550 mm = 21.65 inches.

Millimeters to inches

(B)

1 mm = 0.03937 in

	0	1	2	3	4	5	6	7	8	9
0	0	0.039	0.079	0.118	0.157	0.197	0.236	0.276	0.315	0.354
10	0.394	0.433	0.472	0.512	0.551	0.591	0.630	0.669	0.709	0.748
20	0.787	0.827	0.866	0.906	0.945	0.984	1.024	1.063	1.102	1.142
30	1.181	1.220	1.260	1.299	1.339	1.378	1.417	1.457	1.496	1.536
40	1.575	1.614	1.654	1.693	1.732	1.772	1.811	1.850	1.890	1.929
						(C)				
50	1.969	2.008	2.047	2.087	2.126	2.165	2.205	2.244	2.283	2.323
60	2.362	2.402	2.441	2.480	2.520	2.559	2.598	2.638	2.677	2.717
70	2.756	2.795	2.835	2.874	2.913	2.953	2.992	3.032	3.071	3.110
80	3.150	3.189	3.228	3.268	3.307	3.346	3.386	3.425	3.465	3.504
90	3.543	3.583	3.622	3.661	3.701	3.740	3.780	3.819	3.858	3.898

(A)

Millimeters to inches

1 mm = 0.03937 in

	0	1	2	3	4	5	6	7	8	9
0	0	0.039	0.079	0.118	0.157	0.197	0.236	0.276	0.315	0.354
10	0.394	0.433	0.472	0.512	0.551	0.591	0.630	0.669	0.709	0.748
20	0.787	0.827	0.866	0.906	0.945	0.984	1.024	1.063	1.102	1.142
30	1.181	1.220	1.260	1.299	1.339	1.378	1.417	1.457	1.496	1.536
40	1.575	1.614	1.654	1.693	1.732	1.772	1.811	1.850	1.890	1.929
50	1.969	2.008	2.047	2.087	2.126	2.165	2.205	2.244	2.283	2.323
60	2.362	2.402	2.441	2.480	2.520	2.559	2.598	2.638	2.677	2.717
70	2.756	2.795	2.835	2.874	2.913	2.953	2.992	3.032	3.071	3.110
80	3.150	3.189	3.228	3.268	3.307	3.346	3.386	3.425	3.465	3.504
90	3.543	3.583	3.622	3.661	3.701	3.740	3.780	3.819	3.858	3.898

Kilogram to pound

1 kg = 2.2046 lb

	0	1	2	3	4	5	6	7	8	9
0	0	2.20	4.41	6.61	8.82	11.02	13.23	15.43	17.64	19.84
10	22.05	24.25	26.46	28.66	30.86	33.07	35.27	37.48	39.68	41.89
20	44.09	46.30	48.50	50.71	51.91	55.12	57.32	59.53	61.73	63.93
30	66.14	68.34	70.55	72.75	74.96	77.16	79.37	81.57	83.78	85.98
40	88.18	90.39	92.59	94.80	97.00	99.21	101.41	103.62	105.82	108.03
50	110.23	112.44	114.64	116.85	119.05	121.25	123.46	125.66	127.87	130.07
60	132.28	134.48	136.69	138.89	141.10	143.30	145.51	147.71	149.91	152.12
70	154.32	156.53	158.73	160.94	163.14	165.35	167.55	169.76	171.96	174.17
80	176.37	178.57	180.78	182.98	185.19	187.39	189.60	191.80	194.01	196.21
90	198.42	200.62	202.83	205.03	207.24	209.44	211.64	213.85	216.05	218.26

Liters to U.S. Gallons

1 ℓ = 0.2642 U.S.Gal

	0	1	2	3	4	5	6	7	8	9
0	0	0.264	0.528	0.793	1.057	1.321	1.585	1.849	2.113	2.378
10	2.642	2.906	3.170	3.434	3.698	3.963	4.227	4.491	4.755	5.019
20	5.283	5.548	5.812	6.076	6.340	6.604	6.869	7.133	7.397	7.661
30	7.925	8.189	8.454	8.718	8.982	9.246	9.510	9.774	10.039	10.303
40	10.567	10.831	11.095	11.359	11.624	11.888	12.152	12.416	12.680	12.944
50	13.209	13.473	13.737	14.001	14.265	14.529	14.795	15.058	15.322	15.586
60	15.850	16.115	16.379	16.643	16.907	17.171	17.435	17.700	17.964	18.228
70	18.492	18.756	19.020	19.285	19.549	19.813	20.077	20.341	20.605	20.870
80	21.134	21.398	21.662	21.926	22.190	22.455	22.719	22.983	23.247	23.511
90	23.775	24.040	24.304	24.568	24.832	25.096	25.361	25.625	25.889	26.153

Liters to U.K. Gallons

1 ℓ = 0.21997 U.K.Gal

	0	1	2	3	4	5	6	7	8	9
0	0	0.220	0.440	0.660	0.880	1.100	1.320	1.540	1.760	1.980
10	2.200	2.420	2.640	2.860	3.080	3.300	3.520	3.740	3.950	4.179
20	4.399	4.619	4.839	5.059	5.279	5.499	5.719	5.939	6.159	6.379
30	6.599	6.819	7.039	7.259	7.479	7.699	7.919	8.139	8.359	8.579
40	8.799	9.019	9.239	9.459	9.679	9.899	10.119	10.339	10.559	10.778
50	10.998	11.281	11.438	11.658	11.878	12.098	12.318	12.528	12.758	12.978
60	13.198	13.418	13.638	13.858	14.078	14.298	14.518	14.738	14.958	15.178
70	15.398	15.618	15.838	16.058	16.278	16.498	16.718	16.938	17.158	17.378
80	17.598	17.818	18.037	18.257	18.477	18.697	18.917	19.137	19.357	19.577
90	19.797	20.017	20.237	20.457	20.677	20.897	21.117	21.337	21.557	21.777

kgm to ft.lb

1 kgm = 7.233 ft.lb

	0	1	2	3	4	5	6	7	8	9
0	0	7.2	14.5	21.7	28.9	36.2	43.4	50.6	57.9	65.1
10	72.3	79.6	86.8	94.0	101.3	108.5	115.7	123.0	130.2	137.4
20	144.7	151.9	159.1	166.4	173.6	180.8	188.1	195.3	202.5	209.8
30	217.0	224.2	231.5	238.7	245.9	253.2	260.4	267.6	274.9	282.1
40	289.3	296.6	303.8	311.0	318.3	325.5	332.7	340.0	347.2	354.4
50	361.7	368.9	376.1	383.4	390.6	397.8	405.1	412.3	419.5	426.8
60	434.0	441.2	448.5	455.7	462.9	470.2	477.4	484.6	491.8	499.1
70	506.3	513.5	520.8	528.0	535.2	542.5	549.7	556.9	564.2	571.4
80	578.6	585.9	593.1	600.3	607.6	614.8	622.0	629.3	636.5	643.7
90	651.0	658.2	665.4	672.7	679.9	687.1	694.4	701.6	708.8	716.1
100	723.3	730.5	737.8	745.0	752.2	759.5	766.7	773.9	781.2	788.4
110	795.6	802.9	810.1	817.3	824.6	831.8	839.0	846.3	853.5	860.7
120	868.0	875.2	882.4	889.7	896.9	904.1	911.4	918.6	925.8	933.1
130	940.3	947.5	954.8	962.0	969.2	976.5	983.7	990.9	998.2	1005.4
140	1012.6	1019.9	1027.1	1034.3	1041.5	1048.8	1056.0	1063.2	1070.5	1077.7
150	1084.9	1092.2	1099.4	1106.6	1113.9	1121.1	1128.3	1135.6	1142.8	1150.0
160	1157.3	1164.5	1171.7	1179.0	1186.2	1193.4	1200.7	1207.9	1215.1	1222.4
170	1129.6	1236.8	1244.1	1251.3	1258.5	1265.8	1273.0	1280.1	1287.5	1294.7
180	1301.9	1309.2	1316.4	1323.6	1330.9	1338.1	1345.3	1352.6	1359.8	1367.0
190	1374.3	1381.5	1388.7	1396.0	1403.2	1410.4	1417.7	1424.9	1432.1	1439.4

kg/cm^2 to lb/in^2

1 kg/cm^2 = 14.2233 lb/in^2

	0	1	2	3	4	5	6	7	8	9
0	0	14.2	28.4	42.7	56.9	71.1	85.3	99.6	113.8	128.0
10	142.2	156.5	170.7	184.9	199.1	213.4	227.6	241.8	256.0	270.2
20	284.5	298.7	312.9	327.1	341.4	355.6	369.8	384.0	398.3	412.5
30	426.7	440.9	455.1	469.4	483.6	497.8	512.0	526.3	540.5	554.7
40	568.9	583.2	597.4	611.6	625.8	640.1	654.3	668.5	682.7	696.9
50	711.2	725.4	739.6	753.8	768.1	782.3	796.5	810.7	825.0	839.2
60	853.4	867.6	881.8	896.1	910.3	924.5	938.7	953.0	967.2	981.4
70	995.6	1,010	1,024	1,038	1,053	1,067	1,081	1,095	1,109	1,124
80	1,138	1,152	1,166	1,181	1,195	1,209	1,223	1,237	1,252	1,266
90	1,280	1,294	1,309	1,323	1,337	1,351	1,365	1,380	1,394	1,408
100	1,422	1,437	1,451	1,465	1,479	1,493	1,508	1,522	1,536	1,550
110	1,565	1,579	1,593	1,607	1,621	1,636	1,650	1,664	1,678	1,693
120	1,707	1,721	1,735	1,749	1,764	1,778	1,792	1,806	1,821	1,835
130	1,849	1,863	1,877	1,892	1,906	1,920	1,934	1,949	1,963	1,977
140	1,991	2,005	2,020	2,034	2,048	2,062	2,077	2,091	2,105	2,119
150	2,134	2,148	2,162	2,176	2,190	2,205	2,219	2,233	2,247	2,262
160	2,276	2,290	2,304	2,318	2,333	2,347	2,361	2,375	2,389	2,404
170	2,418	2,432	2,446	2,460	2,475	2,489	2,503	2,518	2,532	2,546
180	2,560	2,574	2,589	2,603	2,617	2,631	2,646	2,660	2,674	2,688
190	2,702	2,717	2,731	2,745	2,759	2,773	2,788	2,802	2,816	2,830
200	2,845	2,859	2,873	2,887	2,901	2,916	2,930	2,944	2,958	2,973
210	2,987	3,001	3,015	3,030	3,044	3,058	3,072	3,086	3,101	3,115
220	3,129	3,143	3,158	3,172	3,186	3,200	3,214	3,229	3,243	3,257
230	3,271	3,286	3,300	3,314	3,328	3,343	3,357	3,371	3,385	3,399
240	3,414	3,428	3,442	3,456	3,470	3,485	3,499	3,513	3,527	3,542

Temperature

Fahrenheit-Centigrade conversion: A simple way to convert a Fahrenheit temperature reading into a Centigrade temperature reading or vice versa is to enter the accompanying table in the center (boldface column) of figures. These figures refer to the temperature in either Fahrenheit or Centigrade degrees.
If it is desired to convert from Fahrenheit to Centigrade degrees, consider the center column to be a table of Fahrenheit temperatures and read the corresponding Centigrade temperature in the column at the left.
If it is desired to convert from Centigrade to Fahrenheit degrees, consider the center column to be a table of Centigrade values, and read the corresponding Fahrenheit temperature on the right.

1°C = 33.8°F

°C		°F	°C		°F	°C		°F	°C		°F
–40.4	**–40**	–40.0	–11.7	**11**	51.8	7.8	**46**	114.8	27.2	**81**	177.8
–37.2	**–35**	–31.0	–11.1	**12**	53.6	8.3	**47**	116.6	27.8	**82**	179.6
–34.4	**–30**	–22.0	–10.6	**13**	55.4	8.9	**48**	118.4	28.3	**83**	181.4
–31.7	**–25**	–13.0	–10.0	**14**	57.2	9.4	**49**	120.2	28.9	**84**	183.2
–28.9	**–20**	–4.0	–9.4	**15**	59.0	10.0	**50**	122.0	29.4	**85**	185.0
–28.3	**–19**	–2.2	–8.9	**16**	60.8	10.6	**51**	123.8	30.0	**86**	186.8
–27.8	**–18**	–0.4	–8.3	**17**	62.6	11.1	**52**	125.6	30.6	**87**	188.6
–27.2	**–17**	1.4	–7.8	**18**	64.4	11.7	**53**	127.4	31.1	**88**	190.4
–26.7	**–16**	3.2	–7.2	**19**	66.2	12.2	**54**	129.2	31.7	**89**	192.2
–26.1	**–15**	5.0	–6.7	**20**	68.0	12.8	**55**	131.0	32.2	**90**	194.0
–25.6	**–14**	6.8	–6.1	**21**	69.8	13.3	**56**	132.8	32.8	**91**	195.8
–25.0	**–13**	8.6	–5.6	**22**	71.6	13.9	**57**	134.6	33.3	**92**	197.6
–24.4	**–12**	10.4	–5.0	**23**	73.4	14.4	**58**	136.4	33.9	**93**	199.4
–23.9	**–11**	12.2	–4.4	**24**	75.2	15.0	**59**	138.2	34.4	**94**	201.2
–23.3	**–10**	14.0	–3.9	**25**	77.0	15.6	**60**	140.0	35.0	**95**	203.0
–22.8	**–9**	15.8	–3.3	**26**	78.8	16.1	**61**	141.8	35.6	**96**	204.8
–22.2	**–8**	17.6	–2.8	**27**	80.6	16.7	**62**	143.6	36.1	**97**	206.6
–21.7	**–7**	19.4	–2.2	**28**	82.4	17.2	**63**	145.4	36.7	**98**	208.4
–21.1	**–6**	21.2	–1.7	**29**	84.2	17.8	**64**	147.2	37.2	**99**	210.2
–20.6	**–5**	23.0	–1.1	**30**	86.0	18.3	**65**	149.0	37.8	**100**	212.0
–20.0	**–4**	24.8	–0.6	**31**	87.8	18.9	**66**	150.8	40.6	**105**	221.0
–19.4	**–3**	26.6	0	**32**	89.6	19.4	**67**	152.6	43.3	**110**	230.0
–18.9	**–2**	28.4	0.6	**33**	91.4	20.0	**68**	154.4	46.1	**115**	239.0
–18.3	**–1**	30.2	1.1	**34**	93.2	20.6	**69**	156.2	48.9	**120**	248.0
–17.8	**0**	32.0	1.7	**35**	95.0	21.1	**70**	158.0	51.7	**125**	257.0
–17.2	**1**	33.8	2.2	**36**	96.8	21.7	**71**	159.8	54.4	**130**	266.0
–16.7	**2**	35.6	2.8	**37**	98.6	22.2	**72**	161.6	57.2	**135**	275.0
–16.1	**3**	37.4	3.3	**38**	100.4	22.8	**73**	163.4	60.0	**140**	284.0
–15.6	**4**	39.2	3.9	**39**	102.2	23.3	**74**	165.2	62.7	**145**	293.0
–15.0	**5**	41.0	4.4	**40**	104.0	23.9	**75**	167.0	65.6	**150**	302.0
–14.4	**6**	42.8	5.0	**41**	105.8	24.4	**76**	168.8	68.3	**155**	311.0
–13.9	**7**	44.6	5.6	**42**	107.6	25.0	**77**	170.6	71.1	**160**	320.0
–13.3	**8**	46.4	6.1	**43**	109.4	25.6	**78**	172.4	73.9	**165**	329.0
–12.8	**9**	48.2	6.7	**44**	111.2	26.1	**79**	174.2	76.7	**170**	338.0
–12.2	**10**	50.0	7.2	**45**	113.0	26.7	**80**	176.0	79.4	**175**	347.0

Engine

107E-1 Series

01 Specification

Specification and technical data

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Specification and technical data

General

1. Applicable machine

Engine	Serial No.	Applicable machine	
SAA6D107E-1		PC200-8, PC200LC-8 PC220-8, PC220LC-8 PC270-8	Hydraulic excavator Hydraulic excavator Hydraulic excavator
		WA380-6	Wheel loader

Specifications

Engine			SAA6D107E-1
Applicable model			PC200-8, PC200LC-8
No. of cylinders – Bore × Stroke		mm	6 – 107 × 124
Total piston displacement		ℓ {cc}	6.69 {6,690}
Firing order		—	1–5–3–6–2–4
Dimensions	Overall length	mm	1,355
	Overall width	mm	790
	Overall height (excluding exhaust pipe)	mm	—
	Overall height (including exhaust pipe)	mm	1,409
Performance	Flywheel horsepower	kW{HP}/rpm	116 {155}/2,000 (Gross)
	Max. torque	Nm{kgm}/rpm	624 {63.6}/1,500 (Gross)
	High idle speed	rpm	2,060 ± 50
	Low idle speed	rpm	1,050 ± 25
	Min. fuel consumption ratio	g/kW·h {g/HP·h}	223 {166}
Dry weight		kg	553
Fuel injection system		—	HPCR
Fuel injection system control			Electronic control type
Lubricating oil amount (refill capacity)		ℓ	25.4 (23.1)
Coolant amount		ℓ	8.4
Alternator		—	24V, 35A
Starting motor		—	24V, 4.5kW
Battery		—	12V, 110Ah × 2
Turbocharger		—	HOLSET HX35W
Air compressor		—	—
Others		—	—

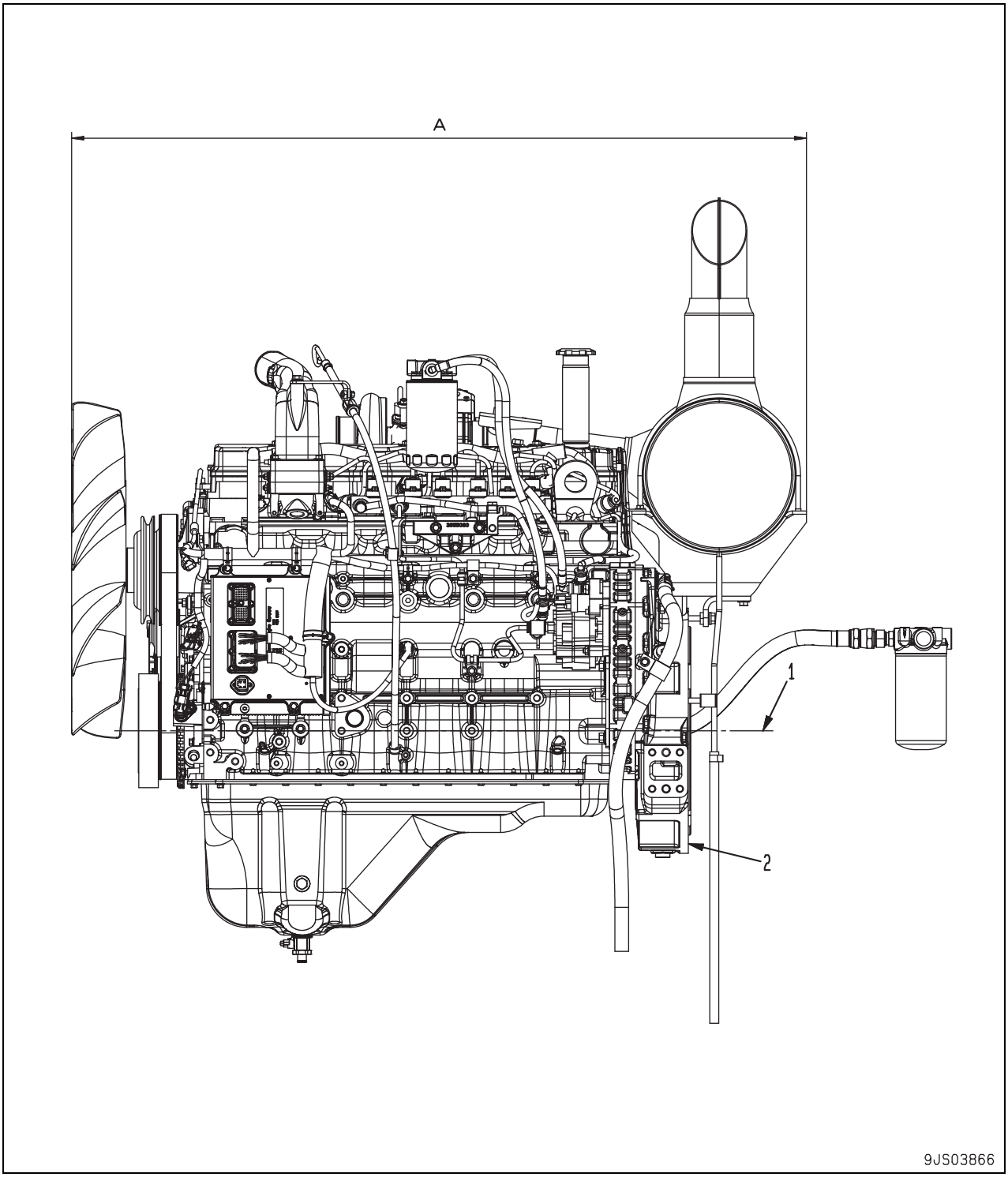
Engine			SAA6D107E-1
Applicable model			PC220-8, PC220LC-8
No. of cylinders – Bore × Stroke		mm	6 – 107 × 124
Total piston displacement		ℓ {cc}	6.69 {6,690}
Firing order		—	1–5–3–6–2–4
Dimensions	Overall length	mm	1,355
	Overall width	mm	790
	Overall height (excluding exhaust pipe)	mm	—
	Overall height (including exhaust pipe)	mm	1,409
Performance	Flywheel horsepower	kW{HP}/rpm	134 {180}/2,000 (Gross)
	Max. torque	Nm{kgm}/rpm	710 {72.4}/1,500 (Gross)
	High idle speed	rpm	2,160 ± 50
	Low idle speed	rpm	1,050 ± 25
	Min. fuel consumption ratio	g/kW·h {g/HP·h}	223 {166}
Dry weight		kg	553
Fuel injection system		—	HPCR
Fuel injection system control			Electronic control type
Lubricating oil amount (refill capacity)		ℓ	25.4 (23.1)
Coolant amount		ℓ	8.4
Alternator		—	24V, 35A
Starting motor		—	24V, 4.5kW
Battery		—	12V, 110Ah × 2
Turbocharger		—	HOLSET HX35W
Air compressor		—	—
Others		—	—

Engine			SAA6D107E-1
Applicable model			PC270-8
No. of cylinders – Bore × Stroke		mm	6 – 107 × 124
Total piston displacement		ℓ {cc}	6.69 {6,690}
Firing order		—	1–5–3–6–2–4
Dimensions	Overall length	mm	1,355
	Overall width	mm	790
	Overall height (excluding exhaust pipe)	mm	—
	Overall height (including exhaust pipe)	mm	1,409
Performance	Flywheel horsepower	kW{HP}/rpm	149 {200}/2,050 (Gross)
	Max. torque	Nm{kgm}/rpm	771 {78.6}/1,500 (Gross)
	High idle speed	rpm	2,180 ± 50
	Low idle speed	rpm	1,050 ± 25
	Min. fuel consumption ratio	g/kW·h {g/HP·h}	223 {166}
Dry weight		kg	553
Fuel injection system		—	HPCR
Fuel injection system control			Electronic control type
Lubricating oil amount (refill capacity)		ℓ	25.4 (23.1)
Coolant amount		ℓ	8.4
Alternator		—	24V, 35A
Starting motor		—	24V, 4.5kW
Battery		—	12V, 110Ah × 2
Turbocharger		—	HOLSET HX35W
Air compressor		—	—
Others		—	—

Engine			SAA6D107E-1
Applicable model			WA380-6
No. of cylinders – Bore × Stroke		mm	6 – 107 × 124
Total piston displacement		ℓ {cc}	6.69 {6,690}
Firing order		—	1–5–3–6–2–4
Dimensions	Overall length	mm	1,101
	Overall width	mm	882
	Overall height (excluding exhaust pipe)	mm	1,592
	Overall height (including exhaust pipe)	mm	—
Performance	Flywheel horsepower	kW{HP}/rpm	143.4 {192}/2,100 (Gross)
	Max. torque	Nm{kgm}/rpm	941 {96}/1,450 (Gross)
	High idle speed	rpm	2,250 ± 50
	Low idle speed	rpm	850 ± 25
	Min. fuel consumption ratio	g/kW·h {g/HP·h}	234 {174}
Dry weight		kg	588
Fuel injection system		—	HPCR
Fuel injection system control			Electronic control type
Lubricating oil amount (refill capacity)		ℓ	25.4 (23.1)
Coolant amount		ℓ	8.4
Alternator		—	24V, 60A
Starting motor		—	24V, 5.5kW
Battery		—	12V, 136Ah × 2
Turbocharger		—	HOLSET HX35W
Air compressor		—	—
Others		—	—

General view

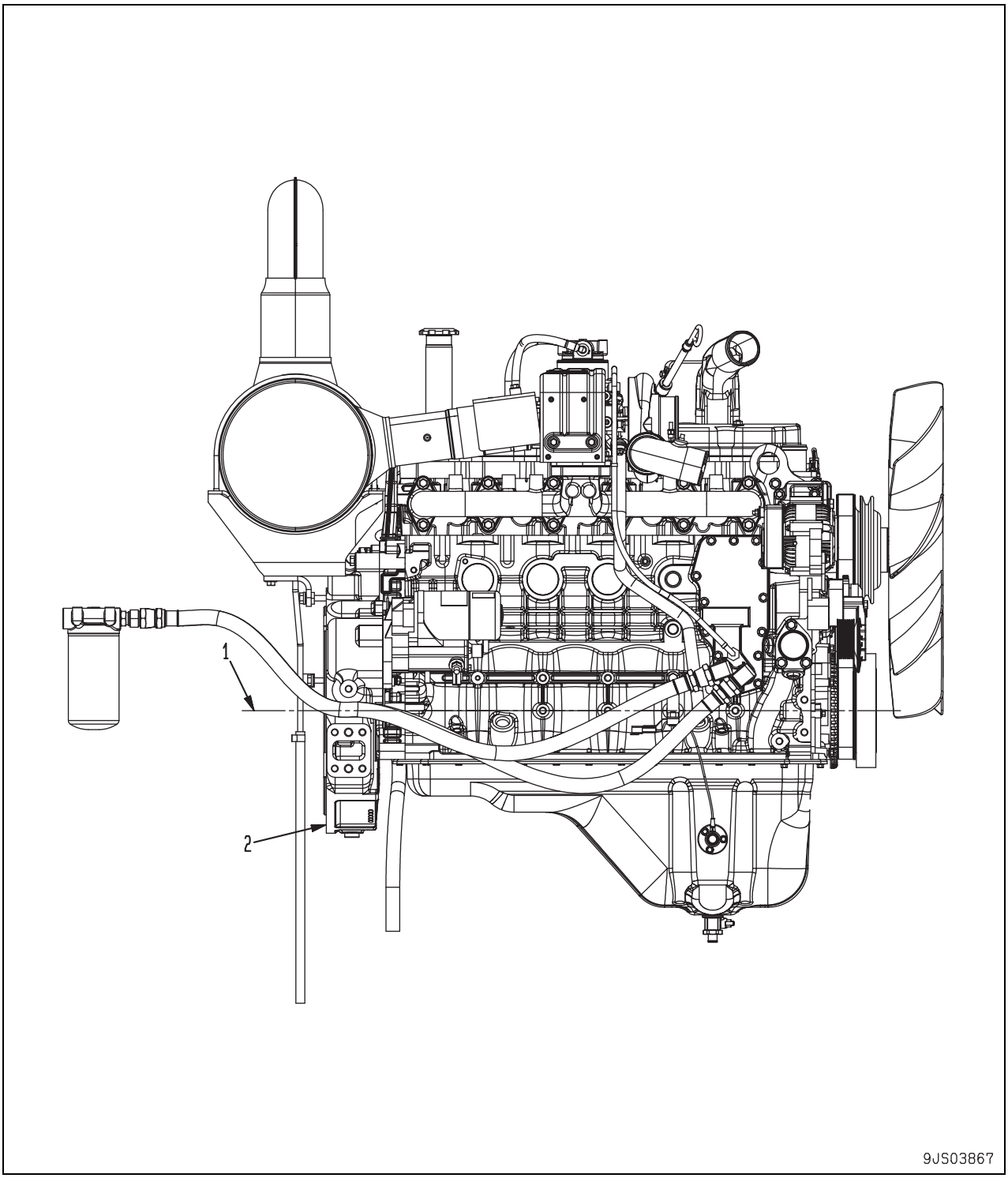
SAA6D107E-1 (Left side view of engine)
Machine model: PC200-8, PC200LC-8



★ The shape may differ according to the machine model.

1. Crankshaft center
2. Flywheel housing rear surface

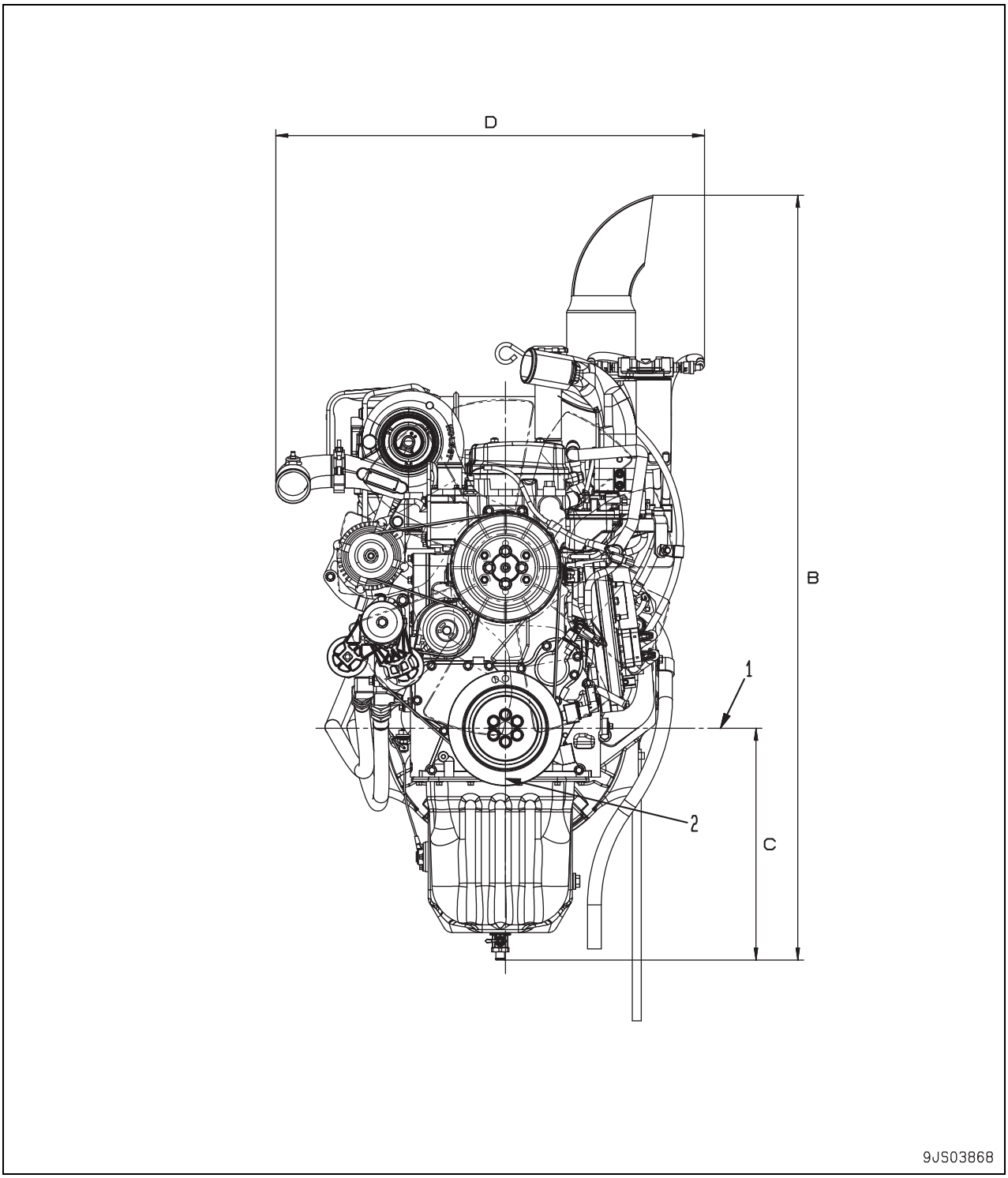
SAA6D107E-1 (Right side view of engine)
Machine model: PC200-8, PC200LC-8



★ The shape may differ according to the machine model.

1. Crankshaft center
2. Flywheel housing rear surface

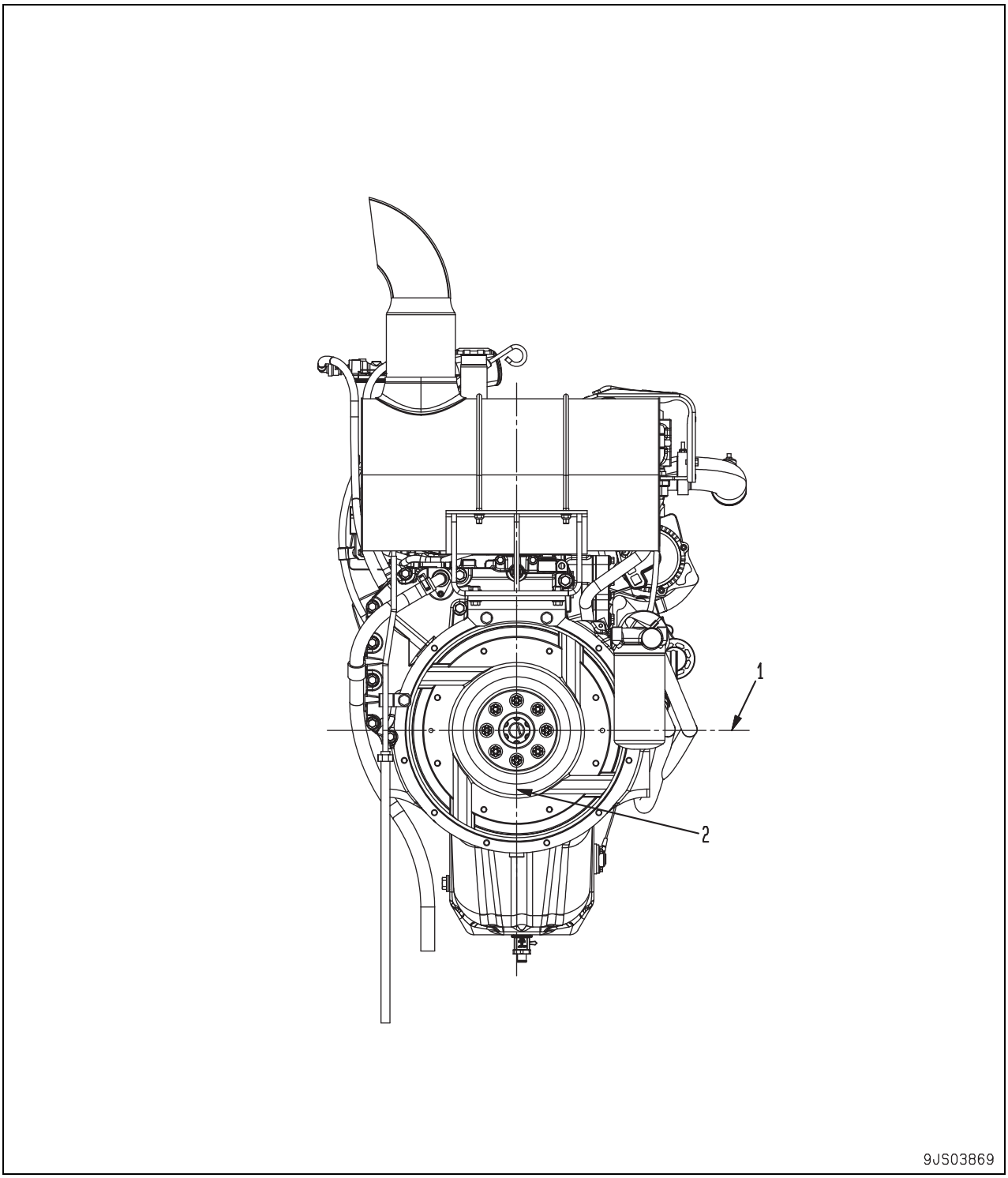
SAA6D107E-1 (Front view of engine)
Machine model: PC200-8, PC200LC-8



★ The shape may differ according to the machine model.

1. Crankshaft center
2. Cylinder center

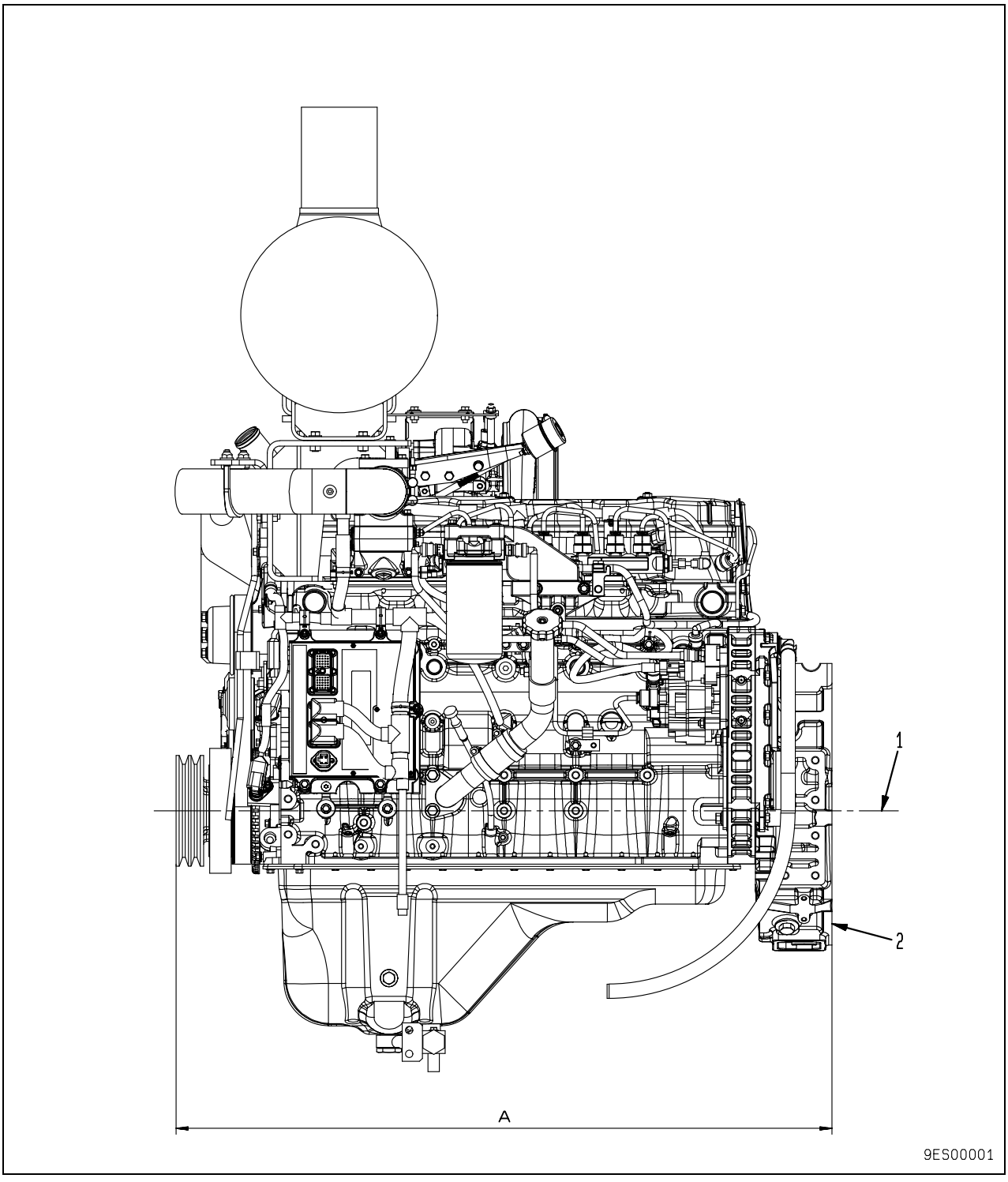
SAA6D107E-1 (Rear view of engine)
Machine model: PC200-8, PC200LC-8



★ The shape may differ according to the machine model.

1. Crankshaft center
2. Cylinder center

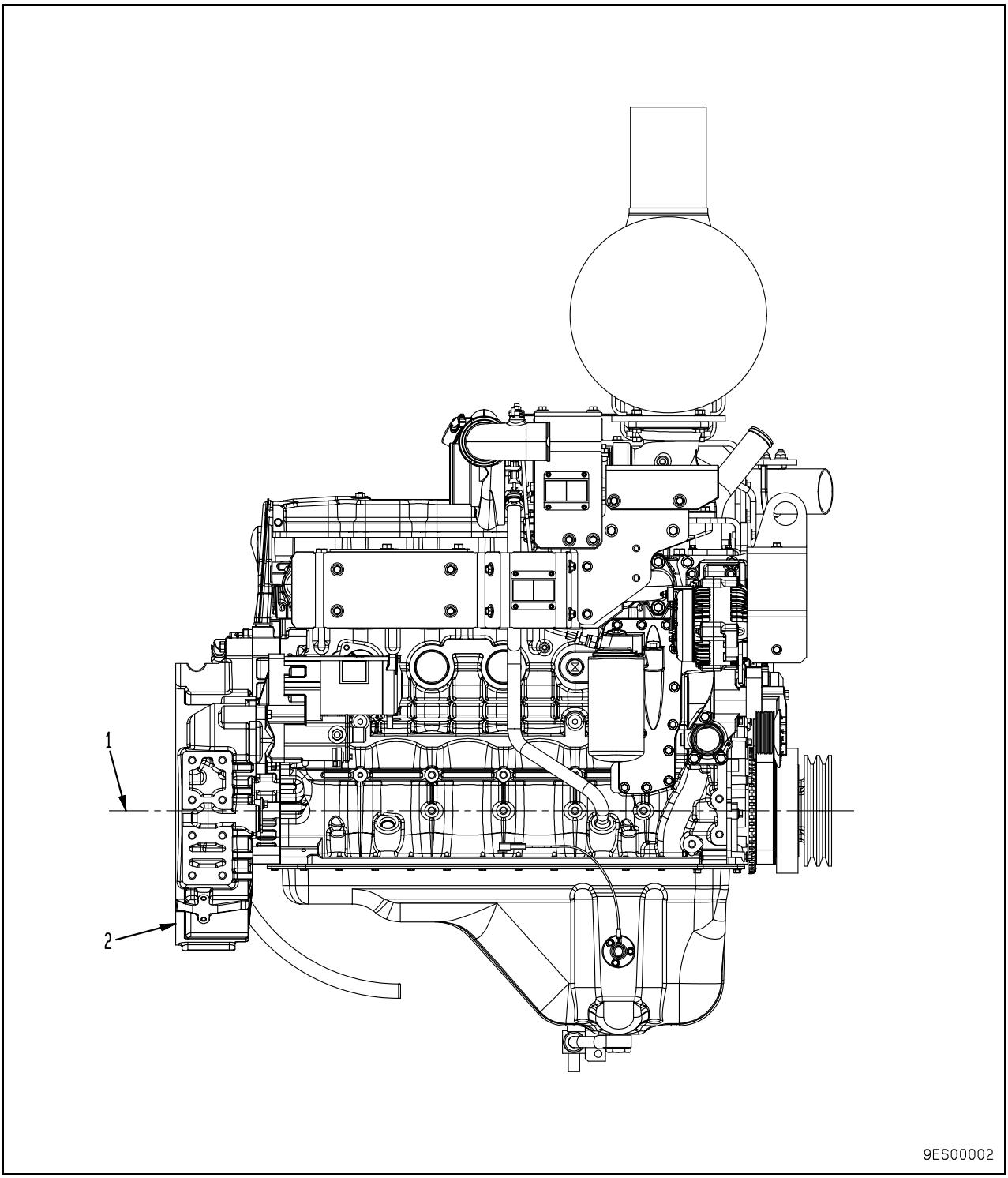
SAA6D107E-1 (Left side view of engine)
Machine model: WA380-6



★ The shape may differ according to the machine model.

1. Crankshaft center
2. Flywheel housing rear surface

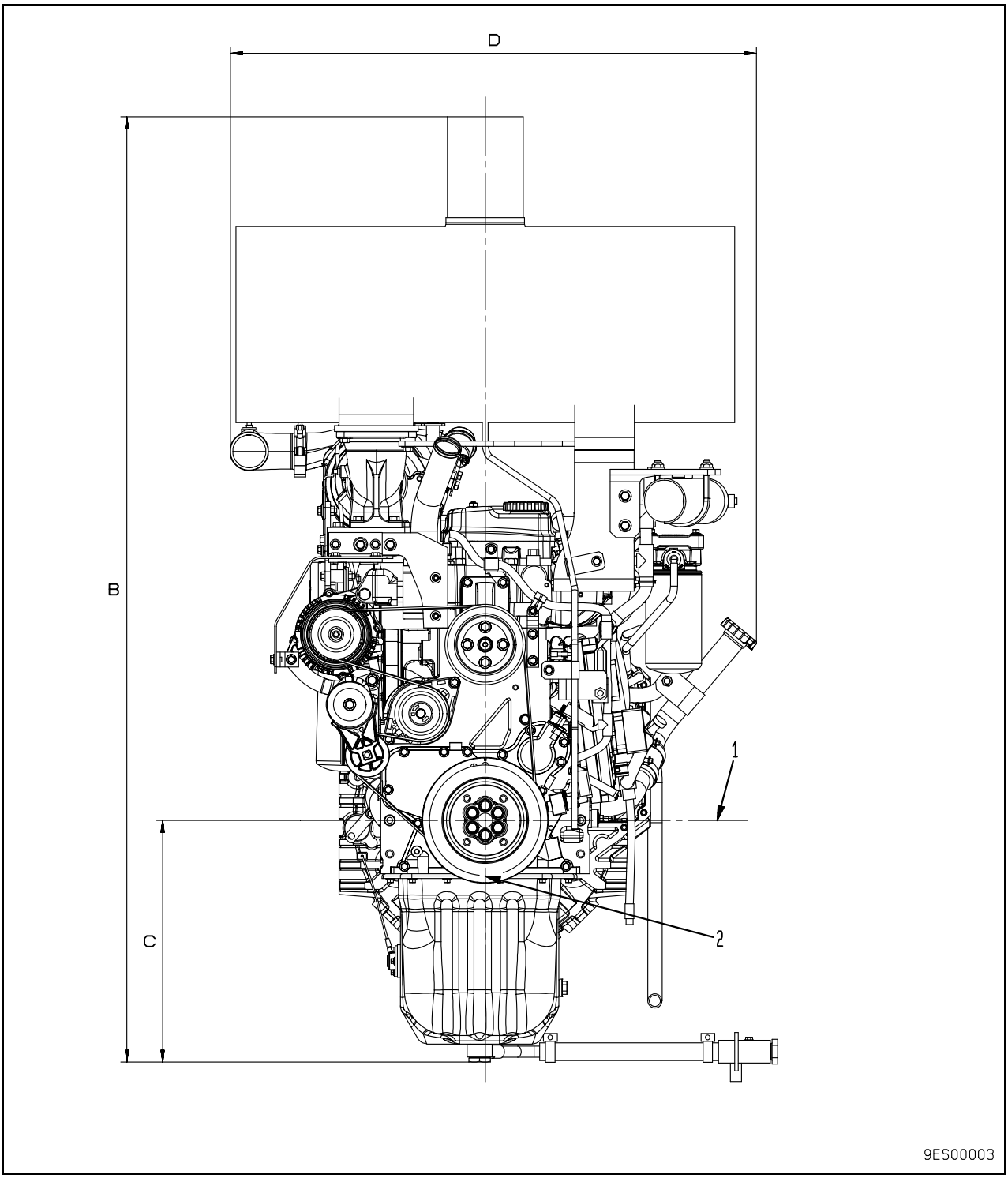
SAA6D107E-1 (Right side view of engine)
Machine model: WA380-6



★ The shape may differ according to the machine model.

1. Crankshaft center
2. Flywheel housing rear surface

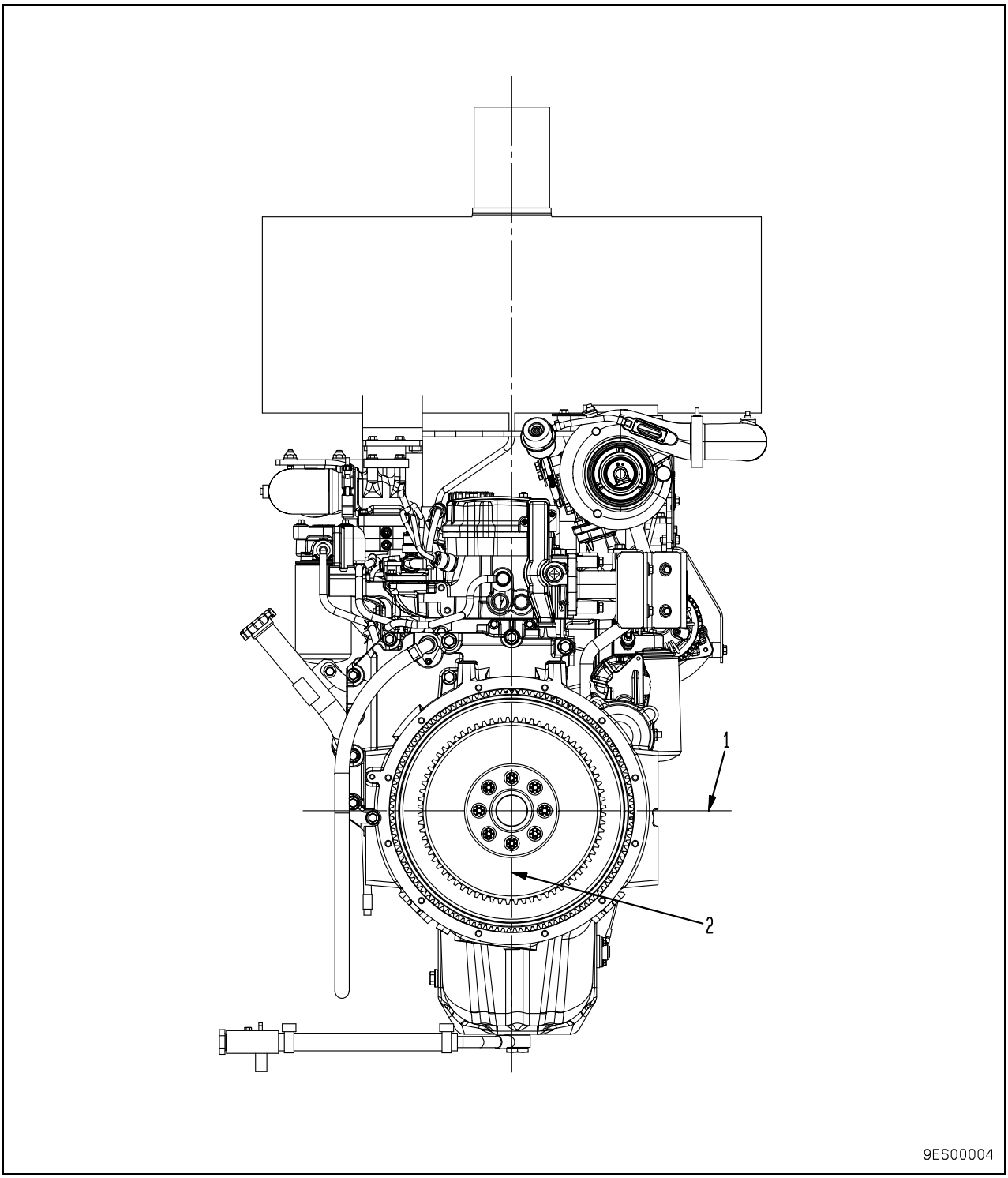
SAA6D107E-1 (Front view of engine)
Machine model: WA380-6



★ The shape may differ according to the machine model.

1. Crankshaft center
2. Cylinder center

SAA6D107E-1 (Rear view of engine)
Machine model: WA380-6



★ The shape may differ according to the machine model.

1. Crankshaft center
2. Cylinder center

Dimensions table

Unit: mm

Engine	Machine model	Dimension of each part							
		A	B	C	D				
SAA6D107E-1	PC200-8	1,355	1,409	427	790				
	PC220-8	1,355	1,409	427	790				
	PC270-8	1,355	1,409	427	790				
	WA380-6	1,101	1,592	407	882				

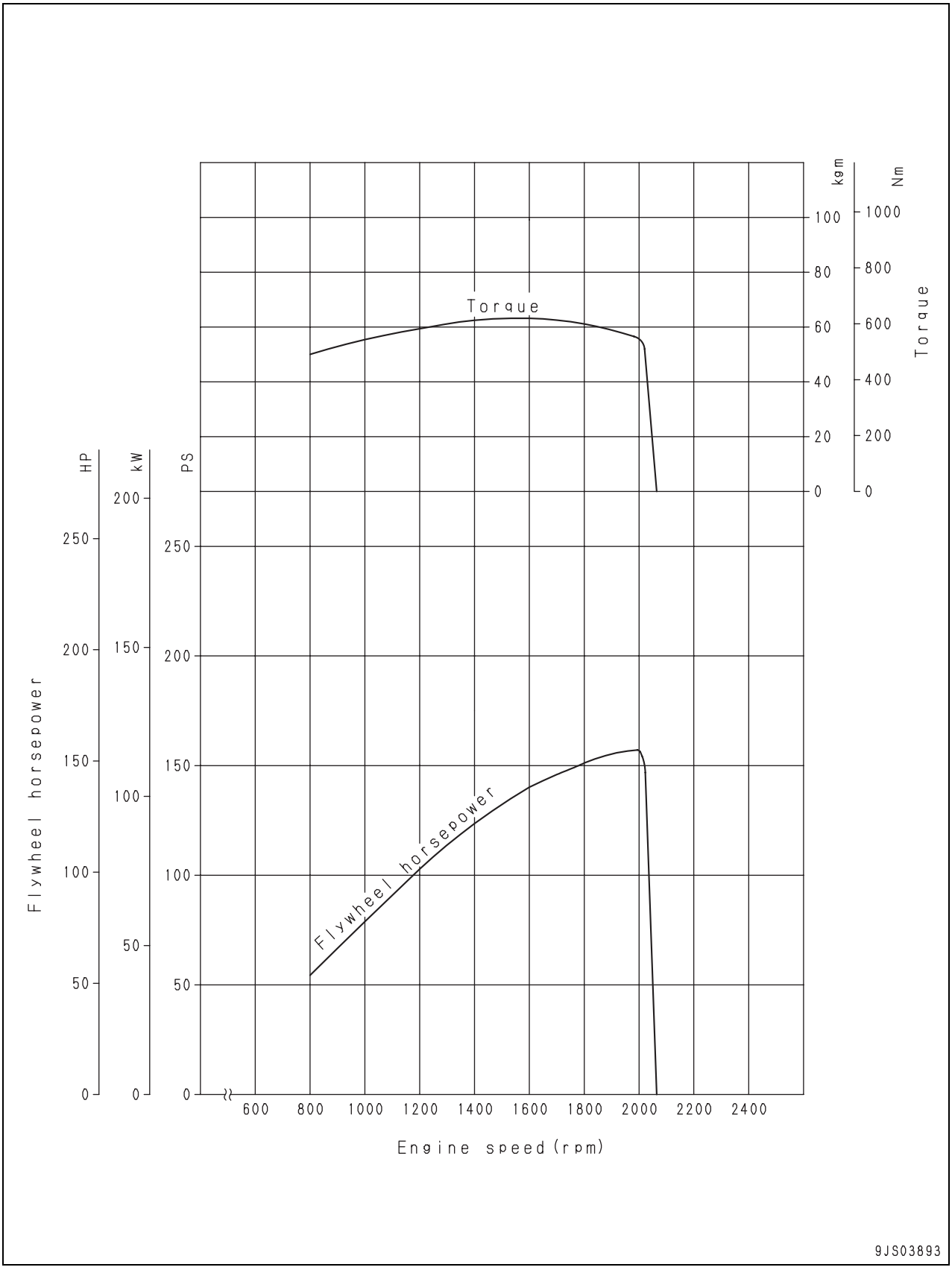
★ These dimensions are given for reference when the engine is set on a test bench.

Engine performance curves

Engine	Engine serial No.	Machine model	Page
SAA6D107E-1		PC200-8, PC200LC-8	17
		PC220-8, PC220LC-8	18
		PC270-8	19
		WA380-6	20

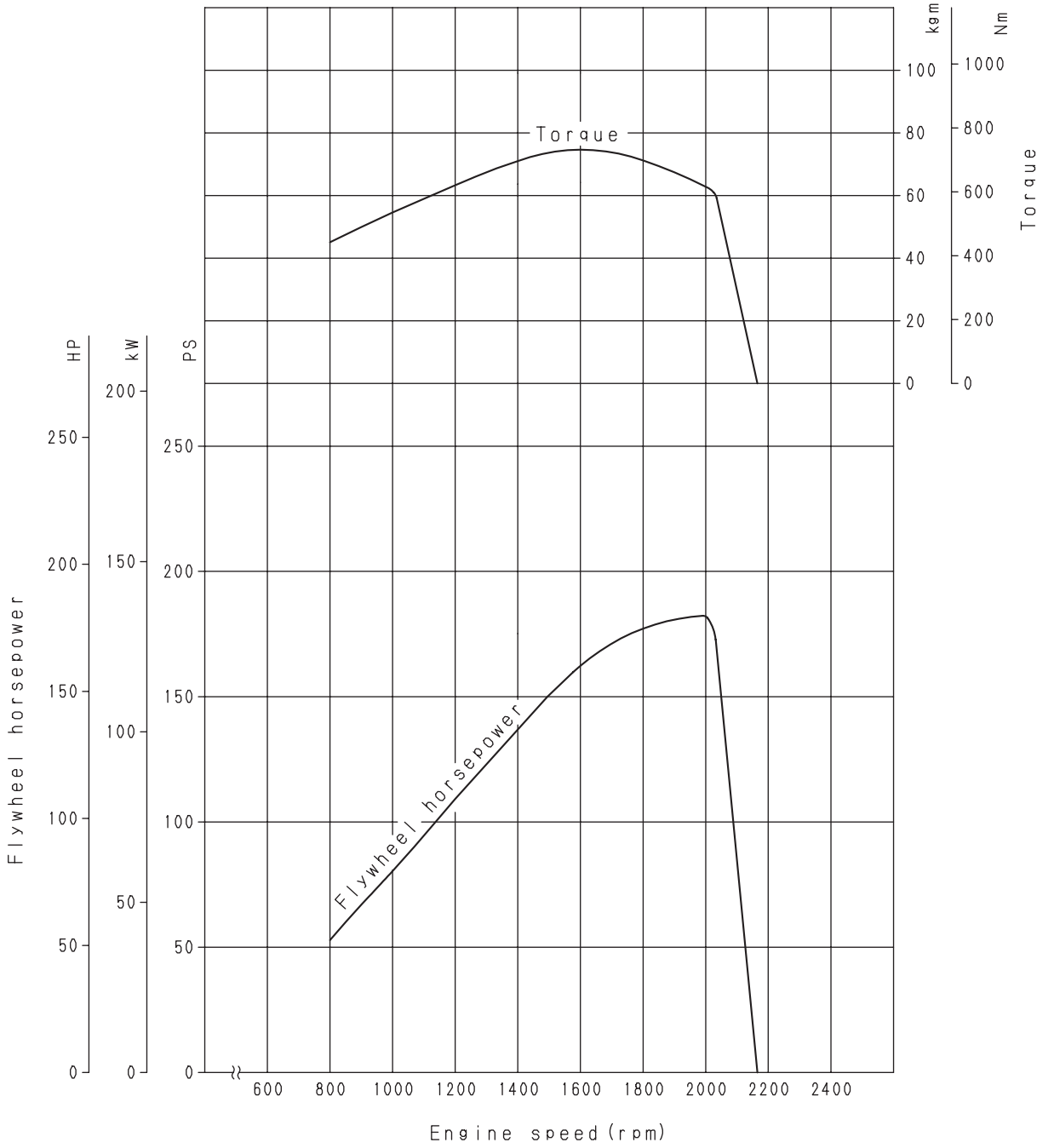
SAA6D107E-1 (Machine model: PC200-8, PC200LC-8)

Rated output: 116 kW {155 HP}/2,000 rpm (Gross)
Max. torque: 624 Nm {63.6 kgm}/1,500 rpm (Gross)



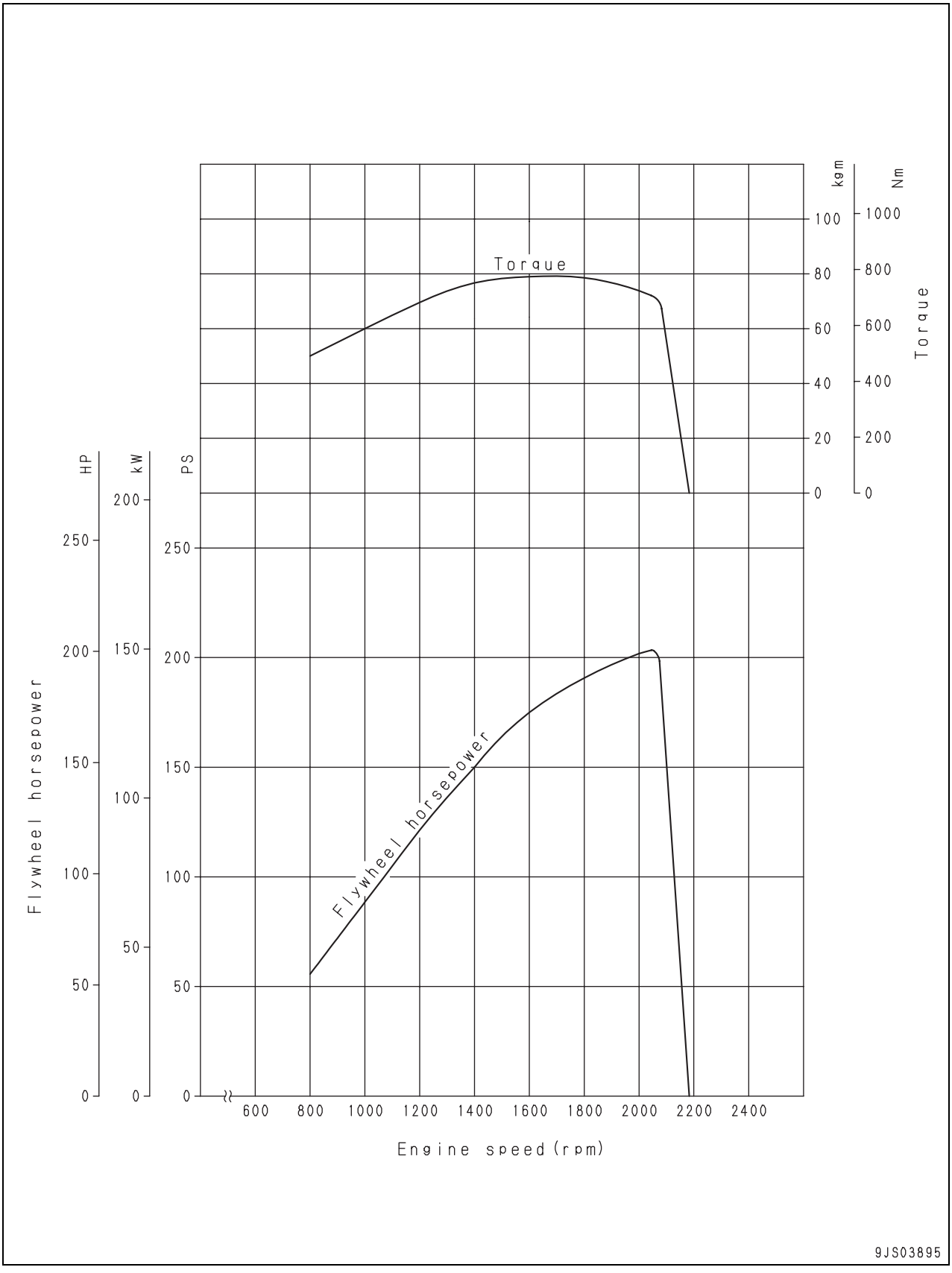
SAA6D107E-1 (Machine model: PC220-8, PC220LC-8)

Rated output: 134 kW {180 HP}/2,000 rpm (Gross)
Max. torque: 710 Nm {72.4 kgm}/1,500 rpm (Gross)



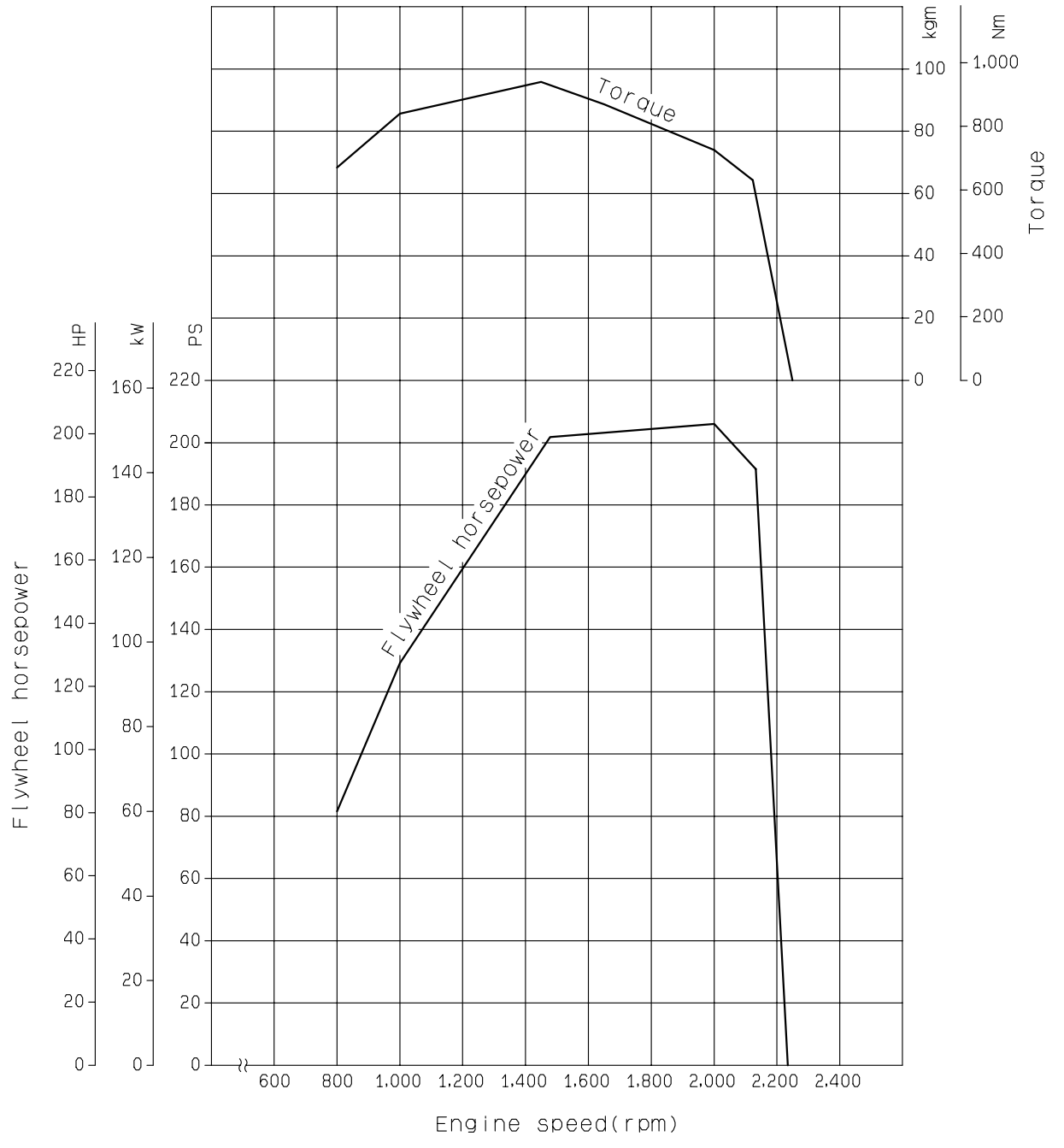
SAA6D107E-1 (Machine model: PC270-8)

Rated output: 149 kW {200 HP}/2,050 rpm (Gross)
Max. torque: 771 Nm {78.6 kgm}/1,500 rpm (Gross)



SAA6D107E-1 (Machine model: WA380-6)

Rated output: 143.4 kW {192 HP}/2,100 rpm (Gross)
Max. torque: 941 Nm {96 kgm}/1,450 rpm (Gross)



Engine

107E-1 Series

10 Structure, function and maintenance standard

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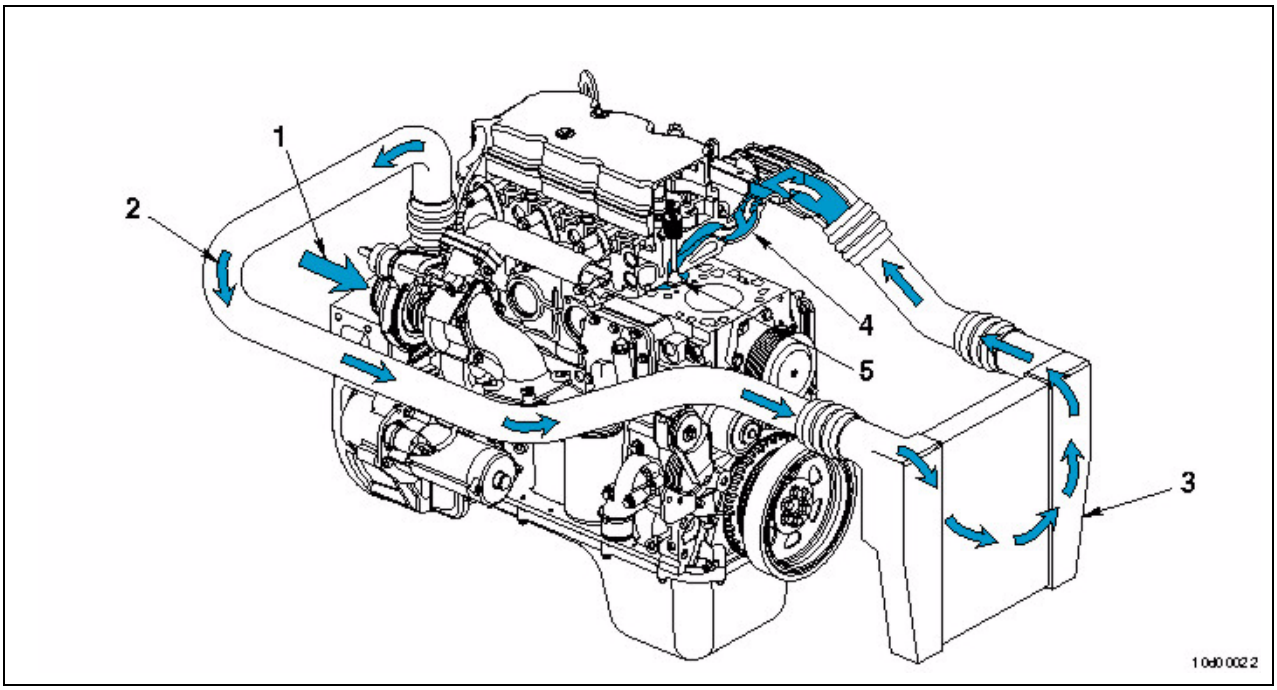
Intake, exhaust system

Intake system

General Information

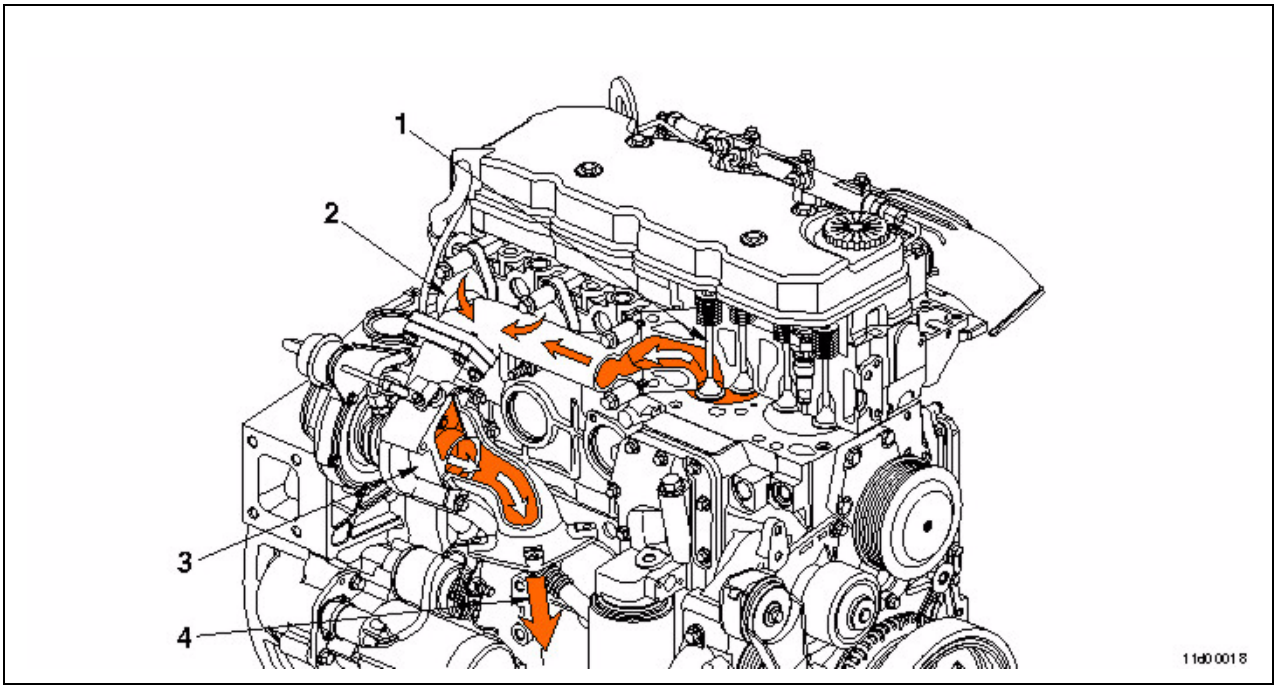


1. Intake air inlet to turbocharger
2. Turbocharger air to charge air cooler
3. Charge air cooler
4. Intake manifold
   (integral part of the cylinder head)
5. Intake valve.

Air is drawn through the air cleaner into the compressor side of the turbocharger. It is then forced through the charge air cooler piping to the charge air cooler, the intake air heater (if applicable), and into the intake manifold. From the intake manifold, air is forced into the cylinders and used for combustion.

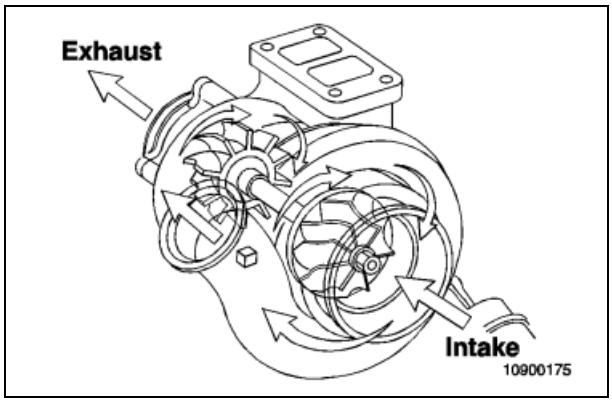
Exhaust system

General Information

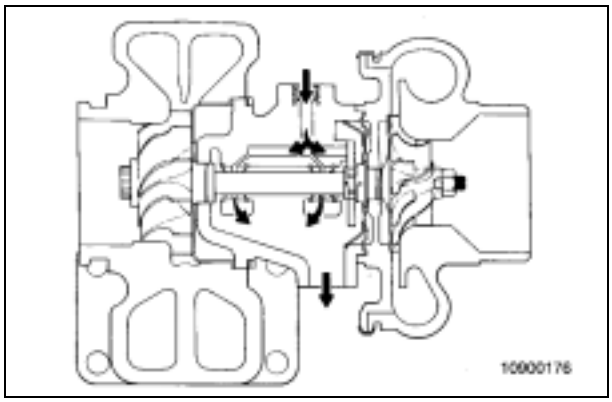


1. Exhaust valve
2. Exhaust manifold
3. Turbocharger
4. Turbocharger exhaust outlet

The turbocharger uses exhaust gas energy to turn the turbine wheel. The turbine wheel drives the compressor impeller, which provides pressurized air to the engine for combustion. The additional air provided by the turbocharger allows more fuel to be injected to increase the power output from the engine.

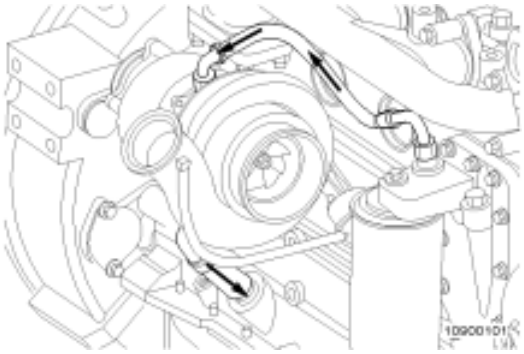


The turbine and compressor wheels share a common shaft (referred to as a rotor assembly), which is supported by two rotating bearings in the bearing housing.
Passages in the bearing housing direct filtered, pressurized engine oil to the shaft bearings and thrust bearings. The oil is used to lubricate and cool the rotating components. Oil then drains from the bearing housing to the engine sump through the oil drain line.

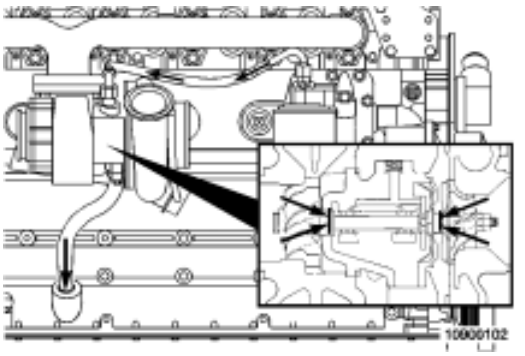


Lubricating Oil Consumption and Leaks

Engine lubricating oil is used to lubricate the bearings and provide some cooling for the turbocharger. The lubricating oil supplied to the turbocharger through the supply line is at engine operating pressure. A return line connected to the bottom of the turbocharger routes the lubricating oil back to the engine lubricating oil pan.



Seal rings are used on each end of the rotor assembly. The primary function of the seals is to prevent exhaust gases and compressed air from entering the turbocharger housing. Lubricating oil leakage from the seals is rare, but it can occur.

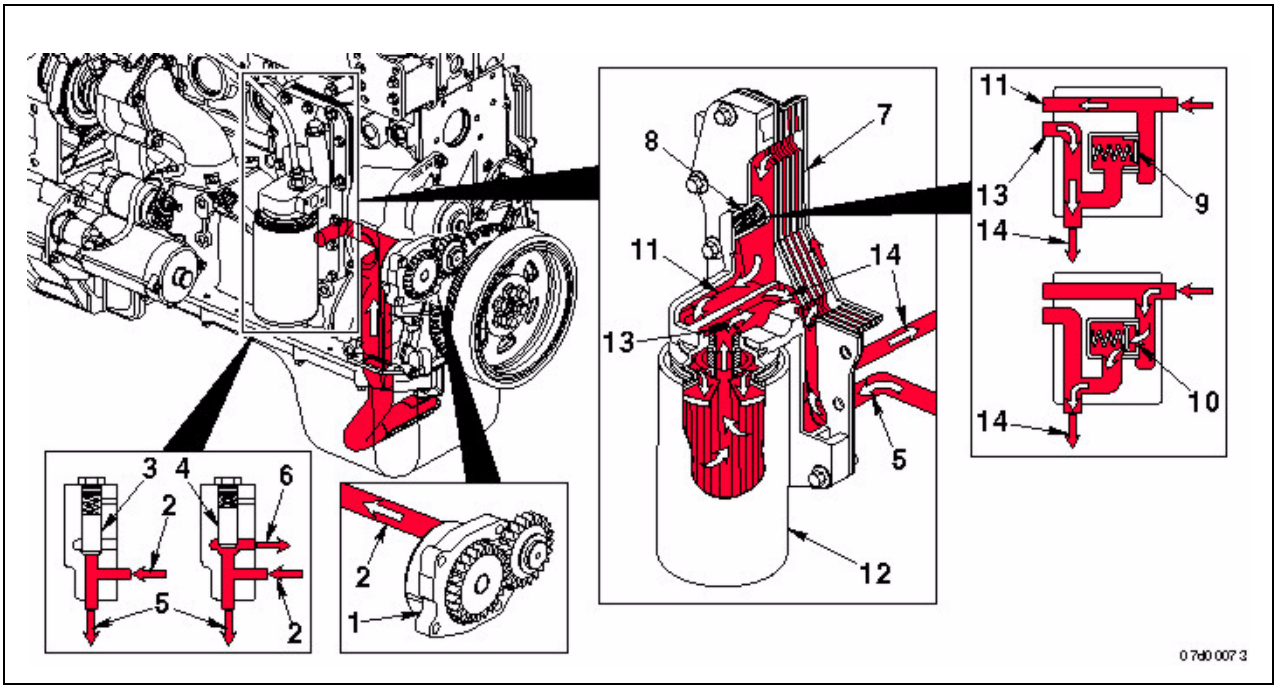


NOTE:
Excessive crankcase pressure will not allow the oil to drain from the turbocharger. This will load the bearing housing and allow lubricating oil to leak past the seal rings and into the engine and exhaust.

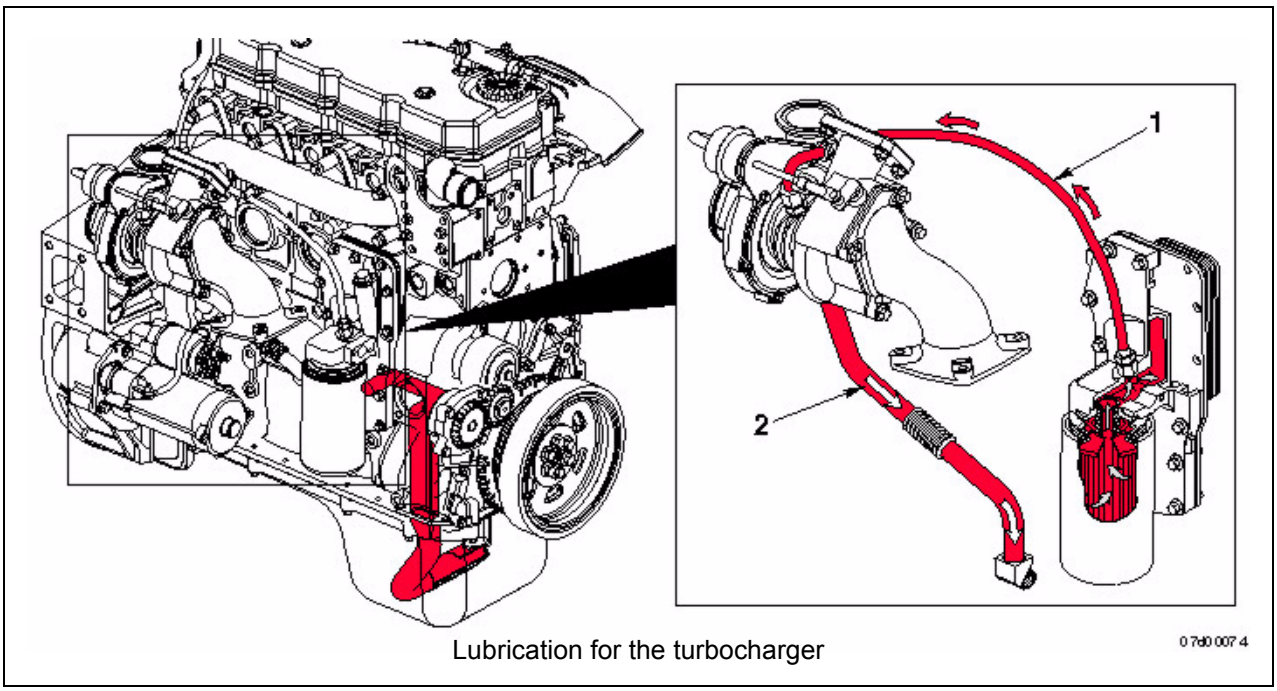
If turbine seal leakage into the exhaust occurs on engines with a catalyst, check the exhaust restriction during the repair.

Lubricating oil system

General Information

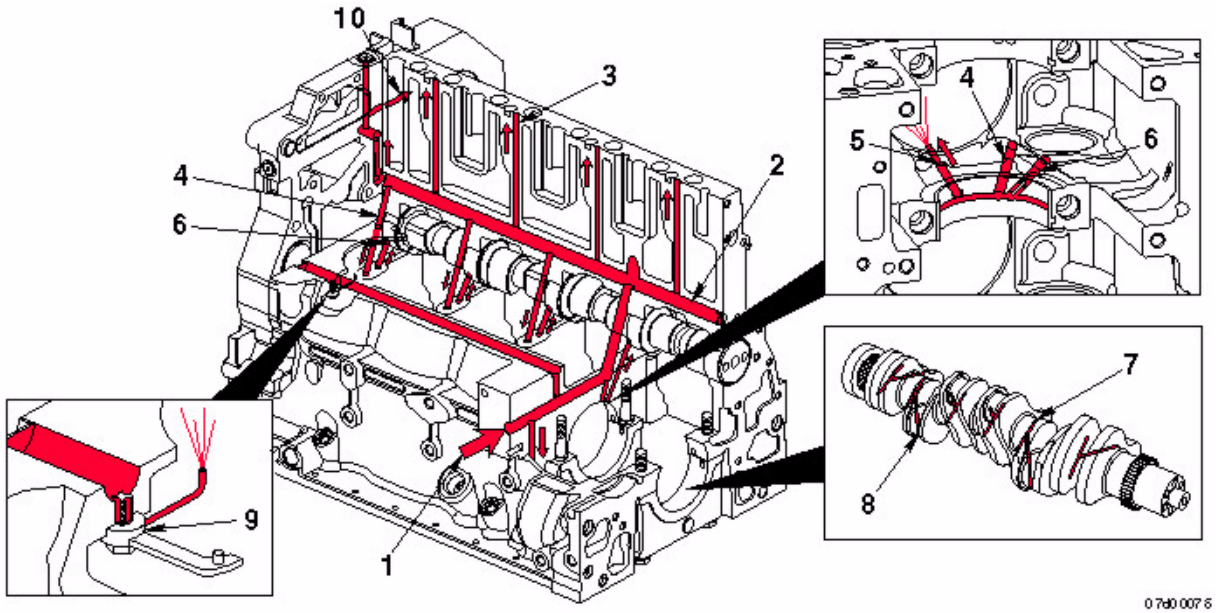


1. Gerotor lubricating oil pump
2. From lubricating oil pump
3. Pressure regulating valve closed
4. Pressure regulating valve open
5. To lubricating oil cooler
6. To oil pan
7. Lubricating oil cooler
8. Filter bypass valve
9. Filter bypass valve closed
10. Filter bypass valve open
11. To lubricating oil filter
12. Full-flow lubricating oil filter
13. From lubricating oil filter
14. Main lubricating oil rifle



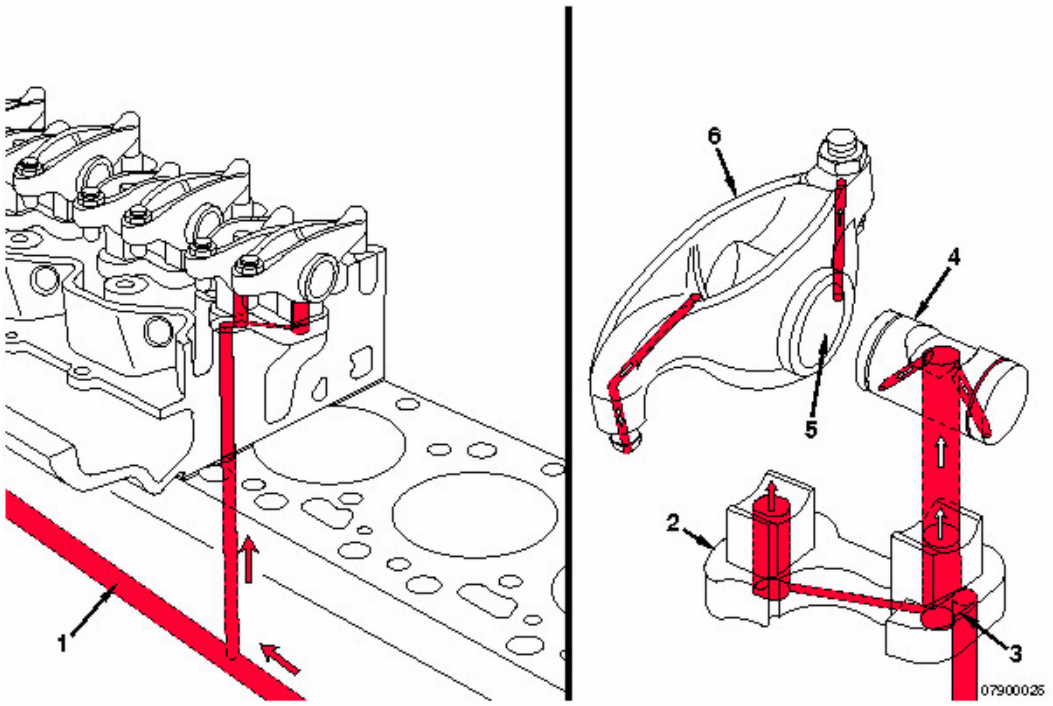
Lubrication for the turbocharger

1. Turbocharger lubricating oil supply
2. Turbocharger lubricating oil drain



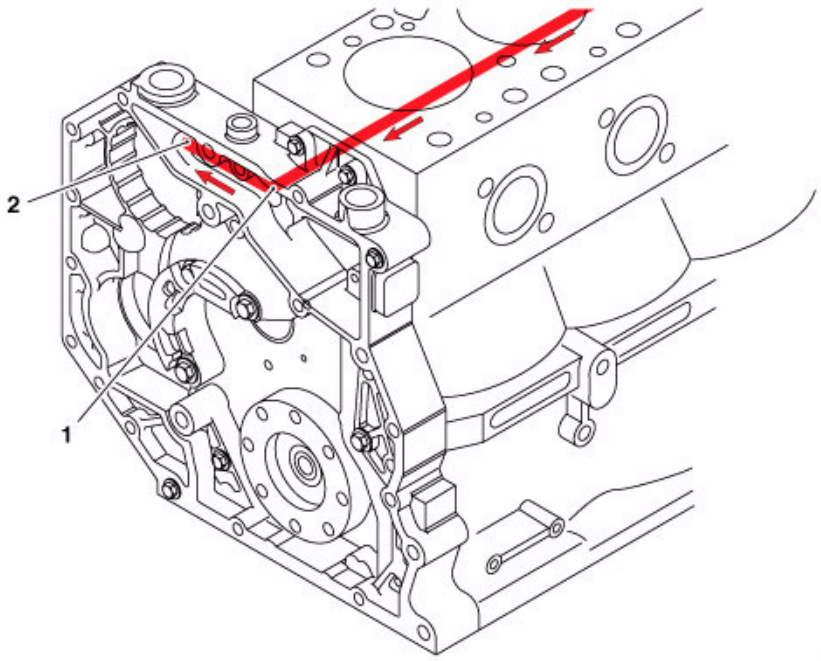
Lubrication for the power components

1. From lubricating oil cooler
2. Main lubricating oil rifle
3. To valve train
4. From main lubricating oil rifle
5. To piston-cooling nozzle
6. To camshaft
7. Crankshaft main journal
8. Oil supply to rod bearings
9. Directed piston-cooling nozzle
10. To internal lubrication of air compressor



Lubrication for the overhead components

1. Main lubricating oil rifle
2. Rocker lever support
3. Transfer slot
4. Rocker lever shaft
5. Rocker lever bore
6. Rocker lever



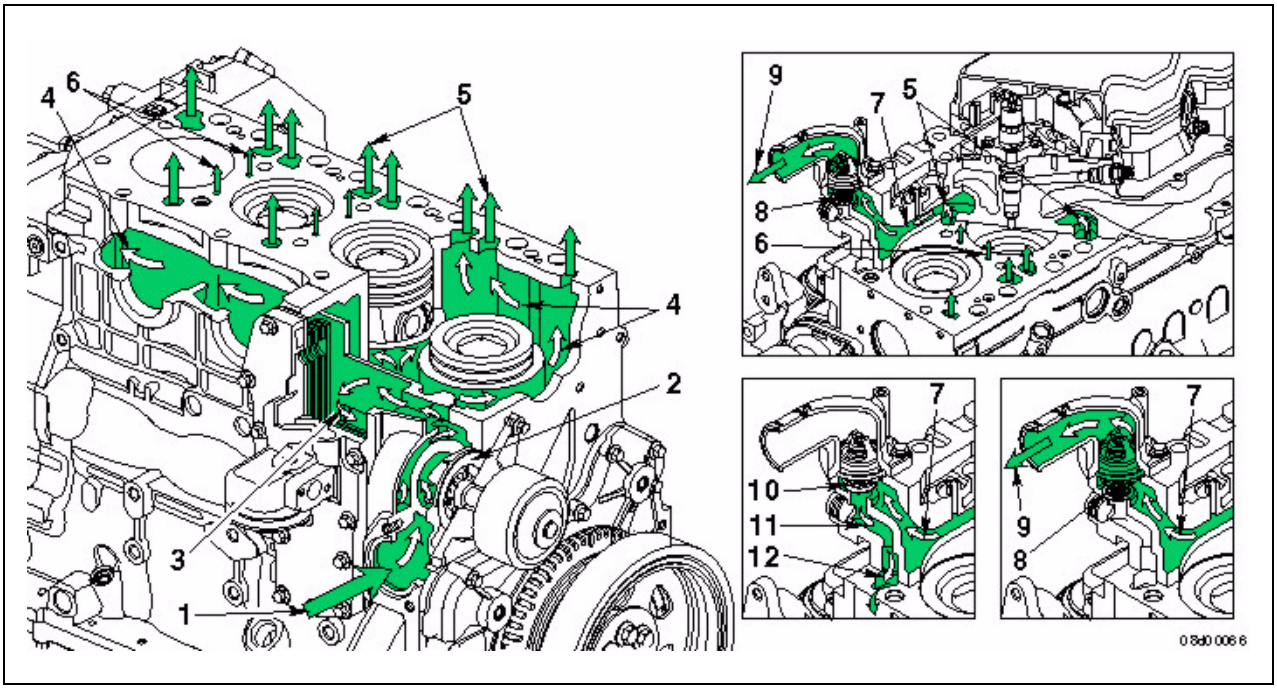
Lubrication for the accessory drive

1. Oil feed from block
2. Oil supply to accessory drive

NOTE:
Oil returns to pan through the gear housing.

Cooling system

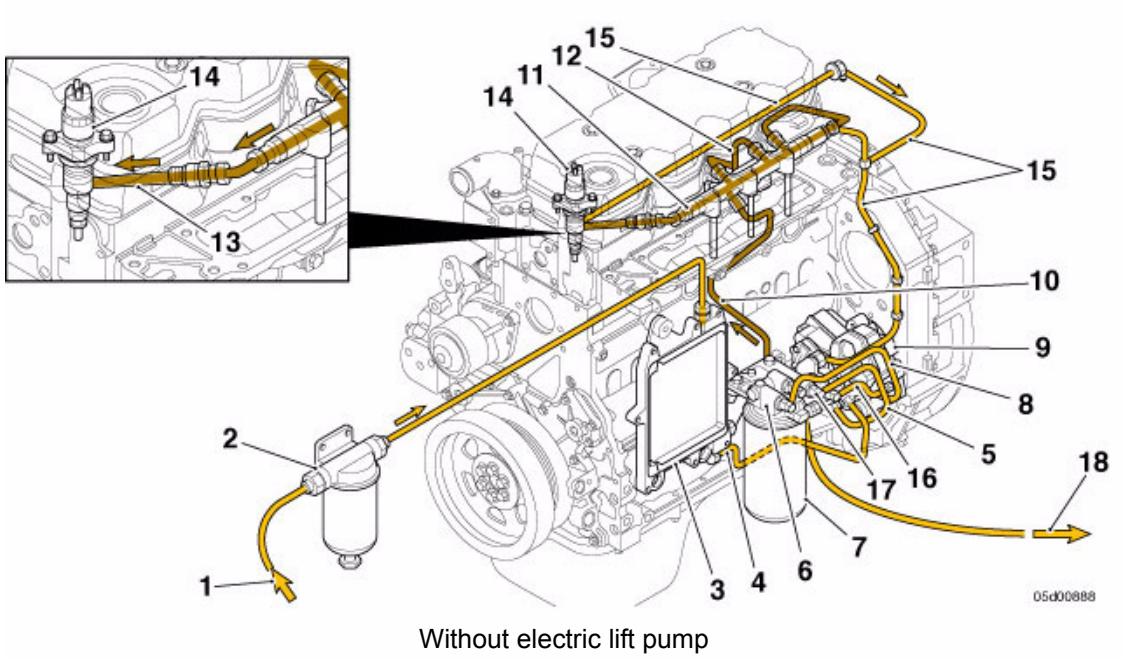
Flow Diagram



1. Coolant inlet
2. Pump impeller
3. Coolant flow past lubricating oil cooler
4. Coolant flow past cylinders
5. Coolant flow from cylinder block to cylinder head
6. Coolant flow between cylinders
7. Coolant flow to thermostat housing
8. Coolant bypass passage
9. Coolant flow back to radiator
10. Bypass open
11. Coolant bypass in cylinder head
12. Coolant flow to water pump inlet

Fuel system

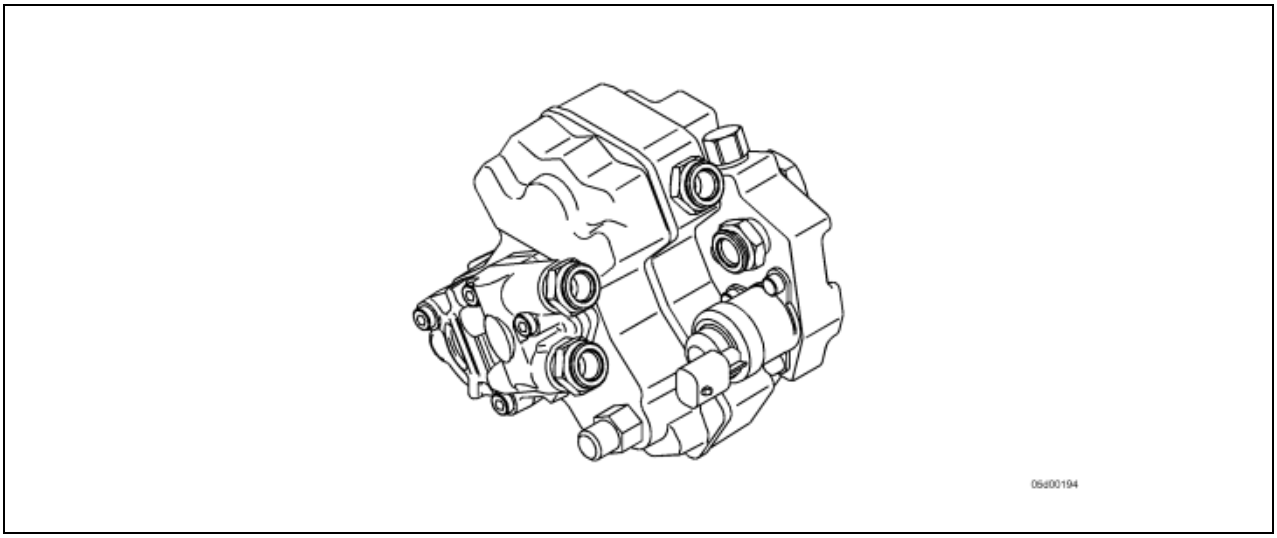
General Information



Without electric lift pump

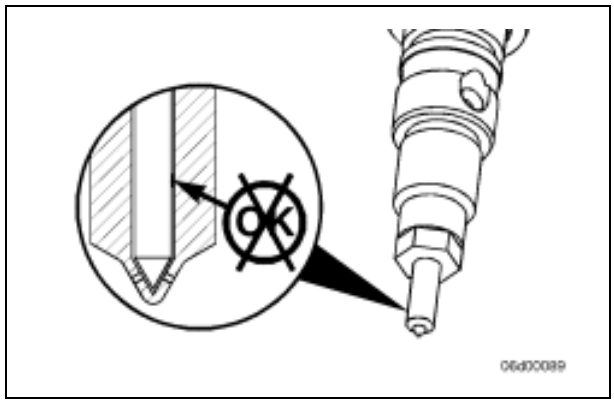
1. From fuel supply tank
2. Water/fuel separator (not mounted on engine)
3. ECM cooling plate
4. To fuel gear pump
5. To fuel filter
6. Fuel filter head
7. Fuel filter
8. To high-pressure pump
9. High-pressure pump
10. To fuel rail
11. Fuel rail
12. To injectors
13. High-pressure connector
14. Injector
15. Fuel return from injectors and fuel rail to fuel filter head
16. Fuel return from high-pressure pump to fuel filter head
17. Fuel return manifold
18. To fuel supply tank.

Injectors and fuel lines - overview
General Information

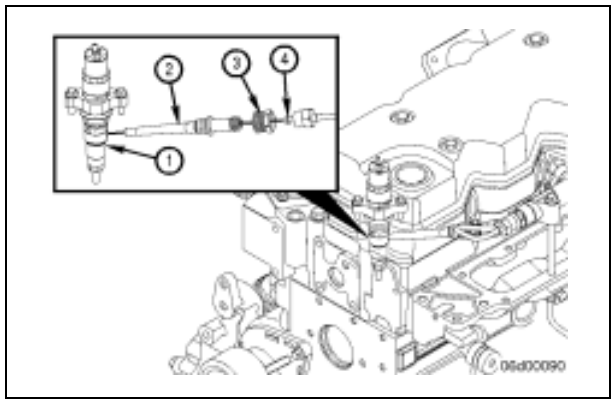


- The fuel system is a high-pressure common rail electronically controlled fuel system. The high-pressure common rail system consists of four main components: Fuel pump gear pump, high-pressure pump, fuel rail, and injectors. The high-pressure pump supplies high-pressure fuel to the fuel rail independent of engine speed. The high-pressure fuel is then accumulated in the fuel rail. High-pressure fuel is constantly supplied to the injectors by the fuel rail. The electronic control module (ECM) controls the fueling and timing of the engine by actuating the injectors.
- A suction side fuel/water separating filter is required. This filter is located off of the engine and contains a hand priming pump. Fuel enters the engine at the ECM cooling plate, if equipped, and then flows to the fuel pump gear pump. Using fuel pump gear pump pressure (approximately 0.3 – 1.3 MPa {3.0 – 13.3 kg/cm^2}), fuel is filtered using an on-engine or remotely mounted fuel filter prior to entering the high-pressure fuel pump.
- The fuel that enters the high-pressure fuel pump is pressurized between 25 and 180 MPa [255 to 1,837 kg/cm^2] by three radial pumping chambers. An M-Prop valve, or EFC valve, at the inlet to the three radial pumping chambers regulates the volume of fuel that is allowed to enter the pumping chambers. By regulating the volume of fuel that is pressurized, the M-Prop valve uses signals from the ECM to maintain the pressure in the fuel rail at a desired level. Fuel that is not allowed to enter the three radial pumping chambers is directed through the Cascade Overflow Valve. The Cascade Overflow Valve directs a certain amount of pressure to the lube channels of the high pressure pump and then returns the fuel to the fuel tank.
- The fuel rail acts as a fuel manifold accumulating and distributing fuel to each of the injector supply lines. Within the fuel rail, there is a fuel rail pressure sensor that monitors the pressure provided to the fuel rail from the high-pressure fuel pump.
  Pressure measured by the fuel rail pressure sensor is used by the ECM to adjust the fuel output of the high-pressure pump. The fuel rail also contains a fuel rail pressure relief valve. The fuel rail pressure relief valve is a safety valve used to bleed off excess pressure if the rail pressure exceeds a preset threshold. Fuel bled off by the fuel rail pressure relief valve is returned to the fuel tank through a fuel drain line connected to the fuel rail.

- High-pressure common rail fuel systems use solenoid-actuated injectors. High-pressure fuel flows into the side of the injector. When the solenoid is activated, an internal needle lifts and fuel is injected. The clearances in the nozzle bore are extremely small and any dirt or contaminants will cause the injector to stick. This is why it is important to clean around all fuel connections before servicing the fuel system. Also, cap or cover any open fuel connections before a fuel system repair is performed.

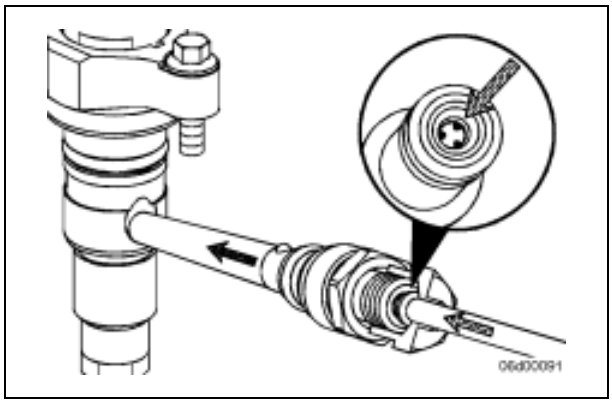


- High-pressure fuel is supplied to the injector from the fuel rail by an injector supply line and a fuel connector. The fuel connector pushes against the injector body when the fuel connector nut is tightened. The injector supply line is then connected to the fuel connector.
- The torque on this fuel connector and the injector supply lines is critical. If the nut or line is undertightened, the surfaces will not seal and a high-pressure fuel leak will result. If the nut is overtightened, the connector and injector will deform and cause a high-pressure fuel leak. This leak will be inside the head and will not be visible. The result will be a fault code, low power, or no-start.

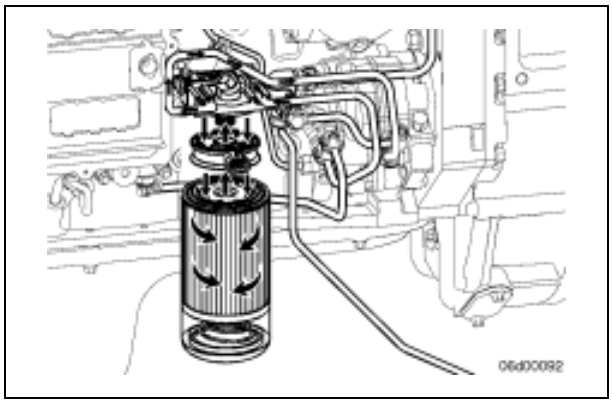


- The fuel connector contains an edge filter that breaks up small contaminants that enter the fuel system. The edge filter uses the pulsating high pressure to break up any particles so that they are small enough to pass through the injector.

  NOTE:
  The edge filters are not a substitute for cleaning and covering all fuel system connections during repair. Edge filters are not a substitute for maintaining the recommended engine mounted fuel filter.

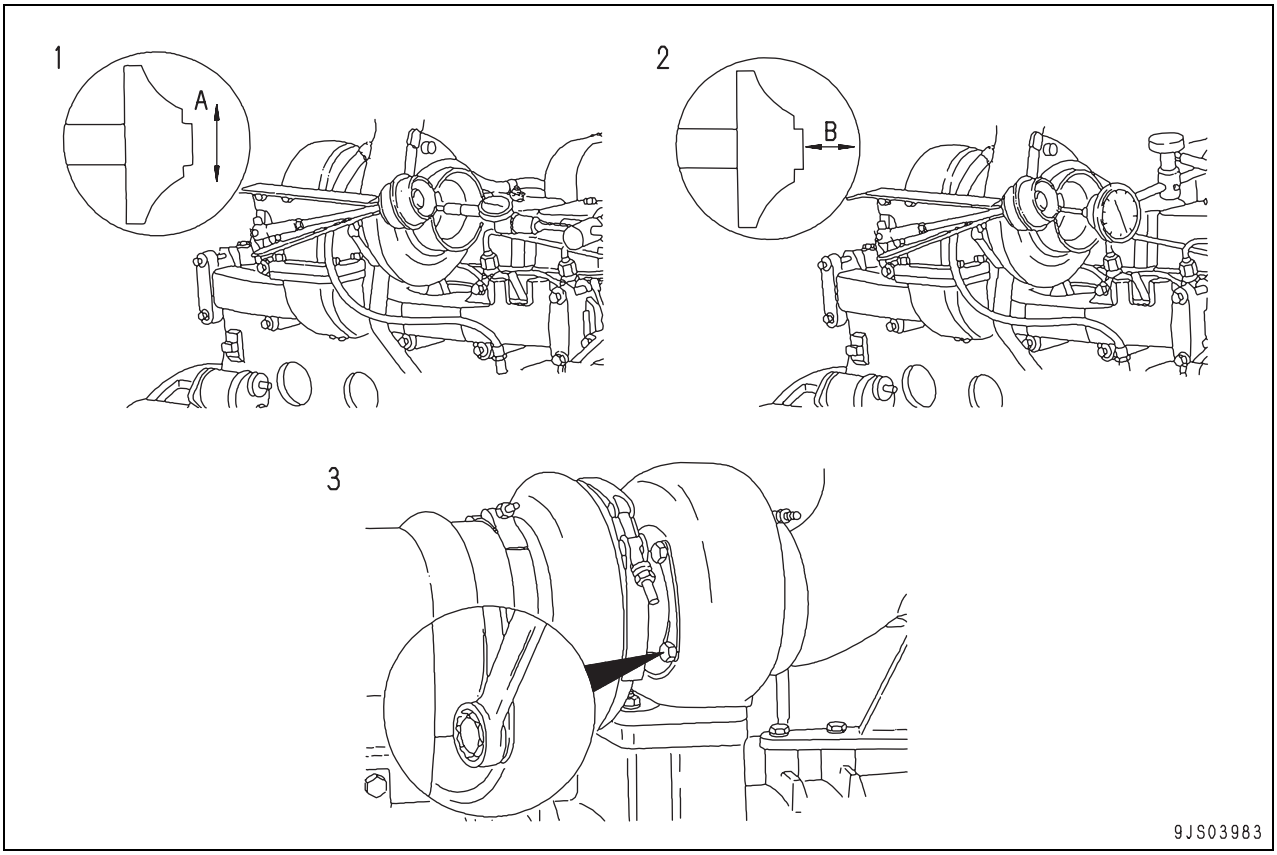


- All injectors feed into a common return circuit contained within the cylinder head. Any excess fuel is returned to the tank via this drilling in the cylinder head and return line attached to the rear of the cylinder head. A back-pressure valve is located on the back of the cylinder head where the drain line attaches. The fuel drain lines may have either a quick-disconnect fitting or a banjo fitting at ends of the fuel drain lines.
  Each of the fuel drain lines combine together at the fuel return manifold.
- The electronic control module (ECM) controls the fueling and timing of the engine by actuating the solenoids on the injector. An electronic pulse is sent to the solenoids to lift the needle and start the injection event. By electronically controlling the injectors, there is a more precise and accurate control of fueling quantity and timing. Also, multiple injection events can be achieved by electronically controlling the injectors.



Maintenance standard

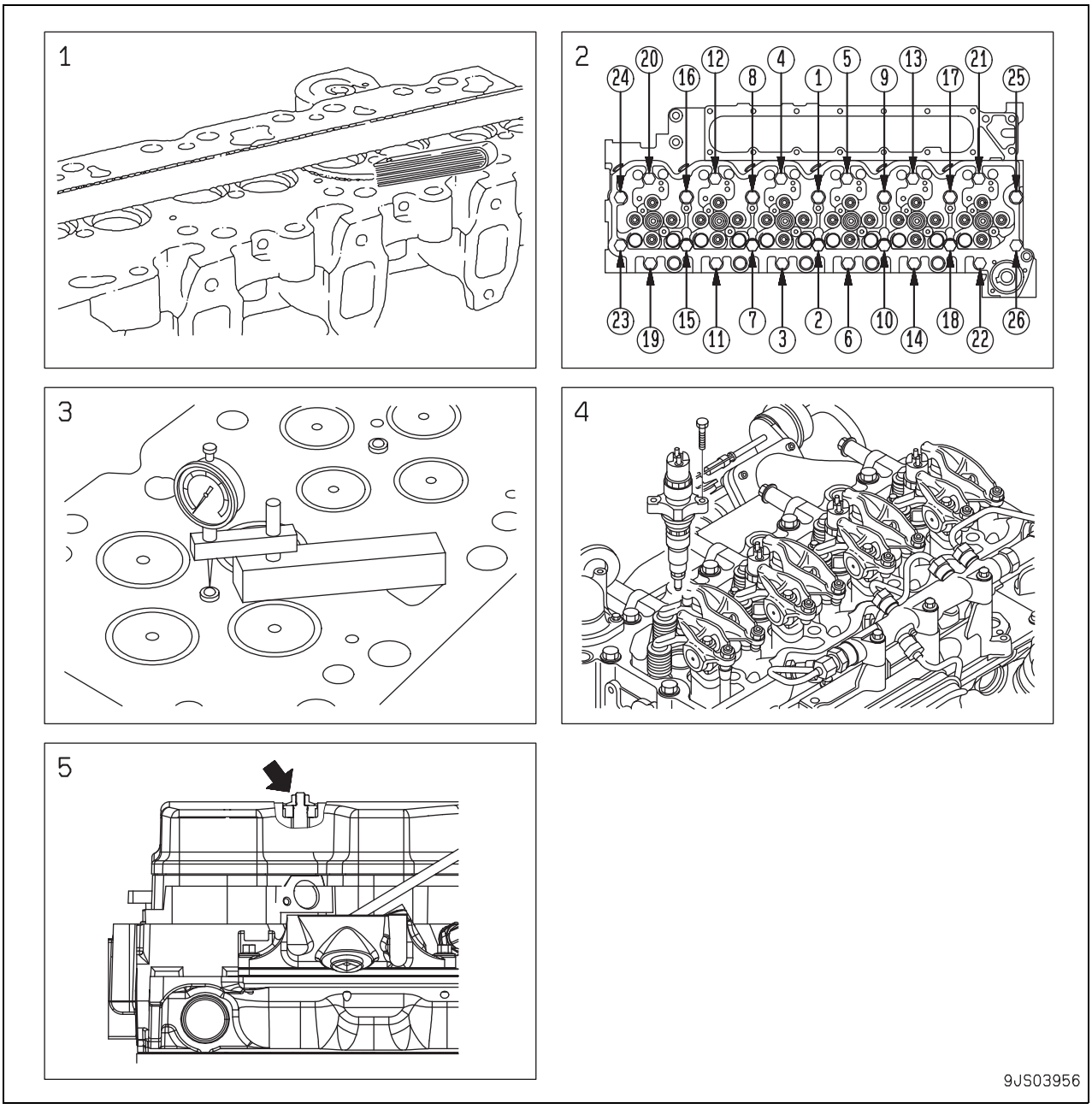
Turbocharger



Unit: mm

No.	Check item	Criteria	Remedy
1	Radial play (Play in radial direction)	0.329 – 0.501	Replace bearing parts
2	End play (Play in axial direction)	0.038 – 0.093	Replace thrust parts
3	Tightening torque of turbine housing bolt	Target (Nm {kgm})	Tighten
		26 {2.6}	

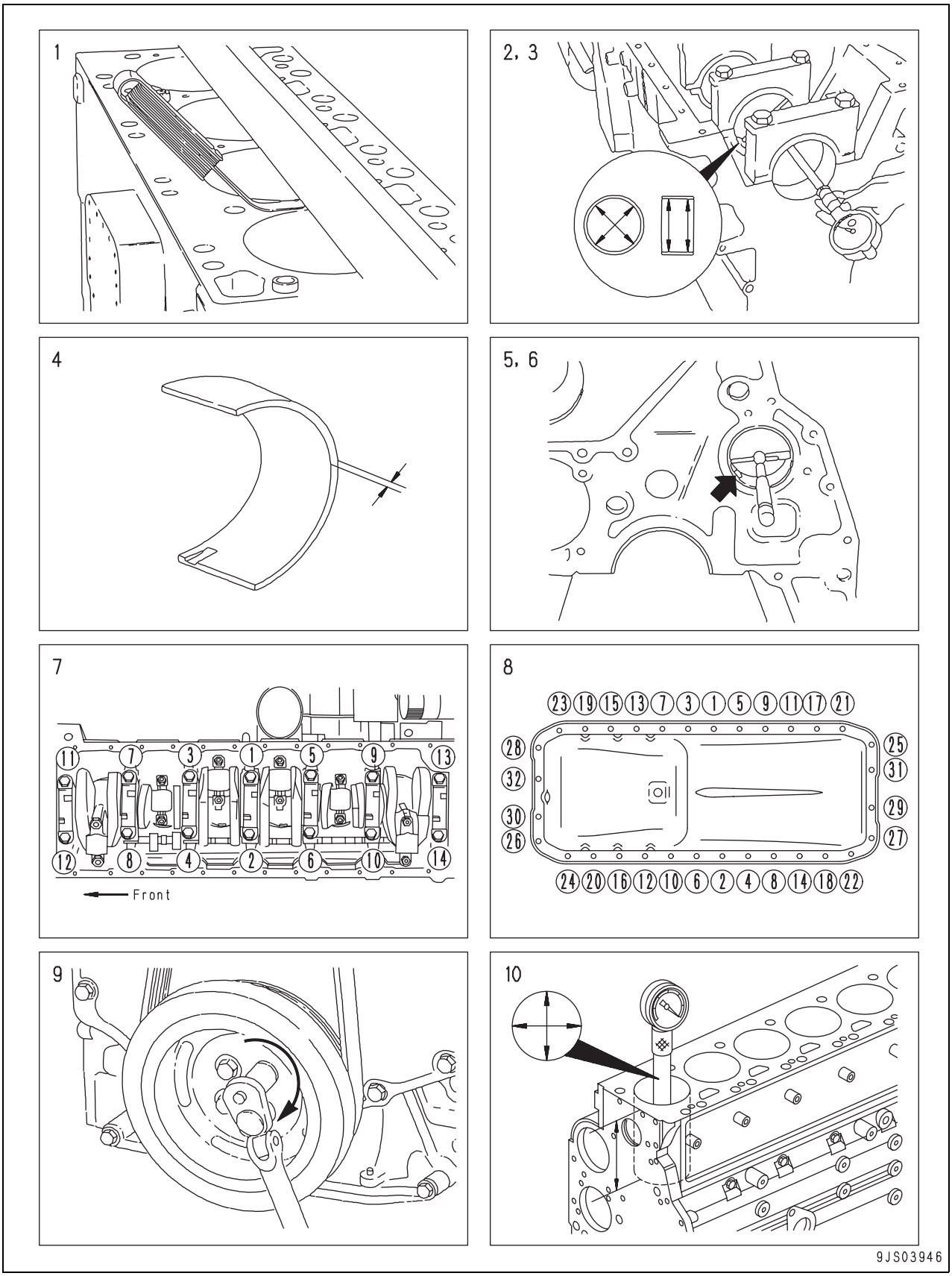
Cylinder head



Unit: mm

No.	Check item	Criteria		Remedy
1	Strain of cylinder head mounting face	End-to-End	Max. 0.305	Correct by grinding or replace
		Side-to-Side	Max. 0.076	
2	Tightening torque for cylinder head mounting bolt (Apply molybdenum disulfide or engine oil to threaded part)	Procedure	Target (Nm {kgm})	Tighten and retighten
		1st stage	90 {9.2}	
		2nd stage	90 {9.2}	
		3rd stage	Retighten 90°	
3	Projection of nozzle	2.45 – 3.15		Replace nozzle or gasket
4	Tightening torque for injector holder mounting bolt	Target (Nm {kgm})		Tighten
		8 {0.8}		
5	Tightening torque for head cover mounting bolt	24 {2.4}		

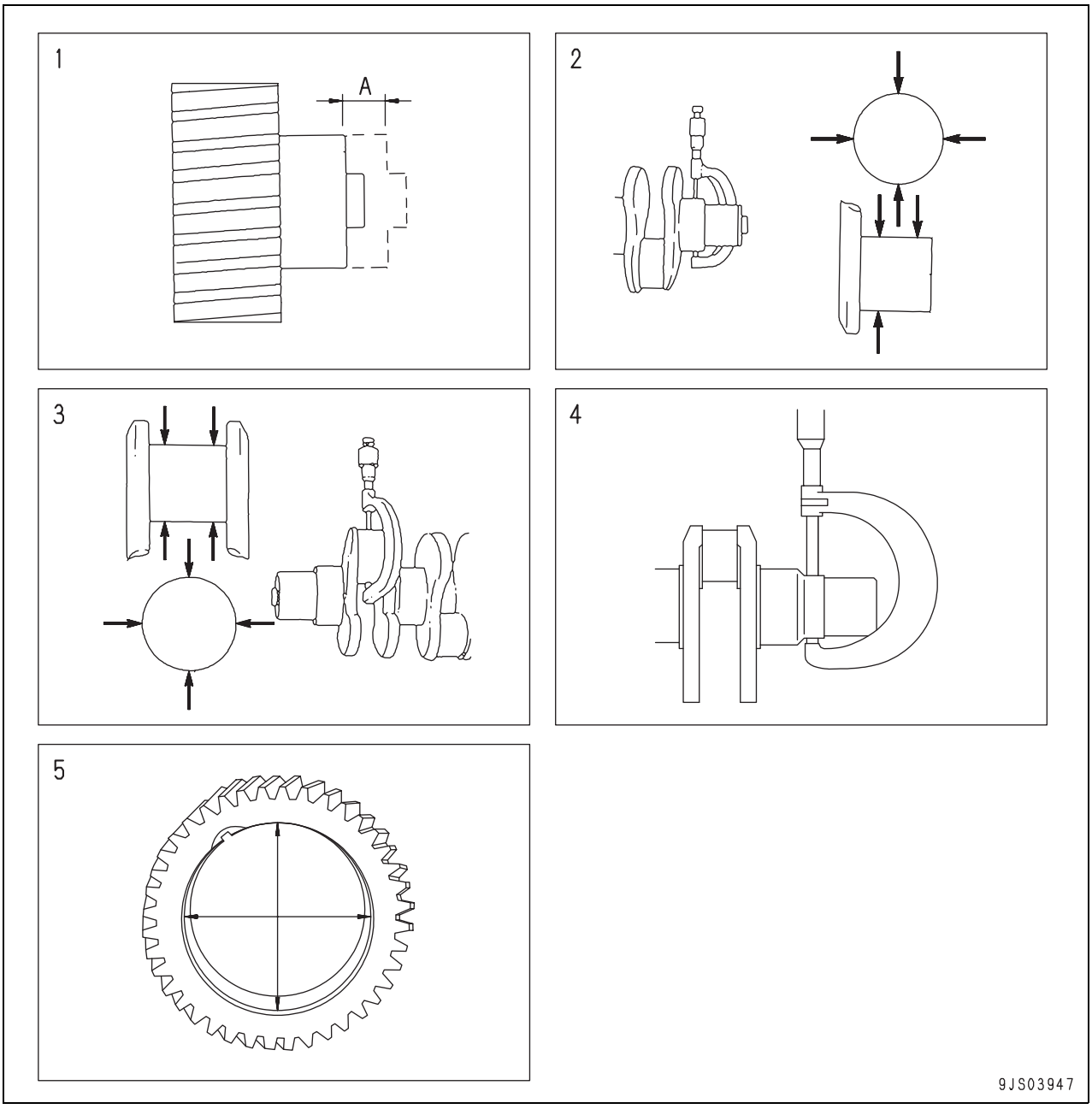
Cylinder block



Unit: mm

No.	Check item	Criteria		Remedy
1	Strain of cylinder head mounting face	End-to-End	Max. 0.075	Correct by grinding or replace
		Side-to-Side	Max. 0.075	
2	Diameter of main bearing metal mounting hole	87.982 – 88.008		Replace main bearing metal cap
3	Inside diameter of main bearing metal	83.041 – 83.109		Replace bearing metal
4	Thickness of main bearing metal	2.456 – 2.464		
5	Diameter of cam bushing mounting hole	Max.: 59.248		Correct or replace block
6	Inside diameter of cam bushing	Max.: 54.164		Replace cam bushing
7	Tightening torque for main cap mounting bolt (Apply engine oil to threads)	Procedure	Target (Nm {kgm})	Tighten and retighten
		1st stage	60 {6.1}	
		2nd stage	80 {8.2}	
		3rd stage	Retighten 90°	
8	Tightening torque for oil pan mounting bolt	Target (Nm {kgm})		Tighten
		28 {2.8}		
9	Tightening torque for crankshaft pulley mounting bolt	125 {12.7}		
10	Inside diameter of cylinder	106.990 – 107.010		Correct by oversize or replace cylinder block
	Roundness of inside of cylinder	Repair limit: 0.038		
	Taper of inside of cylinder	Repair limit: 0.076		

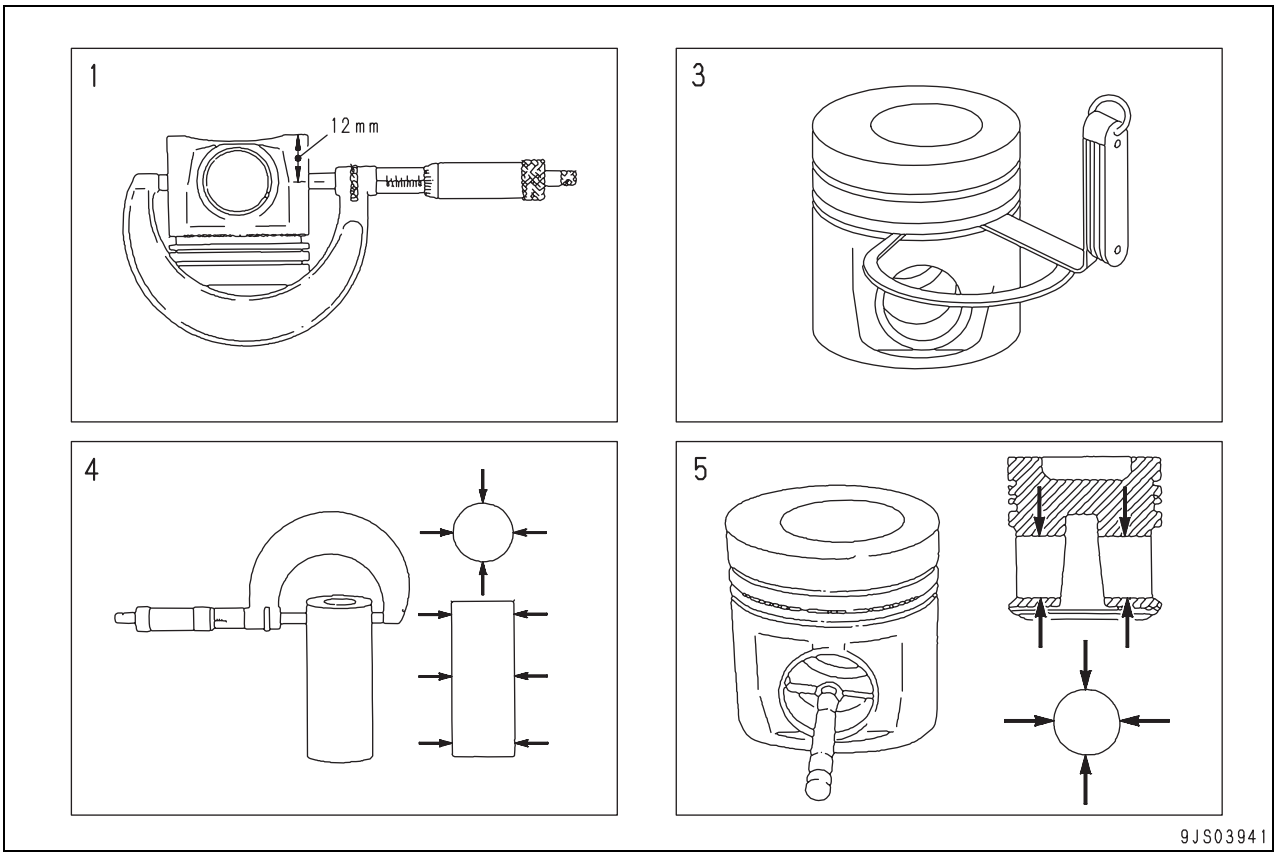
Crankshaft



Unit: mm

No.	Check item	Criteria	Remedy
1	End play (A)	0.065 – 0.415	Replace thrust bearing metal or use oversize metal
2	Outside diameter of main journal	82.962 – 83.013	Use undersize journal or replace
	Roundness of main journal	Repair limit: 0.050	
	Taper of main journal	Repair limit: 0.013	
	Clearance of main journal	0.04 – 0.12	Replace main bearing metal
3	Outside diameter of crank pin journal	68.962 – 69.013	Use undersize journal or replace
	Roundness of crank pin journal	Repair limit: 0.050	
	Clearance of crank pin journal	0.04 – 0.12	Replace connecting rod bearing metal
4	Outside diameter of crankshaft gear journal	70.59 – 70.61	Use undersize crankshaft or replace
5	Inside diameter of crankshaft gear journal	70.51 – 70.55	

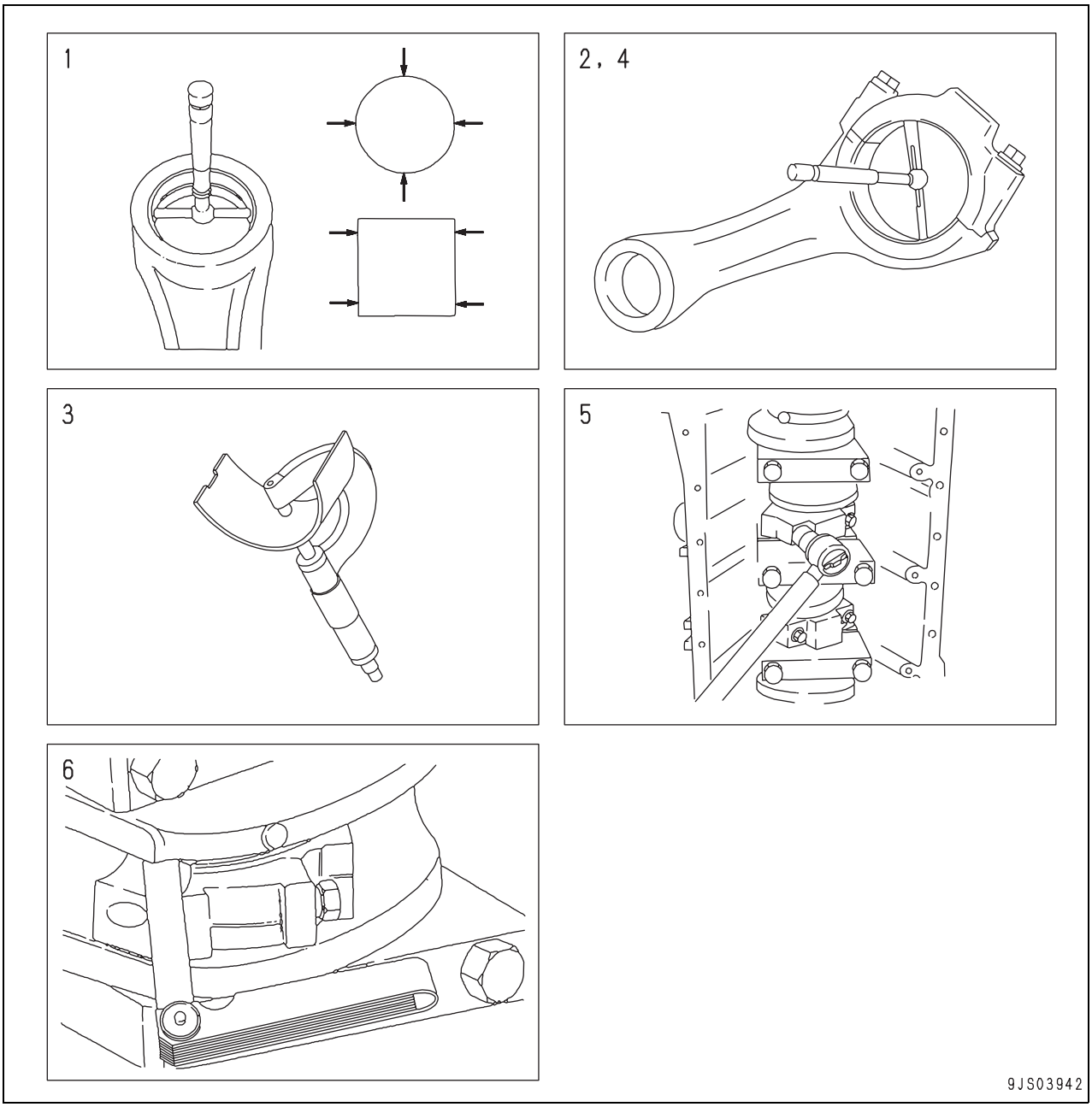
Piston, piston ring and piston pin



Unit: mm

No.	Check item	Criteria		Remedy
1	Outside diameter of piston (at right angles to boss)	107.255 – 107.273		Replace piston
2	Clearance at piston ring end gap	Top ring	0.33 – 0.43	Replace piston ring or piston
		2nd ring	0.55 – 0.85	
		Oil ring	0.25 – 0.50	
3	Clearance of piston ring	2nd ring	0.040 – 0.110	
		Oil ring	0.040 – 0.085	
4	Outside diameter of piston pin	39.997 – 40.003		Peplace piston or piston pin
5	Inside diameter of piston pin bore	40.006 – 40.012		

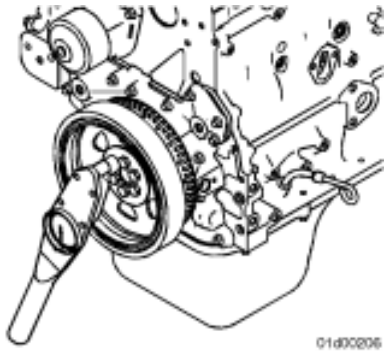
Connecting rod



Unit: mm

No.	Check item	Criteria		Remedy
1	Inside diameter of connecting rod bushing (when bushing is installed)	40.019 – 40.042		Replace bushing (spare part is semi-finished part)
2	Inside diameter of connecting rod bearing	69.05 – 69.10		Replace connecting rod bearing
3	Thickness of connecting rod bearing	1.955 – 1.968		
4	Inside diameter of connecting rod bearing mounting hole	72.99 – 73.01		Replace connecting rod
5	Tightening torque for connecting rod cap mounting bolt (Coat thread of bolt nut with engine oil)	Procedure	Target (Nm {kgm})	Tighten and retighten
		1st stage	60 {6.1}	
		2nd stage	Retighten 60°	
6	Side clearance of connecting rod	0.125 – 0.275		Replace connecting rod

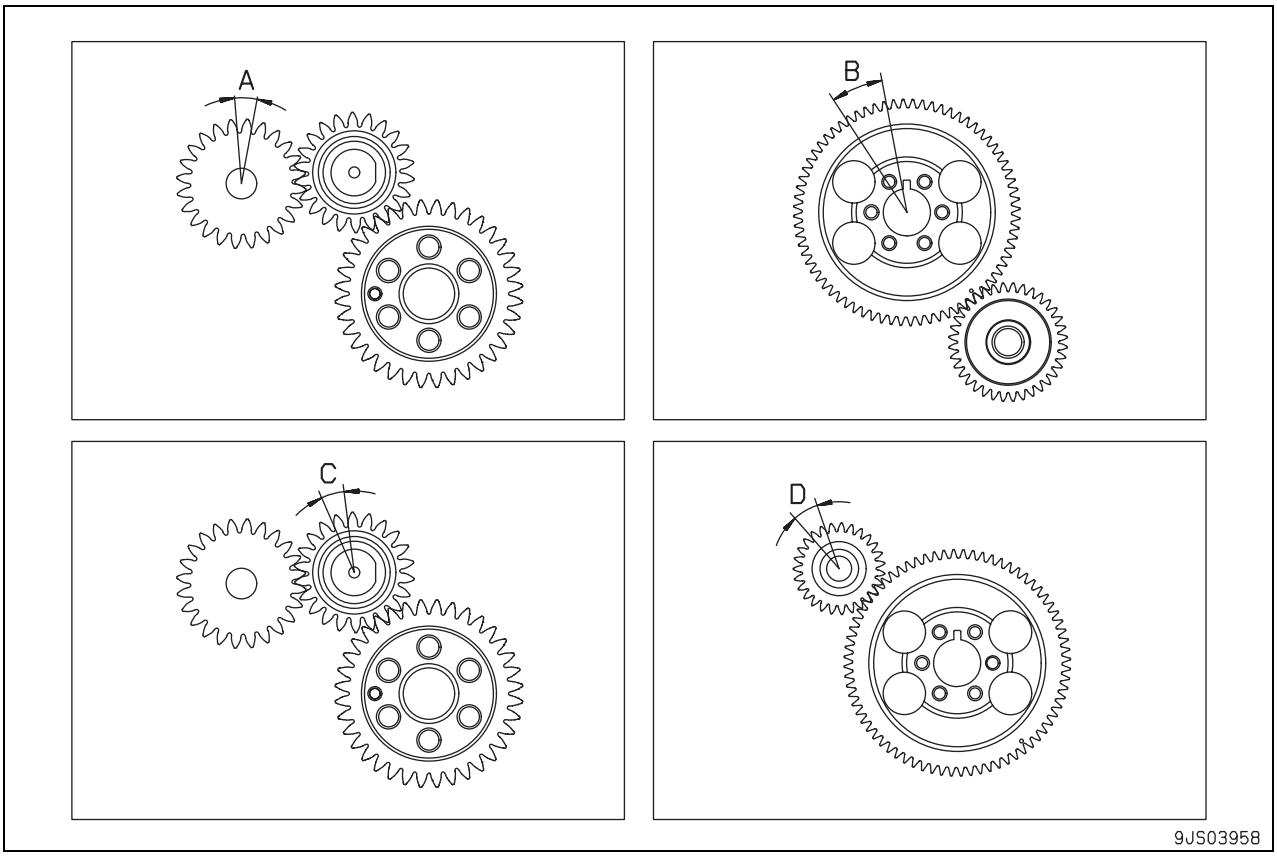
Vibration damper



Unit: mm

No.	Check item	Criteria		Remedy
1	Tightening torque of vibration damper	Procedure	Target (Nm {kgm})	Tighten and retighten
		1st stage	50 {5.1}	
		2nd stage	Retighten 90°	

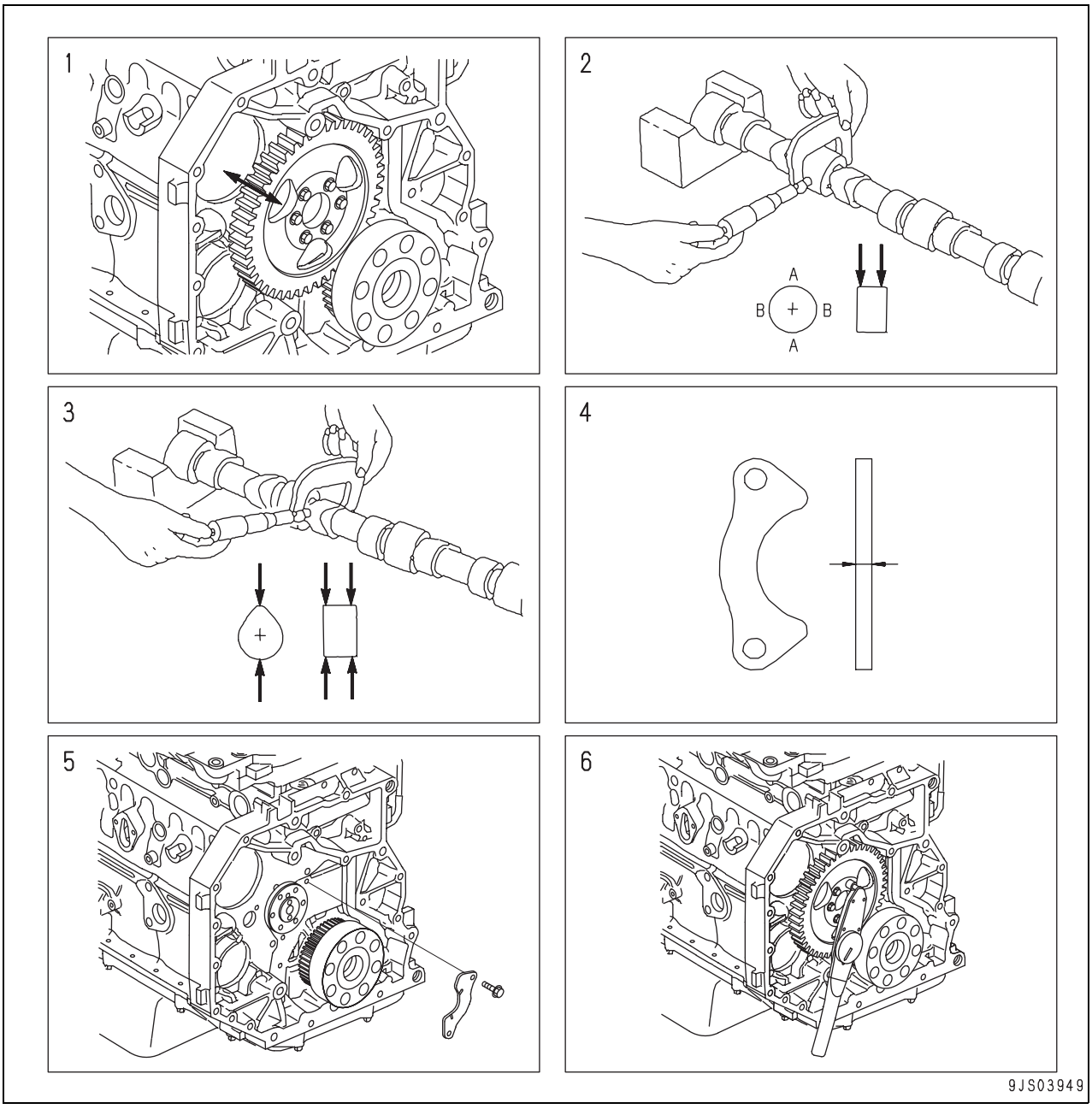
Timing gear



Unit: mm

No.	Check item	Criteria			Remedy
–	Backlash of each gear	A	Backlash of oil pump gear	0.170 – 0.300	Replace
		B	Backlash of camshaft gear	0.076 – 0.280	
		C	Backlash of oil pump idler gear	0.170 – 0.300	
		D	Backlash of fuel injection pump gear	0.146 – 0.222	

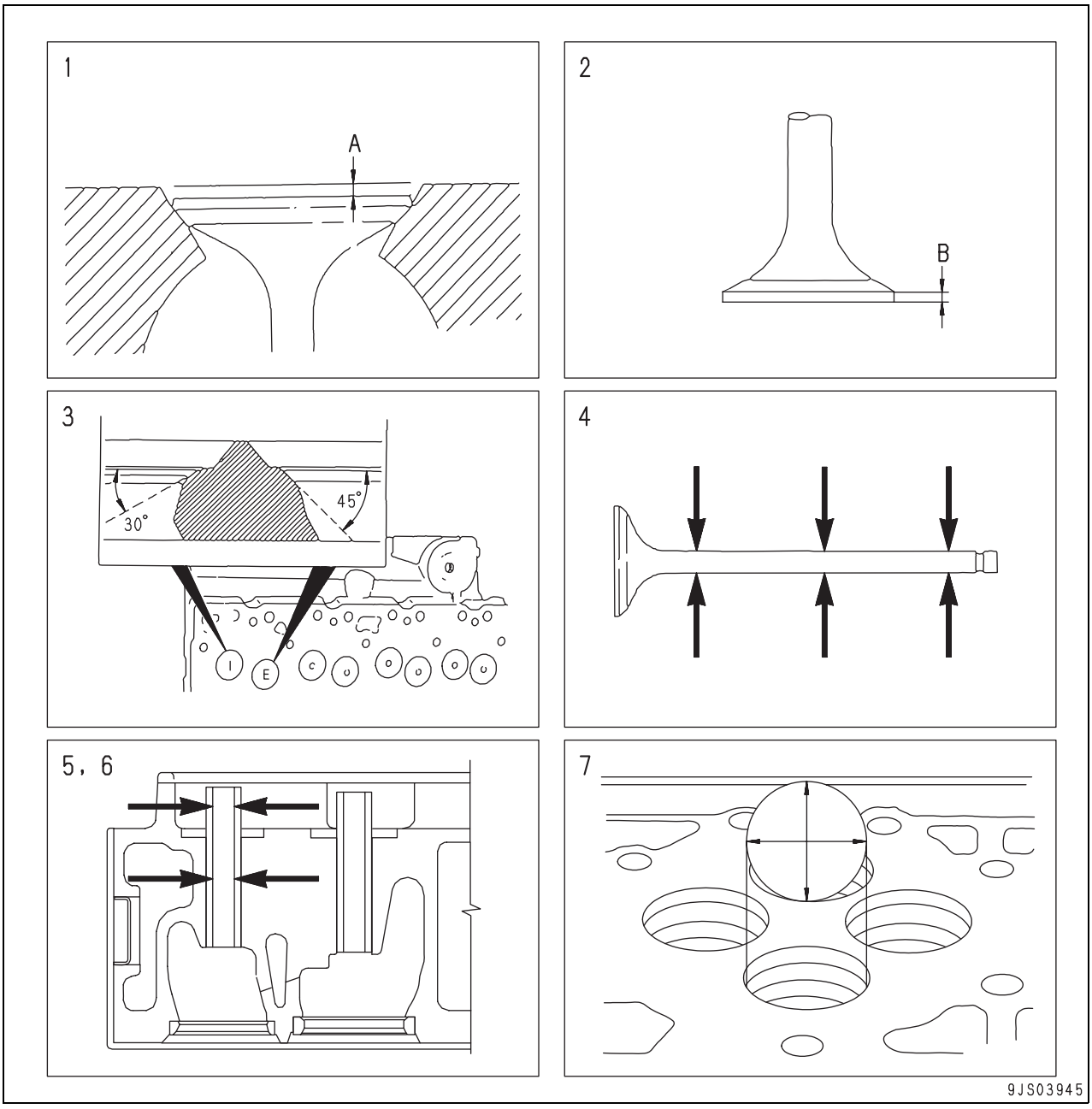
Camshaft



Unit: mm

No.	Check item	Criteria		Remedy
1	End play	0.100 – 0.360		Replace thrust plate
2	Outside diameter of camshaft bearing journal	53.095 – 54.045		Correct or replace
3	Height of cam	Intake	47.175 – 47.855	Replace
		Exhaust	45.632 – 46.312	
4	Thickness of camshaft thrust plate	9.40 – 9.60		
5	Tightening torque of camshaft thrust plate mounting bolt	Target (Nm {kgm})		Tighten
		24 {2.4}		
6	Tightening torque of camshaft gear mounting bolt	36 {3.7}		

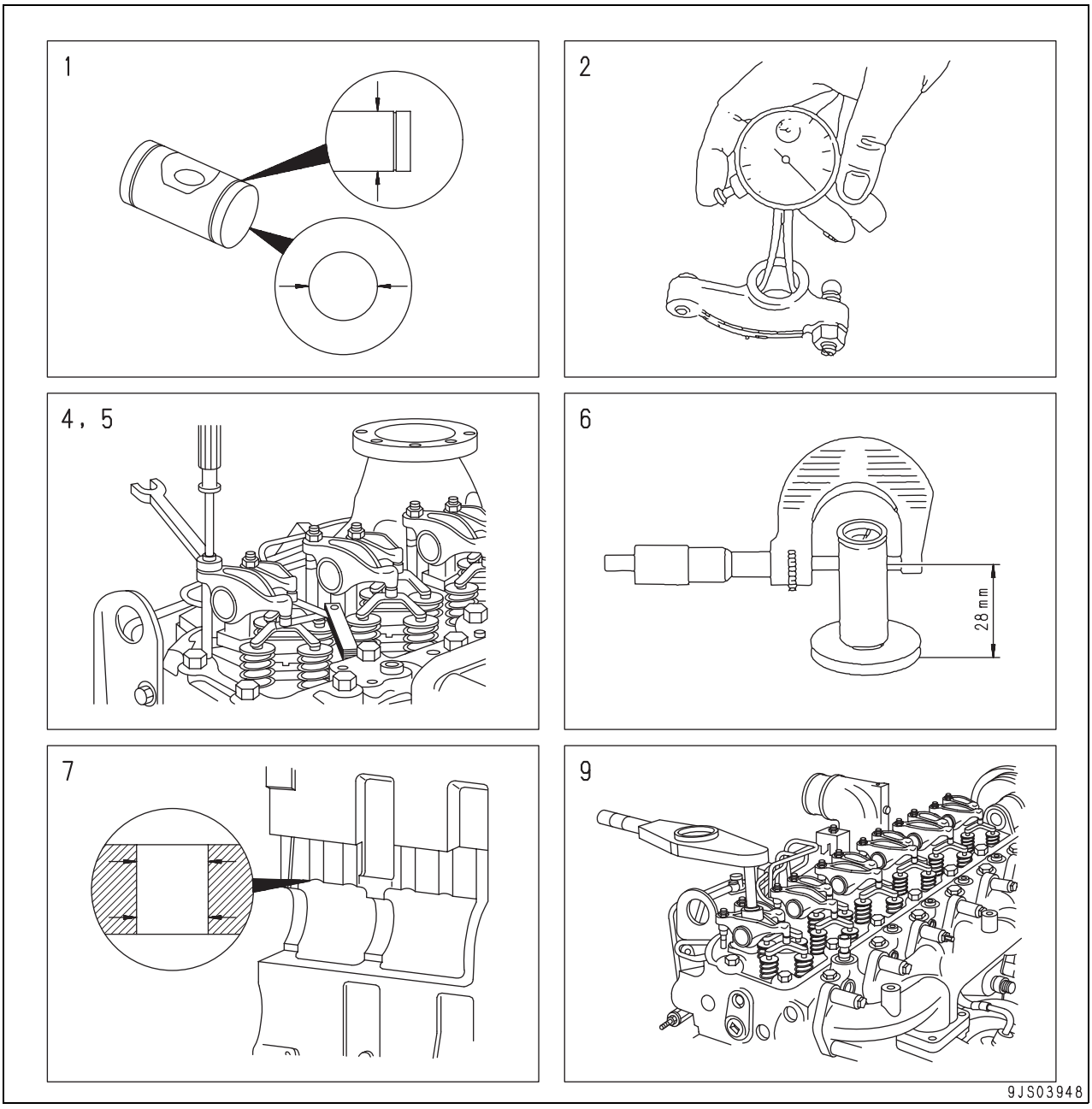
Valve and valve guide



Unit: mm

No.	Check item	Criteria			Remedy
1	Sunk depth (A) of valve	Intake	0.584 – 1.092		Replace valve or valve seat
		Exhaust	0.965 – 1.473		
2	Thickness (B) of valve head	Min. 0.79			Replace
3	Angle of valve seat	Valve	Angle	Repair limit	Correct or replace valve, valve seat
		Intake	30°	Judge condition of contact surface with vacuum test	
		Exhaust	45°		
4	Outside diameter of valve stem	6.96 – 7.01			Replace
5	Diameter of valve guide bore	7.027 – 7.077			
6	Clearance between valve guide and valve stem	0.017 – 0.117			Replace valve or valve guide
7	Depth of valve insert bore (standard insert)	34.847 – 34.863			Replace

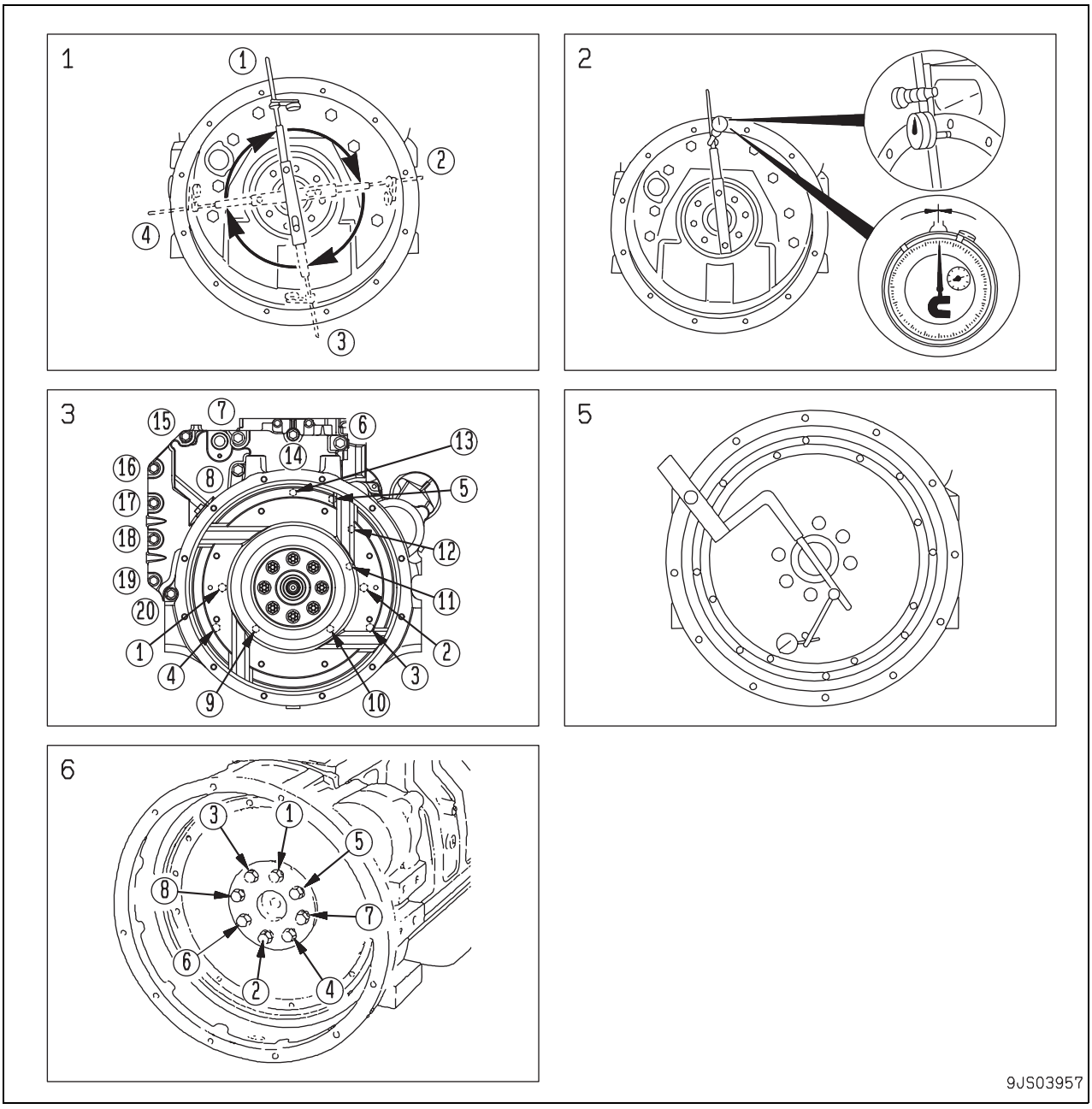
Rocker arm, shaft and tappet



Unit: mm

No.	Check item	Criteria		Remedy
1	Outside diameter of rocker arm shaft	Min. 21.965		Replace rocker arm shaft
2	Inner diameter of rocker arm shaft hole	Max. 22.027		Replace rocker arm
3	Clearance between rocker arm shaft and rocker arm	Max. 0.062		Replace rocker arm or shaft
4	Tightening torque of locknut for rocker arm adjustment screw	Target (Nm {kgm})		Tighten
		24 {2.4}		
5	Valve clearance (cold)	Valve	Repair limit	Adjust
		Intake	0.25	
		Exhaust	0.51	
6	Outside diameter of tappet	15.936 – 15.977		Replace tappet
7	Inside diameter of tappet hole	16.000 – 16.055		Replace cylinder block
8	Clearance between tappet and tappet hole	0.023 – 0.119		Replace tappet or cylinder block
9	Tightening torque of rocker arm mounting bolt	Target (Nm {kgm})		Tighten
		36 {3.7}		

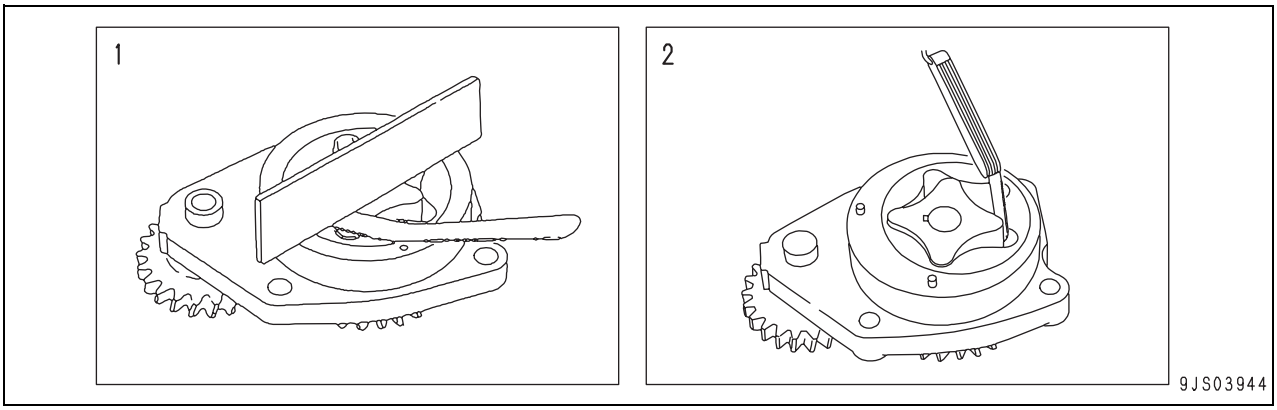
Flywheel and flywheel housing



Unit: mm

No.	Check item	Criteria		Remedy
1	Face runout of flywheel housing	Repair limit: 0.20		Reassemble, correct
2	Radial runout of flywheel housing	Repair limit: 0.20		
3	Tightening torque of flywheel housing mounting bolt	Target (Nm {kgm})		Tighten
		M10: 49 {5.0}		
		M12: 85 {8.7}		
4	Face runout of flywheel	Flywheel	Repair limit	Reassemble, correct
		For clutch	0.013 / ø25.4	
		For torque converter	Diameter (actual measurement) × 0.0005	
5	Radial runout of flywheel	Repair limit: 0.13		
6	Tightening torque of flywheel mounting bolt (Coat bolt thread with engine oil)	Procedure	Target (Nm {kgm})	Tigthen and retighten
		1st stage	30 {3.0}	
		2nd stage	Retighten: 60°	

Oil pump



Unit: mm

No.	Check item	Criteria		Remedy
1	Clearance of rotor in axial direction	Max. 0.127		Replace oil pump
2	Clearance between outer rotor and inner rotor	Max. 0.178		
3	Tightening torque of oil pump mounting bolt	Procedure	Target (Nm {kgm})	Tigthen and retighten
		1st stage	8 {0.8}	
		2nd stage	24 {2.4}	

Engine

107E-1 Series

20 Standard value table
Standard service value table

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Standard service value table

Standard value table for testing, adjusting and troubleshooting

Engine model				SAA6D107E-1	
Applicable machine				PC200, 200LC-8	
	Item	Condition, etc	Unit	Standard value	Permissible value
Prerformance	Engine speed	High idle speed	rpm	2,060 ± 50	2,060 ± 50
		Low idle speed	rpm	1,050 ± 25	1,050 ± 25
	Necessary starting speed	At 0°C	rpm	—	—
		At –20°C	rpm	—	—
Intake and exhaust system	Intake resistance	At all speed	kPa {mmH2O}	Max. 3.73 {Max. 380}	7.45 {760}
	Intake pressure (Air connector inlet)	At rated horsepower	kPa {mmHg}	Min. 140 {Min. 1,050}	113 {850}
	Exhaust pressure (Turbine inlet press.)	At rated horsepower	kPa {mmHg}	—	—
	Exhaust temperature (Turbine outlet temp.)	At all speed (at 20°C)	°C	Max. 550	600
	Exhaust gas color	At rated horsepower	%	Max. 12	22
		Quick acceleration (Low idle → high idle)		Max. 25	35
	Valve clearance	Intake valve	mm	0.25	0.152 – 0.381
		Exhaust valve	mm	0.51	0.381 – 0.762
Engine body	Compression pressure	Oil temperature: 40 – 60°C Engine speed: 250 – 280 rpm	MPa {kg/cm2}	Min. 2.41 {Min. 24.6}	1.69 {17.2}
	Blow-by pressure	At rated horsepower	kPa {mmH2O}	Max. 0.98 {Max. 100}	1.96 {200}
Lubrication system	Oil pressure	At rated horsepower (Oil temperature: Min. 80°C) SAE15W-40	MPa {kg/cm2}	Min. 0.29 {Min. 3.0}	0.25 {2.5}
		At low idle (Oil temperature: Min. 80°C) SAE15W-40	MPa {kg/cm2}	Min. 0.10 {Min. 1.0}	0.07 {0.7}
	Oil temperature	At all speed (Oil in oil pan)	°C	80 – 110	120
	Oil consumption ratio	At continuous rated (Ratio to fuel consumption)	%	Max. 0.15	0.3
Cooling system	Radiator pressure valve	Opening pressure (Differential pressure)	kPa {kg/cm2}	—	—
	Fan speed	At rated engine speed	rpm	1,740	1,740
	Fan and alternator belt tension	Deflects when pushed with a force of 98 N {10 kg}	mm	Auto-tensioner	Auto-tensioner

Engine model				SAA6D107E-1	
Applicable machine				PC220, 220LC-8	
	Item	Condition, etc	Unit	Standard value	Permissible value
Prerformance	Engine speed	High idle speed	rpm	2,160 ± 50	2,160 ± 50
		Low idle speed	rpm	1,050 ± 25	1,050 ± 25
	Necessary starting speed	At 0°C	rpm	—	—
		At –20°C	rpm	—	—
Intake and exhaust system	Intake resistance	At all speed	kPa {mmH_2O}	Max. 3.73 {Max. 380}	7.45 {760}
	Intake pressure (Air connector inlet)	At rated horsepower	kPa {mmHg}	Min. 140 {Min. 1,050}	113 {850}
	Exhaust pressure (Turbine inlet press.)	At rated horsepower	kPa {mmHg}	—	—
	Exhaust temperature (Turbine outlet temp.)	At all speed (at 20°C)	°C	Max. 550	600
	Exhaust gas color	At rated horsepower	%	Max. 12	22
		Quick acceleration (Low idle → high idle)		Max. 25	35
	Valve clearance	Intake valve	mm	0.25	0.152 – 0.381
		Exhaust valve	mm	0.51	0.381 – 0.762
Engine body	Compression pressure	Oil temperature: 40 – 60°C Engine speed: 250 – 280 rpm	MPa {kg/cm^2}	Min. 2.41 {Min. 24.6}	1.69 {17.2}
	Blow-by pressure	At rated horsepower	kPa {mmH_2O}	Max. 0.98 {Max. 100}	1.96 {200}
Lubrication system	Oil pressure	At rated horsepower (Oil temperature: Min. 80°C) SAE15W-40	MPa {kg/cm^2}	Min. 0.29 {Min. 3.0}	0.25 {2.5}
		At low idle (Oil temperature: Min. 80°C) SAE15W-40	MPa {kg/cm^2}	Min. 0.10 {Min. 1.0}	0.07 {0.7}
	Oil temperature	At all speed (Oil in oil pan)	°C	80 – 110	120
	Oil consumption ratio	At continuous rated (Ratio to fuel consumption)	%	Max. 0.15	0.3
Cooling system	Radiator pressure valve	Opening pressure (Differential pressure)	kPa {kg/cm^2}	—	—
	Fan speed	At rated engine speed	rpm	1,920	1,920
	Fan and alternator belt tension	Deflects when pushed with a force of 98 N {10 kg}	mm	Auto-tensioner	Auto-tensioner

Engine model				SAA6D107E-1	
Applicable machine				PC270-8	
	Item	Condition, etc	Unit	Standard value	Permissible value
Prerformance	Engine speed	High idle speed	rpm	2,180 ± 50	2,180 ± 50
		Low idle speed	rpm	1,050 ± 25	1,050 ± 25
	Necessary starting speed	At 0°C	rpm	—	—
		At –20°C	rpm	—	—
Intake and exhaust system	Intake resistance	At all speed	kPa {mmH_2O}	Max. 3.73 {Max. 380}	7.45 {760}
	Intake pressure (Air connector inlet)	At rated horsepower	kPa {mmHg}	Min. 147 {Min. 1,100}	120 {900}
	Exhaust pressure (Turbine inlet press.)	At rated horsepower	kPa {mmHg}	—	—
	Exhaust temperature (Turbine outlet temp.)	At all speed (at 20°C)	°C	Max. 550	600
	Exhaust gas color	At rated horsepower	%	Max. 12	22
		Quick acceleration (Low idle → high idle)		Max. 25	35
	Valve clearance	Intake valve	mm	0.25	0.152 – 0.381
		Exhaust valve	mm	0.51	0.381 – 0.762
Engine body	Compression pressure	Oil temperature: 40 – 60°C Engine speed: 250 – 280 rpm	MPa {kg/cm^2}	Min. 2.41 {Min. 24.6}	1.69 {17.2}
	Blow-by pressure	At rated horsepower	kPa {mmH_2O}	Max. 0.98 {Max. 100}	1.96 {200}
Lubrication system	Oil pressure	At rated horsepower (Oil temperature: Min. 80°C) SAE15W-40	MPa {kg/cm^2}	Min. 0.29 {Min. 3.0}	0.25 {2.5}
		At low idle (Oil temperature: Min. 80°C) SAE15W-40	MPa {kg/cm^2}	Min. 0.10 {Min. 1.0}	0.07 {0.7}
	Oil temperature	At all speed (Oil in oil pan)	°C	80 – 110	120
	Oil consumption ratio	At continuous rated (Ratio to fuel consumption)	%	Max. 0.15	0.3
Cooling system	Radiator pressure valve	Opening pressure (Differential pressure)	kPa {kg/cm^2}	—	—
	Fan speed	At rated engine speed	rpm	2,112	2,112
	Fan and alternator belt tension	Deflects when pushed with a force of 98 N {10 kg}	mm	Auto-tensioner	Auto-tensioner

Engine model				SAA6D107E-1	
Applicable machine				WA380-6	
	Item	Condition, etc	Unit	Standard value	Permissible value
Prerformance	Engine speed	High idle speed	rpm	2,250 ± 50	2,250 ± 50
		Low idle speed	rpm	850 ± 25	850 ± 25
	Necessary starting speed	At 0°C	rpm	—	—
		At –20°C	rpm	—	—
Intake and exhaust system	Intake resistance	At all speed	kPa {mmH2O}	Max. 3.73 {Max. 380}	7.45 {760}
	Intake pressure (Air connector inlet)	At rated horsepower	kPa {mmHg}	Min. 140 {Min. 1,050}	113 {850}
	Exhaust pressure (Turbine inlet press.)	At rated horsepower	kPa {mmHg}	—	—
	Exhaust temperature (Turbine outlet temp.)	At all speed (at 20°C)	°C	Max. 550	600
	Exhaust gas color	At rated horsepower	%	Max. 12	22
		Quick acceleration (Low idle → high idle)		Max. 25	35
	Valve clearance	Intake valve	mm	0.25	0.152 – 0.381
		Exhaust valve	mm	0.51	0.381 – 0.762
Engine body	Compression pressure	Oil temperature: 40 – 60°C Engine speed: 250 – 280 rpm	MPa {kg/cm2}	Min. 2.41 {Min. 24.6}	1.69 {17.2}
	Blow-by pressure	At rated horsepower	kPa {mmH2O}	Max. 0.98 {Max. 100}	1.96 {200}
Lubrication system	Oil pressure	At rated horsepower (Oil temperature: Min. 80°C) SAE15W-40	MPa {kg/cm2}	Min. 0.29 {Min. 3.0}	0.25 {2.5}
		At low idle (Oil temperature: Min. 80°C) SAE15W-40	MPa {kg/cm2}	Min. 0.10 {Min. 1.0}	0.07 {0.7}
	Oil temperature	At all speed (Oil in oil pan)	°C	80 – 110	120
	Oil consumption ratio	At continuous rated (Ratio to fuel consumption)	%	Max. 0.15	0.3
Cooling system	Radiator pressure valve	Opening pressure (Differential pressure)	kPa {kg/cm2}	—	—
	Fan speed	At rated engine speed	rpm	—	—
	Fan and alternator belt tension	Deflects when pushed with a force of 98 N {10 kg}	mm	Auto-tensioner	Auto-tensioner

Run-in standard and performance test criteria

Run-in standard

Engine		SAA6D107E-1				
Applicable machine		PC200, 200LC-8				
Item		Order				
		1	2	3	4	5
Running time	min	2	10	2	3	3
Engine speed	rpm	1,050	1,000	1,200	1,600	2,000
Dynamometer load	N {kg}	0 {0}	98 {10}	245 {25}	440 {45}	770 {78.6}
Output	kW {HP}	0 {0}	7.4 {10}	22 {29.5}	53 {71}	116 {155}

★ The table gives the standard values for machines without fan.
★ The loads for the dynamometer are at an arm's length of 716 mm.

Performance test criteria

Engine		SAA6D107E-1			
Applicable machine		PC200, 200LC-8			
Test item		Rated horsepower	Max. torque	High idle speed	Low idle speed
Specification value (Gross)	—	116 ± 5.81 kW/ 2,000 rpm {155 ± 8 HP/ 2,000 rpm}	$624^{+19.6}_{-39.2}$ Nm/ 1,500±100 rpm {63.6^{+2}_{-4} kgm/ 1,500 ± 100 rpm}	2,060 ± 50 rpm	1,050 ± 25 rpm
Speed	rpm	2,000	1,500 ± 100	2,060 ± 50	1,050 ± 25
Dynamometer load	N {kg}	771 {78.6}	871 {88.8}	—	—
Output (Gross)	kW {HP}	116 {155}	—	—	—
Torque (Gross)	Nm {kgm}	—	624 {63.6}	—	—
Fuel consumption	sec/ 300cc	35	—	—	—
Coolant temperature	°C	75 – 94	75 – 94	75 – 94	75 – 94
Lubricating oil temperature	°C	80 –110	80 – 110	80 – 110	80 – 110
Lubricating oil pressure	kPa {kg/cm^2}	Min. 261.8 {Min. 2.67}	Min. 193.2 {Min. 1.97}	Min. 261.8 {Min. 2.67}	Min. 54.9 {Min. 0.56}
Exhaust temperature	°C	Max. 650	Max. 650	—	—

★ This table shows the standard values using the JIS correction factor.
★ The output and torque values in the table are the standard values with the fan removed, so they are different from the specification values.
★ This table shows the standard values with an air cleaner installed, muffler installed, alternator under no load.
★ The dynamometer load shows the value for an arm length of 716 mm.
★ Use ASTM D975 diesel oil as the fuel.
★ Use SAE15W-40 as the lubricating oil.

Run-in standard

Engine		SAA6D107E-1				
Applicable machine		PC220, 220LC-8				
Item		Order				
		1	2	3	4	5
Running time	min	2	10	2	3	3
Engine speed	rpm	1,050	1,000	1,200	1,600	2,000
Dynamometer load	N {kg}	0 {0}	98 {10}	245 {25}	440 {45}	893 {91}
Output	kW {HP}	0 {0}	7.4 {10}	22 {29.5}	53 {71}	134 {180}

★ The table gives the standard values for machines without fan.
★ The loads for the dynamometer are at an arm's length of 716 mm.

Performance test criteria

Engine		SAA6D107E-1			
Applicable machine		PC220, 220LC-8			
Test item		Rated horsepower	Max. torque	High idle speed	Low idle speed
Specification value (Gross)	—	134±6.69 kW/2,000 rpm {180±9 HP/2,000 rpm}	710±34.3 Nm/1,500±100 rpm {72.4±3.5 kgm/1,500±100 rpm}	2,160 ± 50 rpm	1,050 ± 25 rpm
Speed	rpm	2,000	1,500 ± 100	2,160 ± 50	1,050 ± 25
Dynamometer load	N {kg}	892 {91.0}	991 {101.0}	—	—
Output (Gross)	kW {HP}	134 {180}	—	—	—
Torque (Gross)	Nm {kgm}	—	710 {72.4}	—	—
Fuel consumption	sec/ 300cc	29	—	—	—
Coolant temperature	°C	75 – 94	75 – 94	75 – 94	75 – 94
Lubricating oil temperature	°C	80 –110	80 – 110	80 – 110	80 – 110
Lubricating oil pressure	kPa {kg/cm^2}	Min. 261.8 {Min. 2.67}	Min. 193.2 {Min. 1.97}	Min. 261.8 {Min. 2.67}	Min. 54.9 {Min. 0.56}
Exhaust temperature	°C	Max. 650	Max. 650	—	—

★ This table shows the standard values using the JIS correction factor.
★ The output and torque values in the table are the standard values with the fan removed, so they are different from the specification values.
★ This table shows the standard values with an air cleaner installed, muffler installed, alternator under no load.
★ The dynamometer load shows the value for an arm length of 716 mm.
★ Use ASTM D975 diesel oil as the fuel.
★ Use SAE15W-40 as the lubricating oil.

Run-in standard

Engine		SAA6D107E-1				
Applicable machine		PC270-8				
Item		Order				
		1	2	3	4	5
Running time	min	2	10	2	3	3
Engine speed	rpm	1,050	1,000	1,200	1,600	2,050
Dynamometer load	N {kg}	0 {0}	98 {10}	245 {25}	440 {45}	971 {99}
Output	kW {HP}	0 {0}	7.4 {10}	22 {29.5}	53 {71}	149 {200}

★ The table gives the standard values for machines without fan.
★ The loads for the dynamometer are at an arm's length of 716 mm.

Performance test criteria

Engine		SAA6D107E-1			
Applicable machine		PC270-8			
Test item		Rated horsepower	Max. torque	High idle speed	Low idle speed
Specification value (Gross)	—	149±7.43 kW/2,050 rpm {200±10 HP/2,050 rpm}	771±38.2 Nm/1,500±100 rpm {78.6±3.9 kgm/1,500±100 rpm}	2,180 ± 50 rpm	1,050 ± 25 rpm
Speed	rpm	2,050	1,500 ± 100	2,180 ± 50	1,050 ± 25
Dynamometer load	N {kg}	971 {99}	1,077 {109.8}	—	—
Output (Gross)	kW {HP}	149 {200}	—	—	—
Torque (Gross)	Nm {kgm}	—	771 {78.6}	—	—
Fuel consumption	sec/ 300cc	26	—	—	—
Coolant temperature	°C	75 – 94	75 – 94	75 – 94	75 – 94
Lubricating oil temperature	°C	80 –110	80 – 110	80 – 110	80 – 110
Lubricating oil pressure	kPa {kg/cm^2}	Min. 261.8 {Min. 2.67}	Min. 193.2 {Min. 1.97}	Min. 261.8 {Min. 2.67}	Min. 54.9 {Min. 0.56}
Exhaust temperature	°C	Max. 650	Max. 650	—	—

★ This table shows the standard values using the JIS correction factor.
★ The output and torque values in the table are the standard values with the fan removed, so they are different from the specification values.
★ This table shows the standard values with an air cleaner installed, muffler installed, alternator under no load.
★ The dynamometer load shows the value for an arm length of 716 mm.
★ Use ASTM D975 diesel oil as the fuel.
★ Use SAE15W-40 as the lubricating oil.

Run-in standard

Engine		SAA6D107E-1				
Applicable machine		WA380-6				
Item		Order				
		1	2	3	4	5
Running time	min	2	10	2	3	3
Engine speed	rpm	1,050	1,000	1,200	1,600	2,100
Dynamometer load	N {kg}	0 {0}	98 {10}	245 {25}	440 {45}	911 {92.9}
Output	kW {HP}	0 {0}	7.4 {10}	22 {29.5}	53 {71}	143.4 {192}

★ The table gives the standard values for machines without fan.
★ The loads for the dynamometer are at an arm's length of 716 mm.

Performance test criteria

Engine		SAA6D107E-1			
Applicable machine		WA380-6			
Test item		Rated horsepower	Max. torque	High idle speed	Low idle speed
Specification value (Gross)	—	143.4 ± 7.2 kW/2,100 rpm {192 ± 9.7 HP/2,100 rpm}	941 ± 47 Nm/1,450 ± 100 rpm {96 ± 4.8 kgm/1,450 ± 100 rpm}	2,250 ± 50 rpm	850 ± 25 rpm
Speed	rpm	2,100	1,450 ± 100	2,250 ± 50	850 ± 25
Dynamometer load	N {kg}	911 {92.9}	1,315 {134.1}	—	—
Output (Gross)	kW {HP}	143.4 {192}	—	—	—
Torque (Gross)	Nm {kgm}	—	94 {96}	—	—
Fuel consumption	sec/ 300cc	26	—	—	—
Coolant temperature	°C	75 – 94	75 – 94	75 – 94	75 – 94
Lubricating oil temperature	°C	80 –110	80 – 110	80 – 110	80 – 110
Lubricating oil pressure	kPa {kg/cm^2}	Min. 29.4 {Min. 3.0}	Min. 193.2 {Min. 1.97}	Min. 294 {Min. 3.0}	Min. 98 {Min. 1.0}
Exhaust temperature	°C	Max. 700	Max. 700	—	—

★ This table shows the standard values using the JIS correction factor.
★ The output and torque values in the table are the standard values with the fan removed, so they are different from the specification values.
★ This table shows the standard values with an air cleaner installed, muffler installed, alternator under no load.
★ The dynamometer load shows the value for an arm length of 716 mm.
★ Use ASTM D975 diesel oil as the fuel.
★ Use SAE15W-40 as the lubricating oil.

Engine

107E-1 Series

30 Testing and adjusting

Testing and adjusting

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Testing and adjusting

Testing and adjusting tools list

Testing and adjusting item	Symbol		Part No.	Part name	Q'ty	Remarks
Testing air boost pressure	A		799-201-2202	Boost gauge kit	1	–101 – 200 kPa {–760 – 1,500 mmHg}
			799-401-2220	Hose	1	I-coupler type (if necessary)
Adjusting valve clearance	C	1	795-799-1131	Gear	1	
		2	Commercially available	Clearance gauge	1	
Testing compression pressure	D	1	795-799-6700	Puller	1	
		2	795-502-1590	Gauge assembly	1	
		3	795-790-4410	Adapter	1	
		4	6754-11-3130	Gasket	1	
Testing blow-by pressure	E		799-201-1504	Blow-by checker	1	0 – 5 kPa {0 – 500 mmH_2O}
Testing oil pressure	F	1	799-101-5002	Hydraulic tester	1	Pressure gauge: 2.5, 5.9, 39.2, 58.8 MPa {25, 60, 400, 600 kg/cm^2}
		2	790-301-1220	Nipple	1	
		3	799-101-5160	Nipple	1	
		4	799-401-2320	Gauge	1	Pressure gauge: 0.98 MPa {10 kg/cm^2}
Testing fuel pressure	G	1	799-101-5002	Hydraulic tester	1	Pressure gauge: 2.5, 5.9, 39.2, 58.8 MPa {25, 60, 400, 600 kg/cm^2}
			790-261-1204	Digital hydraulic tester	1	Pressure gauge: 58.8 MPa {600 kg/cm^2}
		2	6732-81-3170	Adapter	1	10 × 1.0 mm → PT1/8
			6215-81-9710	O-ring	1	
		3	799-401-2320	Hydraulic tester	1	Pressure gauge: 2.58 MPa {25 kg/cm^2}
Testing leakage from pressure limiter and return rate from injector	H	1		Supply pump testing tool	1	
		2		Pressure limiter testing tool	1	
		3		Injector testing tool	1	
		4	Commercially available	Measuring cylinder	1	

Testing air boost pressure

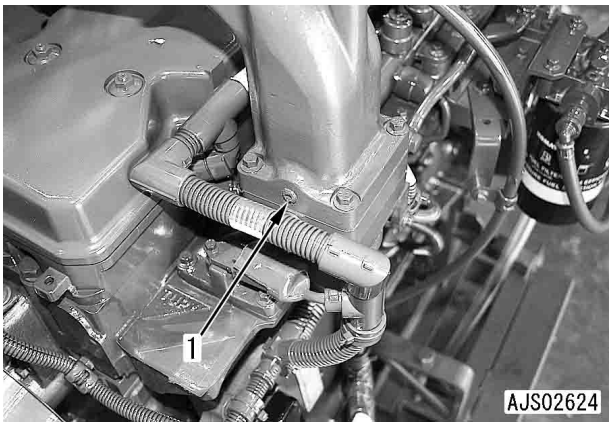
★ Testing tools for air boost pressure

Symbol	Part No.	Part name
A	799-201-2202	Boost gauge kit
	799-401-2220	Hose

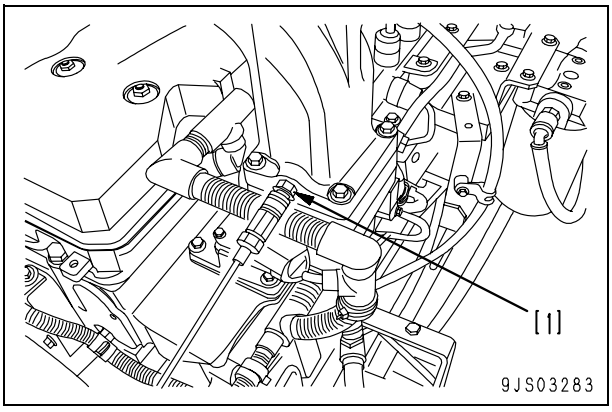
⚠ **Be careful not to touch any hot part when removing or installing the testing tools.**

★ The test point is subject to machine models.

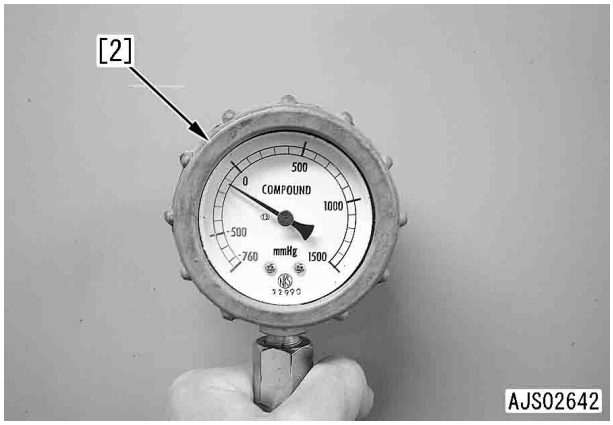
1. Remove air boost pressure pickup plug (1).



2. Install nipple [1] of boost gauge kit **A** and connect it to gauge [2].



3. Run the engine at the rated output and test the air boost pressure.
   ★ When testing with the engine mounted on the machine, test on the condition described in the shop manual for the machine.



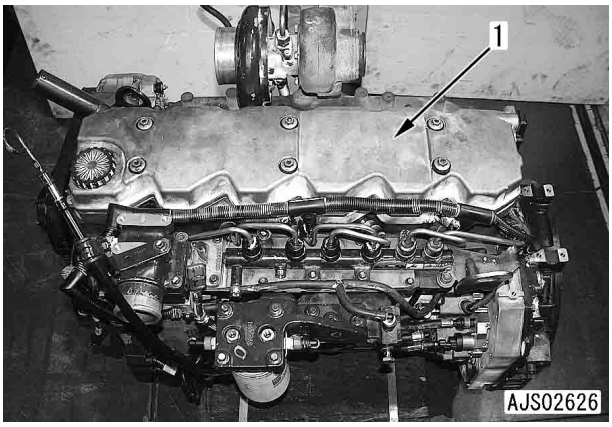
4. After finishing testing, remove the measuring tools and return the removed parts.

Adjusting valve clearance

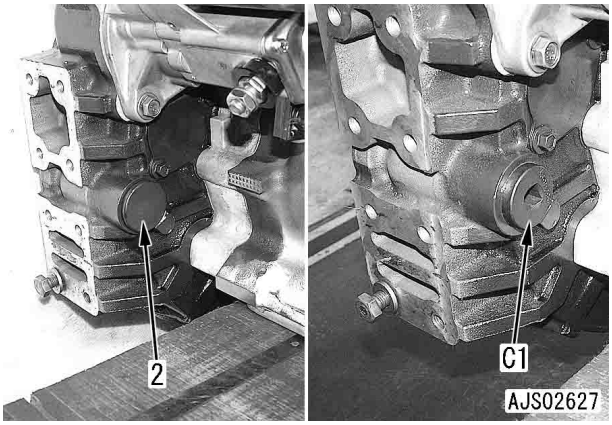
★ Adjusting tools for valve clearance

Symbol		Part No.	Part name
C	1	795-799-1131	Gear
	2	Commercially available	Clearance gauge

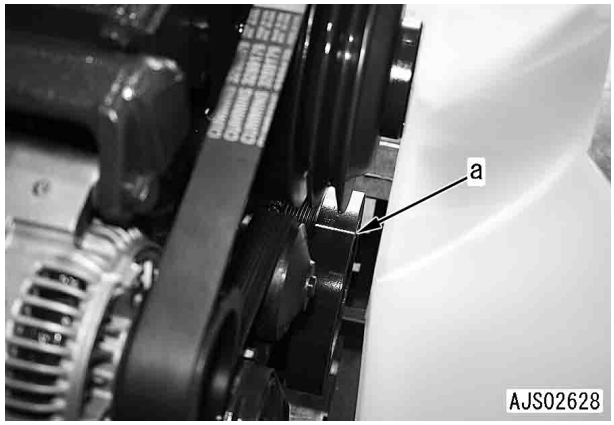
1. Remove cylinder head cover (1).



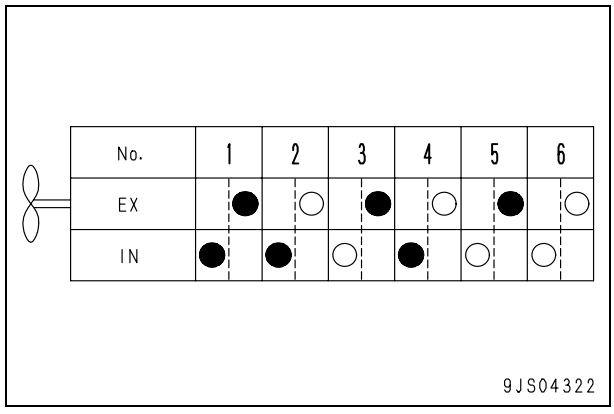
2. Remove cap (2) of flywheel housing and install gear **C1**.



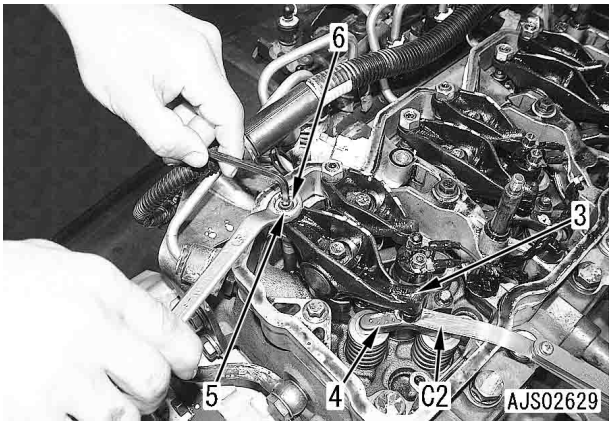
3. Rotate the crankshaft forward using gear **C1** until the stamped "1.6TOP" line (**a**) of the crankshaft pulley faces upward and set the No. 1 cylinder to the compression top dead center.
   ★ When the No. 1 cylinder is at the compression top dead center, the rocker arm of the No. 1 cylinder can be moved by the valve clearance with the hand. If the rocker arm cannot be moved, rotate the crankshaft one more turn. (Alternatively, adjust the valve clearance indicated by ○ of the valve layout plan in the next section without rotating the crankshaft.)



4. Adjust the valve clearance of No. 1 cylinder according to the following procedure.
   ★ When the No. 1 cylinder is at the compression top dead center, the valve clearance indicated by ● of the valve layout plan can be adjusted.
   ● Valve layout plan



   1) Insert clearance gauge **C2** into the clearance between rocker arm (3) and crosshead (4).
   2) Loosen locknut (5) and adjust valve clearance using adjustment screw (6).
      ★ With the filler gauge inserted, turn the adjustment screw to a degree that you can move the filler gauge lightly.
   3) Fix adjustment screw (6) and tighten locknut (5).
      Locknut:
      24 ± 4 Nm {2.45 ± 0.41 kgm}
      ★ After tightening the locknut, check the valve clearance again.



5. In the same manner as step 3, rotate the crankshaft forward using gear **C1** until the stamped “1.6TOP“ line (**a**) of the crankshaft pulley faces upward.

6. Adjust the valve clearance of No. 6 cylinder.
   ★ The adjustment method is the same as step 4.
   ★ When the No. 6 cylinder is at the compression top dead center, the valve clearance indicated by ○ of the valve layout plan can be adjusted.

7. After finishing adjustment, return the removed parts.
   Cylinder head cover mounting bolt:
   24 ± 4 Nm {2.45 ± 0.41 kgm}

Testing compression pressure

★ Testing tools for compression pressure

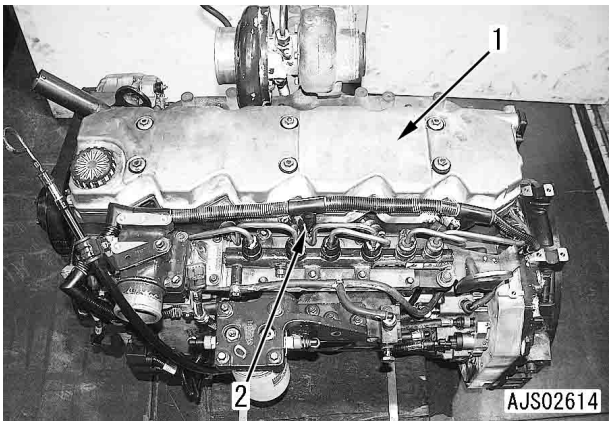
Symbol		Part No.	Part name
D	1	795-799-6700	Puller
	2	795-502-1590	Gauge assembly
	3	795-790-4410	Adapter
	4	6754-11-3130	Gasket

⚠ **When measuring the compression pressure, take care not to burn yourself on the exhaust manifold, muffler, etc. or get caught in a rotating part.**

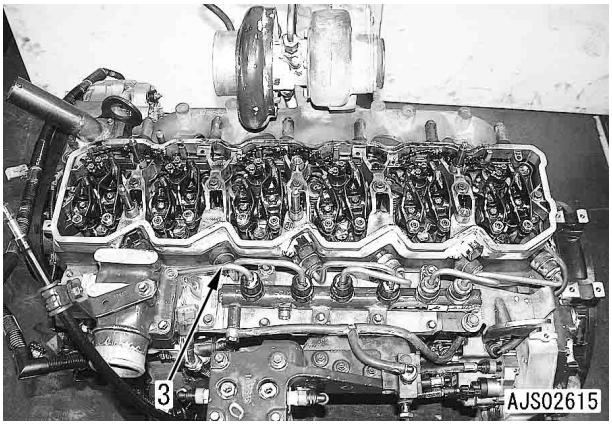
★ Measure the compression pressure after the engine is warmed up.
(Engine oil temperature: 40 – 60°C).

1. Remove cylinder head cover (1).

2. Disconnect injector harness connectors (2) (at 6 places) and move the harnesses.



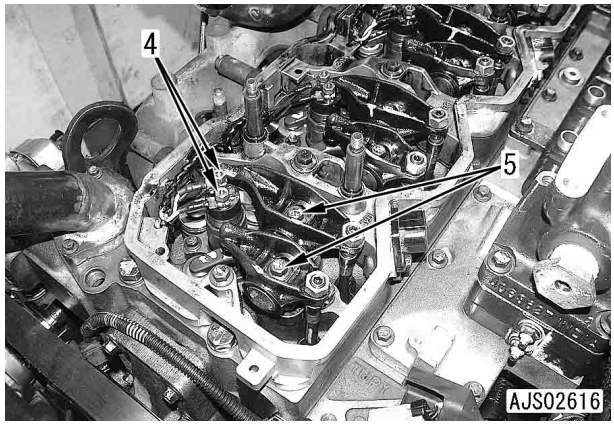
3. Disconnect fuel high-pressure tube (3).



4. Loosen injector terminal nut (4) and remove the terminal from the injector.

5. Move the cylinder to be tested to the compression top dead center.
★ See “Adjusting valve clearance”.

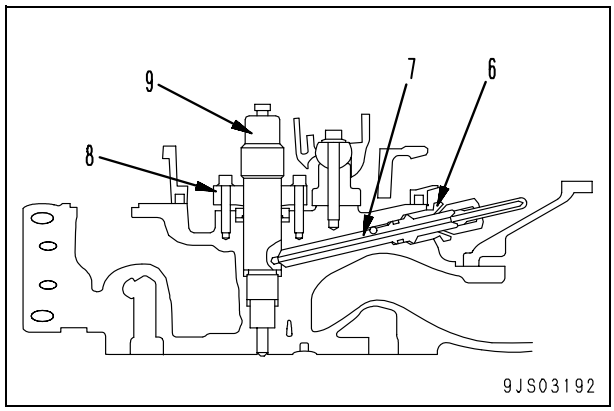
6. Remove rocker arm assembly (5).

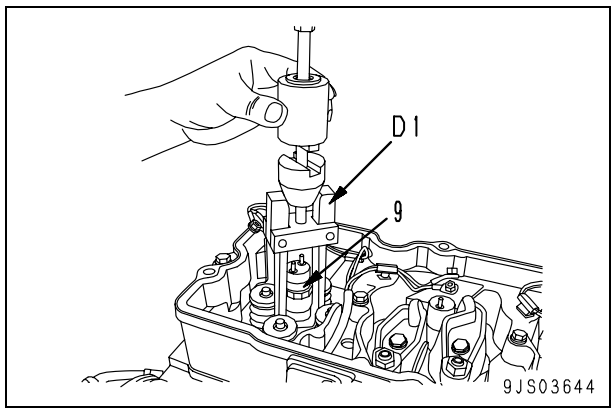


7. Remove retainer (6) and disconnect fuel inlet connector (7).

8. Remove holder (8).

9. Remove injector (9).
★ Remove the injector using puller **D1** and the impact of slide hammer.
★ Do not unclench the upper part of the injector.





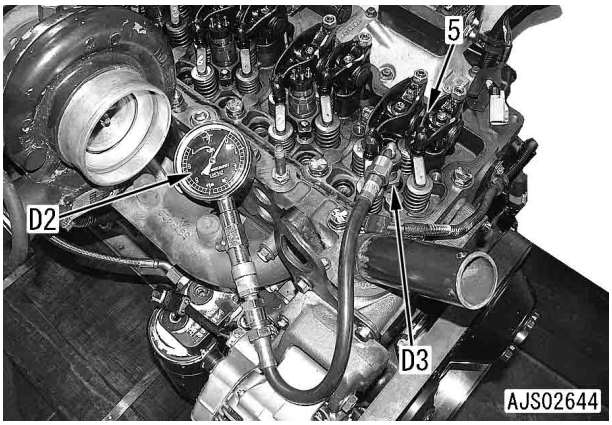
10. Install gasket **D4** to the tip of adapter **D3** and connect it to the injector mount.

11. Fix adapter **D3** with the injector holder.
    ★ Tighten the holder mounting bolts alternately.
    Holder mounting bolt:
    8 ± 0.8 Nm {0.8 ± 0.08 kgm}

12. Connect gauge assembly **D2** to adaptor **D3**.
    ★ Apply a little amount of engine oil to the connecting parts of the adapter and gauge so that air will not leak easily.

13. Install rocker arm assembly (5).
    Rocker arm assembly mounting bolt:
    36 ± 5 Nm {3.67 ± 0.51 kgm}

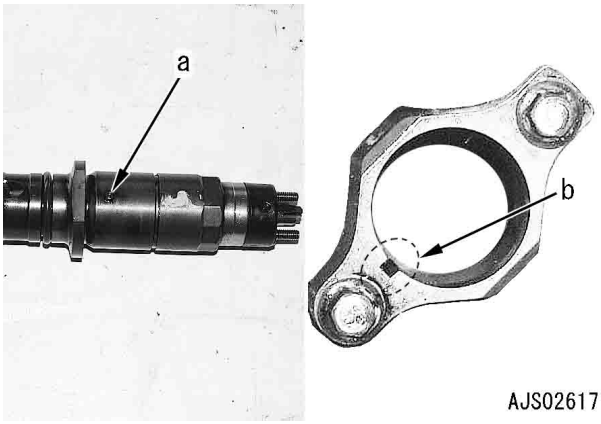
14. Adjust valve clearance.
    ★ See “Adjusting valve clearance”.



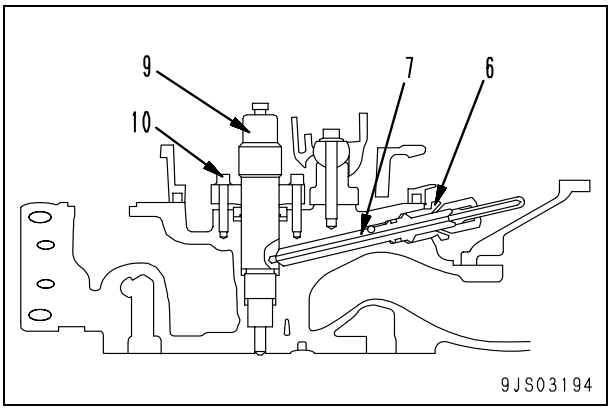
15. Connect INSITE (troubleshooting kit) to the engine and set the engine in the no-injection cranking mode.
    ⚠ **If the engine is not set in the no-injection cranking mode, it will start during testing and will be dangerous. Accordingly, be sure to set the engine in this mode.**

16. Rotate the engine with the starting motor and measure the compression pressure.
    ★ Read the pressure gauge pointer when it is stabilized.

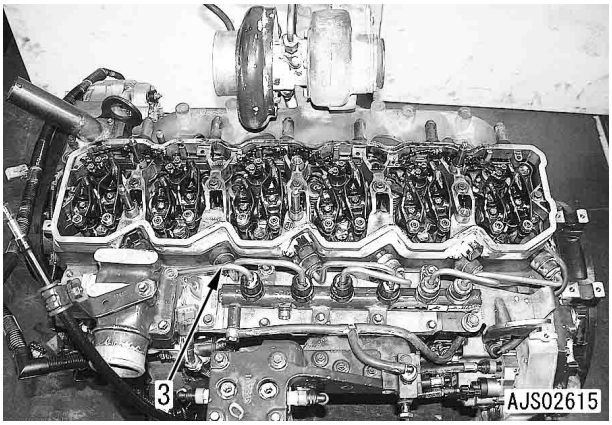
17. After finishing testing, remove the testing tools and return the removed parts.
    ★ Install the injector and fuel high-pressure tube according to the following procedure.
    1) Mate protrusion (**a**) of the injector with notch (**b**) of the holder, and set the injector to the cylinder head.
       ★ Set the injector with the above mating position toward the fuel inlet connector insertion side.



2) Tighten injector holder mounting bolt (10) 3 to 4 threads.
3) Set fuel inlet connector (7).
   ★ Press it in until you feel the O-ring clicks in.
4) Alternately tighten injector holder mounting bolt (10).
   Holder mounting bolt:
   8 ± 0.8 Nm {0.8 ± 0.08 kgm}
5) Tighten retainer (6) of the fuel inlet connector.
   Retainer: **50 ± 5 Nm {5.1 ± 0.5 kgm}**



6) Tighten the sleeve nut of the fuel high-pressure tube in order of the head side and common rail side.
   Sleeve nut:
   35 ± 3.9 Nm {3.6 ± 0.4 kgm}



★ After installing the rocker arm assembly, adjust the valve clearance. For details, see “Adjusting valve clearance”.
Rocker arm assembly mounting bolt:
36 ± 5 Nm {3.67 ± 0.51 kgm}

★ Tighten the injector terminal nut with the following torque.
Terminal nut:
1.5 ± 0.25 Nm {0.15 ± 0.025 kgm}

★ Tighten the head cover with the following torque.
Head cover mounting nut:
24 ± 4 Nm {2.45 ± 0.41 kgm}

Testing blow-by pressure

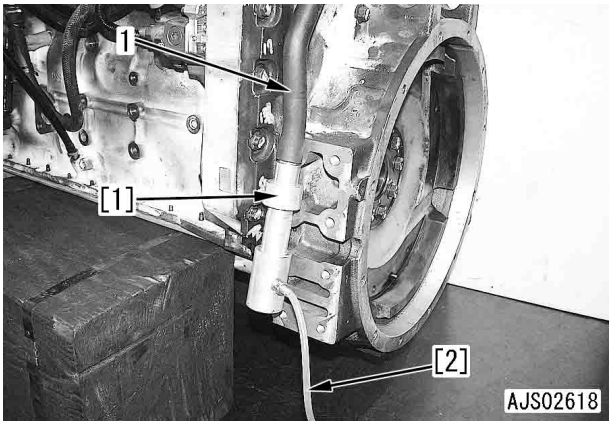
★ Testing tool for blow-by pressure

Symbol	Part No.	Part name
E	799-201-1504	Blow-by checker

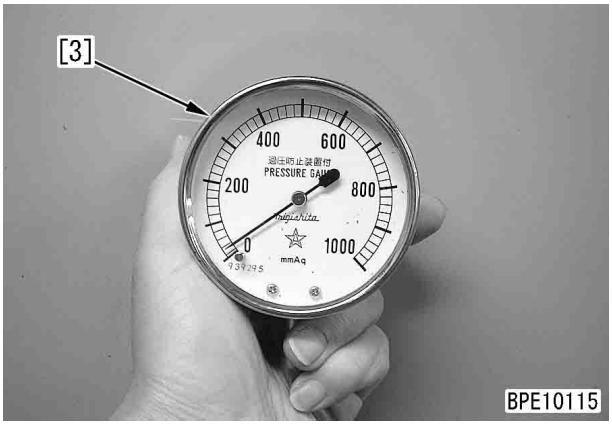
★ The test point is subject to machine models.

1. Install adaptor [1] of blow-by checker **E** to the tip of blow-by hose (1).

2. Connect hose [2], and then connect it to adaptor [1] and gauge [3].



3. Run the engine at the rated output and test the blow-by pressure.
   ★ When testing with the engine mounted on the machine, test on the condition described in the shop manual for the machine.



4. After finishing testing, remove the measuring tools and return the removed parts.

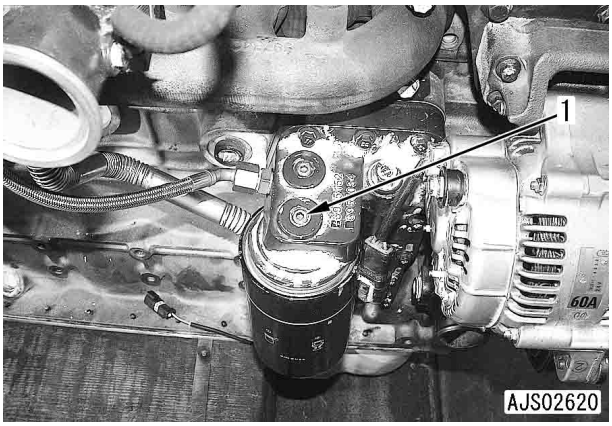
Testing oil pressure

★ Testing tools for oil pressure

Symbol		Part No.	Part name
F	1	799-101-5002	Hydraulic tester
	2	790-301-1220	Nipple
	3	799-101-5160	Nipple
	4	799-401-2320	Gauge

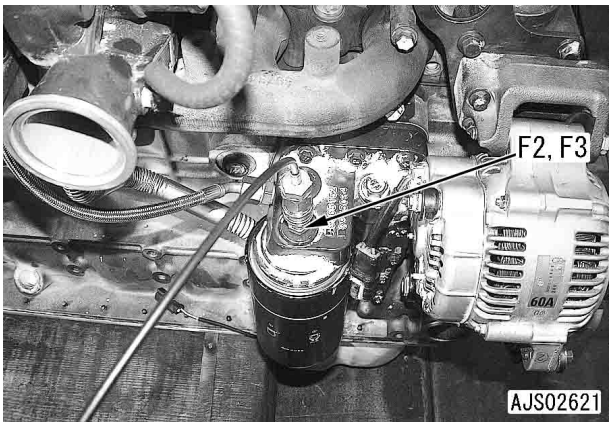
★ The test point is subject to machine models.

1. Remove oil pressure pickup plug (1).

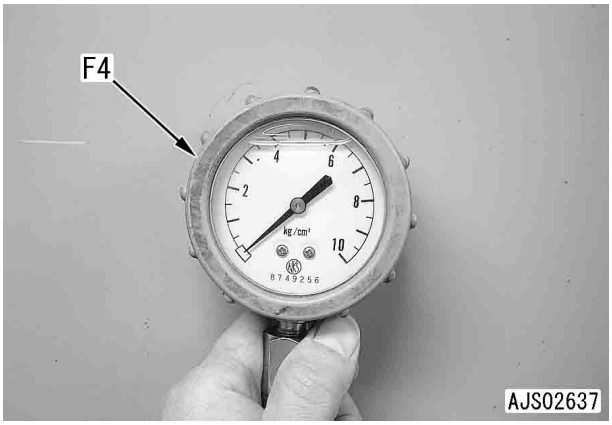


2. Install the nipples to the plug mount in order of **F2** and **F3**.

3. Connect the hose of the hydraulic tester **F1** to the nipples and gauge **F4**.



4. Run the engine at the rated output and low idle and test the oil pressure.
   ★ When testing with the engine mounted on the machine, test on the condition described in the shop manual for the machine.



5. After finishing testing, remove the measuring tools and return the removed parts.

Handling fuel system parts

★ Precautions for testing and maintaining fuel system
The common rail fuel injection system (CRI) consists of more precise parts than the conventional fuel injection pump and nozzle. If foreign matter enters this system, it can cause a trouble.
When testing and maintaining the fuel system, take care more than the past. If dust, etc. sticks to any part, wash that part thoroughly with clean fuel.

★ Precautions for replacing fuel filter cartridge
Be sure to use the Komatsu genuine fuel filter cartridge.
Since the common rail fuel injection system (CIS) consists of more precise parts than the conventional fuel injection pump and nozzle, it employs a high-efficiency special filter to prevent foreign matter from entering it. If a filter other than the genuine one is used, the fuel system may have a trouble. Accordingly, never use such a filter.

Releasing residual pressure in fuel system

★ Pressure is generated in the low-pressure circuit and high-pressure circuit of the fuel system while the engine is running.
Low-pressure circuit:
Feed pump – Fuel filter – Fuel supply pump
High-pressure circuit:
Fuel supply pump – Common rail – Fuel injector

★ The pressure in both low-pressure circuit and high-pressure circuit lowers to a safety level automatically 30 seconds after the engine is stopped.

★ Before the fuel circuit is tested and its parts are removed, the residual pressure in the fuel circuit must be released completely. Accordingly, observe the following.

⚠ **Before testing the fuel system or removing its parts, wait at least 30 seconds after stopping the engine until the residual pressure in the fuel circuit is released. (Do not start the work just after stopping the engine since there is residual pressure.)**

Testing fuel pressure

★ Testing tools for fuel pressure

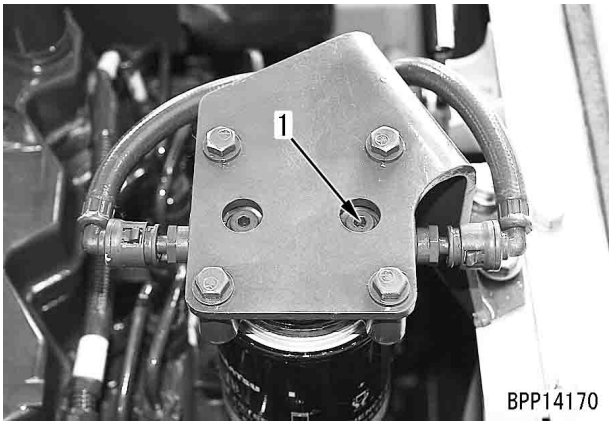
Symbol		Part No.	Part name
G	1	799-101-5002	Hydraulic tester
		790-261-1204	Digital hydraulic tester
	2	6732-81-3170	Adapter (10 × 1.0 mm → PT1/8)
		6215-81-9710	O-ring
	3	799-401-2320	Hydraulic tester

★ Test only the fuel pressure in the low-pressure circuit from the feed pump through the fuel main filter to the supply pump and in the return circuit from the supply pump through the common rail and the injector to the fuel tank.

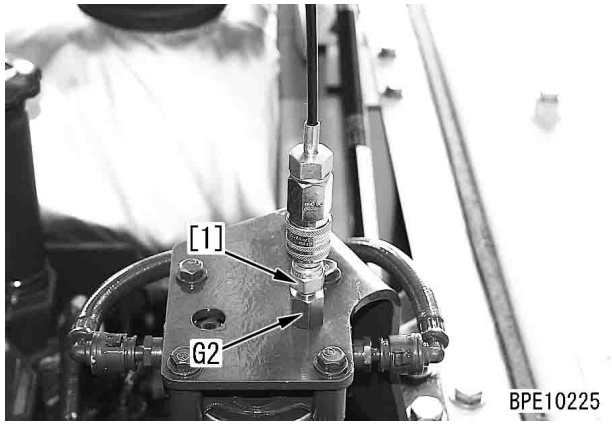
⚠ **Since the pressure in the high-pressure circuit from the fuel supply pump through the common rail to the injector is very high, it cannot be measured.**

1. Testing fuel low-pressure circuit pressure

1) Remove fuel pressure pickup plug (1) from the fuel main filter.



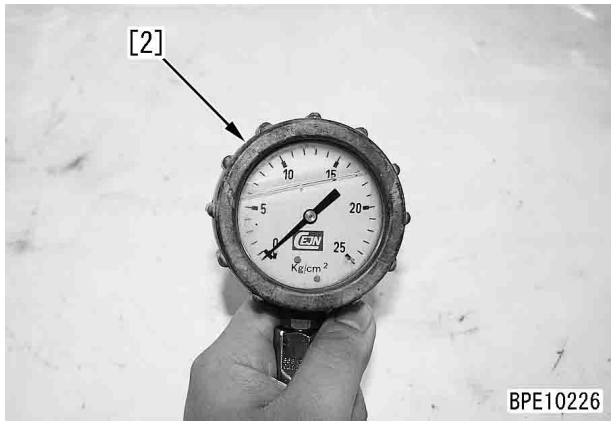
2) Install adapter **G2** and nipple [1] of hydraulic tester **G1** and connect it to oil pressure gauge [2].
   ★ Use an oil pressure gauge of 2.5 MPa {25 kg/cm^2}.



3) Run the engine at low idle and test the fuel low-pressure circuit pressure.
   ★ If the fuel low-pressure circuit pressure is in the following range, it is normal.

Low idle	0.5 – 1.3 MPa {5.1 – 13.3 kg/cm^2}
Cranking	0.3 – 1.1 MPa {3.1 – 11.3 kg/cm^2}

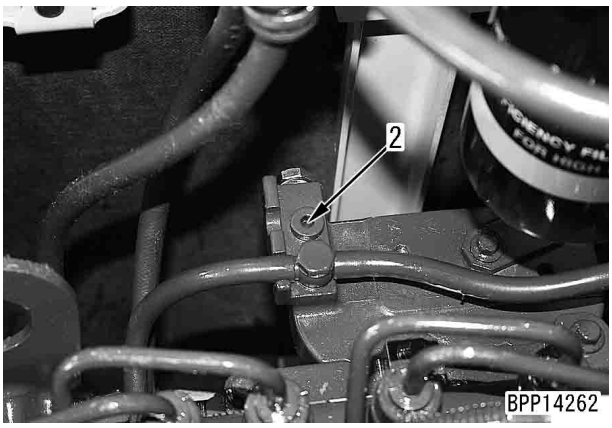
⚠ **When the engine cannot run, you may test it by cranking the engine with the starting motor. To protect the starting motor, do not continuously crank the engine 20 seconds or more.**



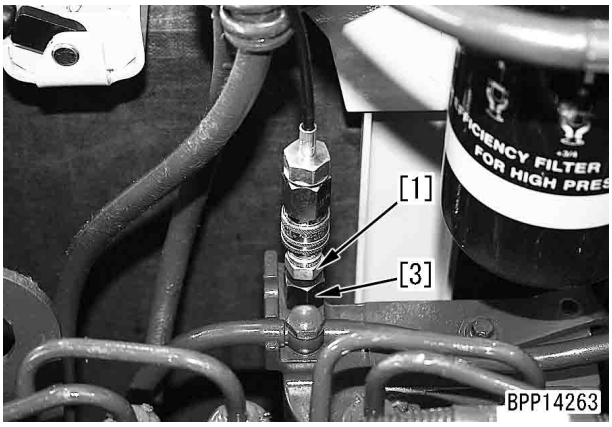
4) After finishing testing, remove the testing tools and return the removed parts.
   Fuel pressure pickup plug:
   10 ± 2 Nm {1 ± 0.2 kgm}

2. Check of fuel circuit pressure

1) Remove fuel pressure pickup plug (2) from the fuel return block.



2) Connect nipples [3] and [1] of hydraulic tester **G1** there and connect hydraulic tester **G3**.
   ★ Nipple [3]: 790-301-1181, 07002-11223



3) Run the engine at low idle and test the fuel return circuit pressure.
   ★ If the fuel return circuit pressure is in the following range, it is normal.

Low idle	Max. 0.02 MPa {Max. 0.19 kg/cm^2}
Cranking	

⚠ **When the engine cannot run, you may test it by cranking the engine with the starting motor. To protect the starting motor, do not continuously crank the engine 20 seconds or more.**

4) After finishing testing, remove the testing tools and return the removed parts.
   Fuel pressure pickup plug:
   24 ± 4 Nm {2.4 ± 0.4 kgm}

Reduced cylinder mode operation

★ Reduced cylinder mode operation means to run the engine with the fuel injectors of 1 or more cylinders disabled electrically to reduce the number of effective cylinders. The purposes and effects of this operation are as follows.

1. This operation is used to find out a cylinder which does not output power normally (or, combustion in it is abnormal).
2. When a cylinder is selected for the reduced cylinder mode operation, if the engine speed and output do not change from the normal operation (all-cylinder operation), that cylinder has 1 or more defects.
   The possible defects are as follows.
   - Leakage through cylinder head gasket
   - Defective injection
   - Defective piston or piston ring
   - Defective valve mechanism (Moving valve system)
   - Defect in electrical system
3. Since the common rail fuel injection system controls the injector of each cylinder electronically, the operator can perform the reduced cylinder mode operation easily with switches to find out a defective cylinder.
   ★ Perform the reduced cylinder mode operation with the engine mounted on the machine or connected to INSITE (troubleshooting kit).
   Before performing, see Testing method in the shop manual for the machine or INSITE (troubleshooting kit).

No-injection cranking

★ No-injection cranking means to crank the engine with the starting motor while all the injections are stopped electrically. The purpose and effect of this operation are as follows. Before the engine is started after it or the machine has been stored for a long period, the no-injection cranking is performed to lubricate the engine parts and protect them from seizure.

★ Perform the no-injection cranking mode operation with the engine mounted on the machine or connected to INSITE (troubleshooting kit).
Before performing, see Testing method in the shop manual for the machine or INSITE (troubleshooting kit).

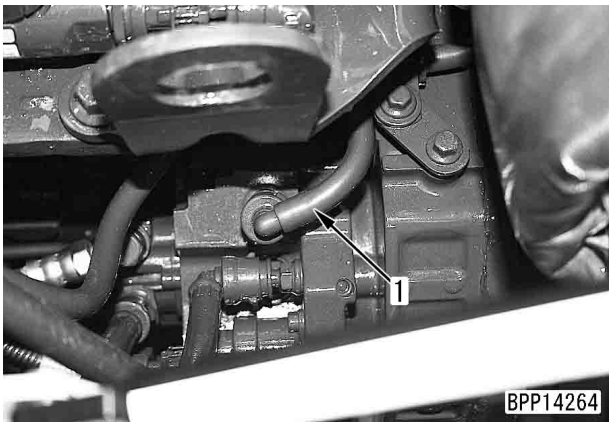
Testing leakage from pressure limiter and return rate from injector

★ Testing tools for leakage from pressure limiter and return rate from injector

Symbol		Part No.	Part name
H	1		Supply pump testing tool
	2		Pressure limiter testing tool
	3		Injector testing tool
	4	Commercially available	Measuring cylinder

1. Testing supply pump return rate

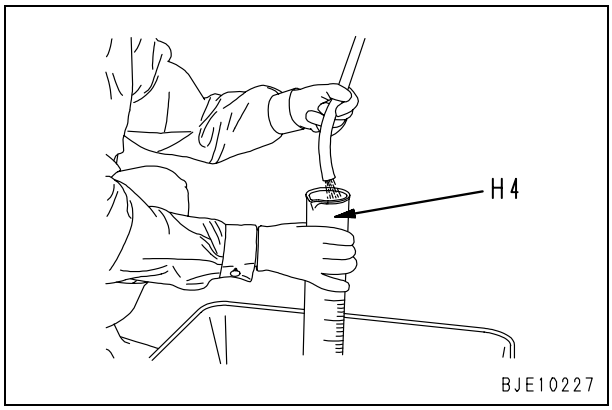
1) Disconnect return hose (1) of the supply pump.



2) Connect testing tool **H1** to the supply pump side, and insert its tip into measuring cylinder **H4**.
3) Connect blocking tool **H1** to the return hose side to block fuel leakage.
4) Run the engine at low idle and test the return rate from the supply pump.
   ★ If the return rate from the supply pump is in the following range, it is normal.

Low idle	Max. 1,000 cc/min.
Cranking	Max. 140 cc/min.

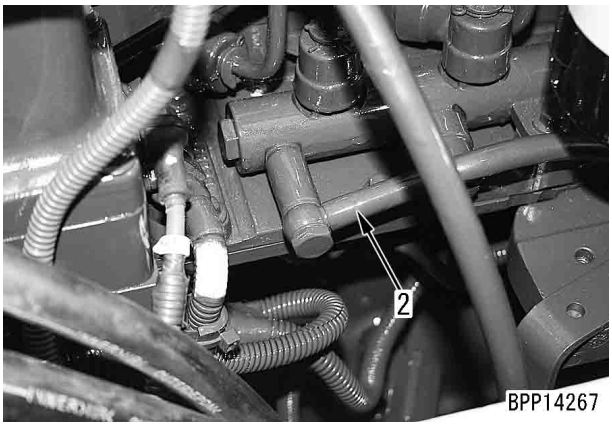
⚠ **When the engine cannot run, you may test it by cranking the engine with the starting motor. To protect the starting motor, do not continuously crank the engine 20 seconds or more.**



5) After finishing testing, remove the testing tools and return the removed parts.

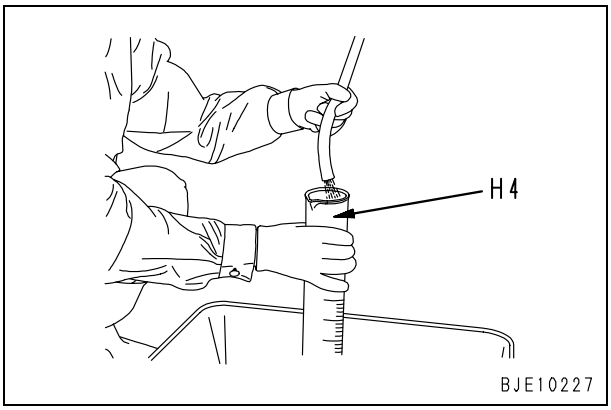
2. Testing leakage from pressure limiter

1) Disconnect return hose (2) of the pressure limiter.



2) Connect testing tool **H2** to the pressure limiter side, and insert its tip into measuring cylinder **H4**.
3) Connect blocking tool **H2** to the return hose side to block fuel leakage.
4) Run the engine at low idle and test the return rate from the pressure limiter.
   ★ If the leakage from the pressure limiter is in the following range, it is normal.

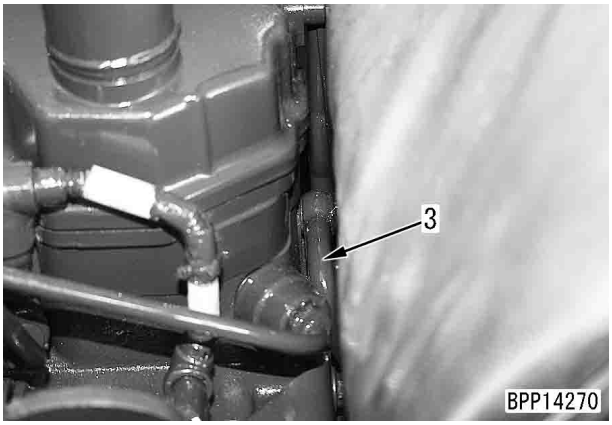
Low idle	0 cc (No leakage)



5) After finishing testing, remove the testing tools and return the removed parts.
   Joint bolt:
   24 ± 4 Nm {2.4 ± 0.4 kgm}

3. Testing return rate from injector

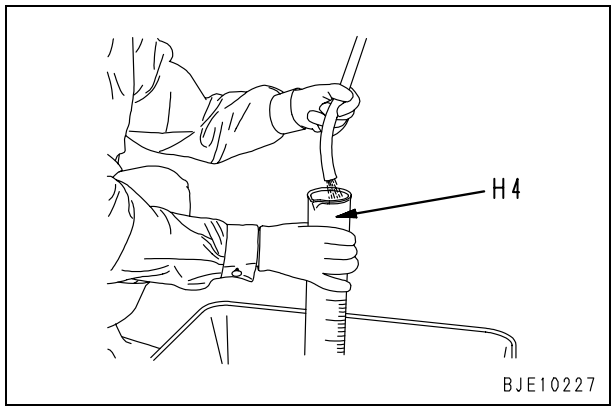
1) Disconnect return hose (3) from the cylinder head.



2) Connect testing tool **H3** to the cylinder head side, and insert its tip into measuring cylinder **H4**.
3) Connect blocking tool **H3** to the return hose side to block fuel leakage.
4) Run the engine at low idle and test the return rate from the injector.
   ★ If the return rate from the injector is in the following range, it is normal.

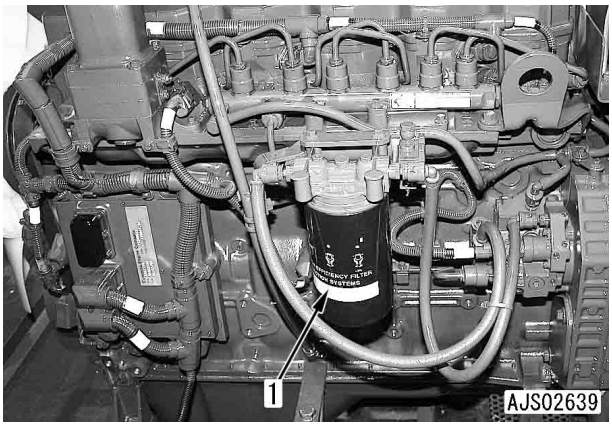
Low idle	Max. 180 cc/min.
Cranking	Max. 90 cc/min.

⚠ When the engine cannot run, you may test it by cranking the engine with the starting motor. To protect the starting motor, do not continuously crank the engine 20 seconds or more.



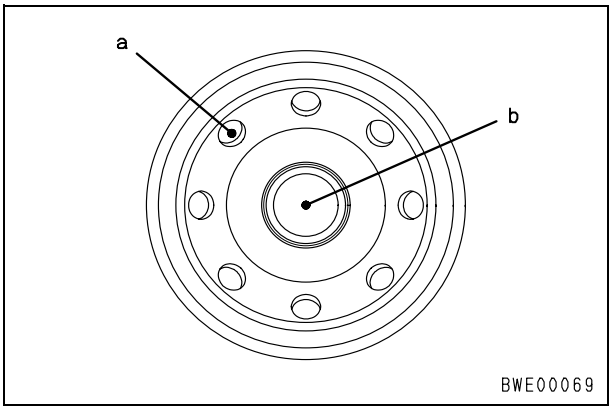
5) After finishing testing, remove the testing tools and return the removed parts.
   Joint bolt:
   24 ± 4 Nm {2.4 ± 0.4 kgm}

Bleeding air from fuel circuit



★ If fuel is used up or if a fuel circuit part is removed and installed, bleed air from the fuel circuit according to the following procedure.
★ When the priming pump and fuel filter is remote mounted, bleed air by operating the priming pump at the remote side.

1. Remove fuel filter (1), fill it with fuel, and install it back.
   ★ Fill the fuel filter with clean fuel and take care that dirt will not enter it.
   ★ Fill fuel through inlet port (a) of the filter (at 8 places). Do not fill fuel through (b), which is an outlet port (clean side) after filtration.
   ★ If clean fuel is not available, do not remove the filter but fill it with the fuel by operating the priming pump.



2. Loosen the knob of the priming pump (remote mount side) and move it back and forth.
   ★ Continue moving the knob until it becomes hard to move.
3. Start the engine with the starting motor.
   ★ The air in the high-pressure circuit is bled automatically if the engine is cranked.
   ★ If the engine does not start, there may be still air in the low-pressure circuit. In this case, repeat the above procedure from step 2.

Testing fuel system for leakage

⚠ **Very high pressure is generated in the high-pressure circuit of the fuel system. If fuel leaks while the engine is running, it is dangerous since it can catch fire.**
After testing the fuel system or removing its parts, test it for fuel leakage according to the following procedure.

★ Clean and degrease the engine and the parts around it in advance so that you can test it easily for fuel leakage.

1. Spray color checker (developer) over the fuel supply pump, common rail, fuel injector, and joints of the high-pressure piping.

2. Run the engine at speed below 1,000 rpm and stop it after its speed is stabilized.

3. Inspect the fuel piping and devices for fuel leakage.
   ★ Check mainly around the high-pressure circuit parts coated with the color checker for fuel leakage.
   ★ If any fuel leakage is detected, repair it and inspect again from step 2.

4. Run the engine at low idle.

5. Inspect the fuel piping and devices for fuel leakage.
   ★ Check mainly around the high-pressure circuit parts coated with the color checker for fuel leakage.
   ★ If any fuel leakage is detected, repair it and inspect again from step 2.

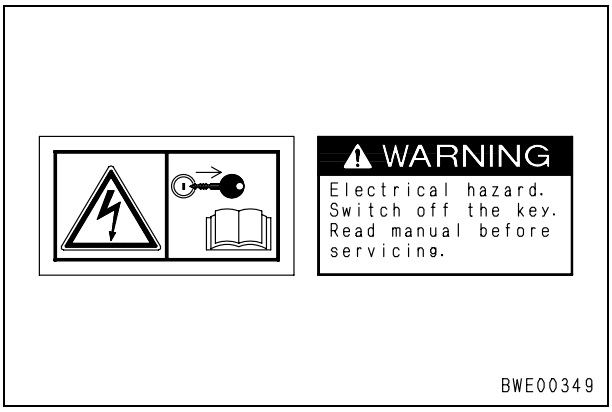
6. Run the engine at high idle.

7. Inspect the fuel piping and devices for fuel leakage.
   ★ Check mainly around the high-pressure circuit parts coated with the color checker for fuel leakage.
   ★ If any fuel leakage is detected, repair it and inspect again from step 2.

8. Run the engine at high idle and load it.
   ★ When checking while the components to be checked are mounted on the machine, stall the torque converter or relieve the hydraulic pump.

9. Inspect the fuel piping and devices for fuel leakage.
   ★ Check mainly around the high-pressure circuit parts coated with the color checker for fuel leakage.
   ★ If any fuel leakage is detected, repair it and inspect again from step 2.
   ★ If no fuel leakage is detected, check is completed.

Handling controller high-voltage circuit



⚠ **The engine controller uses a high-voltage circuit to drive the fuel injector.**
Accordingly, the high-voltage circuit is connected to the wiring harnesses and connectors between the engine controller and injector.

★ Normally, the engine controller keeps outputting the high voltage to the injector only while the engine is running and stops outputting when the engine stops.

⚠ **If you touch the high-voltage circuit directly, you may get an electric shock. To avoid this, observe the following precautions when testing.**

1. The following connectors are used in the high-voltage circuit.
   - Engine controller connector: **J1**
   - Injector intermediate connector:
     INJ CYL 1&2, INJ CYL 3&4, INJ CYL 4&5
   - Injector head terminal (in head cover)

2. When disconnecting or connecting a connector related to the high-voltage circuit, be sure to turn the starting switch OFF.

3. If a T-adapter is inserted in or connected to a connector related to the high-voltage circuit for troubleshooting, do not start the engine.
   ★ You may turn the starting switch to the OFF or ON position but must not turn it to the START position.

Engine

107E-1 Series

40 Troubleshooting

General information on troubleshooting

General information on troubleshooting 2
Points on troubleshooting 2
Error and failure code table 3

General information on troubleshooting

Points on troubleshooting

- ⚠ **Stop the machine in a level place, and check that the safety pin, blocks, and parking brake are securely fitted.**
- ⚠ **When carrying out the operation with 2 or more workers, keep strictly to the agreed signals, and do not allow any unauthorized person to come near.**
- ⚠ **If the radiator cap is removed when the engine is hot, hot water may spurt out and cause burns, so wait for the engine to cool down before starting troubleshooting.**
- ⚠ **Be extremely careful not to touch any hot parts or to get caught in any rotating parts.**
- ⚠ **When disconnecting wiring, always disconnect the negative (–) terminal of the battery first.**
- ⚠ **When removing the plug or cap from a location which is under pressure from oil, water, or air, always release the internal pressure first. When installing measuring equipment, be sure to connect it properly.**

The aim of troubleshooting is to pinpoint the basic cause of the failure, to carry out repairs swiftly, and to prevent reoccurrence of the failure. When carrying out troubleshooting, and important point is of course to understand the structure and function. However, a short cut to effective troubleshooting is to ask the operator various questions to form some idea of possible causes of the failure that would produce the reported symptoms.

1. **Do not rush at disassembling the machine even if a failure occurs.**
   Otherwise,
   - Parts that have no connection with the failure or other unnecessary parts will be disassembled.
   - It will become impossible to find the cause of the failure.
     It will also cause a waste of manhours, parts, or oil or grease, and at the same time, will also lose the confidence of the user or operator. For this reason, when carrying out troubleshooting, it is necessary to carry out thorough prior investigation and to carry out troubleshooting in accordance with the fixed procedure.
2. **Questions to users and operators**
   1) Have any other problems occurred apart from the problem that has been reported?
   2) Was there anything strange about the machine before the failure occurred?
   3) Did the failure occur suddenly, or were there problems with the machine condition before this?
   4) Under what conditions did the failure occur?
   5) Had any repairs been carried out before the failure? When were these repairs carried out?
   6) Has the same kind of failure occurred before?
3. **Inspections before troubleshooting**
   1) Is there any sign of irregularities of the machine?
   2) Make checks before starting day's work.
   3) Make checks of other items.
   4) Check other maintenance items which can be visually checked and are considered necessary.
4. **Confirmation of failure**
   Confirm the degree of failure on your own to verify if it is a real failure and the machine is handled or operated improperly.
   ★ When operating the machine to reenact the troubleshooting symptoms, do not carry out any investigation or measurement that may make the problem worse.
5. **Troubleshooting**
   Pinpoint high possibility causes according to the results of examinations and inspections of items 2 to 4, and find the failure location based on the troubleshooting table or troubleshooting flow (chart).
   ★ The basic procedure for troubleshooting is as follows.
      1] Start from the simple points.
      2] Start from the most likely points.
      3] Investigate other related parts or information.
6. **Radical countermeasures against failure cause**
   Even if the immediate problem is removed, a similar failure will reoccur unless its root cause is removed. To prevent reoccurence, examine the reason of the failure cause occurrence and remove the root cause.

Error and failure code table

Error code of INSITE	Failure code of applicable machine	Trouble
111	CA111	Abnormality in engine controller
115	CA115	Abnormality in engine NE and Bkup speed sensors
122	CA122	Abnormally high level in charge pressure sensor
123	CA123	Abnormally low level in charge pressure sensor
131	CA131	Abnormally high level in throttle sensor
132	CA132	Abnormally low level in throttle sensor
144	CA144	Abnormally high level in coolant temperature sensor
145	CA145	Abnormally low level in coolant temperature sensor
153	CA153	Abnormally high level in charge temperature sensor
154	CA154	Abnormally low level in charge temperature sensor
155	CA155	Derating of speed by abnormally high charge temperature
187	CA187	Abnormally low level in sensor power supply 2 circuit
221	CA221	Abnormally high level in atmospheric pressure sensor
222	CA222	Abnormally low level in atmospheric pressure sensor
227	CA227	Abnormally high level in sensor power supply 2 circuit
234	CA234	Engine overspeed
238	CA238	Abnormality in Ne speed sensor power supply
271	CA271	Short circuit in PCV1
272	CA272	Disconnection in PCV1
322	CA322	Disconnection or short circuit in injector #1 (L/B #1) system
323	CA323	Disconnection or short circuit in injector #5 (L/B #5) system
324	CA324	Disconnection or short circuit in injector #3 (L/B #3) system
325	CA325	Disconnection or short circuit in injector #6 (L/B #6) system
331	CA331	Disconnection or short circuit in injector #2 (L/B #2) system
332	CA332	Disconnection or short circuit in injector #4 (L/B #4) system
342	CA342	Matching error in engine controller data
351	CA351	Abnormality in injector drive circuit
352	CA352	Abnormally low level in sensor power supply 1 circuit
386	CA386	Abnormally high level in sensor power supply 1 circuit
428	CA428	Abnormally high level in water sensor
429	CA429	Abnormally low level in water sensor
431	CA431	Abnormality in idle validation switch
432	CA432	Abnormality in processing of idle validation
435	CA435	Abnormality in engine oil pressure switch
441	CA441	Abnormally low power supply voltage
442	CA442	Abnormally high power supply voltage
449	CA449	Abnormally high common rail pressure 2
451	CA451	Abnormally high level in common rail pressure sensor
452	CA452	Abnormally low level in common rail pressure sensor
488	CA488	Derating of torque by abnormally high charge temperature
553	CA553	Abnormally high common rail pressure 1
559	CA559	No-pressure feed by supply pump 1

Error code of INSITE	Failure code of applicable machine	Trouble
689	CA689	Abnormality in engine Ne speed sensor
731	CA731	Abnormality in engine Bkup speed sensor phase
757	CA757	Loss of all data in engine controller
778	CA778	Abnormality in engine Bkup speed sensor
1117	CA1117	Loss of partial data in engine controller
1633	CA1633	Abnormality in KOMNET
2185	CA2185	Abnormally high level in throttle sensor power supply
2186	CA2186	Abnormally low level in throttle sensor power supply
2249	CA2249	No-pressure feed by supply pump 2
2311	CA2311	Abnormality in IMV solenoid
2555	CA2555	Abnormally low level intake air heater relay power supply voltage
2556	CA2556	Abnormally high level intake air heater relay power supply voltage
—	B@BAZG	Derating of speed by engine oil pressure reduction
—	B@BAZG	Derating of torque by engine oil pressure reduction
—	B@BCNS	Engine overheat

★ Error code of INSITE:
3- or 4-digit error code displayed on INSITE when a failure occurrence condition is confirmed using INSITE.

★ Failure code of applicable machine:
5- or 6-digit failure code displayed on the machine monitor when a failure occurrence condition is confirmed using the machine monitor at the applicable machine side.

Engine

107E-1 Series

40 Troubleshooting

Troubleshooting of mechanical system (S-mode)

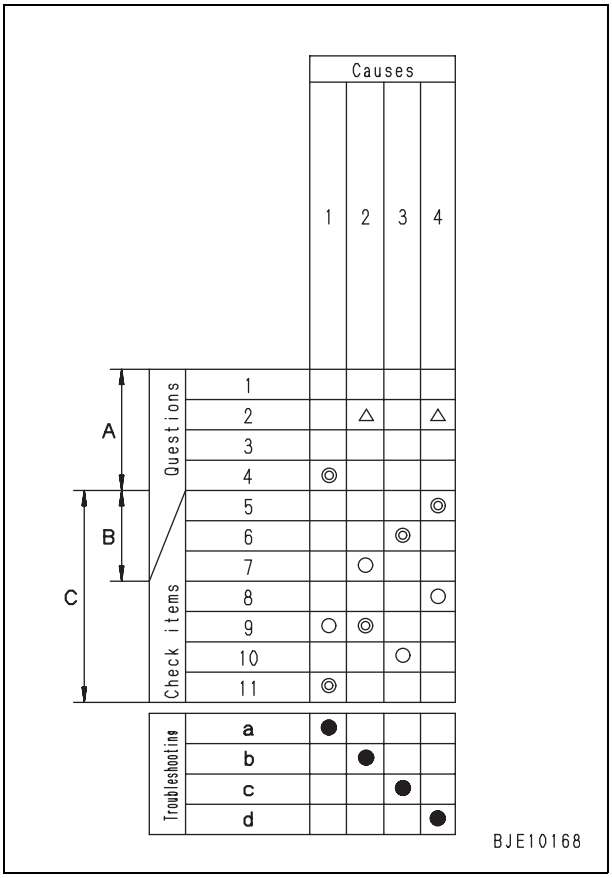
Troubleshooting of mechanical system (S-mode) 4
Method of using troubleshooting charts 4
S-1 Starting performance is poor 8
S-2 Engine does not start 9
S-3 Engine does not pick up smoothly 12
S-4 Engine stops during operations 13
S-5 Engine does not rotate smoothly 14
S-6 Engine lacks output (or lacks power) 15
S-7 Exhaust smoke is black (incomplete combustion) 16
S-8 Oil consumption is excessive (or exhaust smoke is blue) 17
S-9 Oil becomes contaminated quickly 18
S-10 Fuel consumption is excessive 19
S-11 Oil is in coolant (or coolant spurts back or coolant level goes down) 20
S-12 Oil pressure drops 21

S-13 Oil level rises (Entry of coolant or fuel) 22
S-14 Coolant temperature becomes too high (overheating) 23
S-15 Abnormal noise is made 24
S-16 Vibration is excessive 25

Troubleshooting of mechanical system (S-mode)

Method of using troubleshooting charts

The troubleshooting chart consists of the “questions“, “check items“, “causes“, and “troubleshooting“ blocks.
The questions and check items are used to pinpoint high probability causes by simple inspection or from phenomena without using troubleshooting tools.
Next, troubleshooting tools or direct inspection are applied to check the narrowed causes in order from the most probable one to make final confirmation according to the troubleshooting procedure.
[Questions]:
Items to be drawn from the user or operator.
They correspond to A and B in the chart on the right.
The items in A are basic ones.
The items in B can be drawn from the user or operator, depending on their level.
[Check items]:
Items to be simply checked by the serviceman to narrow down causes.
They correspond to C in the chart on the right.
[Causes]:
Items to be narrowed from the questions and check items.
The serviceman narrows down the probable causes from A, B, and C.
[Troubleshooting]:
Items to finally verify whether the narrowed down causes are undoubtedly the real causes using the troubleshooting tools and direct inspections.



Items listed in the [Questions] and [Check items] and related to the [Causes] are marked with △, ○, or ◎.

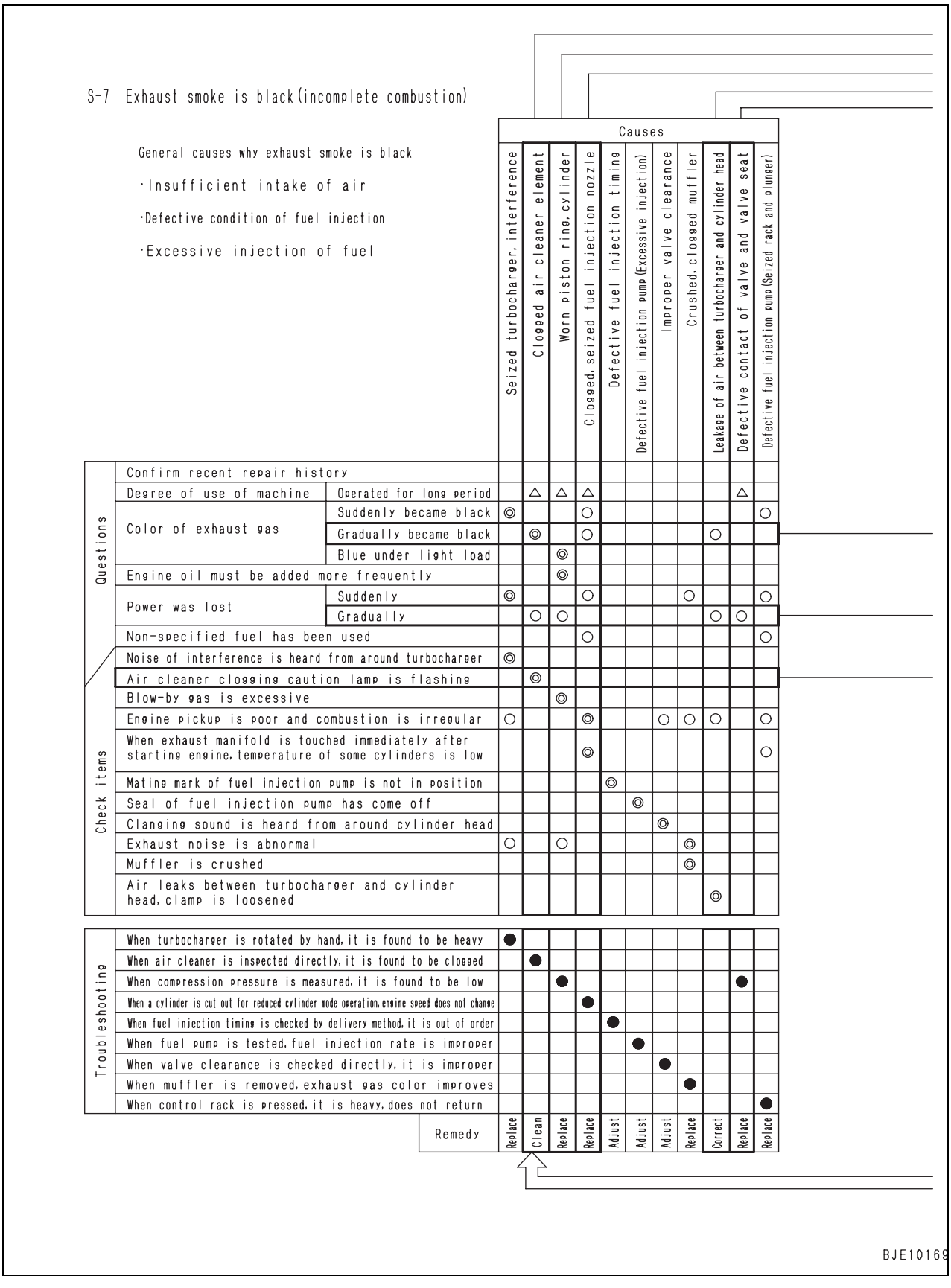
△ : Causes to be referred to for questions and check items

○ : Causes related to questions and check items

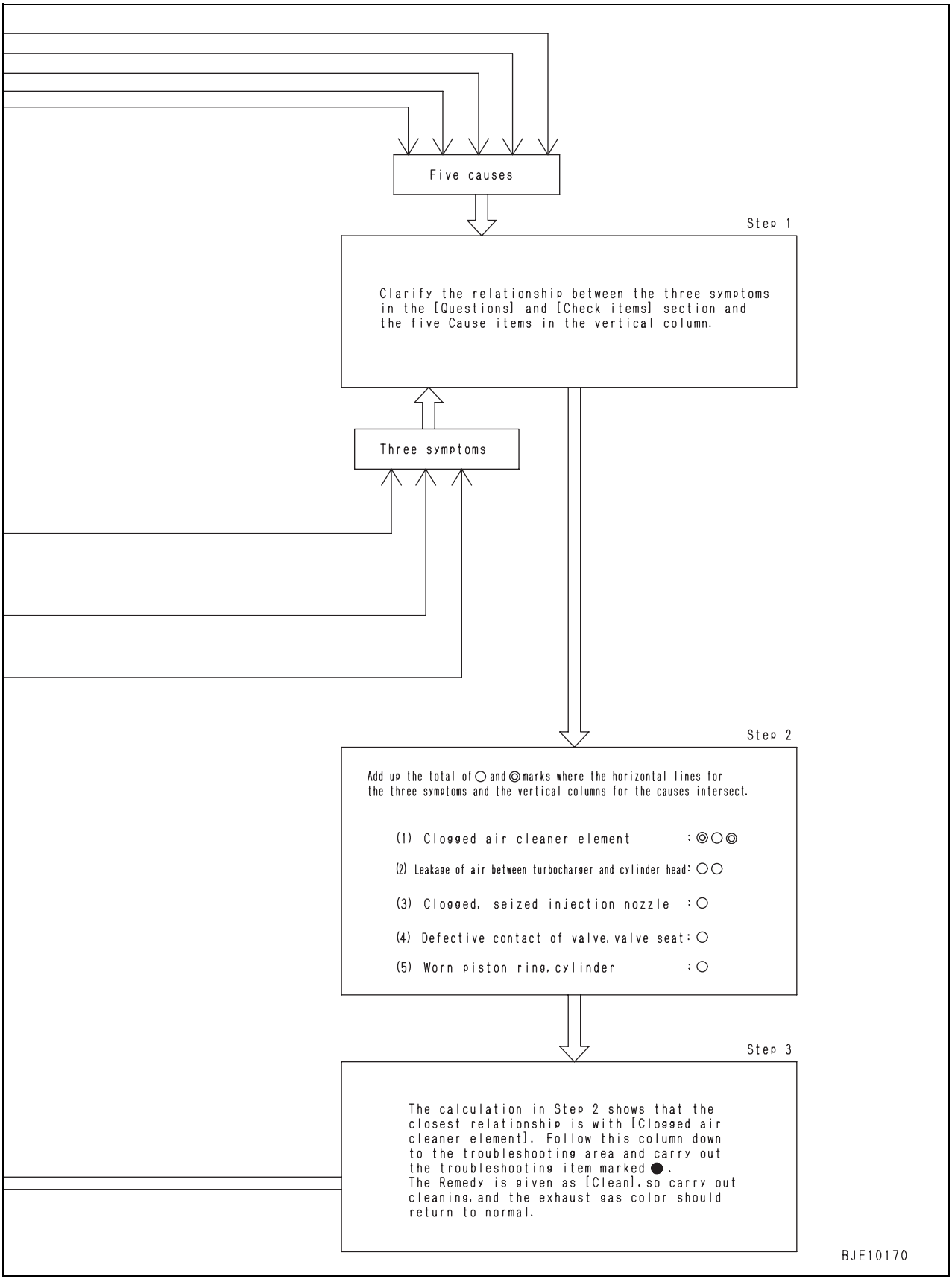
◎ : Causes highly probable among ones marked with ○

★ When narrowing the “causes“, apply the items marked with ◎ before those marked with ○.
When narrowing the causes, do not apply the items marked with △. (If no items have other marks and the causes cannot be narrowed, however, you may apply them.)

<Example of troubleshooting> Exhaust gas is black
Let us assume that [Clogged air cleaner] is taken to be the cause of black exhaust gas. 3 symptoms have causal relationship with this problem: [Color of exhaust gas gradually became black], [Power was lost gradually], and [Air cleaner clogging caution lamp is flashing].



If we look from these 3 symptoms to find the causes, we find that there is a relationship with 5 causes. Let us explain here the method of using this causal relationship to pinpoint the most probable cause.



S-1 Starting performance is poor

General causes why starting performance is poor
- Defective electrical system
- Insufficient supply of fuel
- Insufficient intake of air
- Improper selection of fuel

★ The common rail fuel injection system (CRI) recognizes the fuel injection timing electrically. Accordingly, even if the starting operation is carried out, the engine may not start until the crankshaft revolves 2 turns at maximum. This phenomenon does not indicate a trouble, however.

			Causes												
			Clogged air cleaner element	Defective contact of valve and valve seat	Worn piston ring, cylinder	Clogged air breather hole of fuel tank cap	Leaking or clogged fuel piping, entry of air	Clogged fuel filter, element	Stuck, seized supply pump plunger	Defective injector	Defective intake air heater system	Defective alternator (regulator section)	Defective alternator (generator section)	Defective or deteriorated battery
Questions	Confirm recent repair history													
	Degree of use of machine	Operated for long period	△					△						△
	Starting performance	Became worse gradually	○	◎	◎			○						
		Engine starts easily when warm									◎			◎
	Non-specified fuel is being used							○	○	○				
	Replacement of filters has not been carried out according to Operation and Maintenance Manual		◎					◎	○	○				
	Oil must be added more frequently				◎									
	When engine is preheated or when temperature is low, preheating monitor does not indicate normally (if monitor is installed)										◎			
	During operation, charge level monitor indicates abnormal charge (if monitor is installed)											◎	◎	
Check items	Dust indicator is red (if indicator is installed)		◎											
	Air breather hole of fuel tank cap is clogged					◎								
	Fuel is leaking from fuel piping						◎		○					
	When priming pump is operated, it makes no reaction or it is heavy						◎	○						
	Starting motor cranks engine slowly													◎
	While engine is cranked with starting motor,	If air bleeding plug of fuel filter is removed, fuel does not flow out						◎						
		If spill hose from injector is disconnected, little fuel spills							◎					
	When exhaust manifold is touched immediately after starting engine, temperature of some cylinders is low									◎				
	Engine does not pick up smoothly and combustion is irregular			○	○					◎				
	There is hunting from engine (rotation is irregular)					○	◎	○						
	Blow-by gas is excessive				◎									
Troubleshooting	Inspect air cleaner directly		●											
	When compression pressure is measured, it is found to be low			●	●									
	When air is bled from fuel system, air comes out						●							
	Inspect fuel filter, strainer directly							●						
	Carry out troubleshooting according to “No-pressure feed by supply pump (*1)“ in E-mode								●					
	When a cylinder is cut out for reduced cylinder mode operation, engine speed does not change									●				
	When starting switch is turned to HEAT, intake air heater mount does not become warm										●			
	Is voltage 20 – 30 V between alternator terminal B and terminal E with engine at low idle?	Yes										●		
		No											●	
	When specific gravity of electrolyte and voltage of battery are measured, they are low													●
		Remedy	Clean	Replace	Replace	Clean	Correct	Clean	Replace	Replace	Replace	Replace	Replace	Replace

*1: Code [559/CA559] and code [2249/CA2249] in E-mode

S-2 Engine does not start

a) Engine does not turn

General causes why engine does not turn

- Seized parts inside engine
  → See “S-4 Engine stops during operations”
- Defective electrical system
- Problem in drive devices on applicable machine side
  → Carry out troubleshooting for devices on applicable machine

			Causes							
			Broken flywheel ring gear	Defective or deteriorated battery	Defective connection of battery terminal	Defective battery relay	Defective starting switch	Defective safety relay	Defective starting motor (motor section)	Defective steering circuit wiring
Questions	Confirm recent repair history									
	Degree of use of machine	Operated for long period	△	△						
	Condition of horn when starting switch is turned ON	Horn does not sound			○		○			◎
		Horn volume is low		◎						
Check items	Battery electrolyte is low			◎						
	Battery terminal is loose				◎					
	When starting switch is turned ON, there is no operating sound from battery relay			○		◎				
	When starting switch is turned to START, starting pinion does not move out			○			○			◎
	When starting switch is turned to START, starting pinion moves out, but	Speed of rotation is low		◎						
		Makes grating noise	◎						◎	
		Soon disengages again						◎		
		Makes rattling noise and does not turn		○				○	○	
Troubleshooting	Inspect flywheel ring gear directly		●							Carry out troubleshooting on applicable machine
	When specific gravity of electrolyte and voltage of battery are measured, they are low			●						
	Turn starting switch OFF, connect cord, and carry out troubleshooting at ON	There is not voltage (20 – 30 V) between battery relay terminal B and terminal E				●				
		When terminal B and terminal C of starting switch are connected, engine starts					●			
		When terminal B and terminal C at safety relay outlet are connected, engine starts						●		
		Even if terminal B and terminal C at safety relay outlet are connected, engine does not start							●	
		When terminal at safety switch and terminal B at starting motor are connected, engine starts							●	
		Remedy	Replace	Replace	Correct	Replace	Replace	Replace	Replace	—

b) Engine turns but no exhaust smoke comes out

General causes why engine turns but no exhaust smoke comes out

- Fuel is not being supplied
- Supply of fuel is extremely small
- Improper selection of fuel (particularly in winter)

			Causes												
			Use of improper fuel	Insufficient fuel in tank	Clogged air breather hole of fuel tank cap	Leaking or clogged fuel piping, entry of air	Clogged fuel filter element	Seized, abnormally worn feed pump	Broken supply pump shaft	Stuck, seized supply pump plunger	Defective supply pump MPROP	Defective operation of overflow valve (Does not close)	Defective common rail pressure limiter	Defective fuel injector
Questions	Confirm recent repair history													
Questions	Degree of use of machine	Operated for long period				△	△							
Questions	Exhaust smoke suddenly stopped coming out (when starting again)							◎	○	◎	◎			△
Questions	Replacement of filters has not been carried out according to Operation and Maintenance Manual						◎			△				○
Check items	When fuel tank is inspected, it is found to be empty			◎										
Check items	Air breather hole of fuel tank cap is clogged			○	◎									
Check items	Rust and water are found when fuel tank is drained						○	△		△	△			
Check items	When fuel filter is removed, there is not fuel in it		◎			◎								
Check items	Fuel is leaking from fuel piping					◎								
Check items	When priming pump is operated, it makes no reaction or it is heavy					◎	○							
Check items	While engine is cranked with starting motor,	If air bleeding plug of fuel filter is removed, fuel does not flow out		○		○		◎	○					
Check items	While engine is cranked with starting motor,	If spill hose from injector is disconnected, little fuel spills				○			◎	◎	◎			○
Troubleshooting	When air is bled from fuel system, air comes out					●								
Troubleshooting	Inspect fuel filter directly						●							
Troubleshooting	Inspect feed pump directly							●						
Troubleshooting	Carry out troubleshooting according to “No-pressure feed by supply pump (*1)“ in E-mode								●	●				
Troubleshooting	Carry out troubleshooting according to “Abnormality in supply pump PCV (*2)“ in E-mode										●			
Troubleshooting	Inspect overflow valve directly											●		
Troubleshooting	Engine can be started in reduced cylinder mode													●
Troubleshooting	Fuel flows out when pressure limiter return piping is disconnected												●	
		Remedy	Replace	Add	Correct	Correct	Replace	Replace	Replace	Replace	Replace	Replace	Replace	Replace

*1: Code [559/CA559] and code [2249/CA2249] in E mode
*2: Code [271/CA271] and code [272/CA272] in E mode

c) Exhaust smoke comes out but engine does not start (fuel is being injected)

General causes why exhaust smoke comes out but engine does not start

- Lack of rotating force due to defective electrical system
- Insufficient supply of fuel
- Insufficient intake of air
- Improper selection of fuel

			Causes											
			Clogged air cleaner element	Worn dynamic valve system (Valve, rocker lever, etc.)	Worn piston ring, cylinder	Use of improper fuel	Clogged air breather hole of fuel tank cap	Leaking or clogged fuel system, entry of air	Clogging of fuel filter	Stuck, seized supply pump plunger	Clogged injector, defective spray	Defective, deteriorated battery	Defective coolant temperature sensor, wiring harness	Defective intake air heater system
Questions	Confirm recent repair history													
	Degree of use of machine	Operated for long period			△				△		△			
	Suddenly failed to start			◎						◎			○	
	Non-specified fuel is being used									○	○			
	Replacement of filters has not been carried out according to Operation and Maintenance Manual		◎						◎					
	Oil must be added more frequently				◎									
	When engine is preheated or when temperature is low, preheating monitor does not indicate normally (if monitor is installed)													◎
Check items	Dust indicator is red (if indicator is installed)		◎											
	Air breather hole of fuel tank cap is clogged						○							
	Rust and water are found when fuel tank is drained								◎					
	When fuel filter is removed, there is not fuel in it				◎									
	Fuel is leaking from fuel piping							◎						
	When priming pump is operated, it makes no reaction or it is heavy							◎	○					
	Starting motor cranks engine slowly											◎		
	When engine is cranked, abnormal sound is generated around cylinder head			◎										
	While engine is cranked with starting motor,	If air bleeding plug of fuel filter is removed, fuel does not flow out				○			◎					
		If spill hose from injector is disconnected, little fuel spills								◎				
	When exhaust manifold is touched immediately after starting engine, temperature of some cylinders is low										◎			
Troubleshooting	Inspect air cleaner directly		●											
	Inspect dynamic valve system directly			●										
	When compression pressure is measured, it is found to be low				●									
	When air is bled from fuel system, air comes out							●						
	Inspect fuel filter directly								●					
	Carry out troubleshooting according to “No-pressure feed by supply pump (*1)“ in E-mode									●				
	Engine can be started in reduced cylinder mode										●			
	When specific gravity of electrolyte and voltage of battery are measured, they are low											●		
	Coolant temperature gauge does not indicate normally (if coolant temperature gauge is installed)												●	
	When starting switch is turned to HEAT, intake air heater mount does not become warm													●
		Remedy	Clean	Replace	Replace	Replace	Clean	Correct	Clean	Replace	Replace	Replace	Replace	Replace

*1: Code [559/CA559] and code [2249/CA2249] in E-mode

S-3 Engine does not pick up smoothly

General causes why engine does not pick up smoothly
- Insufficient intake of air
- Insufficient supply of fuel
- Defective condition of fuel injection
- Improper selection of fuel
- Controller is controlling in derate mode
  (limiting injection rate (output) because of an error in electrical system)

			Causes									
			Clogged air cleaner element	Defective contact of valve and valve seat	Improper valve clearance	Seized turbocharger, interference of turbocharger	Worn piston ring, cylinder	Clogged air breather hole of fuel tank cap	Leaking or clogged fuel piping, entry of air	Clogging of fuel filter	Stuck, seized supply pump plunger	Clogged injector, defective spray
Questions	Confirm recent repair history											
	Degree of use of machine	Operated for long period	△	△			△			△		
	Engine pick-up suddenly became worse					◎		○	○			○
	Non-specified fuel is being used									◎	◎	◎
	Replacement of filters has not been carried out according to Operation and Maintenance Manual		◎							◎		
	Oil must be added more frequently						◎					
Check items	Dust indicator is red (if indicator is installed)		◎									
	Air breather hole of fuel tank cap is clogged							◎				
	Rust and water are found when fuel tank is drained									◎		
	Fuel is leaking from fuel piping								◎			
	When priming pump is operated, it makes no reaction or it is heavy								◎	○		
	When exhaust manifold is touched immediately after starting engine, temperature of some cylinders is low										○	◎
	Color of exhaust gas is	Blue under light load					◎					
		Black	◎	○		◎						◎
	When engine is cranked, abnormal sound is generated around cylinder head				◎							
	When engine is cranked, interference sound is generated around turbocharger					◎						
	High idle speed under no load is normal, but speed suddenly drops when load is applied							○		◎		
	There is hunting from engine (rotation is irregular)							○		○		○
	Blow-by gas is excessive						◎					

		Clean	Replace	Adjust	Replace	Replace	Clean	Correct	Clean	Replace	Replace
		Remedy										
Troubleshooting	Inspect air cleaner directly	●									
	When compression pressure is measured, it is found to be low		●			●					
	Inspect valve clearance directly			●							
	When turbocharger is rotated by hand, it is found to be heavy				●						
	When air is bled from fuel system, air comes out							●			
	Inspect fuel filter, strainer directly								●		
	Carry out troubleshooting according to “No-pressure feed by supply pump (*1)“ in E-mode									●	
	When a cylinder is cut out for reduced cylinder mode operation, engine speed does not change										●

*1: Code [559/CA559] and code [2249/CA2249] in E-mode

S-4 Engine stops during operations

General causes why engine stops during operations
- Internal parts of engine seized
- Insufficient supply of fuel
- There is overheating
- Problem in drive devices on applicable machine side
  → Carry out troubleshooting for devices on applicable machine

			Causes												
			Broken dynamic valve system (valve, rocker arm, etc.)	Broken, seized piston, connecting rod	Broken, seized crankshaft bearing	Broken, seized gear train	Insufficient fuel in tank	Clogged air breather hole of fuel tank cap	Leaking, clogged fuel piping	Clogging of fuel filter	Broken, seized feed pump	Broken supply pump shaft	Stuck, seized supply pump plunger	Broken auxiliary equipment (pump, compressor, etc.)	Problem in drive devices on applicable machine side
Questions	Confirm recent repair history														
	Degree of use of machine	Operated for long period								△					
	Condition when engine stopped	Abnormal noise was heard and engine stopped suddenly	◎	◎	◎	◎					○	◎	○	◎	◎
		Engine overheated and stopped		◎	○									○	
		Engine stopped slowly					◎			○					
		There was hunting and engine stopped					◎	○		○	○				
	Non-specified fuel is being used									○	○		○		
	Replacement of filters has not been carried out according to Operation and Maintenance Manual									◎					
	Fuel level monitor indicates low level (if monitor is installed)						◎								
Check items	When fuel tank is inspected, it is found to be empty						◎								
	Air breather hole of fuel tank cap is clogged							◎							
	Fuel is leaking from fuel piping								◎						
	When priming pump is operated, it makes no reaction or it is heavy								◎	○					
	Rust and water are found when fuel tank is drained									◎					
	Metal particles are found when oil pan is drained		◎	◎	◎						○				
	When engine is cranked by hand	Does not turn at all		◎	◎										
		Turns in opposite direction	◎												
		Moves by amount of gear backlash				◎								◎	
		Supply pump shaft does not turn										◎			
	Engine turns, but stops when load is applied to machine														◎

Troubleshooting	Inspect dynamic valve system directly	●												Carry out troubleshooting on applicable machine
	Inspect piston, connecting rod directly		●											
	Inspect crankshaft bearing directly			●										
	Inspect gear train directly				●									
	Inspect fuel filter, strainer directly								●					
	Inspect feed pump directly									●				
	Carry out troubleshooting according to “No-pressure feed by supply pump (*1)“ in E-mode										●	●		
	Engine rotates when pump auxiliary equipment (pump, compressor, etc.) is removed												●	
	Remedy	Replace	Replace	Replace	Replace	Add	Clean	Correct	Clean	Replace	Replace	Replace	Replace	—

*1: Code [559/CA559] and code [2249/CA2249] in E-mode

S-5 Engine does not rotate smoothly

General causes why engine does not rotate smoothly
- Air in fuel system
- Defective speed sensor (Error at degree that it is not indicated)

			Causes							
			Insufficient fuel in tank	Clogged air breather hole of fuel tank cap	Leaking or clogged fuel piping, entry of air	Clogging of fuel filter	Clogged injector, defective spray (dirt in injector)	Defective Ne speed sensor, wiring harness	Defective G speed sensor, wiring harness
Questions	Confirm recent repair history								
	Degree of use of machine	Operated for long period				△			
	Hunting	Occurs at a certain speed range						○	○
		Occurs at low idle			○	○	○	○	○
		Occurs even when speed is raised		○				○	○
		Occurs on slopes	◎						
	Replacement of filters has not been carried out according to Operation and Maintenance Manual					◎			
Check items	When fuel tank is inspected, it is found to be empty		◎						
	Air breather hole of fuel tank cap is clogged			◎					
	Rust and water are found when fuel tank is drained					○			
	Fuel is leaking from fuel piping				◎				
	When priming pump is operated, it makes no reaction or it is heavy				◎	○			
Troubleshooting	When air is bled from fuel system, air comes out				●				
	Inspect fuel filter, strainer directly					●			
	When a cylinder is cut out for reduced cylinder mode operation, engine speed does not change						●		
	Carry out troubleshooting according to “Abnormality in Ne speed sensor (*1)“ in E-mode							●	
	Carry out troubleshooting according to “Abnormality in G speed sensor (*2)“ in E-mode								●
		Remedy	Add	Clean	Replace	Replace	Replace	Replace	Replace

*1: Code [689/CA689] in E-mode
*2: Code [778/CA778] in E-mode

S-6 Engine lacks output (or lacks power)

General causes why engine lacks output
- Insufficient intake of air
- Insufficient supply of fuel
- Defective condition of fuel injection
- Improper selection of fuel
- There is overheating
  → See “S-14 Coolant temperature becomes too high (Overheating)“
- Controller is controlling in derate mode (limiting injection rate (output) because of an error in electrical system)

			Causes															
			Clogged air cleaner element	Air leakage from air intake piping	Seized turbocharger, interference of turbocharger	Defective contact of valve and valve seat	Improper valve clearance	Worn piston ring, cylinder	Clogged air breather hole of fuel tank cap	Leaking, clogged fuel piping	Clogging of fuel filter	Stuck, seized supply pump plunger	Clogged injector, defective spray (dirt in injector)	Defective drive of injector (signal, solenoid)	Defective installation of boost pressure sensor (air leakage)	Defective boost pressure sensor, wiring harness	Clogged fuel spill piping
Questions	Confirm recent repair history																	
	Degree of use of machine	Operated for long period	△			△		△			△							
	Power was lost	Suddenly		○	◎									○	○	○	○	
		Gradually	○			○		○			○		○		○			
	Non-specified fuel is being used										○	○	○					
	Replacement of filters has not been carried out according to Operation and Maintenance Manual		◎								◎							
	Oil must be added more frequently					○		○										
Check items	Dust indicator is red (if indicator is installed)		◎	◎														
	Air breather hole of fuel tank cap is clogged				◎				◎									
	Fuel is leaking from fuel piping									◎								
	Output becomes insufficient after short stop of operation			◎														
	Color of exhaust gas is	Black			◎		◎											
		Blue under light load						◎										
	When exhaust manifold is touched immediately after starting engine, temperature of some cylinders is low												◎	◎				
	When engine is cranked, interference sound is generated around turbocharger				◎													
	When engine is cranked, abnormal sound is generated around cylinder head						◎											
	High idle speed is too high													○				
	High idle speed under no load is normal, but speed suddenly drops when load is applied								○		◎	○	○					
	Engine does not pick up smoothly and combustion is irregular			◎	◎				○	○			◎					
	There is hunting from engine (rotation is irregular)								○	○	○		○	○				
	Blow-by gas is excessive				◎			◎										
Troubleshooting	Inspect air cleaner directly		●															
	Inspect air intake piping directly			●														
	When boost pressure is measured, it is found to be low		●	●	●													
	When compression pressure is measured, it is found to be low					●		●										
	Inspect valve clearance directly						●											
	Inspect fuel piping									●								
	Inspect fuel filter, strainer directly										●							
	Inspect spill port check valve directly																	●
	Carry out troubleshooting according to “No-pressure feed by supply pump (*1)“ in E-mode											●						
	When a cylinder is cut out for reduced cylinder mode operation, engine speed does not change												●	●				
	Inspect boost pressure sensor mount directly														●			
	Carry out troubleshooting according to “Abnormality in boost pressure sensor (*2)“ in E-mode															●		
		Remedy	Clean	Correct	Replace	Replace	Adjust	Replace	Clean	Correct	Replace	Replace	Replace	Replace	Correct	Replace	Replace	

*1: Code [559/CA559] and code [2249/CA2249] in E-mode
*2: Code [122/CA122] and code [123/CA123] in E-mode

S-7 Exhaust smoke is black (incomplete combustion)

General causes why exhaust smoke is black
- Insufficient intake of air
- Defective condition of fuel injection
- Improper selection of fuel
- There is overheating
  → See “S-14 Coolant temperature becomes too high (Overheating)“
- Controller is controlling in derate mode
  (limiting injection rate (output) because of an error in electrical system)

			Causes													
			Clogged air cleaner element	Seized turbocharger, interference of turbocharger	Defective contact of valve and valve seat	Improper valve clearance	Leakage of air between turbocharger and cylinder head	Crushed, clogged muffler	Worn piston ring, cylinder	Stuck, seized supply pump plunger	Clogged, seized injector	Abnormally worn injector	Improper injection timing	Improper fuel injection pressure	Defective coolant temperature sensor, wiring harness
Questions	Confirm recent repair history															
	Degree of use of machine	Operated for long period	△		△				△		△					
	Color of exhaust gas is	Suddenly became black		◎			◎			○	○	○				
		Gradually became black	◎				○				○					
		Blue under light load							◎							
	Non-specified fuel is being used									○	○					
	Oil must be added more frequently								◎							
	Power was lost	Suddenly		◎				○		○	○					
		Gradually	○		○		○		○							
Check items	Dust indicator is red (if indicator is installed)		◎													
	Muffler is crushed							◎								
	Air leaks between turbocharger and cylinder head, clamp is loosened						◎									
	Engine is operated in low-temperature mode at normal temperature												○	○	○	
	When exhaust manifold is touched immediately after starting engine, temperature of some cylinders is low									○	◎					
	When engine is cranked, interference sound is generated around turbocharger			◎												
	When engine is cranked, abnormal sound is generated around cylinder head					◎										
	Torque converter stall speed or pump relief speed is high (Fuel is injected excessively)											○				
	Exhaust noise is abnormal			○				◎			○					
	Engine does not pick up smoothly and combustion is irregular			○		○	○	○		○	◎					
	Blow-by gas is excessive								◎							
	If spill hose from injector is disconnected, abnormally much fuel spills											◎				
Troubleshooting	Inspect air cleaner directly		●													
	When turbocharger is rotated by hand, it is fount to be heavy			●												
	When compression pressure is measured, it is found to be low				●				●							
	Inspect valve clearance directly					●										
	When muffler is removed, exhaust sound improves							●								
	Carry out troubleshooting according to “No-pressure feed by supply pump (*1)“ in E-mode									●		●				
	When a cylinder is cut out for reduced cylinder mode operation, engine speed does not change										●					
	Carry out troubleshooting according to “Abnormality in coolant temperature sensor (*2)“ in E-mode															●
	Confirm with INSITE or with monitoring function on applicable machine side												●	●		
		Remedy	Clean	Replace	Replace	Adjust	Correct	Replace	Replace	Replace	Replace	Replace	Replace	Replace	Replace	Replace

*1: Code [559/CA559] and code [2249/CA2249] in E-mode
*2: Code [144/CA144] and code [145/CA145] in E-mode

S-8 Oil consumption is excessive (or exhaust smoke is blue)

General causes why oil consumption is excessive
- Abnormal consumption of oil
- Long-time operation of engine at low idle or high idle (Do not run engine at idle for more than 20 minutes continuously)
- External leakage of oil
- Wear of pats in lubrication system

			Causes														
			Dust sucked in from intake system	Worn, damaged valve (stem, guide, seal)	Turbocharger: Worn seal at turbine end	Turbocharger: Worn seal at blower end	Clogged breather, breather hose	Broken piston ring	Worn piston ring, cylinder	Worn, damaged rear oil seal	Broken oil cooler	Oil leakage from oil cooler	Oil leakage from oil filter	Oil leakage from oil piping	Oil leakage from oil drain plug	Oil leakage from oil pan, cylinder head, etc.
Questions	Confirm recent repair history																
Questions	Degree of use of machine	Operated for long period		△	△	△			△								
Questions	Oil consumption suddenly increased							◎			○						
Questions	Oil must be added more frequently								◎		○						
Questions	Oil becomes contaminated quickly						○	○	◎								
Check items	Outside of engine is dirty with oil											◎	◎	◎	◎	◎
Check items	There are loose piping clamps in intake system		◎														
Check items	Inside of turbocharger intake outlet pipe is dirty with oil					◎											
Check items	Inside of turbocharger exhaust outlet pipe is dirty with oil			○	◎												
Check items	There is oil in coolant										◎						
Check items	Oil level in clutch chamber or damper chamber is high									◎							
Check items	Exhaust smoke is blue under light load							◎	◎								
Check items	Amount of blow-by gas	Excessive		○		○		◎	◎								
Check items	Amount of blow-by gas	None					◎										
Troubleshooting	When intake manifold is removed, dust is found inside		●														
Troubleshooting	When intake manifold is removed, inside is found to be dirty abnormally			●													
Troubleshooting	Excessive play of turbocharger shaft				●	●											
Troubleshooting	Check breather and breather hose directly						●										
Troubleshooting	When compression pressure is measured, it is found to be low							●	●								
Troubleshooting	Inspect rear oil seal directly									●							
Troubleshooting	Pressure-tightness test of oil cooler shows there is leakage										●	●					
Troubleshooting	There is external leakage of oil from engine												●	●	●	●	
		Remedy	Correct	Correct	Replace	Replace	Clean	Replace	Replace	Correct	Replace	Replace	Correct	Correct	Correct	Correct	

S-9 Oil becomes contaminated quickly

General causes why oil becomes contaminated quickly
- Entry of exhaust gas into oil due to internal wear
- Clogging of lubrication passage
- Use of improper fuel
- Use of improper oil
- Operation under excessive load

			Causes									
			Defective seal at turbocharger turbine end	Worn valve, valve guide	Worn piston ring, cylinder	Clogged breather, breather hose	Clogged oil cooler	Clogged oil filter	Defective oil filter safety valve	Clogged turbocharger lubrication drain tube	Exhaust smoke is bad	
Questions	Confirm recent repair history											
Questions	Degree of use of machine	Operated for long period	△	△	△							
Questions	Non-specified fuel is being used				○			○				
Questions	Oil must be added more frequently				◎							
Check items	Metal particles are found when oil filter is drained			○	○			◎				
Check items	Inside of exhaust pipe is dirty with oil			◎								
Check items	Engine oil temperature rises quickly						◎					
Check items	Color of exhaust gas is	Blue under light load			◎							
Check items	Color of exhaust gas is	Black									◎	
Check items	Amount of blow-by gas	Excessive	○	○	◎					○		
Check items	Amount of blow-by gas	None				◎						
Troubleshooting	Excessive play of turbocharger shaft		●									See S-7
Troubleshooting	When compression pressure is measured, it is found to be low			●	●							See S-7
Troubleshooting	Check breather and breather hose directly					●						See S-7
Troubleshooting	Inspect oil cooler directly						●					See S-7
Troubleshooting	Inspect oil filter directly							●				See S-7
Troubleshooting	Spring of oil filter safety valve is hitched or broken								●			See S-7
Troubleshooting	Inspect turbocharger lubrication drain tube directly									●		See S-7
		Remedy	Replace	Replace	Replace	Clean	Clean	Replace	Replace	Clean	—	

S-10 Fuel consumption is excessive

General causes why fuel consumption is excessive
- Leakage of fuel
- Defective condition of fuel injection (fuel pressure, injection timing)
- Excessive injection of fuel

			Causes									
			Fuel leakage inside head cover	Fuel leakage from fuel filter, piping, etc.	Defective feed pump oil seal	Defective supply pump plunger	Defective common rail pressure	Defective spray by injector	Defective operation of injector	Improper injection timing	Defective coolant temperature sensor, wiring harness
Questions	Confirm recent repair history										
	Degree of use of machine	Operated for long period			△	△		△			
	Condition of fuel consumption	More than for other machines of same model					○		○	○	○
		Gradually increased				○		○			
		Suddenly increased	○	○							
Check items	There is external leakage of fuel from engine			◎							
	Combustion is irregular							◎			
	Engine oil level rises and oil smells of diesel fuel		◎		◎						
	When exhaust manifold is touched immediately after starting engine, temperature of some cylinders is low							◎			
	Low idle speed is high								○		
	Torque converter stall speed or pump relief speed is high								○		
	Color of exhaust gas is	Black					○	○		○	○
		White	○								
Troubleshooting	Remove head cover and inspect directly		●								
	Inspect feed pump oil seal directly				●						
	Carry out troubleshooting according to “No-pressure feed by supply pump (*1)“ in E-mode					●					
	When a cylinder is cut out for reduced cylinder mode operation, engine speed does not change							●			
	If spill hose from injector is disconnected, much fuel spills								●		
	Carry out troubleshooting according to “Abnormality in coolant temperature sensor (*2)“ in E-mode										●
	Confirm with INSITE or with monitoring function on applicable machine side						●			●	
		Remedy	Correct	Correct	Replace	Replace	Correct	Replace	Replace	Replace	Replace

*1: Code [559/CA559] and code [2249/CA2249] in E-mode
*2: Code [144/CA144] and code [145/CA145] in E-mode

S-11 Oil is in coolant (or coolant spurts back or coolant level goes down)

General causes why oil is in coolant
- Internal leakage in lubrication system
- Internal leakage in cooling system

			Causes				
			Broken cylinder head, head gasket	Cracks inside cylinder block	Holes caused by pitting	Broken oil cooler core, O-ring	Broken hydraulic oil cooler or power train oil cooler on applicable machine side
Questions	Confirm recent repair history						
	Degree of use of machine	Operated for long period			△	△	
	Oil level	Suddenly rose	○			○	○
		Gradually rose		○	○		
	Hard water is being used as coolant				○	○	
Check items	Oil level has risen and oil is milky			○	○	◎	
	There are excessive air bubbles in radiator, coolant spurts back		◎				
	Hydraulic oil or power train oil on applicable machine side is milky						◎
	When hydraulic oil or power train oil is drained, water is found						◎

Trouble-shooting	Pressure-tightness test of cylinder head shows there is leakage		●				Carry out troubleshooting on applicable machine
	Inspect cylinder block, liner directly			●	●		
	Pressure-tightness test of oil cooler shows there is leakage					●	
		Remedy	Replace	Replace	Replace	Replace	—

S-12 Oil pressure drops

General causes why oil pressure drops
- Leakage, clogging, wear or lubrication system
- Defective oil pressure control
- Improper selection of fuel (improper viscosity)
- Deterioration of oil due to overheating

			Causes										
			Worn journal of bearing	Lack of oil in oil pan	Coolant, fuel in oil	Clogged strainer in oil pan	Clogged, broken pipe in oil pan	Defective oil pump	Defective regulator valve	Clogged oil filter	Leaking, crushed, clogged hydraulic piping	Defective oil level sensor, wiring harness
Questions	Confirm recent repair history											
Questions	Degree of use of machine	Operated for long period	△					△		△		
Questions	Oil pressure monitor indicates low oil pressure (if monitor is installed)								○	◎		
Questions	Non-specified oil is being used		○							○		
Questions	Replacement of filters has not been carried out according to Operation and Maintenance Manual									◎		
Check items	Oil pressure monitor (if installed)	Indicates pressure drop at low idle	◎									
Check items	Oil pressure monitor (if installed)	Indicates pressure drop at low, high idle		○		◎	◎	◎	○			
Check items	Oil pressure monitor (if installed)	Indicates pressure drop on slopes		◎								
Check items	Oil pressure monitor (if installed)	Sometimes indicates pressure drop							◎			○
Check items	Oil level monitor indicates oil level drop (if monitor is installed)			◎								◎
Check items	Oil level in oil pan is low			◎								
Check items	External hydraulic piping is leaking, crushed										◎	
Check items	Oil is milky or smells of diesel oil				◎							
Check items	Metal particles are found when oil pan is drained		◎									
Check items	Metal particles are found when oil filter is drained		◎					○				
Troubleshooting	Metal particles are found in oil filter		●		See S-13							
Troubleshooting	Inspect oil pan strainer, pipe directly					●	●					
Troubleshooting	Oil pump rotation is heavy, there is play in oil pump							●				
Troubleshooting	Valve and spring of regulator valve are fatigued, damaged								●			
Troubleshooting	Inspect oil filter directly									●		
Troubleshooting	If oil level sensor is replaced, oil level monitor indicates normally											●
		Remedy	Clean	Add	—	Clean	Clean	Replace	Adjust	Clean	Correct	Replace

S-13 Oil level rises (Entry of coolant or fuel)

General causes why oil level rises
- Coolant in oil (milky)
- Fuel in oil (smells diluted diesel fuel)

★ If oil is in coolant, carry out troubleshooting for “S-11 Oil is in coolant”

			Causes								
			Broken cylinder head, head gasket	Broken injector, O-ring	Cracks inside cylinder block	Holes caused by pitting	Worn, damaged rear oil seal	Broken oil cooler core, O-ring	Defects in supply pump	Defective seal of auxiliary equipment (pump, compressor)
Questions	Confirm recent repair history									
	Degree of use of machine	Operated for long period		△		△	△			△
	Fuel must be added more frequently			◎					◎	
	Coolant must be added more frequently		○		○					
Check items	There is oil in coolant		○	○	○	○		◎		
	Oil smells of diesel fuel			◎					◎	
	Oil is milky		○			○				
	When engine is started, drops of water come from muffler		○							
	When radiator cap is removed and engine is run at low idle, an abnormal number of bubbles appears, or coolant spurts back		◎			○				
	Exhaust smoke is white			○						
	Oil level in clutch chamber or damper chamber is low						◎			
	Oil level in hydraulic tank of machine is low									◎
Troubleshooting	When compression pressure is measured, it is found to be low		●							
	Remove injector and inspect O-ring.			●						
	Inspect cylinder block, liner directly				●	●				
	Inspect rear oil seal directly						●			
	Pressure-tightness test of oil cooler shows there is leakage							●		
	Remove and inspect supply pump directly								●	
	Inspect seal of auxiliary equipment directly									●
		Remedy	Replace	Correct	Replace	Replace	Correct	Replace	Replace	Replace

S-14 Coolant temperature becomes too high (overheating)

General causes why coolant temperature becomes too high
- Lack of cooling air (deformation, damage of fan)
- Drop in heat dissipation efficiency
- Problem in coolant circulation system
- Rise in oil temperature in power train
  → Carry out troubleshooting on applicable machine

			Causes												
			Broken cylinder head, head gasket	Holes caused by pitting	Clogged, broken oil cooler	Lack of coolant	Broken water pump	Defective operation of thermostat	Clogged, crushed radiator fins	Clogged radiator core	Defective radiator cap (pressure valve)	Slipping fan belt, worn fan pulley	Defective coolant temperature gauge	Rise in power train oil temperature on applicable machine side
Questions	Confirm recent repair history													
	Degree of use of machine	Operated for long period	△	△					△	△				
	Condition overheating	Sudden overheated				○	◎					○		
		Always tends to overheat						○	◎	◎		○		
	Coolant temperature gauge (if installed)	Rises quickly				○		◎						
		Does not go down from red range											◎	
Check items	Radiator coolant level monitor indicates drop of coolant level (if monitor is installed)					◎								
	Engine oil level has risen, oil is milky			◎	○									
	Fan belt tension is low											◎		
	When fan pulley is turned, it has play						◎							
	Milky oil is floating on coolant				◎									
	There are excessive air bubbles in radiator, coolant spurts back		◎											
	When light bulb is held behind radiator core, no light passes through								◎					
	Radiator shroud, inside of underguard on applicable machine side are clogged with dirt or mud								◎			◎		
	Coolant is leaking because of cracks in hose or loose clamps					◎								
	Coolant flows out from radiator overflow hose										◎			
	Fan belt whines under sudden acceleration											◎		
	Power train oil temperature enters red range faster than engine coolant temperature (if oil temperature gauge and coolant temperature gauge are installed)													◎
Troubleshooting	When compression pressure is measured, it is found to be low		●											Carry out troubleshooting on applicable machine
	Inspect cylinder liner directly			●										
	Inspect oil cooler directly				●									
	Temperature difference between upper and lower tanks of radiator is large						●							
	When operation of thermostat is carried out, it does not open at cracking temperature							●						
	Temperature difference between upper and lower tanks of radiator is slight								●					
	Inspect radiator core directly									●				
	When operation of radiator cap is carried out, its cracking pressure is low										●			
	Inspect fan belt, pulley directly											●		
	When coolant temperature is measured, it is found to be normal												●	
		Remedy	Replace	Replace	Replace	Add	Replace	Replace	Correct	Correct	Replace	Correct	Replace	—

S-15 Abnormal noise is made

General causes why abnormal noise is made
- Abnormality due to defective parts
- Abnormal combustion
- Air sucked in from intake system

★ Judge if the noise is an internal noise or an external noise before starting troubleshooting.

★ The engine is operated in the low-temperature mode while it is not warmed up sufficiently. Accordingly, the engine sound becomes a little larger. This does not indicate abnormality, however.

★ When the engine is accelerated, it is operated in the acceleration mode and its sound becomes a little larger for up to about 3 seconds. This does not indicate abnormality, however.

			Causes												
			Leakage of air between turbocharger and cylinder head	Interference of turbocharger, seized turbocharger	Broken dynamic valve system (valve, rocker lever)	Defective inside of muffler (dividing board out of position)	Improper valve clearance	Excessive wear of piston ring, cylinder	Improper gear train backlash	Removed, seized bushing	Deformed cooling fan, loose fan belt, interference of fan belt	Clogged, seized injector	Dirt caught in injector	Improper fuel injection timing (abnormality in coolant temperature sensor, boost temperature sensor)
Questions	Confirm recent repair history													
	Degree of use of machine	Operated for long period						△						
	Condition of abnormal noise	Gradually occurred						○			○			
		Suddenly occurred		○	○					○				
	Non-specified fuel is being used											○		
	Oil must be added more frequently							◎						
Check items	Metal particles are found when oil filter is drained							◎		◎				
	Air leaks between turbocharger and cylinder head		◎											
	When engine is cranked, interference sound is generated around turbocharger			◎										
	When engine is cranked, abnormal sound is generated around cylinder head				◎		◎							
	When engine is cranked, beat noise is generated around muffler					◎								
	When exhaust manifold is touched immediately after starting engine, temperature of some cylinders is low											◎	○	
	Color of exhaust gas is	Blue under light load						◎						
		Black	○	◎			○							
	Engine does not pick up smoothly and combustion is irregular											◎		
	Abnormal noise is loud when engine is accelerated						○		○		○	○		
	Blow-by gas is excessive							◎						
Troubleshooting	When turbocharger is rotated by hand, it is found to be heavy			●										
	Inspect dynamic valve system directly				●									
	When muffler is removed, abnormal noise disappears					●								
	Inspect valve clearance directly						●							
	When compression pressure is measured, it is found to be low							●						
	Inspect gear train directly								●	●				
	Inspect fan and fan belt directly										●			
	When a cylinder is cut out for reduced cylinder mode operation, engine speed does not change											●	●	
	Abnormal noise is heard only when engine is started												●	
	Confirm with INSITE or with monitoring function on applicable machine side													●
	Remedy		Replace	Replace	Correct	Replace	Replace	Replace	Replace	Replace	Correct	Replace	Replace	Replace

S-16 Vibration is excessive

General causes why vibration is excessive
- Defective parts (abnormal wear, breakage)
- Misalignment between engine and chassis
- Abnormal combustion

★ If abnormal noise is made and vibration is excessive, carry out troubleshooting for “S-15 Abnormal noise is made“, too.

			Causes							
			Stuck dynamic valve system (valve, rocker lever)	Worn main bearing, connecting rod bearing	Improper gear train backlash	Worn camshaft bushing	Improper fuel injection timing (abnormality in coolant temperature sensor, boost temperature sensor)	Loose engine mounting bolts, broken cushions	Misalignment between engine and devices on applicable machine side	Broken output shaft, parts in damper on applicable machine side
Questions	Confirm recent repair history									
	Degree of use of machine	Operated for long period		△		△		△		
	Condition of vibration	Suddenly increased	○							○
		Gradually increased		○		○		○		
	Non-specified oil is being used			○		○				
Check items	Metal particles are found when oil filter is drained			◎		◎				
	Metal particles are found when oil pan is drained			◎		◎				
	Oil pressure is low at low idle			○		○				
	Vibration occurs at mid-range speed							○		○
	Vibration follows engine speed				○			○	○	○
	Exhaust smoke is black		◎				○			
Troubleshooting	Inspect dynamic valve system directly		●							
	Inspect main bearing and connecting rod bearing directly			●						
	Inspect gear train directly				●					
	Inspect camshaft bushing directly					●				
	Confirm with INSITE or with monitoring function on applicable machine side						●			
	Inspect engine mounting bolts and cushions directly							●		
	When alignment is checked, radial runout or facial runout is detected								●	
	Inspect output shaft or inside of damper directly									●
		Remedy	Replace	Replace	Replace	Replace	Replace	Replace	Adjust	Replace

Engine

107E-1 Series

40 Troubleshooting

Troubleshooting of electrical system (E-mode), Part 1

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Troubleshooting of electrical system (E-mode), Part 1

Information in troubleshooting table

★ The following information is summarized in the troubleshooting table and the related electrical circuit diagram. Before carrying out troubleshooting, understand that information fully.

Error code	Failure code	Trouble	**Problem that appears on engine**
Display on INSITE	**Display on applicable machine**		
Contents of trouble	**Contents of trouble detected by engine controller**		
Action of controller	**Action taken by engine controller to protect system or devices when engine controller detects trouble**		
Problem that appears on machine	**Problem that appears on engine as result of action taken by engine controller (shown above)**		
Related information	**Information related to detected trouble or troubleshooting**		

	Causes		Standard value in normal state/Remarks on troubleshooting
Possible causes and standard value in normal state	1	Possible causes of trouble (Given numbers are reference numbers, which do not indicate priority)	**<Contents of description>** **• Standard value in normal state to judge possible causes** **• Remarks on judgment** **<Troubles in wiring harness>** **• Disconnection** **Connector is connected imperfectly or wiring harness is broken.** **• Ground fault** **Wiring harness which is not connected to chassis ground circuit is in contact with chassis ground circuit.** **• Hot short** **Wiring harness which is not connected to power source (24 V) circuit is in contact with power source (24 V) circuit.** **• Short circuit** **Independent wiring harnesses are in contact with each other abnormally.** **<Precautions for troubleshooting>** **(1) Method of indicating connector No. and handling of T-adapter** **Insert or connect T-adapter as explained below for troubleshooting, unless otherwise specified.** **• If connector No. has no marks of "male" and "female", disconnect connector and insert T-adapters in both male side and female side.** **• If connector No. has marks of "male" and "female", disconnect connector and connect T-adapter to only male side or female side.** **(2) Entry order of pin Nos. and handling of tester leads** **Connect positive (+) lead and negative (–) lead of tester as explained below for troubleshooting, unless otherwise specified.** **• Connect positive (+) lead to pin No. or wiring harness entered on front side.** **• Connect negative (–) lead to pin No. or wiring harness entered on rear side.**
	2		
	3		
	4		

Related circuit diagram

This drawing is a part of the electric circuit diagram related to troubleshooting.
- Connector No.: Indicates (Model – Number of pins) and (Color).

E-1 Code [111/CA111] Abnormality in engine controller

Error code	Failure code	Trouble	Abnormality in engine controller
111	**CA111**		
Contents of trouble	• Abnormality occurred in memory of engine controller or power supply circuit.		
Action of controller	• None in particular.		
Problem that appears on machine	• Engine does not start		
Related information	—		

Possible causes and standard value in normal state	Causes		Standard value in normal state/Remarks on troubleshooting
	1	Defective engine controller	Possible defect in engine controller (Troubleshooting cannot be carried out.)

E-2 Code [115/CA115] Abnormality in engine Ne, Bkup speed sensor

Error code	Failure code	Trouble	Abnormality in engine Ne, Bkup speed sensor
115	**CA115**		
Contents of trouble	• Abnormality occurred in signals of engine Ne speed sensor and engine Bkup speed sensor circuit.		
Action of controller	• None in particular.		
Problem that appears on machine	• Engine stops • Engine does not start		
Related information	• Method of reproducing failure code: Start engine.		

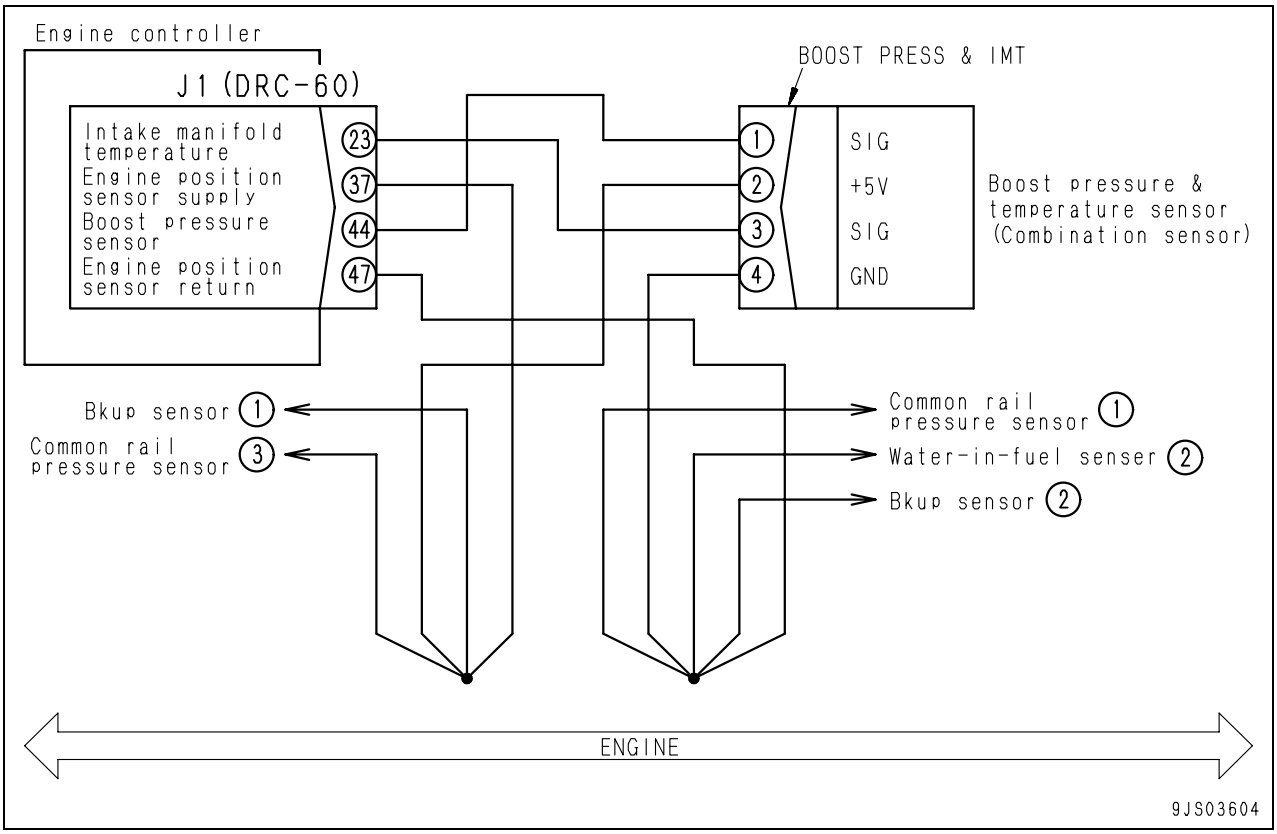
Possible causes and standard value in normal state	Causes		Standard value in normal state/Remarks on troubleshooting
	1	Defective connection of sensor connector	Check Ne speed sensor and Bkup speed sensor for connection directly.

E-3 Code [122/CA122] Abnormally high level in charge pressure sensor

Error code	Failure code	Trouble	Abnormally high level in charge pressure sensor
122	**CA122**		
Contents of trouble	• High voltage is detected in pressure signal circuit of boost pressure sensor and temperature sensor.		
Action of controller	• Fixes charge pressure and continues operation.		
Problem that appears on machine	• Engine output drops.		
Related information	• Method of reproducing failure code: Starting switch ON.		

	Causes		Standard value in normal state/Remarks on troubleshooting		
Possible causes and standard value in normal state	1	Defective sensor power supply 2 circuit	If failure code [**CA227**] is indicated simultaneously, carry out troubleshooting for it first.		
	2	Defective [pressure signal circuit] of boost pressure sensor, temperature sensor	★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting.		
			BOOST PRESS & IMT		Voltage
			Between (2) – (4)	Power supply	4.75 – 5.25 V
			Voltage is measured with wiring harness connected. Accordingly, if voltage is abnormal, check wiring harness and controller, too, for another cause of trouble, and then judge.		
	3	Hot short (Short circuit with 5 V, 24 V circuit) in wiring harness	★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting.		
			Wiring harness between J1 (female) (44) – BOOST PRESS & IMT (female) (1)	Voltage	Max. 1 V
	4	Short circuit in wiring harness (with another wiring harness)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			Wiring harness between J1 (female) (44) – BOOST PRESS & IMT (female) (1) and J1 (female) (37) – BOOST PRESS & IMT (female) (2)	Resistance	Min. 100 kΩ
	5	Defective wiring harness connector	Connecting parts among boost pressure sensor, temperature sensor, engine wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation		
	6	Defective engine controller	★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting.		
			J1		Voltage
			Between (37) – (47)	Power supply	4.75 – 5.25 V

Circuit diagram related to boost pressure, temperature sensor (combination sensor)

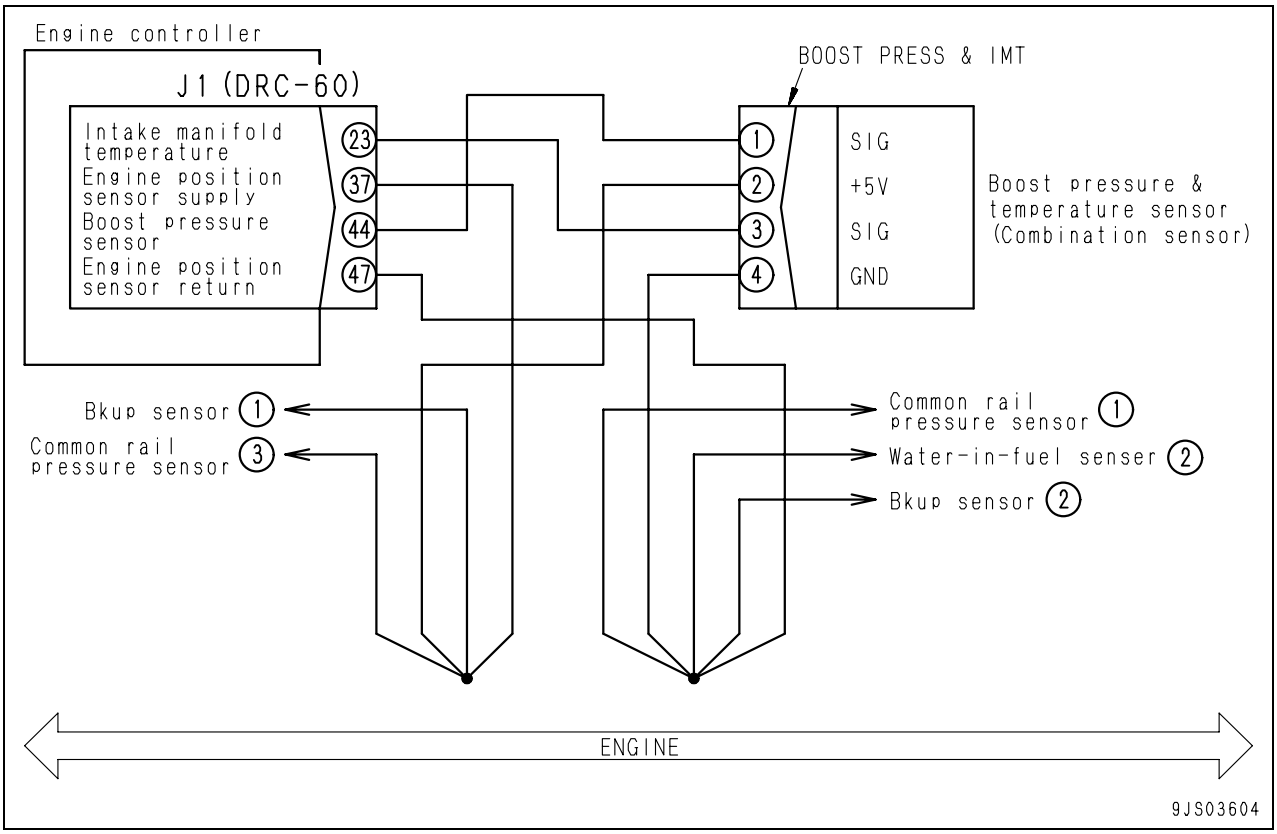


E-4 Code [123/CA123] Abnormally low level in charge pressure sensor

Error code	Failure code	Trouble	Abnormally low level in charge pressure sensor
123	**CA123**		
Contents of trouble	• Low voltage is detected in pressure signal circuit of boost pressure sensor and temperature sensor.		
Action of controller	• Fixes charge pressure and continues operation.		
Problem that appears on machine	• Engine output drops.		
Related information	• Method of reproducing failure code: Starting switch ON.		

	Causes		Standard value in normal state/Remarks on troubleshooting		
Possible causes and standard value in normal state	1	Defective sensor power supply 2 circuit	If failure code [**CA187**] is indicated simultaneously, carry out troubleshooting for it first.		
	2	Defective [pressure signal circuit] of boost pressure sensor, temperature sensor	★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting.		
			BOOST PRESS & IMT		Voltage
			Between (2) – (4)	Power supply	4.75 – 5.25 V
			Voltage is measured with wiring harness connected. Accordingly, if voltage is abnormal, check wiring harness and controller, too, for another cause of trouble, and then judge.		
	3	Ground fault in wiring harness (Short circuit with GND circuit)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			Wiring harness between J1 (female) (44) – BOOST PRESS & IMT (female) (1)	Resistance	Min. 100 kΩ
	4	Short circuit in wiring harness (with another wiring harness)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			Wiring harness between J1 (female) (44) and BOOST PRESS & IMT (female) (1) and J1 (female) (47) and BOOST PRESS & IMT (female) (4)	Resistance	Min. 100 kΩ
	5	Defective wiring harness connector	Connecting parts among boost pressure sensor, temperature sensor, engine wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation		
	6	Defective engine controller	★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting.		
			J1		Voltage
			Between (37) – (47)	Power supply	4.75 – 5.25 V

Circuit diagram related to boost pressure, temperature sensor (combination sensor)

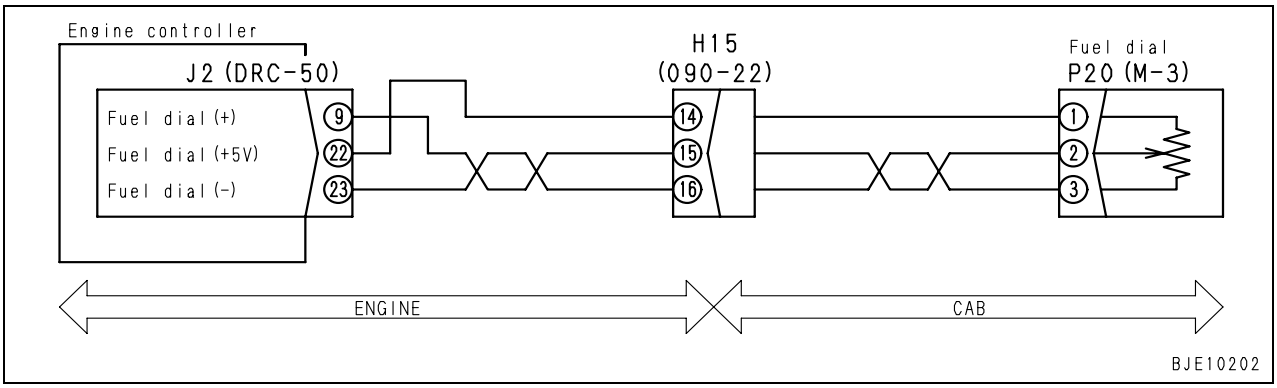


E-5 Code [131/CA131] Abnormally high level in throttle sensor power supply

Error code	Failure code	Trouble	Abnormally high level in throttle sensor
131	**CA131**		
Contents of trouble	• High voltage is detected in fuel control dial signal circuit.		
Action of controller	• Depends on machine model.		
Problem that appears on machine	• Depends on machine model.		
Related information	• Method of reproducing failure code: Starting switch ON.		

	Causes		Standard value in normal state/Remarks on troubleshooting			
Possible causes and standard value in normal state	1	Defective throttle sensor power supply system	If failure code [**CA2185**] is indicated simultaneously, carry out troubleshooting for it first.			
	2	Defective fuel control dial	★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting.			
			P20		Voltage	
			Between (1) – (3)	Power supply	4.75 – 5.25 V	
			Voltage is measured with wiring harness connected. Accordingly, if voltage is abnormal, check wiring harness and controller, too, for another cause of trouble, and then judge.			
	3	Hot short (Short circuit with 5 V, 24 V circuit) in wiring harness	★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting.			
			Wiring harness between J2 (female) (9) – P20 (female) (2)		Voltage	Max. 1 V
	4	Short circuit in wiring harness (with another wiring harness)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness between J2 (female) (9) – P20 (female) (2) and J2 (female) (22) – P20 (female) (1)		Resistance	Min. 100 kΩ
	5	Defective wiring harness connector	Connecting parts among fuel control dial, engine unit wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation			
	6	Defective engine controller	★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting.			
			J2		Voltage	
			Between (22) – (23)	Power supply	4.75 – 5.25 V	

Circuit diagram related to fuel control dial

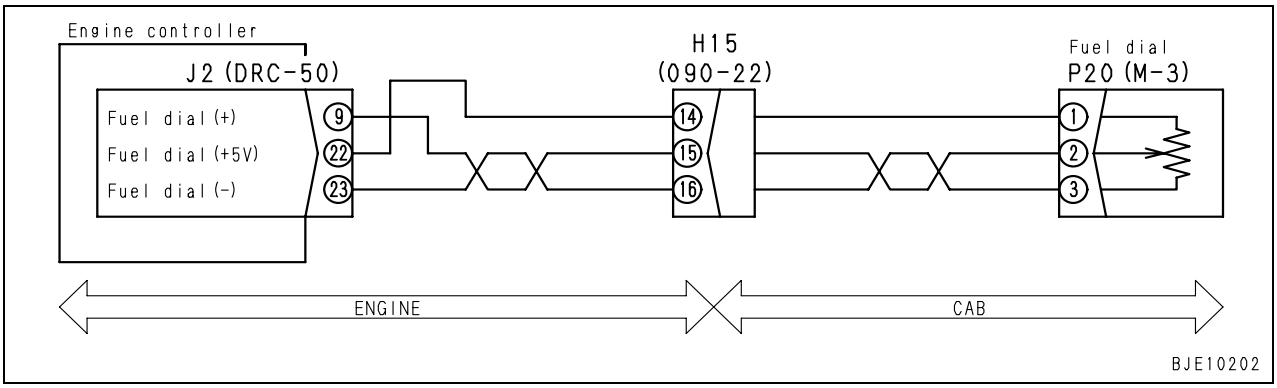


E-6 Code [132/CA132] Abnormally low level in throttle sensor power supply

Error code	Failure code	Trouble	Abnormally low level in throttle sensor
132	**CA132**		
Contents of trouble	• Low voltage is detected in fuel control dial signal circuit.		
Action of controller	• Depends on machine model.		
Problem that appears on machine	• Depends on machine model.		
Related information	• Method of reproducing failure code: Starting switch ON.		

	Causes		Standard value in normal state/Remarks on troubleshooting		
Possible causes and standard value in normal state	1	Defective throttle sensor power supply system	If failure code [**CA2186**] is indicated simultaneously, carry out troubleshooting for it first.		
	2	Defective fuel control dial	★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting.		
			P20		Voltage
			Between (1) – (3)	Power supply	4.75 – 5.25 V
			Voltage is measured with wiring harness connected. Accordingly, if voltage is abnormal, check wiring harness and controller, too, for another cause of trouble, and then judge.		
	3	Ground fault in wiring harness (Short circuit with GND circuit)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			Wiring harness between J2 (female) (9) – P20 (female) (2)	Resistance	Min. 100 kΩ
	4	Short circuit in wiring harness (with another wiring harness)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			Wiring harness between J2 (female) (9) – P20 (female) (2) and J2 (female) (23) – P20 (female) (3)	Resistance	Min. 100 kΩ
	5	Defective wiring harness connector	Connecting parts among fuel control dial, engine unit wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation		
	6	Defective engine controller	★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting.		
			J2		Voltage
			Between (22) – (23)	Power supply	4.75 – 5.25 V

Circuit diagram related to fuel control dial

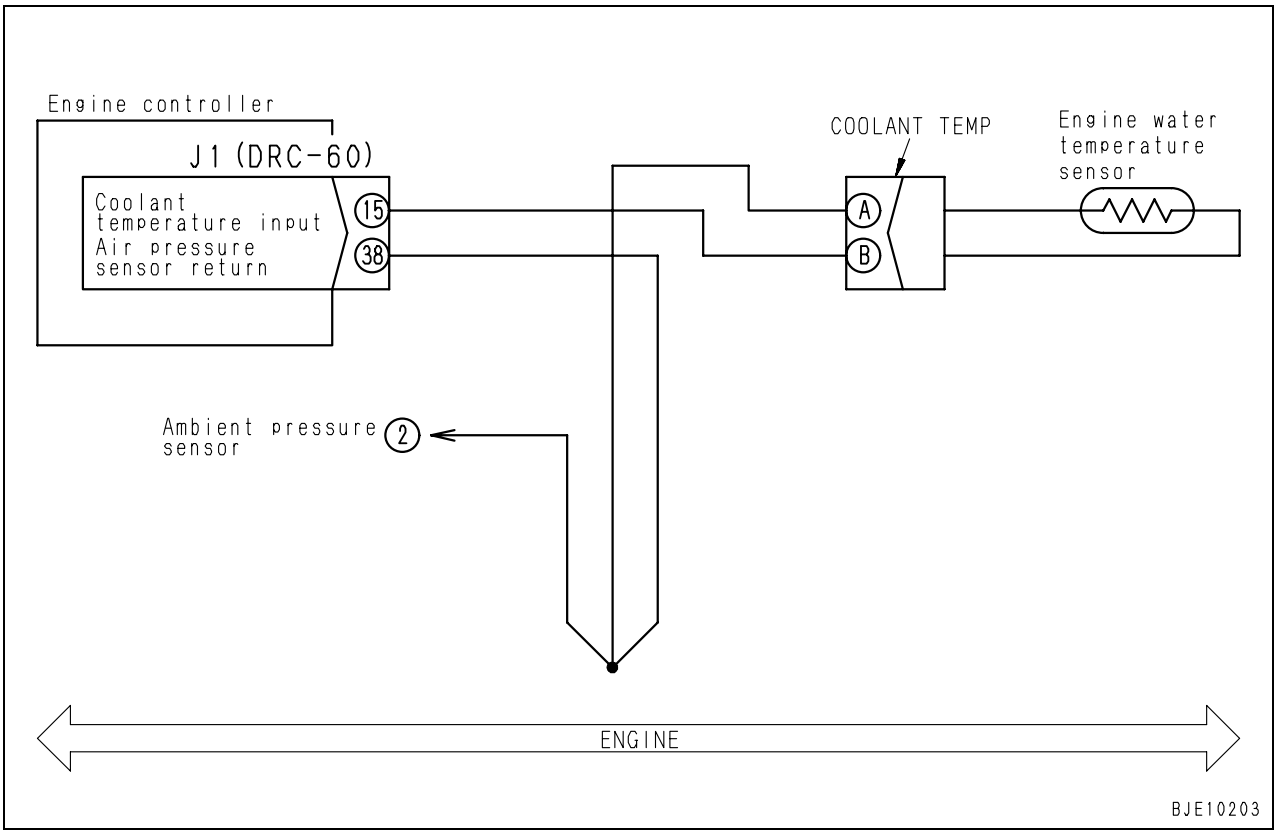


E-7 Code [144/CA144] Abnormally high level in coolant temperature sensor

Error code	Failure code	Trouble	Abnormally high level in coolant temperature sensor
144	**CA144**		
Contents of trouble	• High voltage is detected in coolant temperature sensor signal circuit.		
Action of controller	• Fixes coolant temperature and continues operation.		
Problem that appears on machine	• Exhaust smoke is white • Overheat prevention function does not function.		
Related information	• Method of reproducing failure code: Starting switch ON.		

	Causes		Standard value in normal state/Remarks on troubleshooting			
Possible causes and standard value in normal state	1	Defective coolant temperature sensor	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			COOLANT TEMP (Male)	Resistance		
			Between (A) – (B)	0.18 – 160 kΩ		
	2	Disconnection in wiring harness (Disconnection in wiring or defective contact in connector)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness between J1 (female) (15) – COOLANT TEMP (female) (B)	Resistance	Max. 10 Ω	
			Wiring harness between J1 (female) (38) – COOLANT TEMP (female) (A)	Resistance	Max. 10 Ω	
	3	Short circuit in wiring harness (with another wiring harness)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness among all pins between J1 (female) (15) – J1 (female) (With all connectors of wiring harness disconnected)	Resistance	Min. 100 kΩ	
	4	Defective wiring harness connector	Connecting parts among coolant temperature, engine wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation			
	5	Defective engine controller	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			J1 (Female)	Resistance		
			Between (15) – (38)	0.18 – 160 kΩ		

Circuit diagram related to coolant temperature sensor

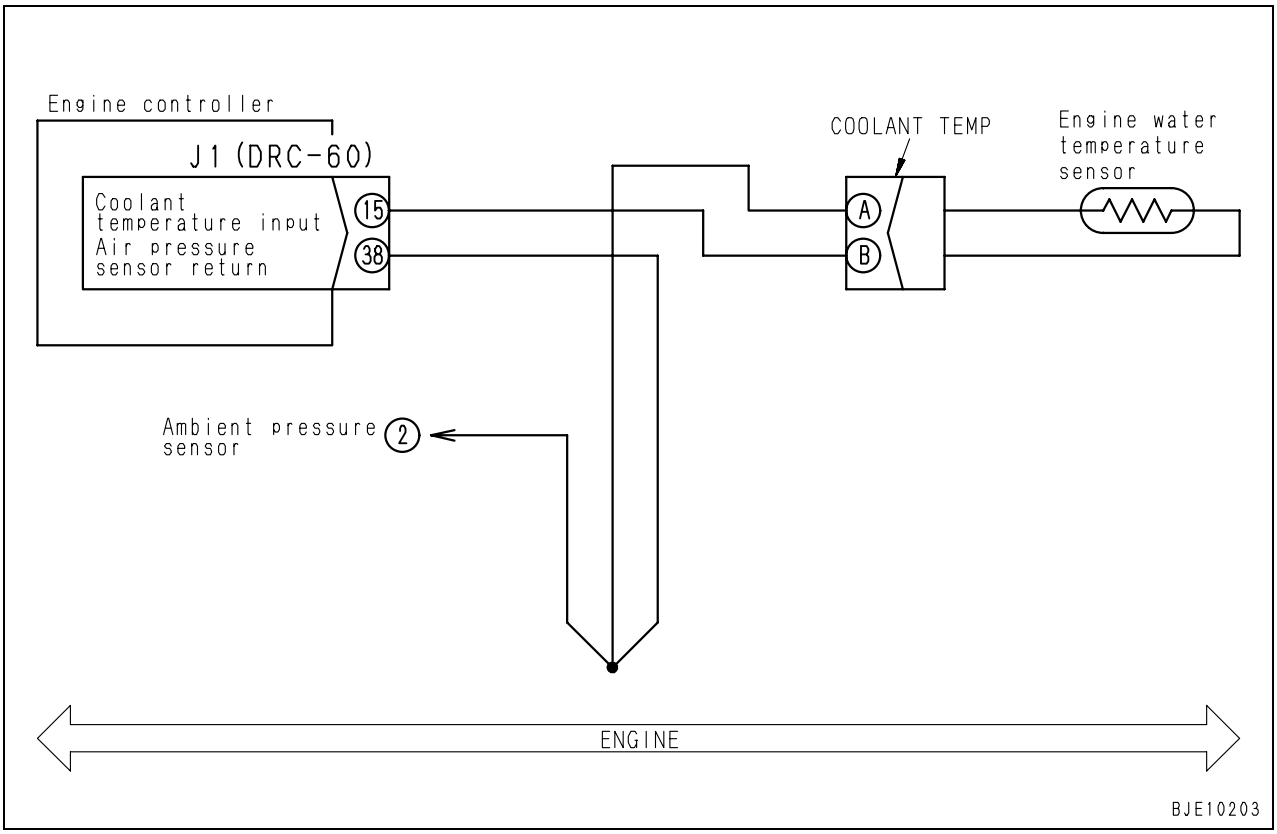


E-8 Code [145/CA145] Abnormally low level in coolant temperature sensor

Error code	Failure code	Trouble	Abnormally low level in coolant temperature sensor
145	**CA145**		
Contents of trouble	• Low voltage is detected in coolant temperature sensor signal circuit.		
Action of controller	• Fixes coolant temperature and continues operation.		
Problem that appears on machine	• Exhaust smoke is white • Overheat prevention function does not function.		
Related information	• Method of reproducing failure code: Starting switch ON.		

	Causes		Standard value in normal state/Remarks on troubleshooting		
Possible causes and standard value in normal state	1	Defective coolant temperature sensor	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			COOLANT TEMP (Male)	Resistance	
			Between (A) – (B)	0.18 – 160 kΩ	
			Between (B) – chassis ground	Min. 100 kΩ	
	2	Ground fault in wiring harness (Short circuit with GND circuit)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			Wiring harness between J1 (female) (15) – COOLANT TEMP (female) (B)	Resistance	Min. 100 kΩ
	3	Short circuit in wiring harness (with another wiring harness)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			Wiring harness among all pins between J1 (female) (15) – J1 (female) (With all connectors of wiring harness disconnected)	Resistance	Min. 100 kΩ
	4	Defective wiring harness connector	Connecting parts among coolant temperature, engine wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation		
	5	Defective engine controller	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			J1 (Female)	Resistance	
			Between (15) – (38)	0.18 – 160 kΩ	
			Between (15) – chassis ground	Min. 100 kΩ	

Circuit diagram related to coolant temperature sensor

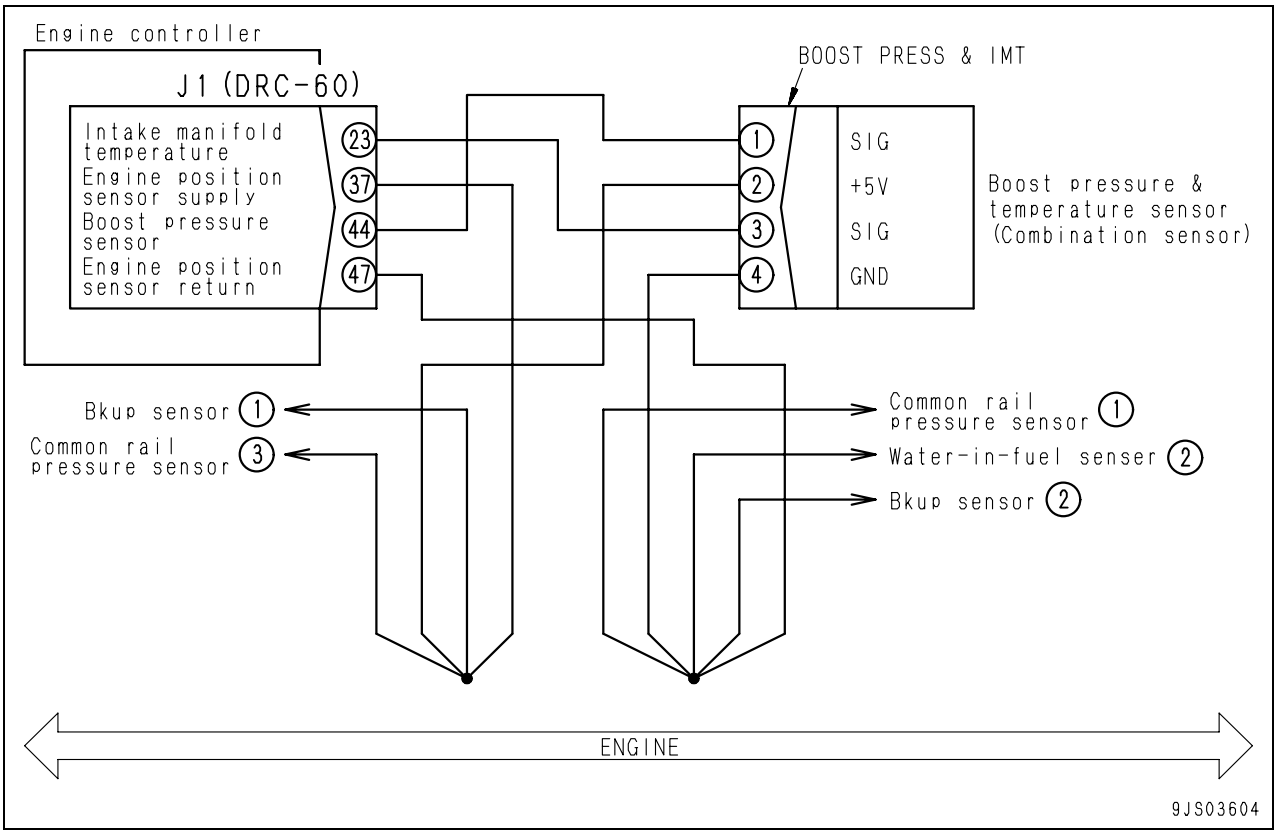


E-9 Code [153/CA153] Abnormally high level in charge temperature sensor

Error code	Failure code	Trouble	Abnormally high level in charge temperature sensor
153	**CA153**		
Contents of trouble	• High voltage is detected in temperature signal circuit of boost pressure sensor and temperature sensor.		
Action of controller	• Fixes charge temperature and continues operation.		
Problem that appears on machine	• Exhaust smoke is white • Boost temperature-based engine protection function does not function.		
Related information	• Method of reproducing failure code: Starting switch ON.		

	Causes		Standard value in normal state/Remarks on troubleshooting			
Possible causes and standard value in normal state	1	Defective [temperature signal circuit] of boost pressure sensor, temperature sensor	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			BOOST PRESS & IMT (Male)	Resistance		
			Between (3) – (4)	0.18 – 160 kΩ		
	2	Disconnection in wiring harness (Disconnection in wiring or defective contact in connector)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness between J1 (female) (23) – BOOST PRESS & IMT (female) (3)	Resistance	Max. 10 Ω	
			Wiring harness between J1 (female) (47) – BOOST PRESS & IMT (female) (4)	Resistance	Max. 10 Ω	
	3	Short circuit in wiring harness (with another wiring harness)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness among all pins between J1 (female) (23) – J1 (female) (With all connectors of wiring harness disconnected)	Resistance	Min. 100 kΩ	
	4	Defective wiring harness connector	Connecting parts among boost pressure sensor, temperature sensor, engine wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation			
	5	Defective engine controller	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			J1 (Female)	Resistance		
			Between (23) – (47)	0.18 – 160 kΩ		

Circuit diagram related to boost pressure, temperature sensor (combination sensor)

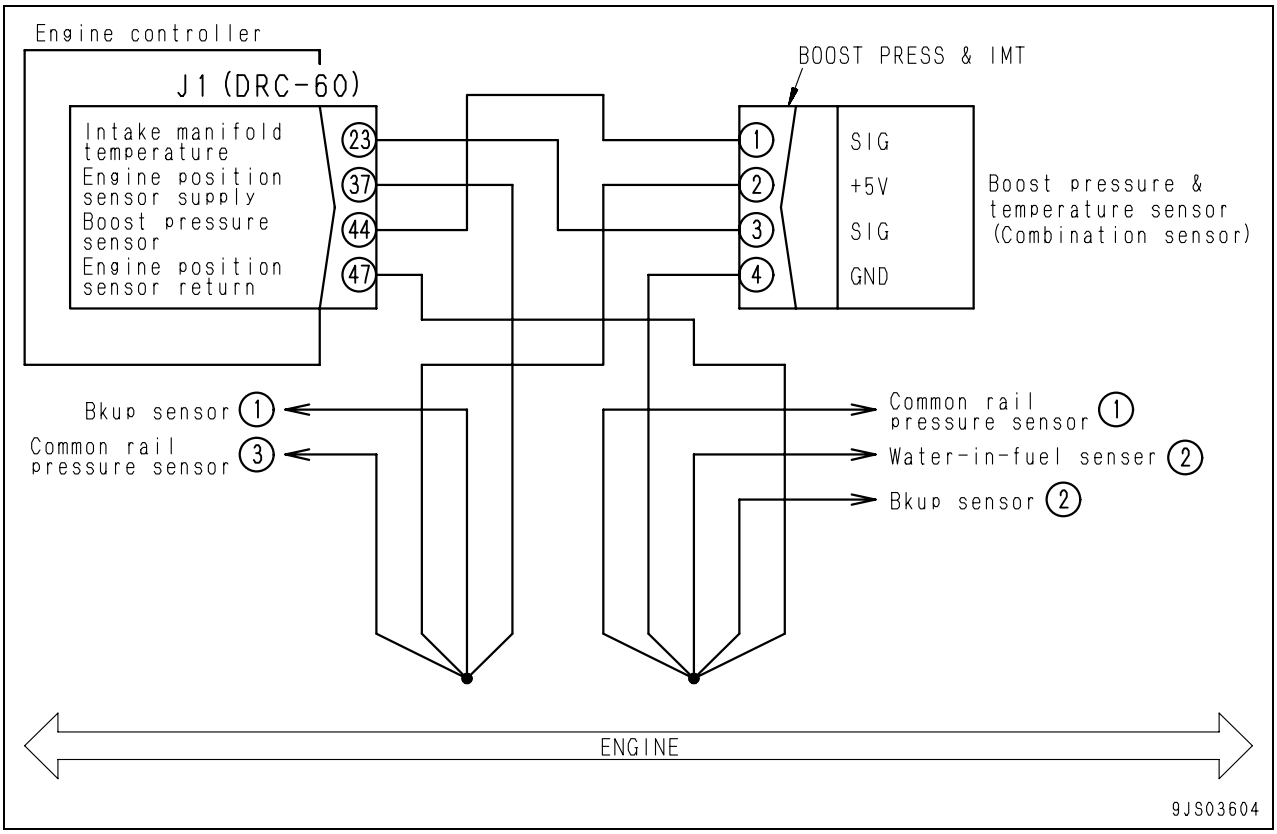


E-10 Code [154/CA154] Abnormally low level in charge temperature sensor

Error code	Failure code	Trouble	Abnormally low level in charge temperature sensor
154	**CA154**		
Contents of trouble	• Low voltage is detected in temperature signal circuit of boost pressure sensor and temperature sensor.		
Action of controller	• Fixes charge temperature and continues operation.		
Problem that appears on machine	• Exhaust smoke is white • Boost temperature-based engine protection function does not function.		
Related information	• Method of reproducing failure code: Starting switch ON.		

	Causes		Standard value in normal state/Remarks on troubleshooting			
Possible causes and standard value in normal state	1	Defective [temperature signal circuit] of boost pressure sensor, temperature sensor	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			BOOST PRESS & IMT (Male)	Resistance		
			Between (3) – (4)	0.18 – 160 kΩ		
			Between (3) – chassis ground	Min. 100 kΩ		
	2	Ground fault in wiring harness (Short circuit with GND circuit)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness between J1 (female) (23) and BOOT PRESS & IMT (female) (3)	Resistance	Min. 100 kΩ	
	3	Short circuit in wiring harness (with another wiring harness)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness among all pins between J1 (female) (23) – J1 (female) (With all connectors of wiring harness disconnected)	Resistance	Min. 100 kΩ	
	4	Defective wiring harness connector	Connecting parts among boost pressure sensor, temperature sensor, engine wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation			
	5	Defective engine controller	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			J1 (Female)	Resistance		
			Between (23) – (47)	0.18 – 160 kΩ		
			Between (23) – chassis ground	Min. 100 kΩ		

Circuit diagram related to boost pressure, temperature sensor (combination sensor)



E-11 Code [155/CA155] Derating of speed by abnormally high charge temperature

Error code	Failure code	Trouble	Derating of speed by abnormally high charge temperature
155	**CA155**		
Contents of trouble	• Temperature signal of boost pressure sensor and temperature sensor is above upper control limit.		
Action of controller	• Limits output and continues operation.		
Problem that appears on machine	• Engine stops • Output drops.		
Related information	—		

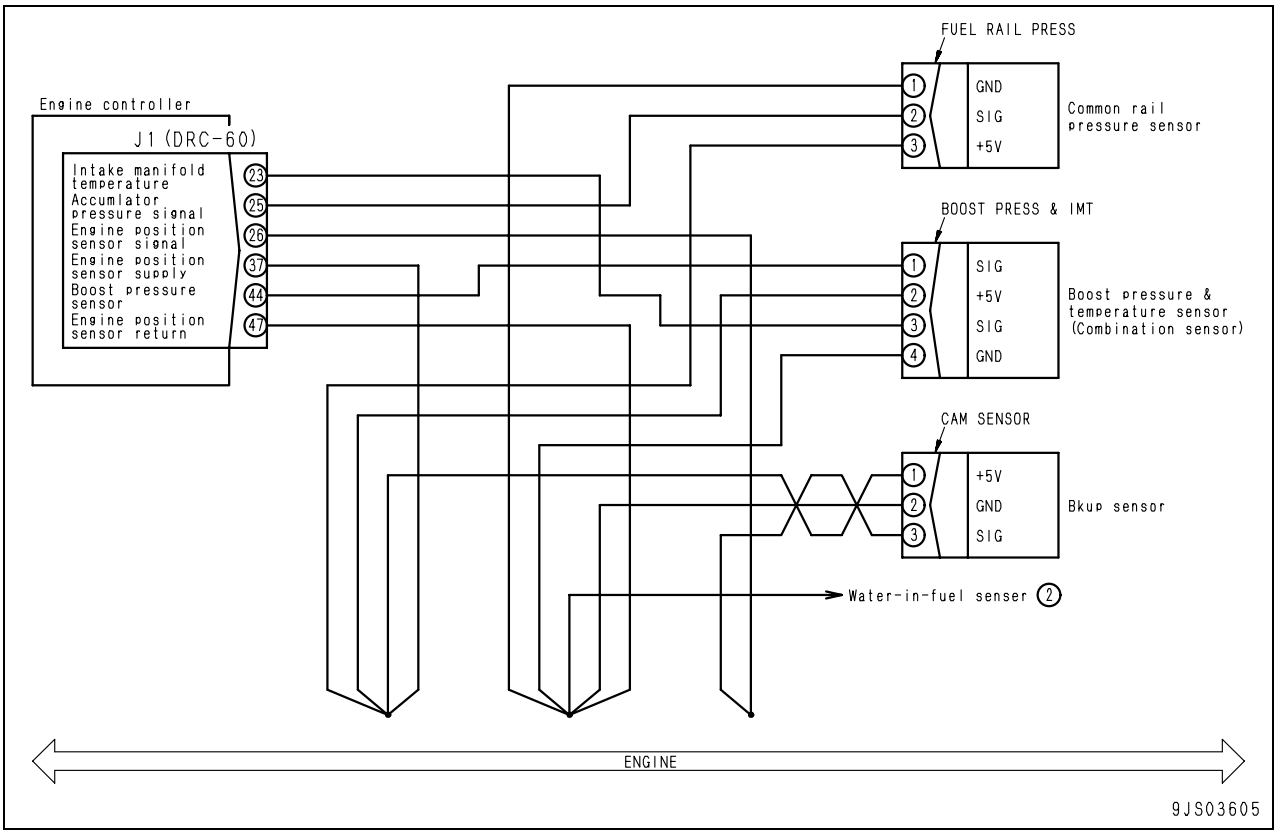
		Causes	Standard value in normal state/Remarks on troubleshooting
Possible causes and standard value in normal state	1	Drop in cooling performance of aftercooler.	Inspect following directly. • Loose, broken fan belt • Lack of cooling air • Clogged aftercooler fin
	2	Abnormally high temperature at turbocharger outlet	Inspect related parts directly.
	3	Defective engine controller	If causes 1 – 2 are not detected, engine controller may be defective. (Since trouble is in system, troubleshooting cannot be carried out.)

E-12 Code [187/CA187] Abnormally low level in sensor power supply 2

Error code	Failure code	Trouble	Abnormally low level in sensor power supply 2 circuit
187	**CA187**		
Contents of trouble	• Low voltage is detected in sensor power supply 2 circuit.		
Action of controller	• Fixes boost pressure and charge temperature, and continues operation. • Limits output and continues operation.		
Problem that appears on machine	• Engine output drops.		
Related information	• Method of reproducing failure code: Starting switch ON.		

	Causes		Standard value in normal state/Remarks on troubleshooting		
Possible causes and standard value in normal state	1	Defective sensor or wiring harness	★ Disconnect connector with starting switch OFF, then turn starting switch ON and carry out troubleshooting.		
			Disconnect devices at right in order. If mark E of failure code disappears after reproduction operation is conducted, that device or wiring harness has a defect in it.	Boost pressure, temperature sensor	BOOST PRESS & IMT
				Common rail pressure sensor	FUEL RAIL PRESS
				Bkup sensor	CAM SENSOR
				Engine wiring harness	J1
	2	Defective wiring harness connector	Connecting parts among each sensor, engine wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation		
	3	Defective engine controller	If causes 1 and 2 are not detected, engine controller may be defective. (Since trouble is in system, troubleshooting cannot be carried out.)		

Circuit diagram related to sensor power supply 2

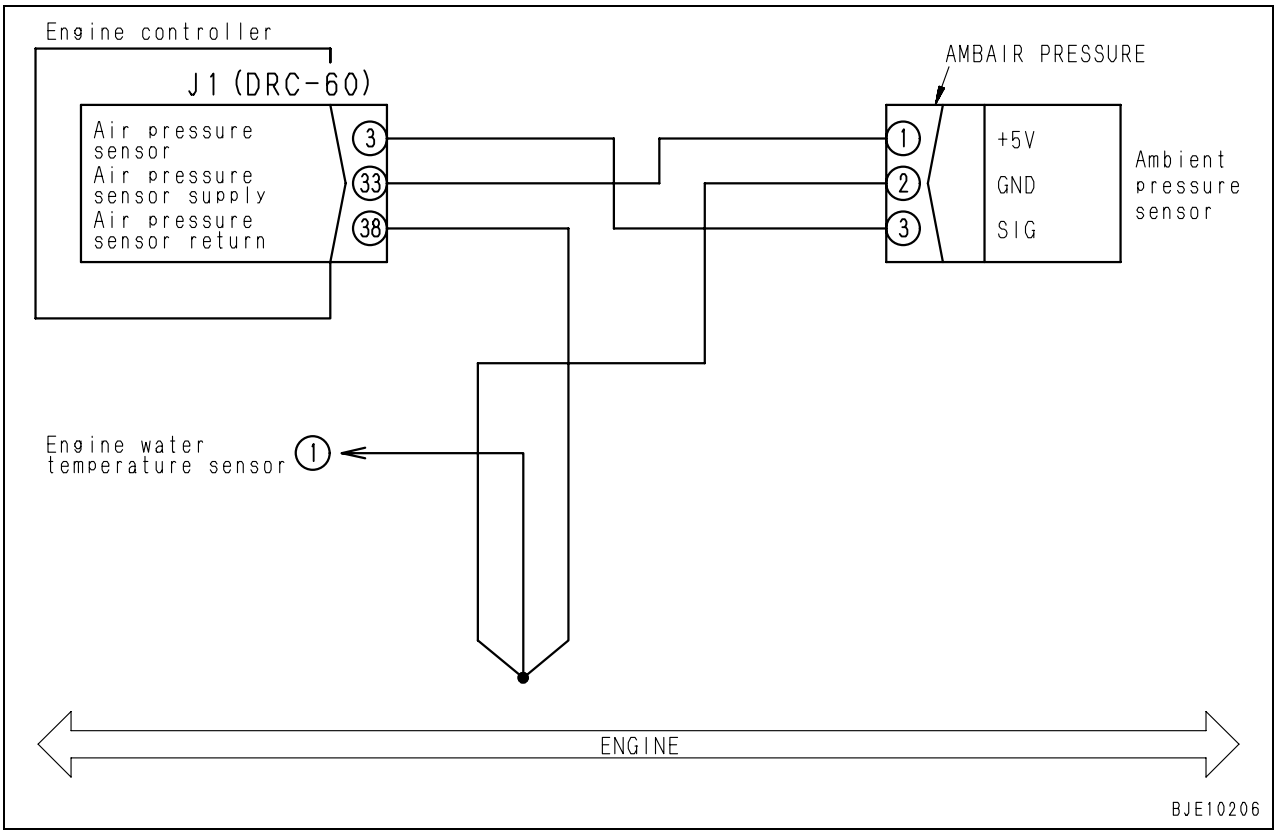


E-13 Code [221/CA221] Abnormally high level in atompspheric pressure sensor

Error code	Failure code	Trouble	Abnormally high level in atmospheric pressure sensor
221	**CA221**		
Contents of trouble	• High voltage is detected in atmospheric pressure sensor		
Action of controller	• Fixes atmospheric pressure and continues operation.		
Problem that appears on machine	• Engine output drops. • Starting performance is poor.		
Related information	• Method of reproducing failure code: Starting switch ON.		

	Causes		Standard value in normal state/Remarks on troubleshooting		
Possible causes and standard value in normal state	1	Defective sensor power supply 1 circuit	If failure code [**CA386**] is indicated simultaneously, carry out troubleshooting for it first.		
	2	Defective atmospheric pressure sensor	★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting.		
			AMBAIR PRESSURE		Voltage
			Between (1) – (2)	Power supply	4.75 – 5.25 V
			Voltage is measured with wiring harness connected. Accordingly, if voltage is abnormal, check wiring harness and controller, too, for another cause of trouble, and then judge.		
	3	Hot short (Short circuit with 5 V, 24 V circuit) in wiring harness	★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting.		
			Wiring harness between J1 (female) (3) – AMBAIR PRESSURE (female) (3)	Voltage	Max. 1 V
	4	Short circuit in wiring harness (with another wiring harness)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			Wiring harness between J1 (female) (3) – AMBAIR PRESSURE (female) (3) and J1(female) (33) – AMBAIR PRESSURE (female) (1)	Resistance	Min. 100 kΩ
	5	Defective wiring harness connector	Connecting parts among atmospheric pressure sensor, engine wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation		
	6	Defective engine controller	★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting.		
			J1		Voltage
			Between (33) – (38)	Power supply	4.75 – 5.25 V

Circuit diagram related to atmospheric pressure sensor

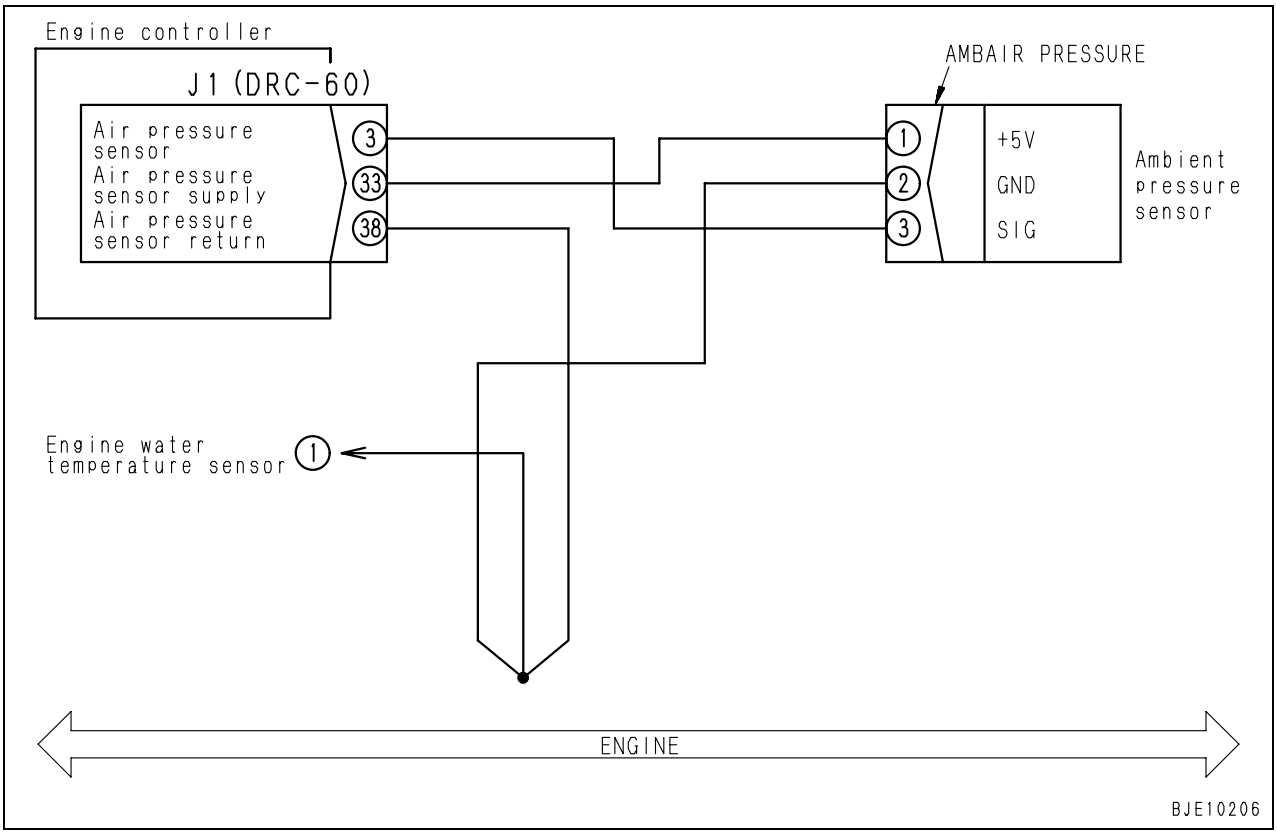


E-14 Code [222/CA222] Abnormally low level in atompspheric pressure sensor

Error code	Failure code	Trouble	Abnormally low level in atmospheric pressure sensor
222	**CA222**		
Contents of trouble	• Low voltage is detected in atmospheric pressure sensor		
Action of controller	• Fixes atmospheric pressure and continues operation.		
Problem that appears on machine	• Engine output drops. • Starting performance is poor.		
Related information	• Method of reproducing failure code: Starting switch ON.		

	Causes		Standard value in normal state/Remarks on troubleshooting			
Possible causes and standard value in normal state	1	Defective sensor power supply 1 circuit	If failure code [**CA352**] is indicated simultaneously, carry out troubleshooting for it first.			
	2	Defective atmospheric pressure sensor	★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting.			
			AMBAIR PRESSURE		Voltage	
			Between (1) – (2)	Power supply	4.75 – 5.25 V	
			Voltage is measured with wiring harness connected. Accordingly, if voltage is abnormal, check wiring harness and controller, too, for another cause of trouble, and then judge.			
	3	Ground fault in wiring harness (Short circuit with GND circuit)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness between J1 (female) (3) – AMBAIR PRESSURE (female) (3)		Resistance	Min. 100 kΩ
	4	Short circuit in wiring harness (with another wiring harness)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness between J1 (female) (3) – AMBAIR PRESSURE (female) (3) and J1 (female) (38) – AMBAIR PRESSURE (female) (2)		Resistance	Min. 100 kΩ
	5	Defective wiring harness connector	Connecting parts among atmospheric pressure sensor, engine wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation			
	6	Defective engine controller	★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting.			
			J1		Voltage	
			Between (33) – (38)	Power supply	4.75 – 5.25 V	

Circuit diagram related to atmospheric pressure sensor

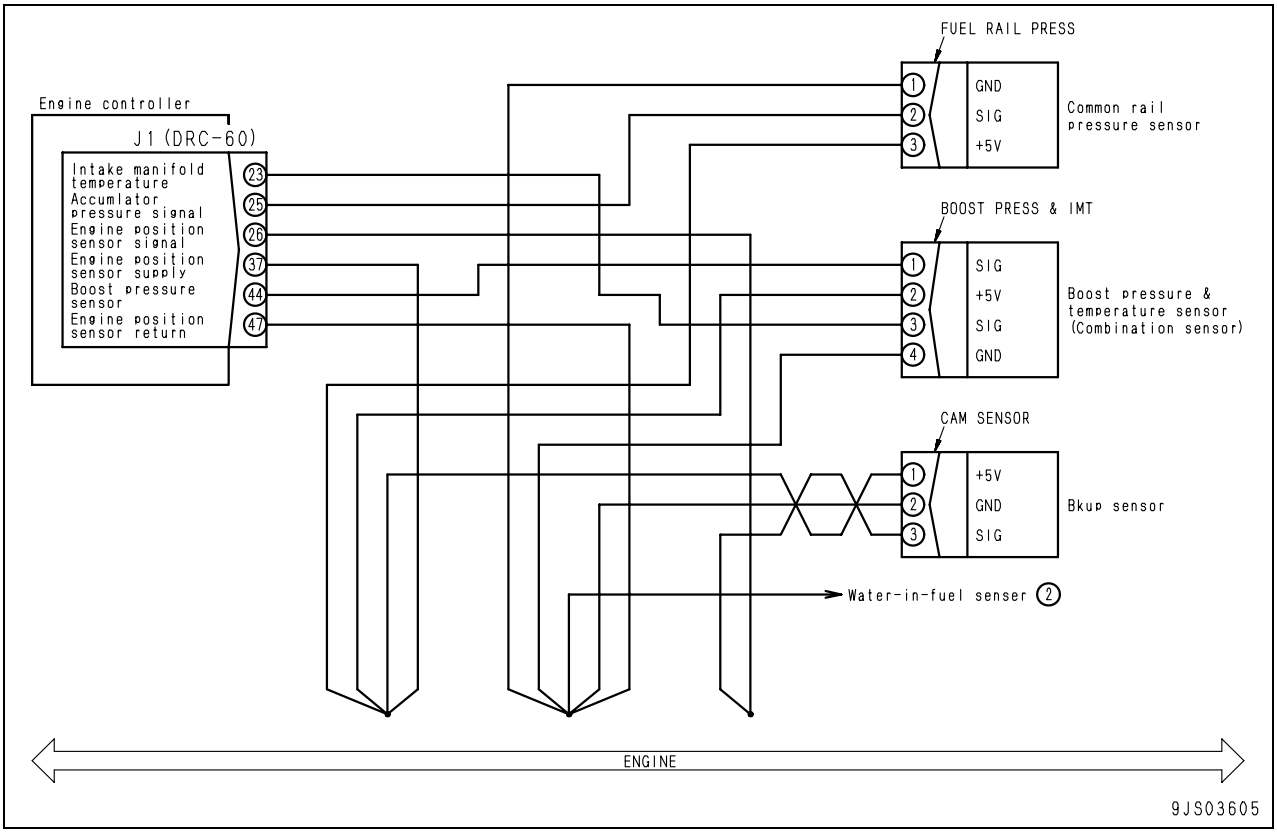


E-15 Code [227/CA227] Abnormally high level in sensor power supply 2

Error code	Failure code	Trouble	Abnormally high level in sensor power supply 2 circuit
227	**CA227**		
Contents of trouble	• High voltage is detected in sensor power supply 2 circuit.		
Action of controller	• Fixes boost pressure and charge temperature, and continues operation. • Limits output and continues operation.		
Problem that appears on machine	• Engine output drops.		
Related information	• Method of reproducing failure code: Starting switch ON.		

	Causes		Standard value in normal state/Remarks on troubleshooting
Possible causes and standard value in normal state	1	Defective wiring harness connector	Connecting parts among each sensor, engine wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation
	2	Defective engine controller	If cause 1 is not detected, engine controller may be defective. (Since trouble is in system, troubleshooting cannot be carried out.)

Circuit diagram related to sensor power supply 2



E-16 Code [234/CA234] Engine overspeed

Error code	Failure code	Trouble	Engine overspeed
234	**CA234**		
Contents of trouble	• Engine speed is above upper control limit.		
Action of controller	• Stops injection of injector until engine speed drops to normal level.		
Problem that appears on machine	• Engine speed fluctuates.		
Related information	• Method of reproducing failure code: Start engine and run it at high idle.		

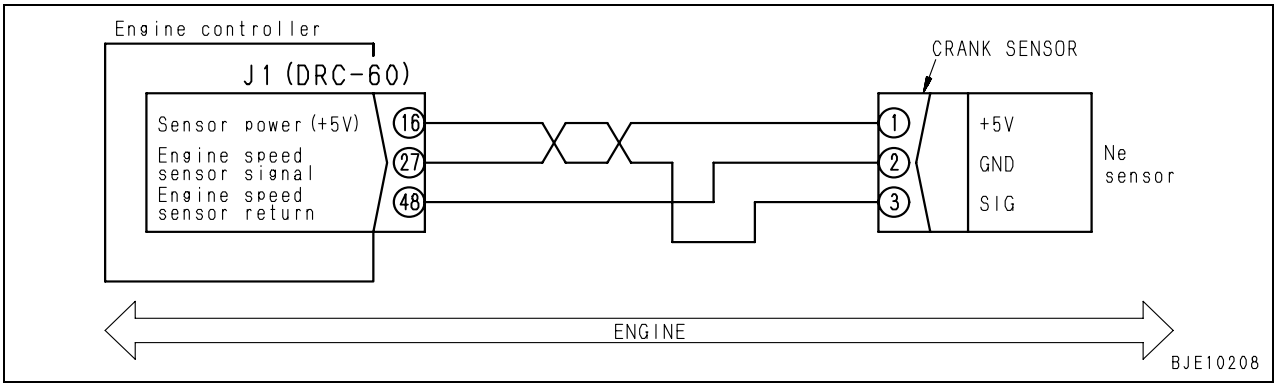
Possible causes and standard value in normal state	Causes		Standard value in normal state/Remarks on troubleshooting
	1	Use of improper fuel	Check fuel used directly (for high viscosity).
	2	Improper way of use	Way of use may be improper. Teach proper way of use to operator.
	3	Defective engine controller	If cause 1 is not detected, engine controller may be defective. (Since trouble is in system, troubleshooting cannot be carried out.)

E-17 Code [238/CA238] Abnormally level in Ne speed sensor power supply

Error code	Failure code	Trouble	Abnormality in Ne speed sensor power supply
238	**CA238**		
Contents of trouble	• Low voltage is detected in engine Ne speed sensor power supply circuit.		
Action of controller	• Controls Ne speed sensor with signals of Bkup speed sensor.		
Problem that appears on machine	• Starting performance is poor • There is hunting from engine.		
Related information	• Method of reproducing failure code: Starting switch ON.		

	Causes		Standard value in normal state/Remarks on troubleshooting		
Possible causes and standard value in normal state	1	Defective sensor or wiring harness	★ Disconnect connector with starting switch OFF, then turn starting switch ON and carry out troubleshooting.		
			Disconnect devices at right in order. If mark E of failure code disappears after reproduction operation is conducted, that device or wiring harness has a defect in it.	Ne speed sensor	CRANK SENSOR
				Engine wiring harness	J1
	2	Defective wiring harness connector	Connecting parts among each sensor, engine wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation		

Circuit diagram related to engine Ne speed sensor

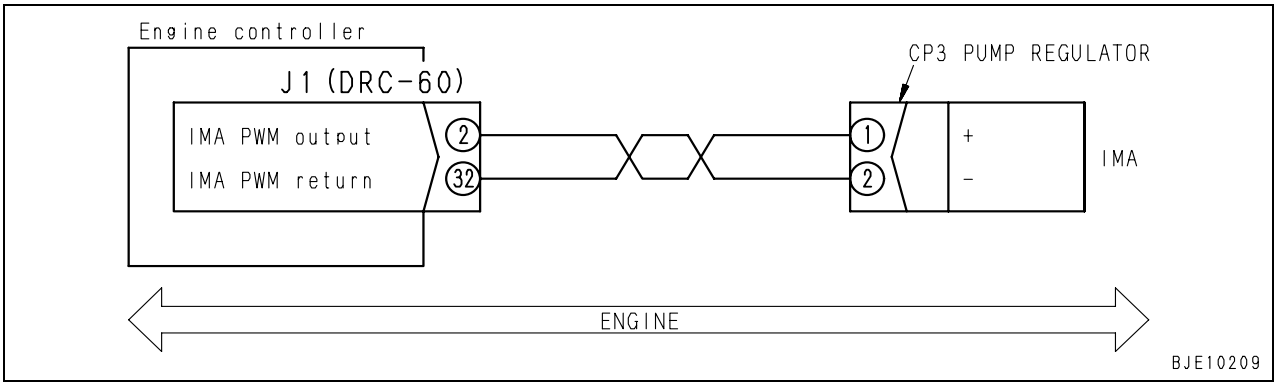


E-18 Code [271/CA271] Short circuit in IMV/PCV1

Error code	Failure code	Trouble	Short circuit in IMV/PCV1
271	**CA271**		
Contents of trouble	• There is short circuit in supply pump actuator drive circuit.		
Action of controller	• None in particular.		
Problem that appears on machine	• Engine speed does not rise from low idle. • Engine output drops. • Common rail fuel pressure is above command value.		
Related information	• Method of reproducing failure code: Starting switch ON.		

		Causes	Standard value in normal state/Remarks on troubleshooting		
Possible causes and standard value in normal state	1	Defective supply pump actuator	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			CP3 PUMP REGULATOR (Male)	Resistance	
			Between (1) – chassis ground	Min. 100 kΩ	
	2	Ground fault in wiring harness (Short circuit with GND circuit)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			Wiring harness between J1 (female) (2) – CP3 PUMP REGULATOR (female) (1)	Resistance	Min. 100 kΩ
	3	Short circuit in wiring harness (with another wiring harness)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			Wiring harness among all pins between J1 (female) (2) – J1 (female) (With all connectors of wiring harness disconnected)	Resistance	Min. 100 kΩ
	4	Defective wiring harness connector	Connecting parts among supply pump actuator, engine wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation		
	5	Defective engine controller	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			J1 (Female)	Resistance	
			Between (2) – chassis ground	Min. 100 kΩ	

Circuit diagram related to supply pump actuator (metering unit)

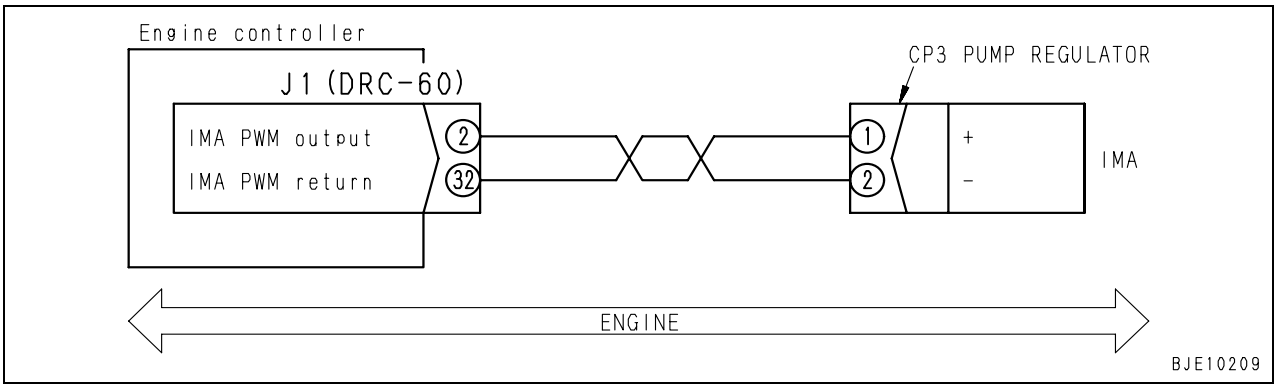


E-19 Code [272/CA272] Disconnection in IMV/PCV1

Error code	Failure code	Trouble	Disconnection in IMV/PCV1
272	**CA272**		
Contents of trouble	• There is disconnection in supply pump actuator drive circuit.		
Action of controller	• None in particular.		
Problem that appears on machine	• Engine speed runs, but in unstable condition. • Common rail fuel pressure is above command value.		
Related information	• Method of reproducing failure code: Starting switch ON.		

	Causes		Standard value in normal state/Remarks on troubleshooting		
Possible causes and standard value in normal state	1	Defective supply pump actuator	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			CP3 PUMP REGULATOR (Male)	Resistance	
			Between (1) – (2)	Max. 5 Ω	
	2	Disconnection in wiring harness (Disconnection in wiring or defective contact in connector)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			Wiring harness between J1 (female) (2) – CP3 PUMP REGULATOR (female) (1)	Resistance	Max. 10 Ω
			Wiring harness between J1 (female) (32) – CP3 PUMP REGULATOR (female) (2)	Resistance	Max. 10 Ω
	3	Hot short (Short circuit with 24V circuit) in wiring harness	★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting.		
			Wiring harness between J1 (female) (2) – CP3 PUMP REGULATOR (female) (1)	Voltage	Max. 3 V
	4	Short circuit in wiring harness (with another wiring harness)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			Wiring harness among all pins between J1 (female) (2) – J1 (female) (With all connectors of wiring harness disconnected)	Resistance	Min. 100 kΩ
	5	Defective wiring harness connector	Connecting parts among supply pump actuator, engine wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation		
	6	Defective engine controller	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			J1 (Female)	Resistance	
			Between (2) – (32)	Max. 5 Ω	

Circuit diagram related to supply pump actuator (metering unit)

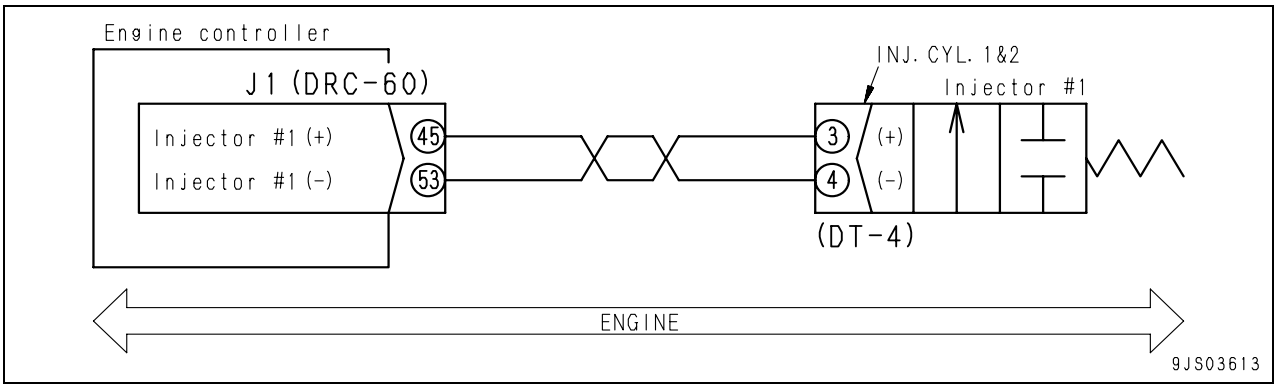


E-20 Code [322/CA322] Disconnection, short circuit in injector No. 1

Error code	Failure code	Trouble	Disconnection, short circuit in injector No. 1
322	**CA322**		
Contents of trouble	• There is disconnection or short circuit in drive circuit of injector No. 1.		
Action of controller	• None in particular.		
Problem that appears on machine	• There is irregular combustion or hunting. • Output drops.		
Related information	• Method of reproducing failure code: Start engine.		

	Causes		Standard value in normal state/Remarks on troubleshooting			
Possible causes and standard value in normal state	1	Defective engine controller	If failure codes ([**CA322**], [**CA324**], [**CA331**]) are indicated simultaneously, engine controller has a defect in it.			
	2	Defective injector No. 1	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			INJ CYL 1 (Male)	Resistance		
			Between (3) – (4)	Max. 2 Ω		
			Between (3) – chassis ground	Min. 100 kΩ		
	3	Disconnection in wiring harness (Disconnection in wiring or defective contact in connector)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness between J1 (female) (45) – INJ CYL 1 (female) (3)	Resistance	Max. 2 Ω	
			Wiring harness between J1 (female) (53) – INJ CYL 1 (female) (4)	Resistance	Max. 2 Ω	
	4	Ground fault in wiring harness (Short circuit with GND circuit)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness between J1 (female) (45) – INJ CYL 1 (female) (3)	Resistance	Max. 2 Ω	
	5	Short circuit in wiring harness (with another wiring harness)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness among all pins between J1 (female) (45) – J1 (female) (With all connectors of wiring harness disconnected)	Resistance	Min. 100 kΩ	
			Wiring harness among all pins between J1 (female) (53) – J1 (female) (With all connectors of wiring harness disconnected)	Resistance	Min. 100 kΩ	
	6	Defective wiring harness connector	Connecting parts among injector No. 1, engine wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation			
	7	Defective engine controller	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			J1 (Female)	Resistance		
			Between (45) – (53)	Max. 2 Ω		
			Between (45) – chassis ground	Min. 100 kΩ		

Circuit diagram related to injector No. 1

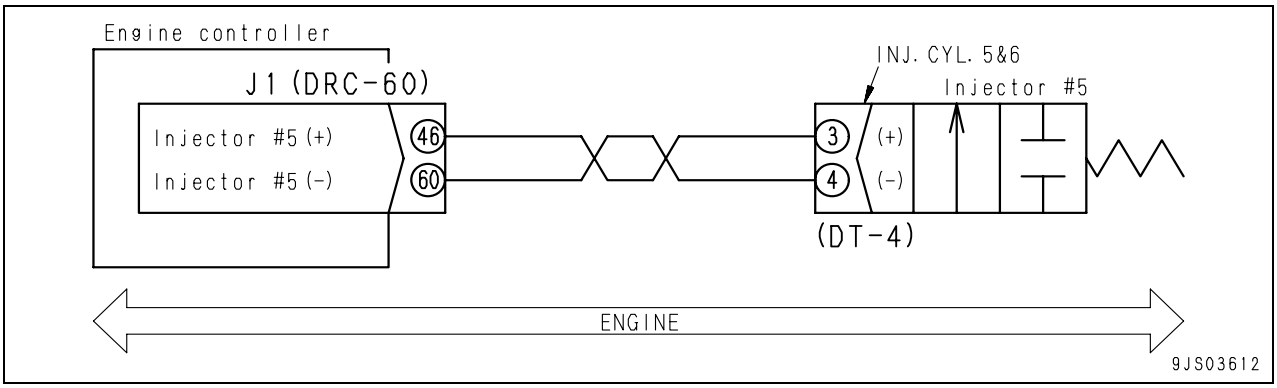


E-21 Code [323/CA323] Disconnection, short circuit in injector No. 5

Error code	Failure code	Trouble	Disconnection, short circuit in injector No. 5
323	**CA323**		
Contents of trouble	• There is disconnection or short circuit in drive circuit of injector No. 5.		
Action of controller	• None in particular.		
Problem that appears on machine	• There is irregular combustion or hunting. • Output drops.		
Related information	• Method of reproducing failure code: Start engine.		

	Causes		Standard value in normal state/Remarks on troubleshooting			
Possible causes and standard value in normal state	1	Defective engine controller	If failure codes ([**CA323**], [**CA325**], [**CA332**]) are indicated simultaneously, engine controller has a defect in it.			
	2	Defective injector No. 5	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			INJ CYL 5 (Male)	Resistance		
			Between (3) – (4)	Max. 2 Ω		
			Between (3) – chassis ground	Min. 100 kΩ		
	3	Disconnection in wiring harness (Disconnection in wiring or defective contact in connector)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness between J1 (female) (46) – INJ CYL 5 (female) (3)	Resistance	Max. 2 Ω	
			Wiring harness between J1 (female) (60) – INJ CYL 5 (female) (4)	Resistance	Max. 2 Ω	
	4	Ground fault in wiring harness (Short circuit with GND circuit)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness between J1 (female) (46) – INJ CYL 5 (female) (3)	Resistance	Max. 2 Ω	
	5	Short circuit in wiring harness (with another wiring harness)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness among all pins between J1 (female) (46) – J1 (female) (With all connectors of wiring harness disconnected)	Resistance	Min. 100 kΩ	
			Wiring harness among all pins between J1 (female) (60) – J1 (female) (With all connectors of wiring harness disconnected)	Resistance	Min. 100 kΩ	
	6	Defective wiring harness connector	Connecting parts among injector No. 5, engine wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation			
	7	Defective engine controller	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			J1 (Female)	Resistance		
			Between (46) – (60)	Max. 2 Ω		
			Between (46) – chassis ground	Min. 100 kΩ		

Circuit diagram related to injector No. 5

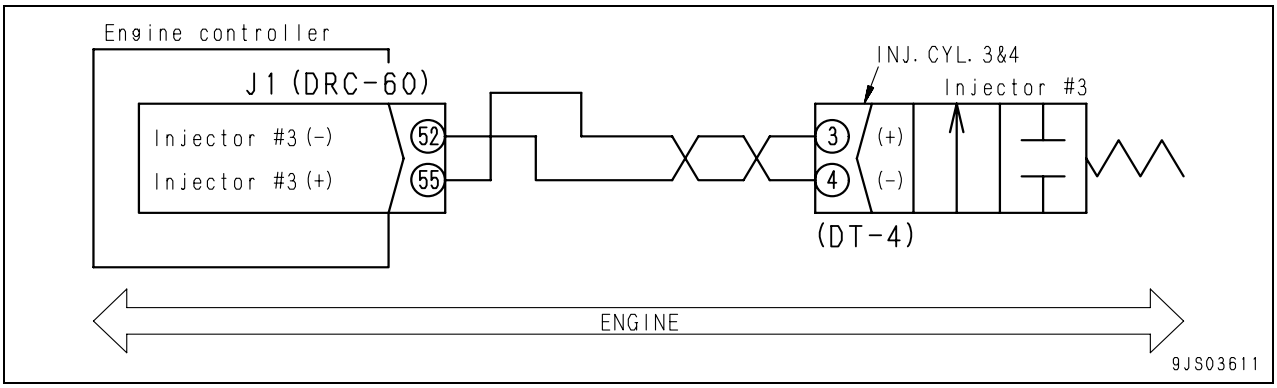


E-22 Code [324/CA324] Disconnection, short circuit in injector No. 3

Error code	Failure code	Trouble	Disconnection, short circuit in injector No. 3
324	**CA324**		
Contents of trouble	• There is disconnection or short circuit in drive circuit of injector No. 3.		
Action of controller	• None in particular.		
Problem that appears on machine	• There is irregular combustion or hunting. • Output drops.		
Related information	• Method of reproducing failure code: Start engine.		

	Causes		Standard value in normal state/Remarks on troubleshooting			
Possible causes and standard value in normal state	1	Defective engine controller	If failure codes ([**CA322**], [**CA324**], [**CA331**]) are indicated simultaneously, engine controller has a defect in it.			
	2	Defective injector No. 3	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			INJ CYL 3 (Male)	Resistance		
			Between (3) – (4)	Max. 2 Ω		
			Between (3) – chassis ground	Min. 100 kΩ		
	3	Disconnection in wiring harness (Disconnection in wiring or defective contact in connector)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness between J1 (female) (55) – INJ CYL 3 (female) (3)	Resistance	Max. 2 Ω	
			Wiring harness between J1 (female) (52) – INJ CYL 3 (female) (4)	Resistance	Max. 2 Ω	
	4	Ground fault in wiring harness (Short circuit with GND circuit)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness between J1 (female) (55) – INJ CYL 3 (female) (3)	Resistance	Max. 2Ω	
	5	Short circuit in wiring harness (with another wiring harness)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness among all pins between J1 (female) (55) – J1 (female) (With all connectors of wiring harness disconnected)	Resistance	Min. 100 kΩ	
			Wiring harness among all pins between J1 (female) (52) – J1 (female) (With all connectors of wiring harness disconnected)	Resistance	Min. 100 kΩ	
	6	Defective wiring harness connector	Connecting parts among injector No. 3, engine wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation			
	7	Defective engine controller	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			J1 (Female)	Resistance		
			Between (55) – (52)	Max. 2 Ω		
			Between (55) – chassis ground	Min. 100 kΩ		

Circuit diagram related to injector No. 3

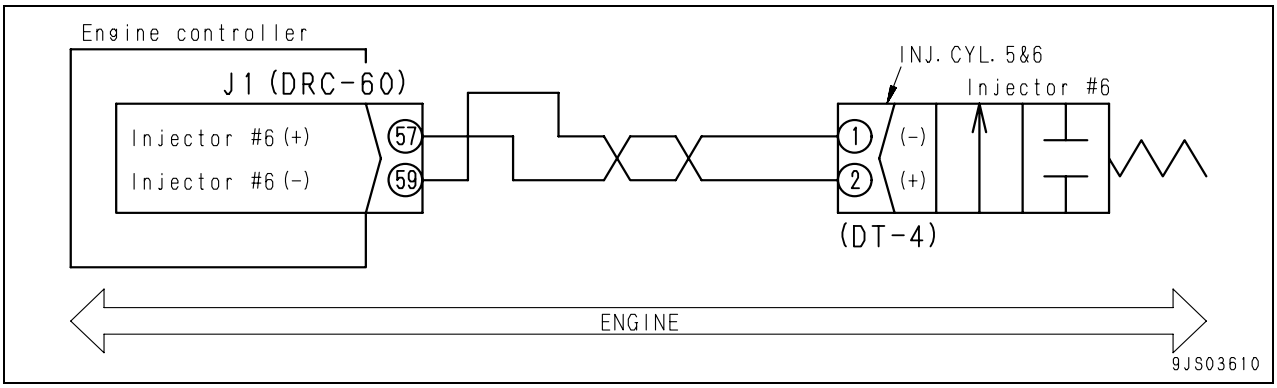


E-23 Code [325/CA325] Disconnection, short circuit in injector No. 6

Error code	Failure code	Trouble	Disconnection, short circuit in injector No. 6
325	**CA325**		
Contents of trouble	• There is disconnection or short circuit in drive circuit of injector No. 6.		
Action of controller	• None in particular.		
Problem that appears on machine	• There is irregular combustion or hunting.		
Related information	• Method of reproducing failure code: Start engine.		

	Causes		Standard value in normal state/Remarks on troubleshooting			
Possible causes and standard value in normal state	1	Defective engine controller	If failure codes ([**CA323**], [**CA325**], [**CA332**]) are indicated simultaneously, engine controller has a defect in it.			
	2	Defective injector No. 6	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			INJ CYL 6 (Male)	Resistance		
			Between (2) – (1)	Max. 2 Ω		
			Between (2) – chassis ground	Min. 100 kΩ		
	3	Disconnection in wiring harness (Disconnection in wiring or defective contact in connector)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness between J1 (female) (57) – INJ CYL 6 (female) (2)	Resistance	Max. 2 Ω	
			Wiring harness between J1 (female) (59) – INJ CYL 6 (female) (1)	Resistance	Max. 2 Ω	
	4	Ground fault in wiring harness (Short circuit with GND circuit)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness between J1 (female) (57) – INJ CYL 6 (female) (2)	Resistance	Max. 2 Ω	
	5	Short circuit in wiring harness (with another wiring harness)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness among all pins between J1 (female) (57) – J1 (female) (With all connectors of wiring harness disconnected)	Resistance	Min. 100 kΩ	
			Wiring harness among all pins between J1 (female) (59) – J1 (female) (With all connectors of wiring harness disconnected)	Resistance	Min. 100 kΩ	
	6	Defective wiring harness connector	Connecting parts among injector No. 6, engine wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation			
	7	Defective engine controller	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			J1 (Female)	Resistance		
			Between (57) – (59)	Max. 2 Ω		
			Between (57) – chassis ground	Min. 100 kΩ		

Circuit diagram related to injector No. 6

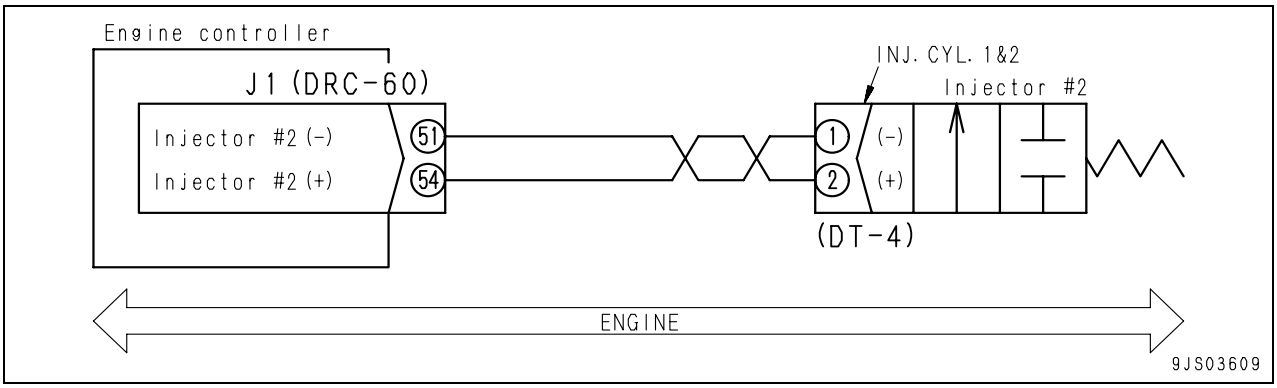


E-24 Code [331/CA331] Disconnection, short circuit in injector No. 2

Error code	Failure code	Trouble	Disconnection, short circuit in injector No. 2
331	**CA331**		
Contents of trouble	• There is disconnection or short circuit in drive circuit of injector No. 2.		
Action of controller	• None in particular.		
Problem that appears on machine	• There is irregular combustion or hunting. • Output drops.		
Related information	• Method of reproducing failure code: Start engine.		

	Causes		Standard value in normal state/Remarks on troubleshooting			
Possible causes and standard value in normal state	1	Defective engine controller	If failure codes ([**CA322**], [**CA324**], [**CA331**]) are indicated simultaneously, engine controller has a defect in it.			
	2	Defective injector No. 2	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			INJ CYL 2 (Male)	Resistance		
			Between (2) – (1)	Max. 2 Ω		
			Between (2) – chassis ground	Min. 100 kΩ		
	3	Disconnection in wiring harness (Disconnection in wiring or defective contact in connector)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness between J1 (female) (54) – INJ CYL 2 (female) (2)	Resistance	Max. 2 Ω	
			Wiring harness between J1 (female) (51) – INJ CYL 2 (female) (1)	Resistance	Max. 2 Ω	
	4	Ground fault in wiring harness (Short circuit with GND circuit)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness between J1 (female) (54) – INJ CYL 2 (female) (2)	Resistance	Max. 2 Ω	
	5	Short circuit in wiring harness (with another wiring harness)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness among all pins between J1 (female) (54) – J1 (female) (With all connectors of wiring harness disconnected)	Resistance	Min. 100 kΩ	
			Wiring harness among all pins between J1 (female) (51) – J1 (female) (With all connectors of wiring harness disconnected)	Resistance	Min. 100 kΩ	
	6	Defective wiring harness connector	Connecting parts among injector No. 2, engine wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation			
	7	Defective engine controller	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			J1 (Female)	Resistance		
			Between (54) – (51)	Max. 2 Ω		
			Between (54) – chassis ground	Min. 100 kΩ		

Circuit diagram related to injector No. 2

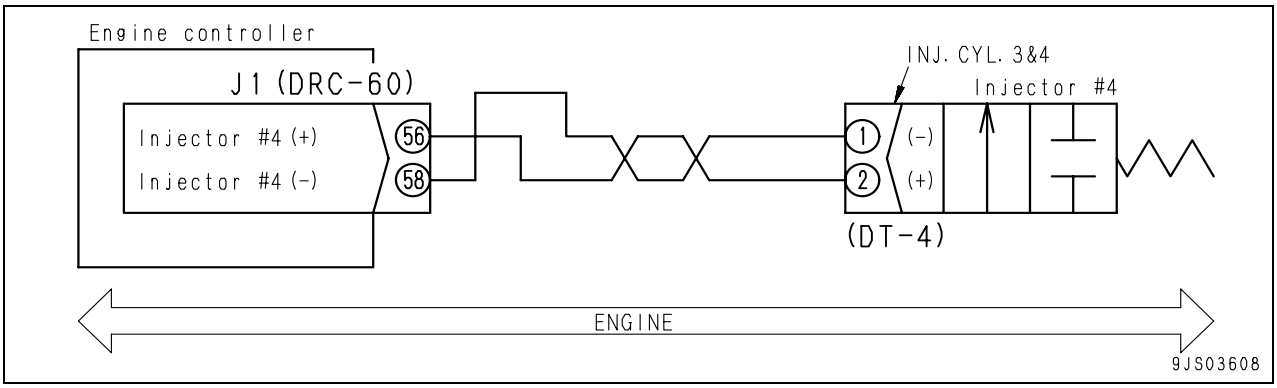


E-25 Code [332/CA332] Disconnection, short circuit in injector No. 4

Error code	Failure code	Trouble	Disconnection, short circuit in injector No. 4
332	**CA332**		
Contents of trouble	• There is disconnection or short circuit in drive circuit of injector No. 4.		
Action of controller	• None in particular.		
Problem that appears on machine	• There is irregular combustion or hunting. • Output drops.		
Related information	• Method of reproducing failure code: Start engine.		

	Causes		Standard value in normal state/Remarks on troubleshooting			
Possible causes and standard value in normal state	1	Defective engine controller	If failure codes ([**CA323**], [**CA325**], [**CA332**]) are indicated simultaneously, engine controller has a defect in it.			
	2	Defective injector No. 4	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			INJ CYL 4 (Male)	Resistance		
			Between (2) – (1)	Max. 2 Ω		
			Between (2) – chassis ground	Min. 100 kΩ		
	3	Disconnection in wiring harness (Disconnection in wiring or defective contact in connector)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness between J1 (female) (56) – INJ CYL 4 (female) (2)	Resistance	Max. 2 Ω	
			Wiring harness between J1 (female) (58) – INJ CYL 4 (female) (1)	Resistance	Max. 2 Ω	
	4	Ground fault in wiring harness (Short circuit with GND circuit)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness between J1 (female) (56) – INJ CYL 4 (female) (2)	Resistance	Max. 2 Ω	
	5	Short circuit in wiring harness (with another wiring harness)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness among all pins between J1 (female) (56) – J1 (female) (With all connectors of wiring harness disconnected)	Resistance	Min. 100 kΩ	
			Wiring harness among all pins between J1 (female) (58) – J1 (female) (With all connectors of wiring harness disconnected)	Resistance	Min. 100 kΩ	
	6	Defective wiring harness connector	Connecting parts among injector No. 4, engine wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation			
	7	Defective engine controller	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			J1 (Female)	Resistance		
			Between (56) – (58)	Max. 2 Ω		
			Between (56) – chassis ground	Min. 100 kΩ		

Circuit diagram related to injector No. 4



E-26 Code [342/CA342] Matching error in engine controller data

Error code	Failure code	Trouble	Matching error in engine controller data
342	**CA342**		
Contents of trouble	• Matching error occurred in engine controller data.		
Action of controller	• None in particular.		
Problem that appears on machine	• Normal operation or engine stops and engine cannot be started.		
Related information	• Method of reproducing failure code: Starting switch ON.		

Possible causes and standard value in normal state	Causes		Standard value in normal state/Remarks on troubleshooting
	1	Defective related system	If another failure code is indicated, carry out troubleshooting for it.
	2	Defective engine controller	Possible defect in engine controller (Troubleshooting cannot be carried out.)

Engine

107E-1 Series

40 Troubleshooting

Troubleshooting of electrical system (E-mode), Part 2

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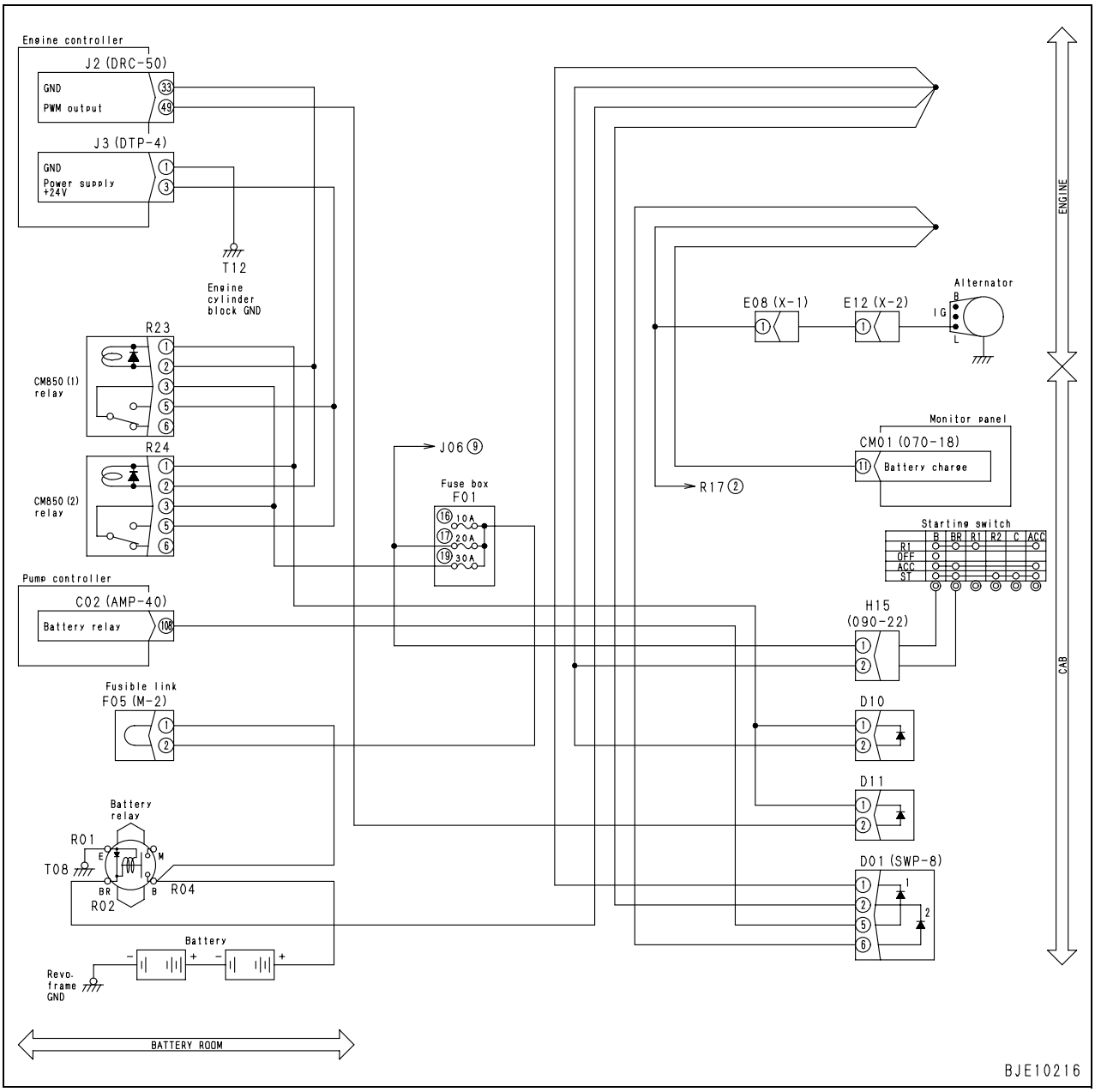
Troubleshooting of electrical system (E-mode), Part 2

E-27 Code [351/CA351] Abnormality in injector drive circuit

Error code	Failure code	Trouble	Abnormality in injector drive circuit
351	**CA351**		
Contents of trouble	• Abnormality occurred in injector drive power supply circuit.		
Action of controller	• Limits output and continues operation.		
Problem that appears on machine	• Exhaust smoke is black • There is irregular combustion. • Engine output drops. • Engine does not start		
Related information	• Method of reproducing failure code: Start engine.		

	Causes		Standard value in normal state/Remarks on troubleshooting		
Possible causes and standard value in normal state	1	Defective related system	If another failure code is indicated, carry out troubleshooting for it.		
	2	Defective fuse No. 19	If fuse is burnt out, circuit probably has ground fault.		
	3	Defective relay for engine controller power supply	★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting.		
			Replace relays for engine controller power supply (R23, R24) with other relays. If mark E of failure code disappears after reproduction operation is conducted, that relay has a defect.		
	4	Disconnection in wiring harness (Disconnection in wiring or defective contact in connector)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			Wiring harness between F01-19 – R23, R24 (female) (3)	Resistance	Max. 0.5 Ω
			Wiring harness between R23, R24 (female) (5) – J3 (female) (3)	Resistance	Max. 0.5 Ω
			Wiring harness between J3 (female) (1) – chassis ground (T12)	Resistance	Max. 10 Ω
	5	Defective engine controller	If causes 1 – 4 are not detected, engine controller may be defective. (Since trouble is in system, troubleshooting cannot be carried out.)		

Circuit diagram related to engine controller power supply

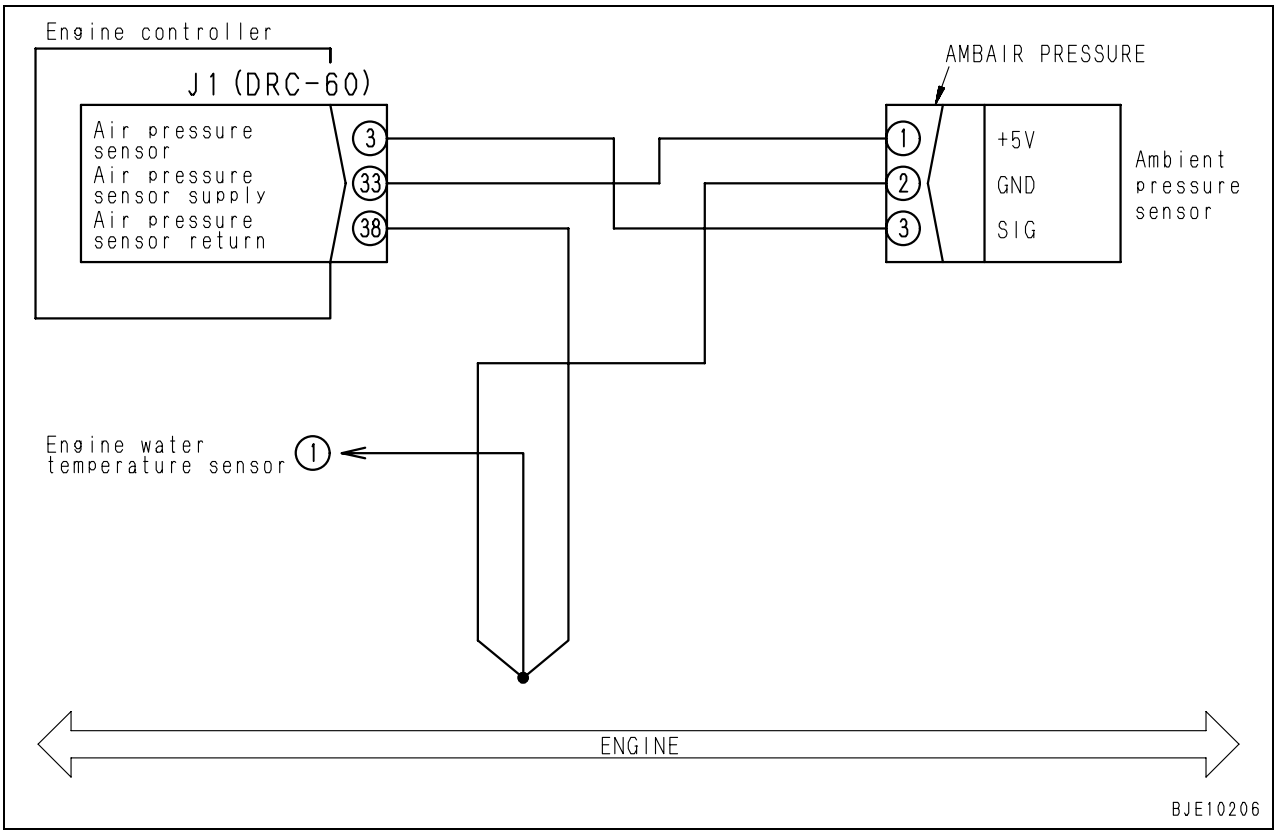


E-28 Code [352/CA352] Abnormally low level in sensor power supply 1

Error code	Failure code	Trouble	Abnormally low level in sensor power supply 1 circuit
352	**CA352**		
Contents of trouble	• Low voltage is detected in sensor power supply 1 circuit.		
Action of controller	• Fixes atmospheric pressure and continues operation.		
Problem that appears on machine	• Engine output drops. • Starting performance is poor.		
Related information	• Method of reproducing failure code: Starting switch ON.		

	Causes		Standard value in normal state/Remarks on troubleshooting		
Possible causes and standard value in normal state			★ Disconnect connector with starting switch OFF, then turn starting switch ON and carry out troubleshooting.		
	1	Defective sensor or wiring harness	Disconnect devices at right in order. If mark E of failure code disappears after reproduction operation is conducted, that device or wiring harness has a defect in it.	Atmospheric pressure sensor	AMBAIR PRESSURE
				Engine wiring harness	J1
	2	Defective wiring harness connector	Connecting parts among atmospheric pressure sensor, engine wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation		
	3	Defective engine controller	If causes 1 and 2 are not detected, engine controller may be defective. (Since trouble is in system, troubleshooting cannot be carried out.)		

Circuit diagram related to sensor power supply 1

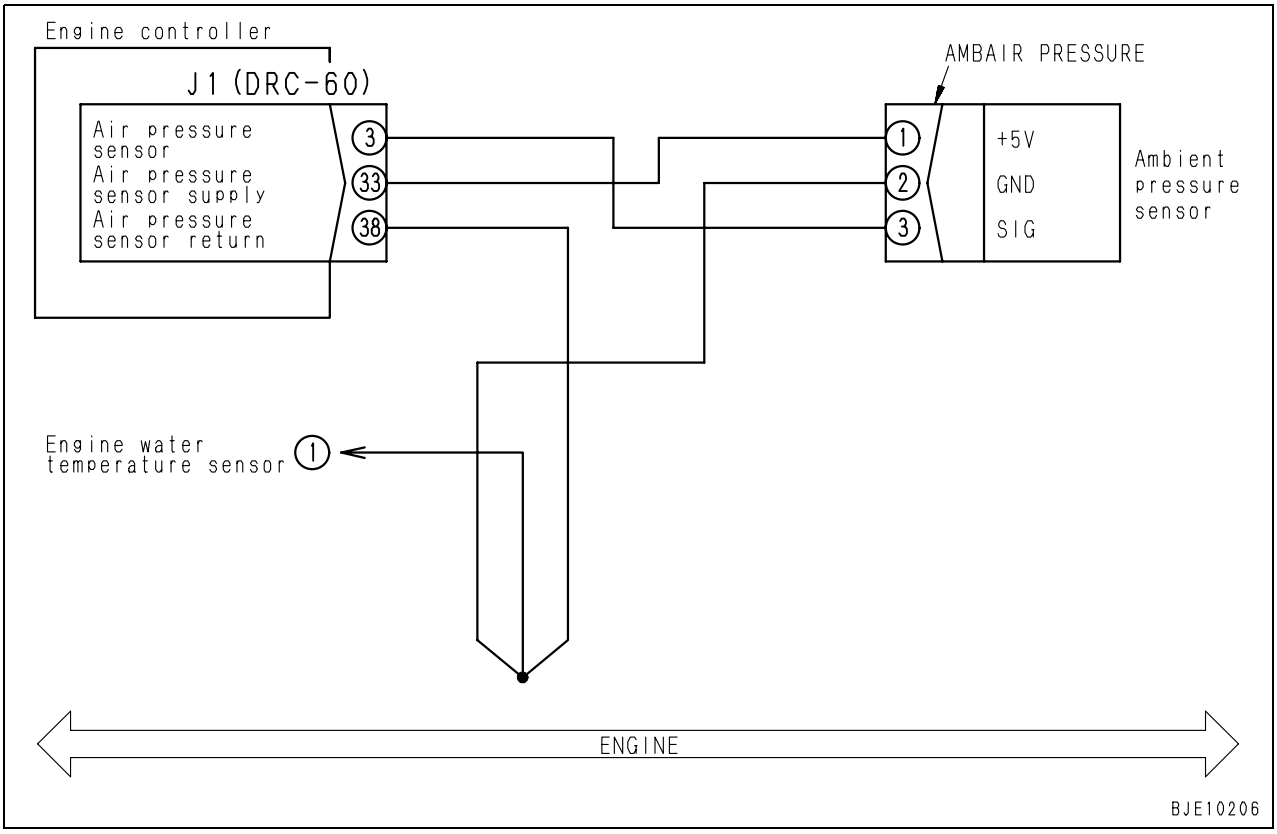


E-29 Code [386/CA386] Abnormally high level in sensor power supply 1

Error code	Failure code	Trouble	Abnormally high level in sensor power supply 1 circuit
386	**CA386**		
Contents of trouble	• High voltage is detected in sensor power supply 1 circuit.		
Action of controller	• Fixes atmospheric pressure and continues operation.		
Problem that appears on machine	• Engine output drops. • Starting performance is poor.		
Related information	• Method of reproducing failure code: Starting switch ON.		

	Causes		Standard value in normal state/Remarks on troubleshooting
Possible causes and standard value in normal state	1	Defective wiring harness connector	Connecting parts among atmospheric pressure sensor, engine wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation
	2	Defective engine controller	If cause 1 is not detected, engine controller may be defective. (Since trouble is in system, troubleshooting cannot be carried out.)

Circuit diagram related to sensor power supply 1

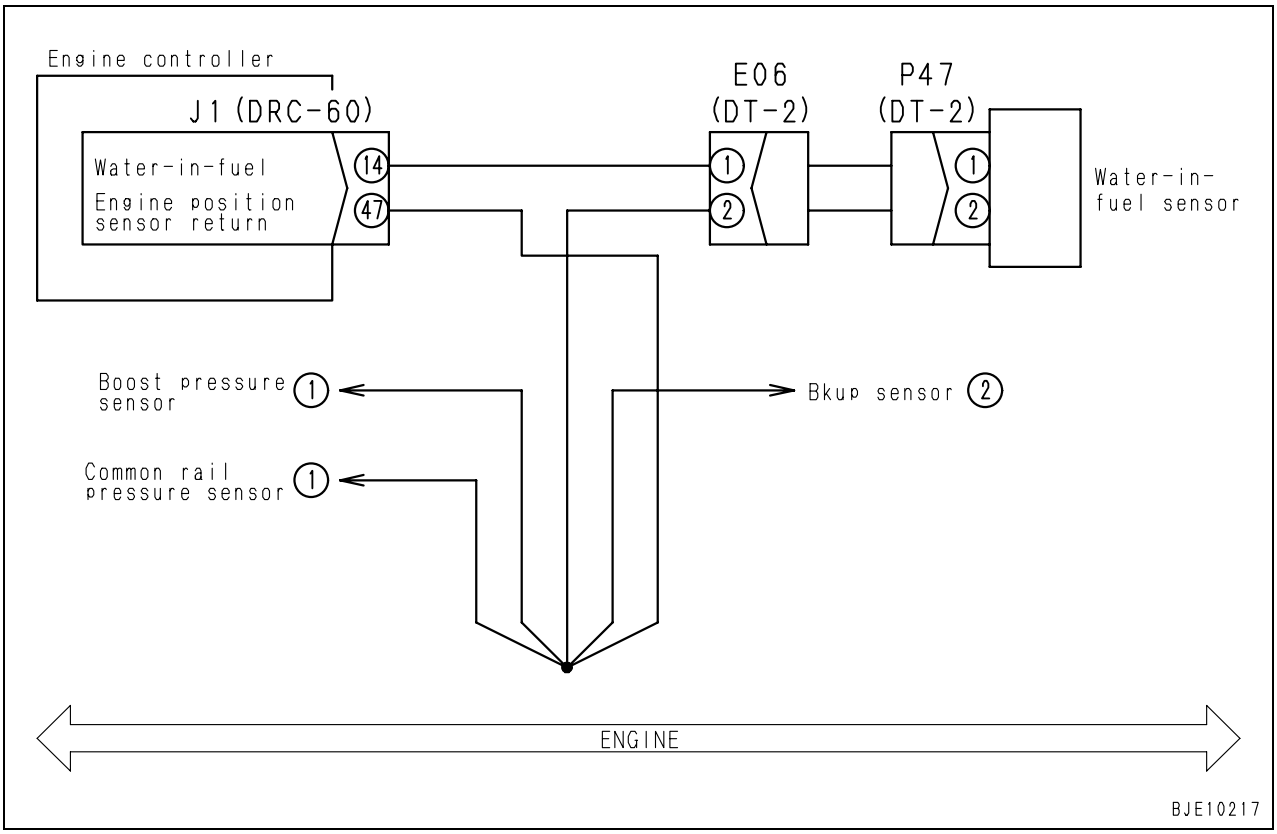


E-30 Code [428/CA428] Abnormally high level in water sensor

Error code	Failure code	Trouble	Abnormally high level in water sensor
428	**CA428**		
Contents of trouble	• High voltage occurred in fuel water sensor		
Action of controller	• None in particular.		
Problem that appears on machine	• Water separator monitor does not indicate normally.		
Related information	• Method of reproducing failure code: Starting switch ON.		

	Causes		Standard value in normal state/Remarks on troubleshooting		
Possible causes and standard value in normal state	1	Defective fuel water sensor	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			P47 (Female)	Resistance	
			Between (1) – (2)	Max. 10 Ω	
	2	Disconnection in wiring harness (Disconnection in wiring or defective contact in connector)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			Wiring harness between J1 (female) (14) – P47 (male) (1)	Resistance	Max. 10 Ω
			Wiring harness between J1 (female) (47) – P47 (male) (2)	Resistance	Max. 10 Ω
	3	Short circuit in wiring harness (With another wiring harness)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			Wiring harness among all pins between J1 (female) (14) – J1 (female) (With all connectors of wiring harness disconnected)	Resistance	Min. 100 kΩ
	4	Defective wiring harness connector	Connecting parts among fuel water sensor, engine wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation		
	5	Defective engine controller	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			J1 (Female)	Resistance	
			Between (14) – (47)	Max. 10 Ω	

Circuit diagram related to fuel water sensor

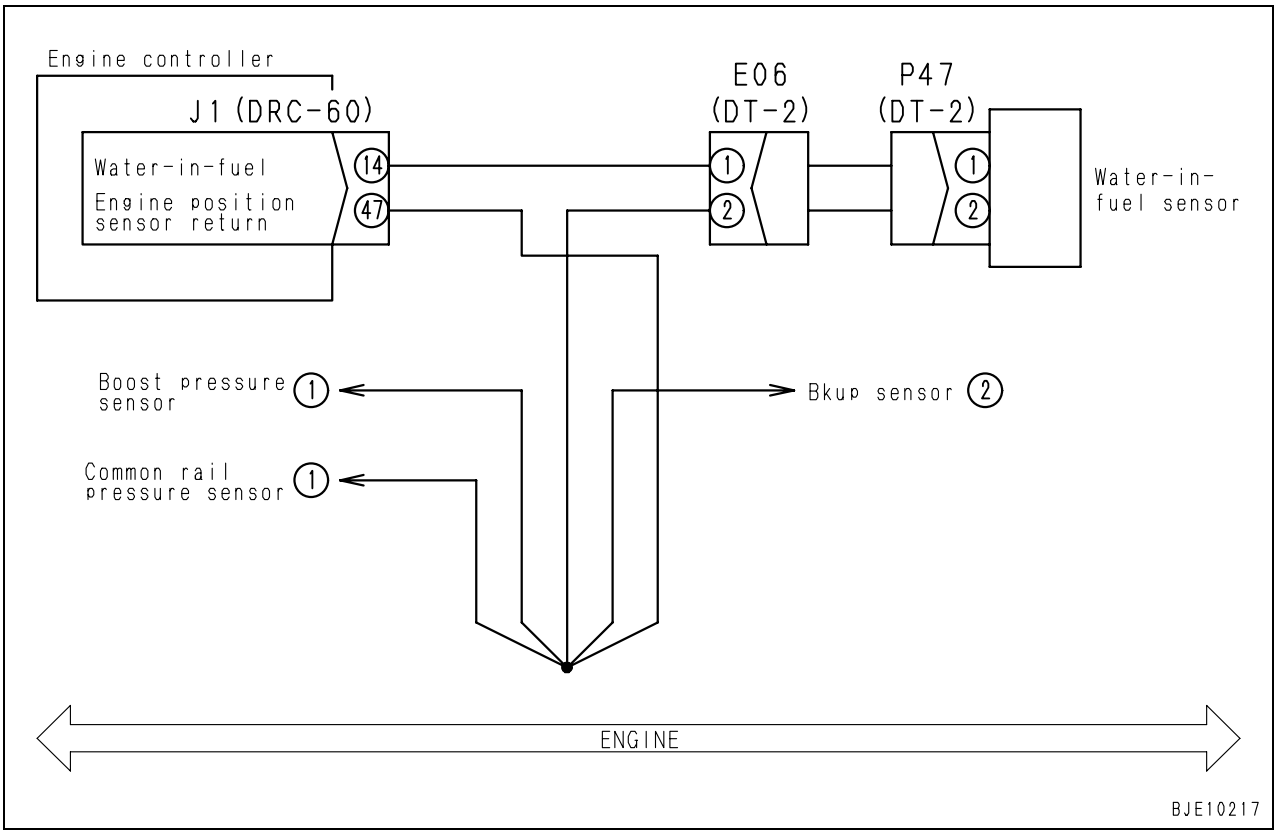


E-31 Code [429/CA429] Abnormally low level in water sensor

Error code	Failure code	Trouble	Abnormally low level in water sensor
429	**CA429**		
Contents of trouble	• Low voltage occurred in fuel water sensor		
Action of controller	• None in particular.		
Problem that appears on machine	• Water separator monitor does not indicate normally.		
Related information	• Method of reproducing failure code: Starting switch ON.		

	Causes		Standard value in normal state/Remarks on troubleshooting			
Possible causes and standard value in normal state	1	Defective fuel water sensor	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			P47 (Female)	Resistance		
			Between (1) – (2)	Max. 10 Ω		
			Between (1) – chassis ground	Min. 100 kΩ		
	2	Ground fault in wiring harness (Short circuit with GND circuit)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness between J1 (female) (14) – P47 (male) (1) and chassis ground	Resistance	Min. 100 kΩ	
	3	Short circuit in wiring harness (With another wiring harness)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness among all pins between J1 (female) (14) – J1 (female) (With all connectors of wiring harness disconnected)	Resistance	Min. 100 kΩ	
	4	Defective wiring harness connector	Connecting parts among fuel water sensor, engine wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation			
	5	Defective engine controller	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			J1 (Female)	Resistance		
			Between (14) – (47)	Max. 10 Ω		
			Between (14) – chassis ground	Min. 100 kΩ		

Circuit diagram related to fuel water sensor

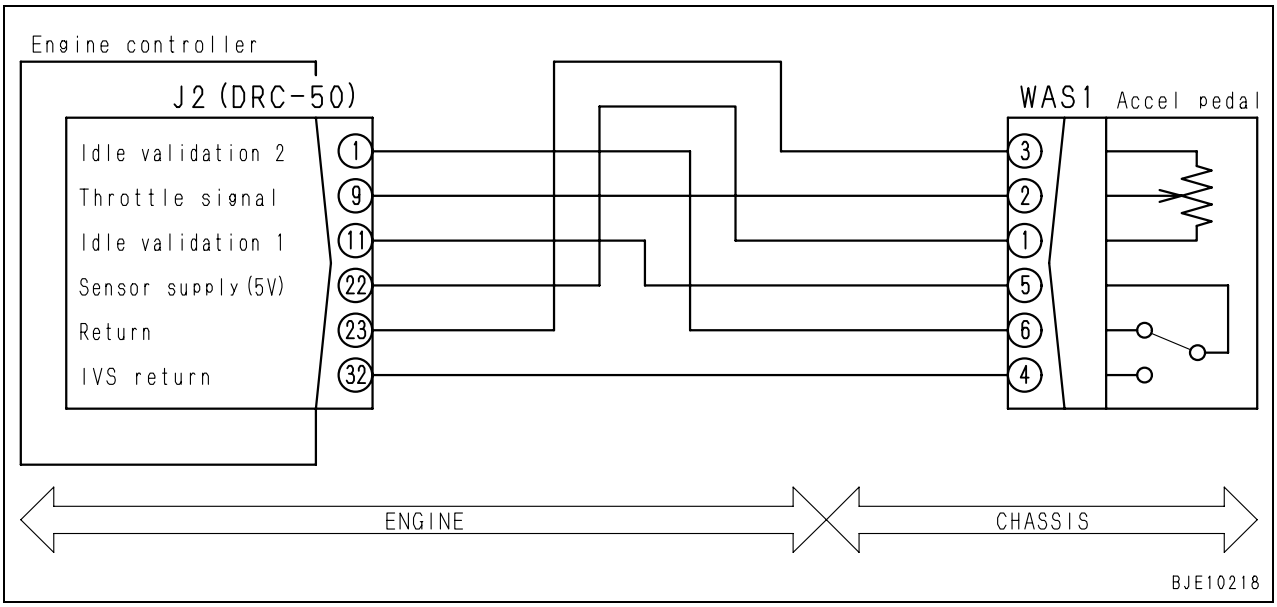


E-32 Code [431/CA431] Abnormality in idle validation switch

Error code	Failure code	Trouble	Abnormality in idle validation switch
431	CA431		
Contents of trouble	• Abnormality occurred in idle validation switch circuit		
Action of controller	• Sets throttle angle to certain value and continues operation (depending on each model).		
Problem that appears on machine	• Engine speed may not rise from low idle.		
Related information	• Method of reproducing failure code: Starting switch ON, accelerator ON.		

	Causes		Standard value in normal state/Remarks on troubleshooting		
Possible causes and standard value in normal state	1	Defective accelerator pedal	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			WAS1 (Male)	Accelerator pedal	Resistance
			Between (5) – (4)	Release	Max. 1 Ω
				Press	Min. 1 MΩ
			Between (6) – (4)	Release	Min. 1 MΩ
				Press	Max. 1 Ω
	2	Disconnection in wiring harness (Disconnection in wiring or defective contact in connector)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			Wiring harness between J2 (female) (11) – WAS1 (female) (5)	Resistance	Max. 10 Ω
			Wiring harness between J2 (female) (1) – WAS1 (female) (6)	Resistance	Max. 10 Ω
			Wiring harness between J2 (female) (32) – WAS1 (female) (4)	Resistance	Max. 10 Ω
	3	Short circuit in wiring harness (With another wiring harness)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			Wiring harness among all pins between J2 (female) (11) – J2 (female) (With also WAS1 disconnected)	Resistance	Min. 100 kΩ
			Wiring harness among all pins between J2 (female) (1) – J2 (female) (With also WAS1 disconnected)	Resistance	Min. 100 kΩ
			Wiring harness among all pins between J2 (female) (11) – J3 (female) (3) (With also WAS1 disconnected)	Resistance	Min. 100 kΩ
			Wiring harness among all pins between J2 (female) (1) – J3 (female) (3) (With also WAS1 disconnected)	Resistance	Min. 100 kΩ
	4	Defective wiring harness connector	Connecting parts among accelerator pedal, engine unit wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation		
	5	Defective engine controller	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			J2 (Female)	Accelerator pedal	Resistance
			Between (11) – (32)	Release	Max. 1 Ω
				Press	Min. 1 MΩ
			Between (1) – (32)	Release	Min. 1 MΩ
				Press	Max. 1 Ω

Circuit diagram related to accelerator pedal

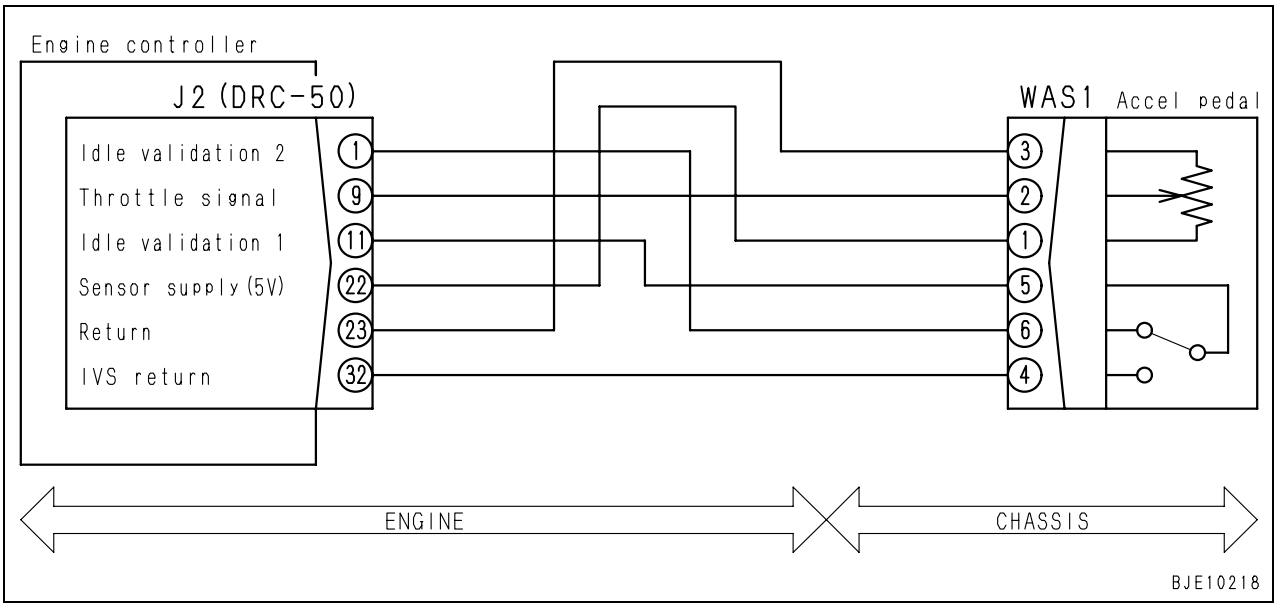


E-33 Code [432/CA432] Abnormality in idle validation processing

Error code	Failure code	Trouble	Abnormality in processing of idle validation processing
432	**CA432**		
Contents of trouble	• Abnormality occurred in processing of idle validation switch circuit.		
Action of controller	• Sets throttle angle to certain value and continues operation (depending on each model).		
Problem that appears on machine	• Engine speed may not rise from low idle.		
Related information	• Method of reproducing failure code: Starting switch ON, accelerator ON.		

	Causes		Standard value in normal state/Remarks on troubleshooting			
Possible causes and standard value in normal state	1	Defective accelerator pedal	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			WAS1 (Male)	Accelerator pedal	Resistance	
			Between (2) – (1)	Release	1.5 – 3.0 kΩ	
				Press	0.25 – 1.5 kΩ	
	2	Disconnection in wiring harness (Disconnection in wiring or defective contact in connector)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness between J2 (female) (22) – WAS1 (female) (1)		Resistance	Max. 10 Ω
			Wiring harness between J2 (female) (9) – WAS1 (female) (2)		Resistance	Max. 10 Ω
	3	Short circuit in wiring harness (With another wiring harness)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness among all pins between J2 (female) (22) – J2 (female) (With also WAS1 disconnected)		Resistance	Min. 100 kΩ
			Wiring harness among all pins between J2 (female) (9) – J2 (female) (With also WAS1 disconnected)		Resistance	Min. 100 kΩ
	4	Defective wiring harness connector	Connecting parts among accelerator pedal, engine unit wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation			
	5	Defective engine controller	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			J2 (Female)	Accelerator pedal	Resistance	
			Between (22) – (9)	Release	1.5 – 3.0 kΩ	
				Press	0.25 – 1.5 kΩ	

Circuit diagram related to accelerator pedal

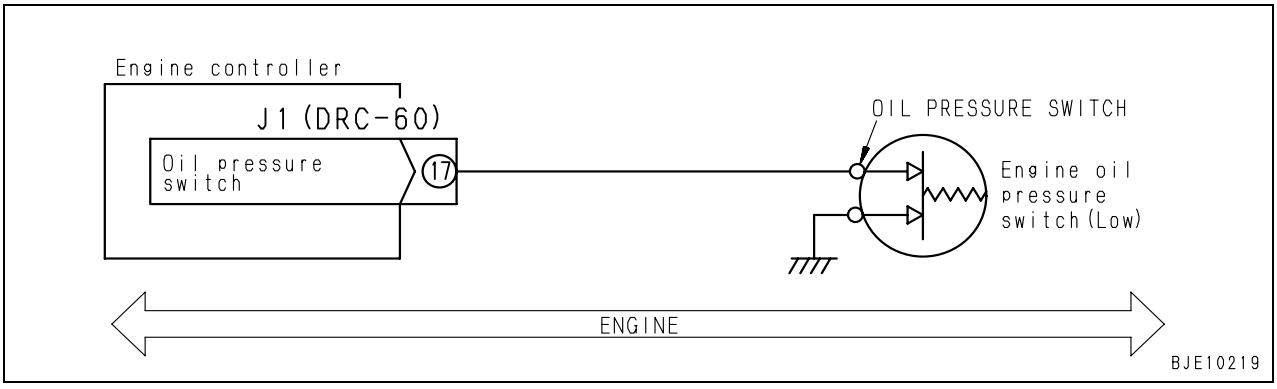


E-34 Code [435/CA435] Abnormality in engine oil pressure switch

Error code	Failure code	Trouble	Abnormality in engine oil pressure switch
435	**CA435**		
Contents of trouble	• Abnormality occurred in engine oil pressure switch signal circuit.		
Action of controller	• None in particular.		
Problem that appears on machine	• Engine oil pressure switch-based engine protection function does not function.		
Related information	• Method of reproducing failure code: Starting switch ON or start engine		

	Causes		Standard value in normal state/Remarks on troubleshooting		
Possible causes and standard value in normal state	1	Defective engine oil pressure switch	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			OIL PRESSURE SWITCH (Male)	Resistance	
			Between (1) – body	Max. 10 Ω	
	2	Disconnection in wiring harness (Disconnection in wiring or defective contact in connector)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			Wiring harness between J1 (female) (17) – OIL PRESSURE SWITCH (female) (1)	Resistance	Max. 10 Ω
	3	Short circuit in wiring harness (With another wiring harness)	★ Troubleshoot with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			Wiring harness among all pins between J1 (female) (17) – J1 (female) (With all connectors of wiring harness disconnected)	Resistance	Min. 100 kΩ
	4	Defective wiring harness connector	Connecting parts among engine oil pressure switch, engine wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation		
	5	Defective engine controller	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			J1 (Female)	Resistance	
			Between (17) – chassis ground	Max. 10 Ω	

Circuit diagram related to engine oil switch



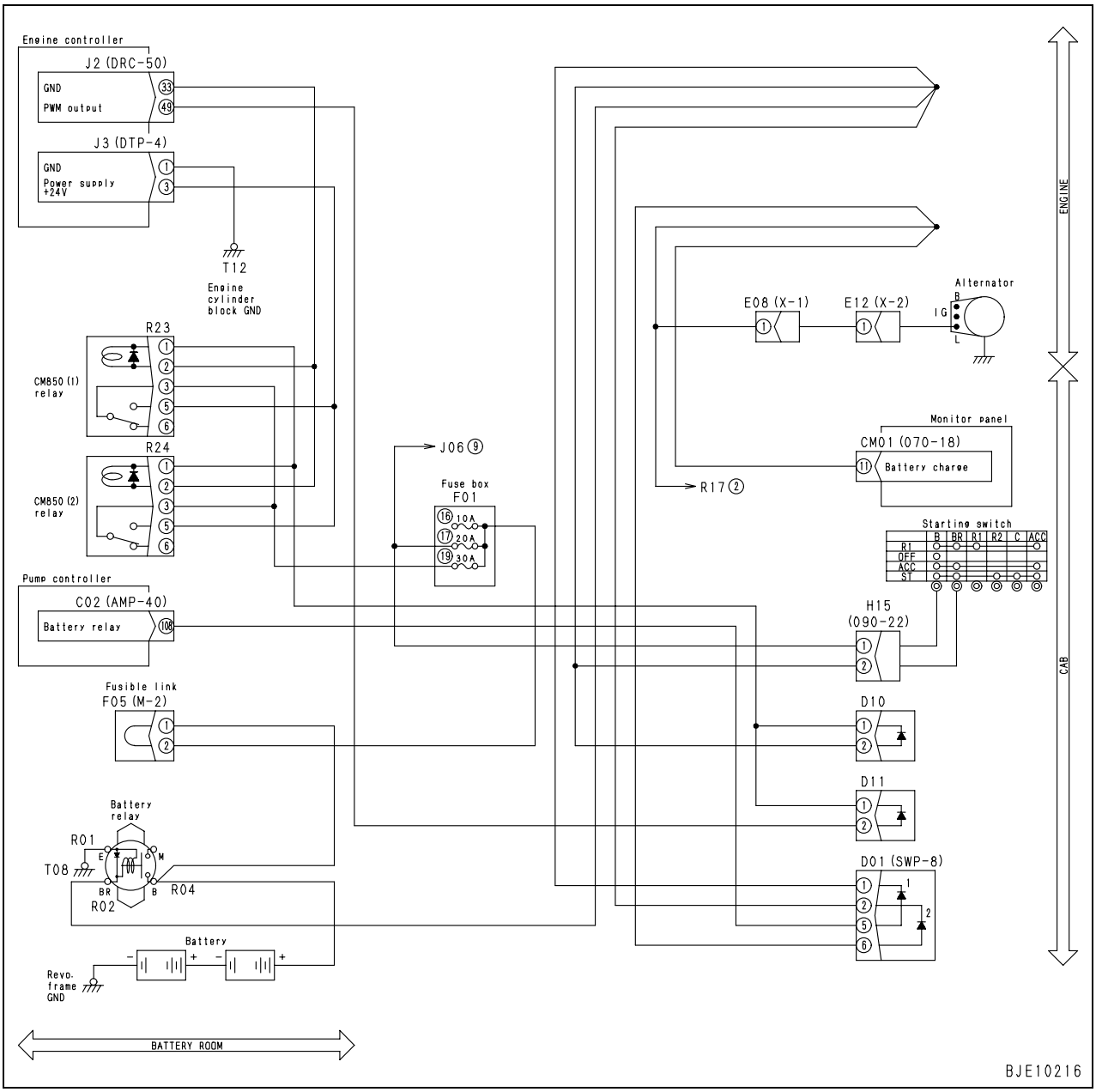
E-35 Code [441/CA441] Abnormally low power supply voltage

Error code	Failure code	Trouble	Abnormally low power supply voltage
441	**CA441**		
Contents of trouble	• Low voltage occurred in controller power supply circuit.		
Action of controller	• None in particular.		
Problem that appears on machine	• Engine stops • Starting performance is poor		
Related information	• Method of reproducing failure code: Starting switch ON.		

	Causes		Standard value in normal state/Remarks on troubleshooting			
Possible causes and standard value in normal state	1	Loose, corroded battery terminal	Inspect battery terminal directly for loose and corrosion.			
	2	Defective battery voltage	★ Prepare with starting switch OFF, then turn starting switch OFF and START and carry out troubleshooting.			
			Battery (1 piece)	Starting switch	Voltage	
			Between (+) – (–) terminals	OFF	Min. 12 V	
				START	Min. 6.2 V	
	3	Defective fuse No. 19	If fuse is burnt out, circuit probably has ground fault. (See cause 6.)			
	4	Defective relay for engine controller power supply	★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting.			
			Replace relays for engine controller power supply (R23, R24) with other relays. If mark E of failure code disappears after reproduction operation is conducted, that relay has a defect.			
	5	Disconnection in wiring harness (Disconnection in wiring or defective contact in connector)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness between F01-19 – R23, R24 (female) (3)		Resistance	Max. 10 Ω
			Wiring harness between R23, R24 (female) (5) – J3 (female) (3)		Resistance	Max. 10 Ω
			Wiring harness between J3 (female) (1) – chassis ground (T12)		Resistance	Max. 10 Ω
	6	Ground fault in wiring harness (Short circuit with GND circuit)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness between F01-19 – R23, R24 (female) (3) and chassis ground		Resistance	Min. 100 kΩ
			Wiring harness between R23, R24 (female) (5) – J3 (female) (3) and chassis ground		Resistance	Max. 10 Ω
	7	Short circuit in wiring harness (With another wiring harness)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness among all pins between J3 (female) (3) – J3 (female) (1) (With battery terminal disconnected)		Resistance	Min. 100 kΩ
			Wiring harness among all pins between J3 (female) (3) – J2 (female) (With battery terminal disconnected)		Resistance	Min. 100 kΩ
			Wiring harness among all pins between J3 (female) (1) – J2 (female) (With battery terminal disconnected)		Resistance	Min. 100 kΩ

	Causes		Standard value in normal state/Remarks on troubleshooting		
Possible causes and standard value in normal state	8	Defective wiring harness connector	Connecting parts among fuse No. 19, engine unit wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation		
	9	Defective engine controller	★ Prepare with starting switch OFF, then turn starting switch ON and START and carry out troubleshooting.		
			J3 (Female)	Starting switch	Voltage
			Between (3) – (1)	ON	Min. 24 V
				START	Min. 12 V

Circuit diagram related to engine controller power supply

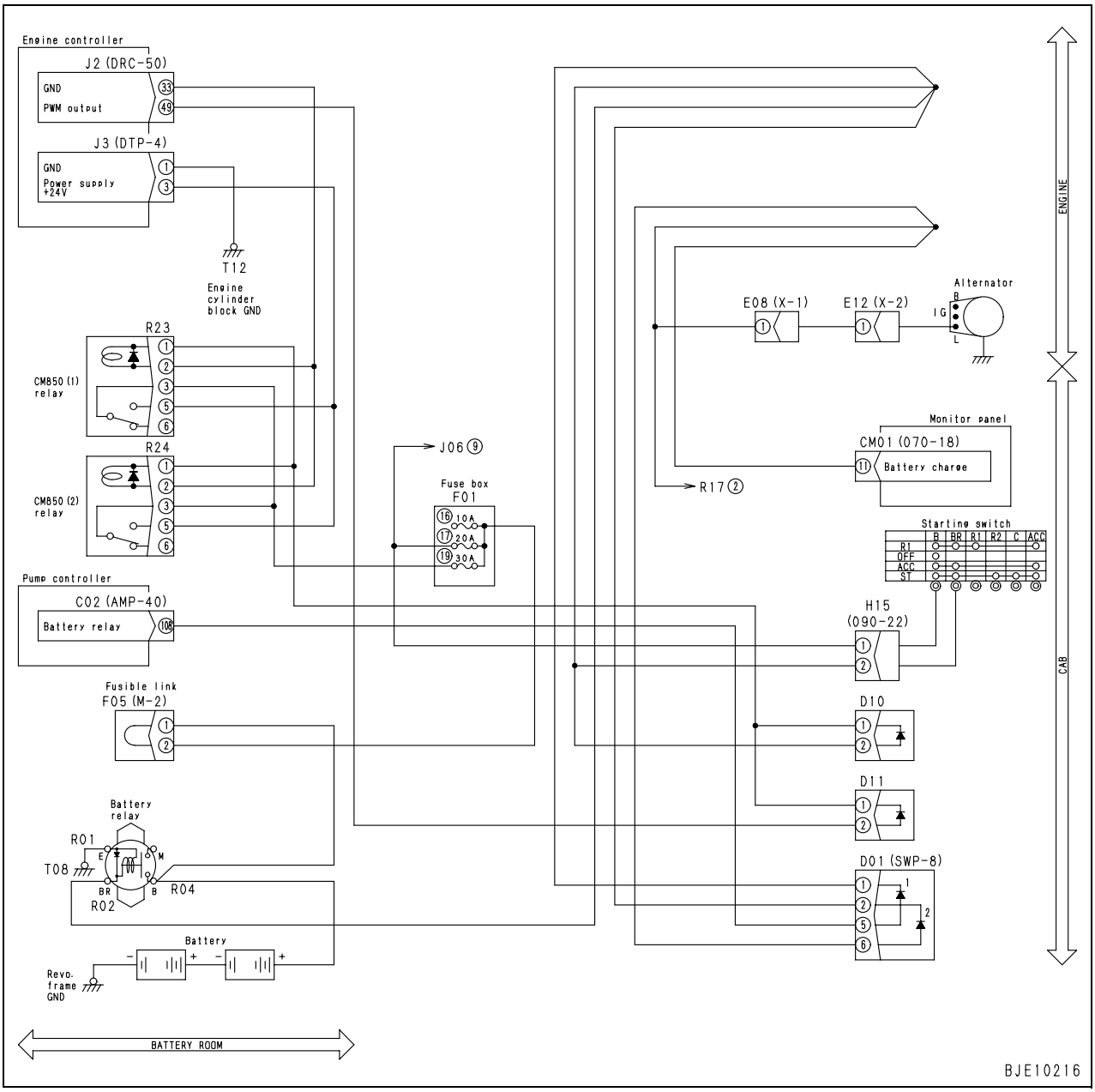


E-36 Code [442/CA442] Abnormally high power supply voltage

Error code	Failure code	Trouble	Abnormally high power supply voltage
442	**CA442**		
Contents of trouble	• High voltage (min. 36V) occurred in controller power supply circuit.		
Action of controller	• None in particular.		
Problem that appears on machine	• Engine may stop.		
Related information	• Method of reproducing failure code: Starting switch ON.		

	Causes		Standard value in normal state/Remarks on troubleshooting		
Possible causes and standard value in normal state	1	Defective battery voltage	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			Battery	Voltage	
			Between (+) – (–) terminals	Max. 32 V	
	2	Defective alternator	★ Prepare with starting switch OFF, then start the engine and carry out troubleshooting.		
			E12 (Male)	Engine speed	Voltage
			Between (1) – chassis ground	Min. medium speed	27.5 – 29.5 V
	3	Defective engine controller	★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting.		
			J3 (Female)	Voltage	
			Between (3) – (1)	Max. 32 V	

Circuit diagram related to engine controller power supply



E-37 Code [449/CA449] Abnormally high common rail pressure 2

Error code	Failure code	Trouble	Abnormally high common rail pressure 2
449	**CA449**		
Contents of trouble	• Pressure (2) in common rail circuit is abnormally high.		
Action of controller	• Limits output and continues operation.		
Problem that appears on machine	• Engine sound becomes larger under no load or light load. • Engine output drops.		
Related information	• Method of reproducing failure code: Start engine.		

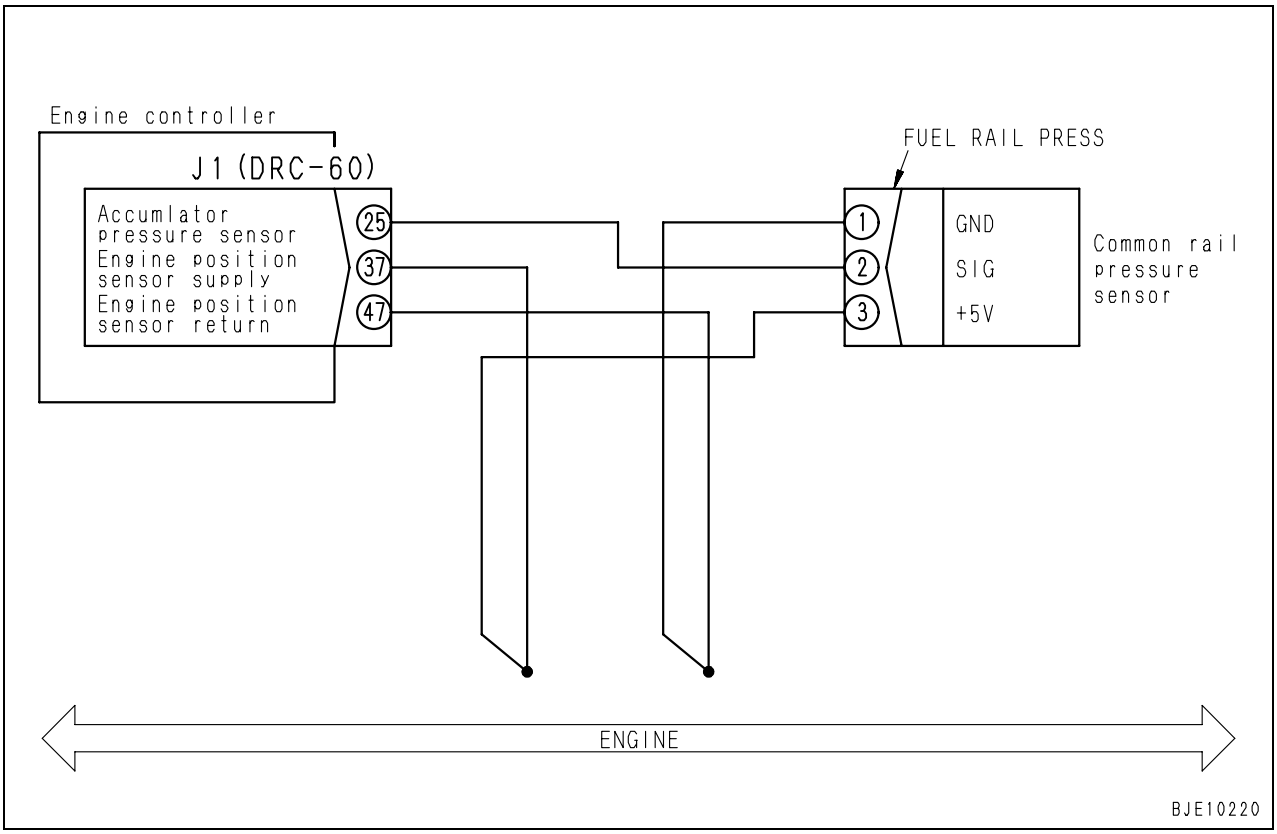
	Causes		Standard value in normal state/Remarks on troubleshooting		
Possible causes and standard value in normal state	1	Defective related system	If another failure code is indicated, carry out troubleshooting for it.		
	2	Air in low-pressure circuit	Inspect it for air entry directly according to the following procedure. 1) Remove pressure pickup plug (outlet side) from fuel main filter. 2) Operate feed pump of fuel pre-fuel filter. 3) Inspect fuel and air from pressure pickup plug for leakage level.		
	3	Defective fuel return circuit device	★ For check of pressure in fuel return circuit, see TESTING AND ADJUSTING, Testing fuel pressure.		
			Fuel return circuit pressure	At low idle or cranking	Max. 0.02 MPa {Max. 0.19 kg/cm^2}
	4	Defective common rail pressure sensor	★ Prepare with starting switch ON, then carry out troubleshooting without turning starting switch OFF.		
			Monitoring on INSITE	Monitoring information	
			Common rail pressure	Condition when engine stopped	0 ± 0.39 MPa {0 ± 4 kg/cm^2}
	5	Defective pressure limiter	★ For testing of leakage from pressure limiter, see TESTING AND ADJUSTING, Testing leakage from pressure limiter and return rate from injector.		
			Leakage from pressure limiter	At low idle	0 cc (No leakage)
	6	Defective supply pump	If causes 1 – 5 are not detected, supply pump may be defective.		

E-38 Code [451/CA451] Abnormally high level in common rail pressure sensor

Error code	Failure code	Trouble	Abnormally high level in common rail pressure sensor
451	**CA451**		
Contents of trouble	• High voltage occurred in common rail pressure sensor.		
Action of controller	• Limits output and continues operation.		
Problem that appears on machine	• Engine speed or output drops. • Engine does not start.		
Related information	• Method of reproducing failure code: Starting switch ON.		

	Causes		Standard value in normal state/Remarks on troubleshooting			
Possible causes and standard value in normal state	1	Defective sensor power supply 2 circuit	If failure code [**CA227**] is indicated simultaneously, carry out troubleshooting for it first.			
	2	Defective common rail pressure sensor	★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting.			
			FUEL RAIL PRESS		Voltage	
			Between (3) – (1)	Power supply	4.75 – 5.25 V	
			Voltage is measured with wiring harness connected. Accordingly, if voltage is abnormal, check wiring harness and controller, too, for another cause of trouble, and then judge.			
	3	Hot short (Short circuit with 5V, 24V circuit) in wiring harness	★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting.			
			Wiring harness between J1 (female) (25) – FUEL RAIL PRESS (female) (2) and chassis ground		Voltage	Max. 1 V
	4	Short circuit in wiring harness (With another wiring harness)	★ Troubleshoot with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness between J1 (female) (25) – FUEL RAIL PRESS (female) (2) and J1 (female) (37) – FUEL RAIL PRESS (female) (3)		Resistance	Min. 100 kΩ
	5	Defective wiring harness connector	Connecting parts among common rail pressure sensor, engine wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation			
	6	Defective engine controller	★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting.			
			J1		Voltage	
			Between (37) – (47)	Power supply	4.75 – 5.25 V	

Circuit diagram related to common rail pressure sensor

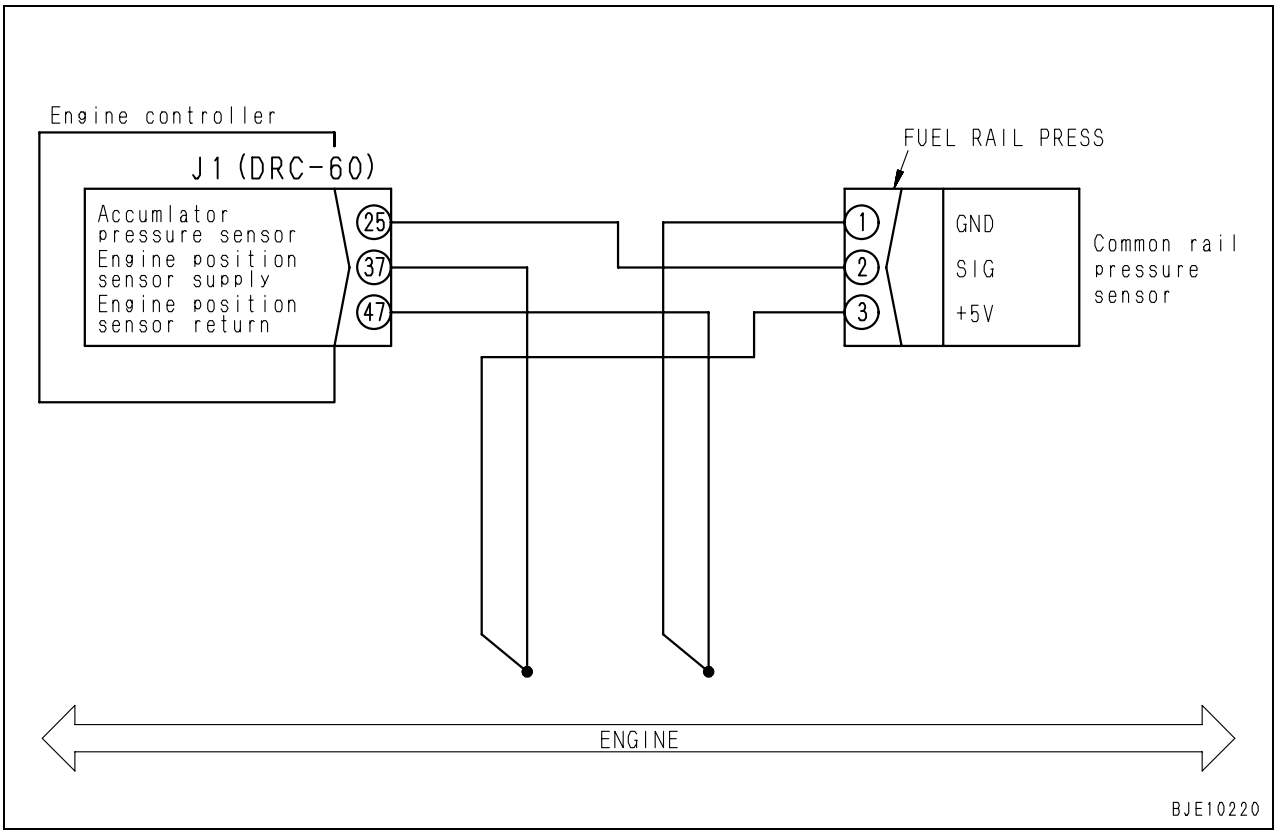


E-39 Code [452/CA452] Abnormally low level in common rail pressure sensor

Error code	Failure code	Trouble	Abnormally low level in common rail pressure sensor
452	**CA452**		
Contents of trouble	• Low voltage occurred in common rail pressure sensor.		
Action of controller	• Limits output and continues operation.		
Problem that appears on machine	• Engine speed or output drops. • Engine does not start		
Related information	• Method of reproducing failure code: Starting switch ON.		

	Causes		Standard value in normal state/Remarks on troubleshooting			
Possible causes and standard value in normal state	1	Defective sensor power supply 2 circuit	If failure code [**CA187**] is indicated simultaneously, carry out troubleshooting for it first.			
	2	Defective common rail pressure sensor	★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting.			
			FUEL RAIL PRESS		Voltage	
			Between (3) – (1)	Power supply	4.75 – 5.25 V	
			Voltage is measured with wiring harness connected. Accordingly, if voltage is abnormal, check wiring harness and controller, too, for another cause of trouble, and then judge.			
	3	Ground fault in wiring harness (Short circuit with GND circuit)	★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting.			
			Wiring harness between J1 (female) (25) – FUEL RAIL PRESS (female) (2) and chassis ground		Resistance	Min. 100 kΩ
	4	Short circuit in wiring harness (With another wiring harness)	★ Troubleshoot with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness between J1 (female) (25) – FUEL RAIL PRESS (female) (2) and J1 (female) (47) – FUEL RAIL PRESS (female) (1)		Resistance	Min. 100 kΩ
	5	Defective wiring harness connector	Connecting parts among common rail pressure sensor, engine wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation			
	6	Defective engine controller	★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting.			
			J1		Voltage	
			Between (37) – (47)	Power supply	4.75 – 5.25 V	

Circuit diagram related to common rail pressure sensor



E-40 Code [488/CA488] Derating of torque by abnormally high charge temperature

Error code	Failure code	Trouble	Derating of torque by abnormally high charge temperature
488	**CA488**		
Contents of trouble	• Temperature signal of boost pressure sensor and temperature sensor is above upper control limit.		
Action of controller	• Limits output and continues operation.		
Problem that appears on machine	• Output drops.		
Related information	• Method of reproducing failure code: Start engine.		

		Causes	Standard value in normal state/Remarks on troubleshooting
Possible causes and standard value in normal state	1	Drop in cooling performance of aftercooler.	Inspect following directly. • Loose, broken fan belt • Lack of cooling air • Clogged aftercooler fin
	2	Abnormally high temperature at turbocharger outlet	Inspect related parts directly.
	3	Defective engine controller	If causes 1 and 2 are not detected, engine controller may be defective. (Since trouble is in system, troubleshooting cannot be carried out.)

E-41 Code [553/CA553] Abnormally high level in common rail pressure sensor 1

Error code	Failure code	Trouble	Abnormally high common rail pressure 1
553	**CA553**		
Contents of trouble	• Voltage (1) in common rail circuit is abnormally high.		
Action of controller	• None in particular.		
Problem that appears on machine	• Engine sound becomes larger under no load or light load. • Engine output drops.		
Related information	• Method of reproducing failure code: Start engine.		

		Causes	Standard value in normal state/Remarks on troubleshooting
Possible causes and standard value in normal state	1	Defective related system	If another failure code is indicated, carry out troubleshooting for it.
	2	Defective connection of ground terminal	Inspect following directly for connection of ground terminal. • Ground terminal (battery (–)) of machine main unit. • Ground terminal of engine • Ground terminal of engine controller • Ground terminal of starting motor
	3	Broken O-ring of supply pump actuator	Inspect O-ring directly for breakage.

E-42 Code [559/CA559] No-pressure feed by supply pump 1

Error code	Failure code	Trouble	No-pressure feed by supply pump 1
559	**CA559**		
Contents of trouble	• No-pressure feed (1) occurred in common rail circuit.		
Action of controller	• None in particular.		
Problem that appears on machine	• Engine does not start or starting performance is poor • Exhaust smoke is black • Engine output drops.		
Related information	• Method of reproducing failure code: Start engine.		

	Causes		Standard value in normal state/Remarks on troubleshooting		
Possible causes and standard value in normal state	1	Fuel leakage to outside	Inspect for fuel leakage directly (visual inspection at engine low idle).		
	2	Defective low-pressure circuit device	★ For check of pressure in fuel low-pressure circuit, see TESTING AND ADJUSTING, Testing fuel pressure.		
			Pressure in fuel low-pressure circuit	At cranking (When engine cannot be started)	0.3 – 1.1 MPa {3.1 – 11.3 kg/cm^2}
				At low idle (When engine can be started)	0.5 – 1.3 MPa {5.1 – 13.3 kg/cm^2}
	3	Defective pressure limiter	★ For testing of leakage from pressure limiter, see TESTING AND ADJUSTING, Testing leakage from pressure limiter and return rate from injector.		
			Leakage from pressure limiter	At low idle	0 cc (No leakage)
	4	Defective injector (High-pressure piping in head included)	★ For testing of return rate from injector, see TESTING AND ADJUSTING, Testing leakage from pressure limiter and return rate from injector.		
			Return rate from injector	At cranking (When engine cannot be started)	Max. 90 cc/min.
				At low idle (When engine can be started)	Max. 180 cc/min.
	5	Defective supply pump	★ For testing of return rate from supply pump, see TESTING AND ADJUSTING, Testing leakage from pressure limiter and return rate from injector.		
			Return rate from supply pump	At cranking (When engine cannot be started)	Max. 140 cc/min.
				At low idle (When engine can be started)	Max. 1,000 cc/min.

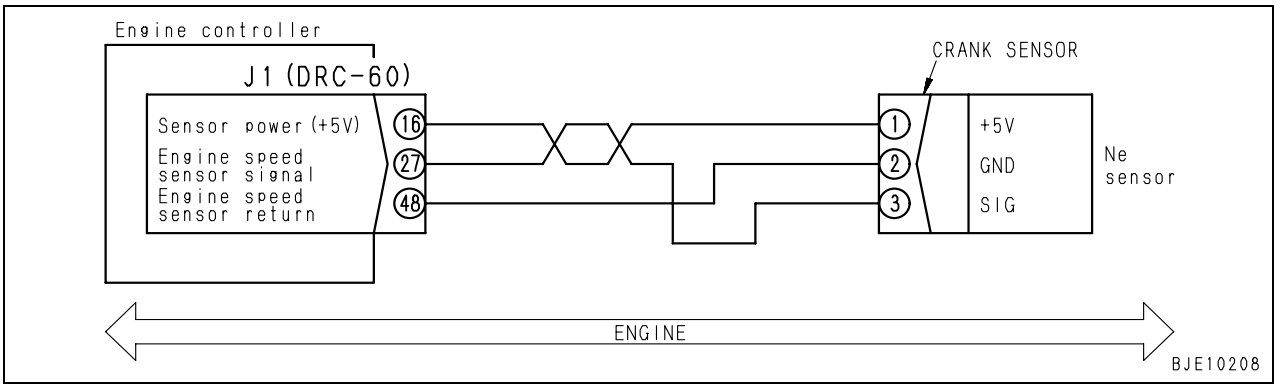
E-43 Code [689/CA689] Abnormality in engine Ne speed sensor

Error code	Failure code	Trouble	Abnormality in engine Ne speed sensor
689	**CA689**		
Contents of trouble	• Abnormality occurred in signals from engine Ne speed sensor.		
Action of controller	• Controls Ne speed sensor with signals of Bkup speed sensor.		
Problem that appears on machine	• There is hunting from engine. • Starting performance is poor. • Engine output drops.		
Related information	• Method of reproducing failure code: Start engine.		

	Causes		Standard value in normal state/Remarks on troubleshooting		
Possible causes and standard value in normal state	1	Defective Ne speed sensor power supply system	If failure code [**CA238**] is indicated simultaneously, carry out troubleshooting for it first.		
	2	Defective engine Ne speed sensor	★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting.		
			CRANK SENSOR		Voltage
			Between (1) – (2)	Power supply	4.75 – 5.25 V
			Voltage is measured with wiring harness connected. Accordingly, if voltage is abnormal, check wiring harness and controller, too, for another cause of trouble, and then judge.		
	3	Broken engine Ne speed sensor or defective clearance	Inspect engine Ne speed sensor for breakage or clearance.		
	4	Broken rotation sensing flywheel	Inspect it for breakage directly.		
	5	Disconnection in wiring harness (Disconnection in wiring or defective contact in connector)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			Wiring harness between J1 (female) (27) – CRANK SENSOR (female) (3)	Resistance	Max. 10 Ω
	6	Ground fault in wiring harness (Short circuit with GND circuit)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			Wiring harness between J1 (female) (27) – CRANK SENSOR (female) (3) and chassis ground	Resistance	Min. 100 kΩ
	7	Hot short (Short circuit with 5 V, 24 V circuit) in wiring harness	★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting.		
			Wiring harness between J1 (female) (27) – CRANK SENSOR (female) (3) and chassis ground	Voltage	Max. 1 V
	8	Short circuit in wiring harness (With another wiring harness)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			Wiring harness between J1 (female) (27) – CRANK SENSOR (female) (3) and J1 (female) (16) – CRANK SENSOR (female) (1)	Resistance	Min. 100 kΩ
			Wiring harness between J1 (female) (27) – CRANK SENSOR (female) (3) and J1 (female) (48) – CRANK SENSOR (female) (2)	Resistance	Min. 100 kΩ

<table>
<tr><td rowspan="4">Possible causes and standard value in normal state</td><td colspan="2">Causes</td><td colspan="3">Standard value in normal state/Remarks on troubleshooting</td></tr>
<tr><td>9</td><td>Defective wiring harness connector</td><td colspan="3">Connecting parts among engine Ne speed sensor, engine wiring harness, and engine controller are suspected. Inspect them directly.
• Loose connector, broken lock, broken seal
• Corrosive, bent, broken, forced-in, or extended pin
• Humidity in connector, entry of dirt or dust, poor insulation</td></tr>
<tr><td rowspan="3">10</td><td rowspan="3">Defective engine controller</td><td colspan="3">★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting.</td></tr>
<tr><td colspan="2">J1</td><td>Voltage</td></tr>
<tr><td>Between (16) – (48)</td><td>Power supply</td><td>4.75 – 5.25 V</td></tr>
</table>

Circuit diagram related to engine Ne speed sensor



E-44 Code [731/CA731] Abnormality in engine Bkup speed sensor phase

Error code	Failure code	Trouble	Abnormality in engine Bkup speed sensor phase
731	**CA731**		
Contents of trouble	• Abnormal phase is detected in signals of engine Ne speed sensor and engine Bkup speed sensor.		
Action of controller	• Controls with signals of engine Ne speed sensor.		
Problem that appears on machine	• Engine does not start or is difficult to start. • Idle speed is unstable. • Exhaust smoke is black.		
Related information	• Method of reproducing failure code: Start engine.		

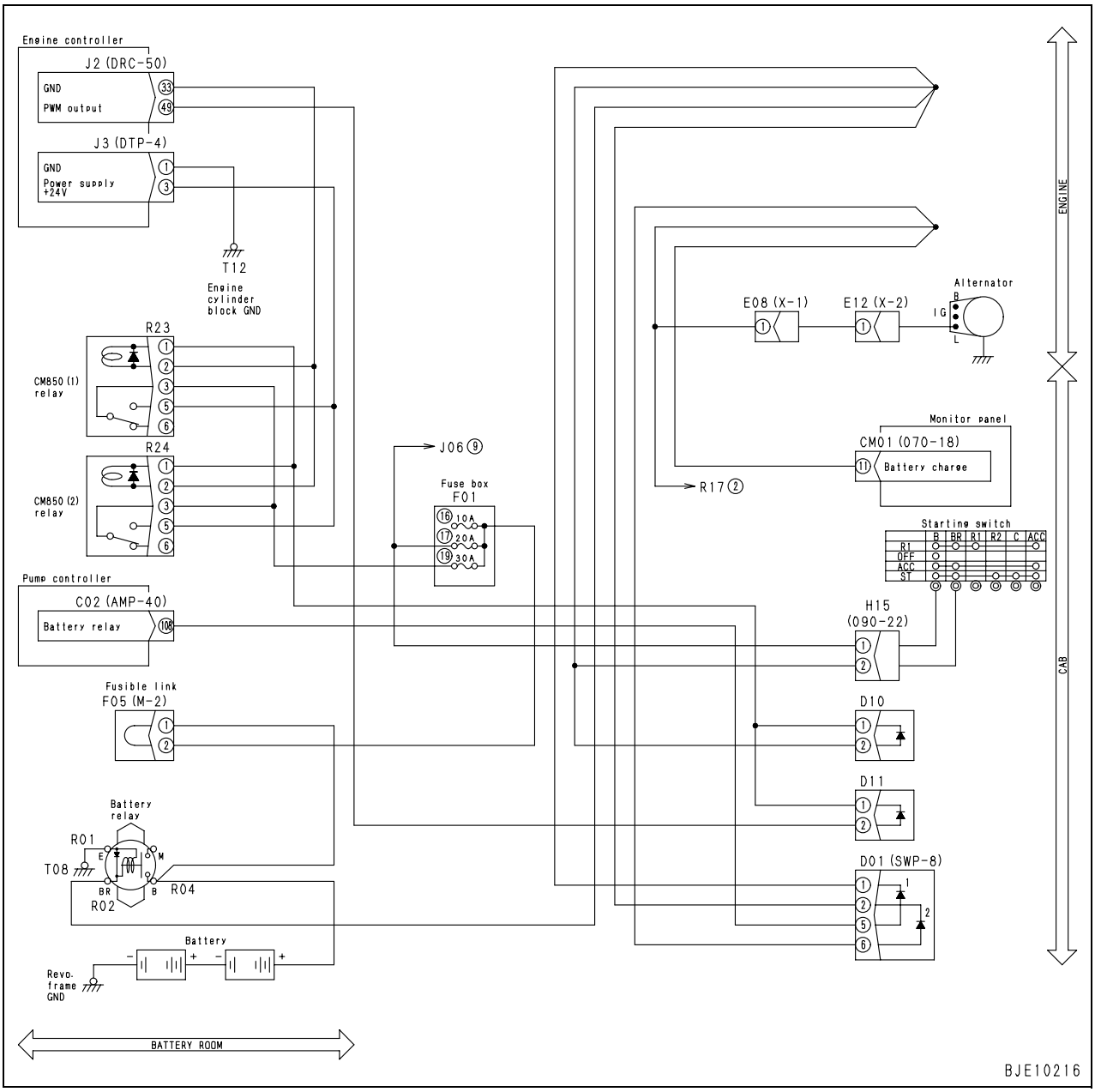
	Causes		Standard value in normal state/Remarks on troubleshooting
Possible causes and standard value in normal state	1	Broken engine Ne speed sensor	Inspect engine Ne speed sensor for breakage directly.
	2	Broken engine Bkup speed sensor	Inspect engine Bkup speed sensor for breakage directly.
	3	Defective or broken mount of rotation sensing flywheel on crankshaft side	Inspect it for defect or breakage according to the following procedure. 1) Set No. 1 cylinder to compression top dead center (align stamping mark). 2) If tip of Ne speed sensor is aligned with center of oblong hole of rotation sensing flywheel, it is mounted normally.
	4	Defective or broken mount of rotation sensing ring on camshaft side	Inspect it for defect or breakage according to the following procedure. 1) Set No. 1 cylinder to compression top dead center (align stamping mark). 2) Remove Bkup speed sensor. 3) If 2 grooves (1 thread) of rotation sensing ring can be seen through sensor mounting hole, it is mounted normally.
	5	Defective timing between crankshaft and camshaft	Inspect crankshaft and camshaft for timing directly.
	6	Defective connection of ground terminal	Inspect following directly for connection of ground terminal. • Ground terminal (battery (–)) of machine main unit. • Ground terminal of engine • Ground terminal of engine controller • Ground terminal of starting motor

E-45 Code [757/CA757] Loss of all data in engine controller

Error code	Failure code	Trouble	Loss of all data in engine controller
757	**CA757**		
Contents of trouble	• All data in engine controller are lost.		
Action of controller	• None in particular.		
Problem that appears on machine	• Engine stops and sometimes cannot be started.		
Related information	• Method of reproducing failure code: Starting switch ON.		

	Causes		Standard value in normal state/Remarks on troubleshooting			
Possible causes and standard value in normal state	1	Defective related system	If another failure code is indicated, carry out troubleshooting for it.			
	2	Loose or corroded battery terminal	Inspect battery terminal directly for loose and corrosion.			
	3	Defective battery voltage	★ Prepare with starting switch OFF, then turn starting switch OFF and START and carry out troubleshooting.			
			Battery (1 piece)	Starting switch	Voltage	
			Between (+) – (–) terminals	OFF	Min. 12 V	
				START	Min. 6.2 V	
	4	Defective fuse No. 19	If fuse is burnt out, circuit probably has ground fault.			
	5	Defective relay for engine controller power supply	★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting.			
			Replace relays for engine controller power supply (R23, R24) with other relays. If mark E of failure code disappears after reproduction operation is conducted, that relay has a defect.			
	6	Disconnection in wiring harness (Disconnection in wiring or defective contact in connector)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness between F01-19 – R23, R24 (female) (3)		Resistance	Max. 10 Ω
			Wiring harness between R23, R24 (female) (5) – J3 (female) (3)		Resistance	Max. 10 Ω
			Wiring harness between J3 (female) (1) – chassis ground (T12)		Resistance	Max. 10 Ω
	7	Defective wiring harness connector	Connecting parts among fuse No. 19, engine unit wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation			
	8	Defective engine controller	★ Prepare with starting switch OFF, then turn starting switch ON and START and carry out troubleshooting.			
			J3 (Female)	Starting switch	Voltage	
			Between (3) – (1)	ON	Min. 24 V	
				START	Min. 12 V	

Circuit diagram related to engine controller power supply



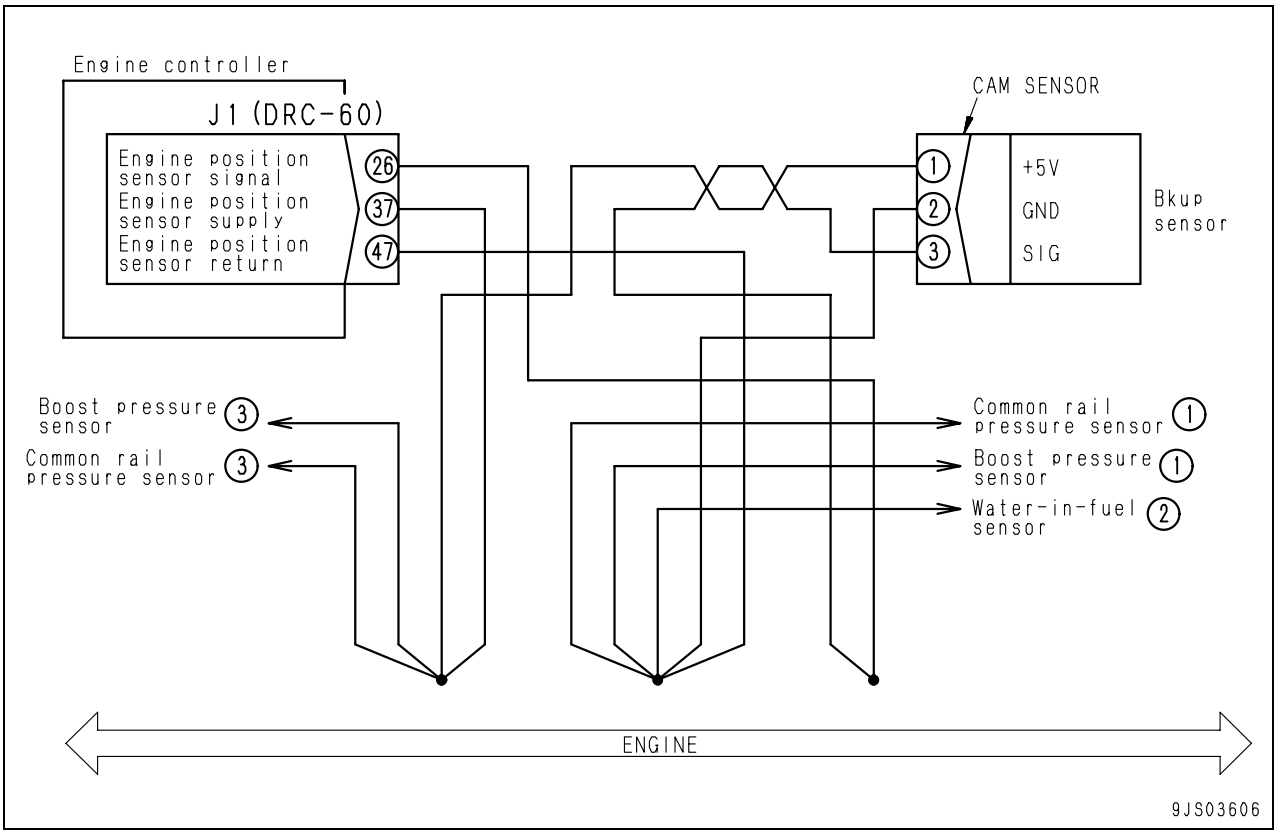
E-46 Code [778/CA778] Abnormality in engine Bkup speed sensor

Error code	Failure code	Trouble	Abnormality in engine Bkup speed sensor
778	**CA778**		
Contents of trouble	• Abnormality is detected in signals of engine Bkup speed sensor.		
Action of controller	• Controls with signals of Ne speed sensor.		
Problem that appears on machine	• Starting performance is poor. • Output drops.		
Related information	• Method of reproducing failure code: Start engine.		

	Causes		Standard value in normal state/Remarks on troubleshooting		
Possible causes and standard value in normal state	1	Defective sensor power supply 2 circuit	If failure code [**CA187**] is indicated simultaneously, carry out troubleshooting for it first.		
	2	Defective engine Bkup speed sensor	★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting.		
			CAM SENSOR		Voltage
			Between (1) – (2)	Power supply	4.75 – 5.25 V
			Voltage is measured with wiring harness connected. Accordingly, if voltage is abnormal, check wiring harness and controller, too, for another cause of trouble, and then judge.		
	3	Broken engine Bkup speed sensor or defective clearance	Inspect engine Bkup speed sensor directly for breakage or clearance.		
	4	Broken rotation sensing ring	Inspect it for breakage directly.		
	5	Disconnection in wiring harness (Disconnection in wiring or defective contact in connector)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			Wiring harness between J1 (female) (26) – CAM SENSOR (female) (3)	Resistance	Max. 10 Ω
	6	Ground fault in wiring harness (Short circuit with GND circuit)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			Wiring harness between J1 (female) (26) – CAM SENSOR (female) (3) and chassis ground	Resistance	Min. 100 kΩ
	7	Hot short (Short circuit with 5 V, 24 V circuit) in wiring harness	★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting.		
			Wiring harness between J1 (female) (26) – CAM SENSOR (female) (3) and chassis ground	Voltage	Max. 1 V
	8	Short circuit in wiring harness (With another wiring harness)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			Wiring harness between J1 (female) (26) – CAM SENSOR (female) (3) and J1 (female) (37) – CAM SENSOR (female) (1)	Resistance	Min. 100 kΩ
			Wiring harness between J1 (female) (26) – CAM SENSOR (female) (3) and J1 (female) (47) – CAM SENSOR (female) (2)	Resistance	Min. 100 kΩ

<table>
<tr><td rowspan="4">Possible causes and standard value in normal state</td><td colspan="2">Causes</td><td colspan="3">Standard value in normal state/Remarks on troubleshooting</td></tr>
<tr><td>9</td><td>Defective wiring harness connector</td><td colspan="3">Connecting parts among engine Bkup speed sensor, engine wiring harness, and engine controller are suspected. Inspect them directly.
• Loose connector, broken lock, broken seal
• Corrosive, bent, broken, forced-in, or extended pin
• Humidity in connector, entry of dirt or dust, poor insulation</td></tr>
<tr><td rowspan="3">10</td><td rowspan="3">Defective engine controller</td><td colspan="3">★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting.</td></tr>
<tr><td colspan="2">J1</td><td>Voltage</td></tr>
<tr><td>Between (37) – (47)</td><td>Power supply</td><td>4.75 – 5.25 V</td></tr>
</table>

Circuit diagram related to engine Bkup speed sensor

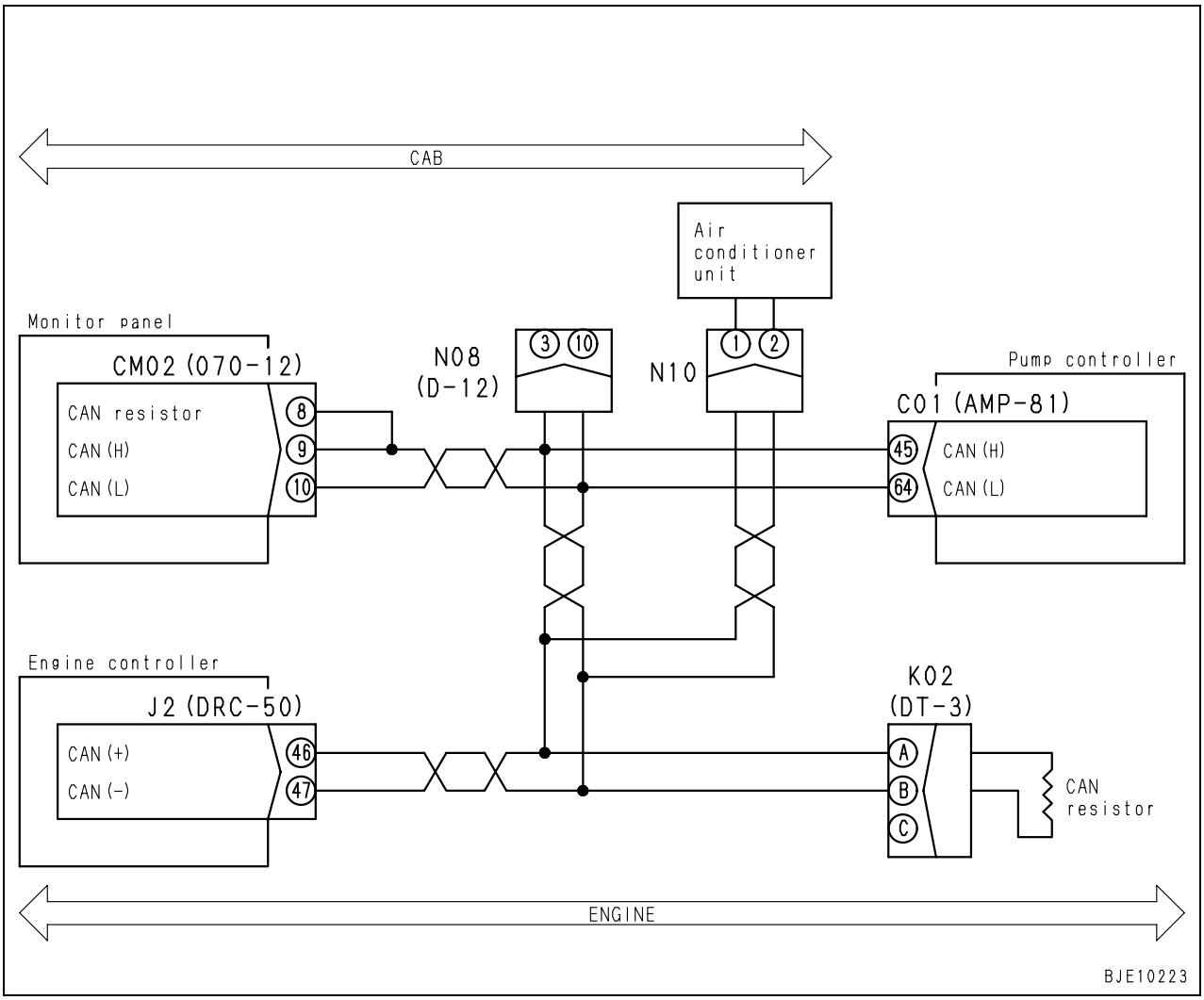


E-47 Code [1633/CA1633] Abnormality in KOMNET

Error code	Failure code	Trouble	Abnormality in KOMNET
1633	**CA1633**		
Contents of trouble	• Engine controller detected abnormality in KOMNET communication circuit to controllers on applicable machine side.		
Action of controller	• Depends on machine model. • If problem is removed, circuit is returned to normal operating state.		
Problem that appears on machine	• KOMNET communication information may not be transmitted correctly to cause incorrect operation of machine. (Problem depends on failure occurrence location.)		
Related information	• Method of reproducing failure code: Starting switch ON.		

	Causes		Standard value in normal state/Remarks on troubleshooting		
Possible causes and standard value in normal state	1	Disconnection in wiring harness (Disconnection in wiring or defective contact in connector)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			Wiring harness between CM02 (female) (8) (9) – C01 (female) (45), CM02 (female) (8) (9) – J2 (female) (46), CM02 (female) (8) (9) – K02 (female) (A)	Resistance	Max. 1 Ω
			Wiring harness between CM02 (female) (10) – C01 (female) (64), CM02 (female) (10) – J2 (female) (47), CM02 (female) (10) – K02 (female) (B)	Resistance	Max. 1 Ω
	2	Ground fault in wiring harness (Short circuit with GND circuit)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			Wiring harness between CM02 (female) (8) (9) – C01 (female) (45), CM02 (female) (8) (9) – J2 (female) (46), CM02 (female) (8) (9) – K02 (female) (A), CM02 (female) (8) (9) – N08 (male) (3)	Resistance	Min. 1 MΩ
			Wiring harness between CM02 (female) (10) – C01 (female) (64), CM02 (female) (10) – J2 (female) (47), CM02 (female) (10) – K02 (female) (B), CM02 (female) (10) – N08 (male) (10)	Resistance	Min. 1 MΩ
	3	Hot short (Short circuit with 24 V circuit) in wiring harness	★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting.		
			Wiring harness between CM02 (female) (8) (9) – C01 (female) (45), CM02 (female) (8) (9) – J2 (female) (46), CM02 (female) (8) (9) – K02 (female) (A), CM02 (female) (8) (9) – N08 (male) (3)	Voltage	Max. 5.5 V
			Wiring harness between CM02 (female) (10) – C01 (female) (64), CM02 (female) (10) – J2 (female) (47), CM02 (female) (10) – K02 (female) (B), CM02 (female) (10) – N08 (male) (10)	Voltage	Max. 5.5 V
	4	Defective terminal resistance (Internal short circuit, disconnection)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			K02 (Male)	Resistance	
			Between (A) – (B)	47 – 67 Ω	
	5	Defective pump controller	If causes 1 – 4 are not detected, engine controller may be defective. (Since trouble is in system, troubleshooting cannot be carried out.)		

Circuit diagram related to KOMNET communication

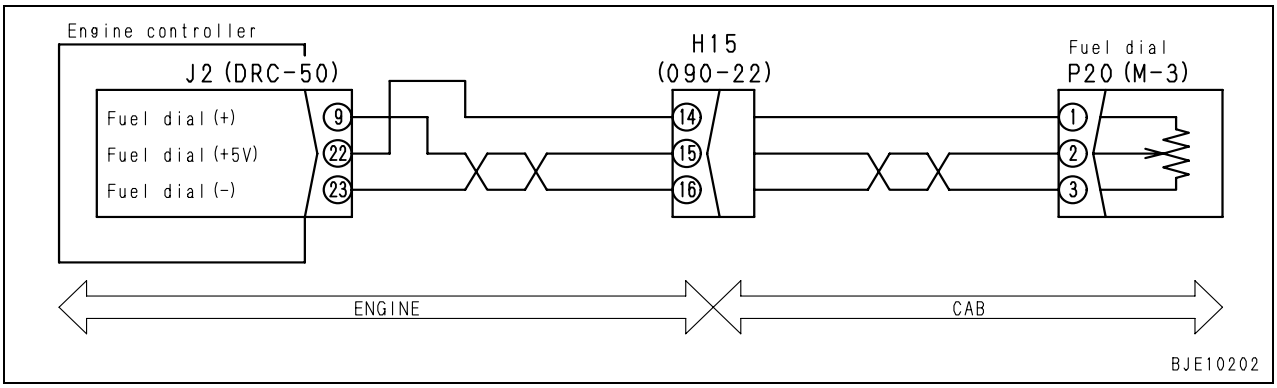


E-48 Code [2185/CA2185] Abnormally high level in throttle sensor power supply

Error code	Failure code	Trouble	Abnormally high level in throttle sensor power supply
2185	**CA2185**		
Contents of trouble	• High voltage (min. 5.25 V) occurred in throttle sensor power supply circuit.		
Action of controller	• Depends on machine model.		
Problem that appears on machine	• Depends on machine model.		
Related information	• Method of reproducing failure code: Starting switch ON.		

	Causes		Standard value in normal state/Remarks on troubleshooting		
Possible causes and standard value in normal state			★ Troubleshoot with starting switch OFF, then carry out trouble-shooting without turning starting switch ON.		
	1	Short circuit in wiring harness (With another wiring harness)	Wiring harness among all pins between J2 (female) (22) – J2 (female) (With also P20 disconnected)	Resis-tance	Min. 100 kΩ
			Wiring harness among all pins between J2 (female) (22) – J3 (female) (3) (With also P20 disconnected)	Resis-tance	Min. 100 kΩ
	2	Defective wiring harness connector	Connecting parts among fuel control dial, engine unit wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation		
	3	Defective engine controller	If causes 1 and 2 are not detected, engine controller may be defective. (Since trouble is in system, troubleshooting cannot be carried out.)		

Circuit diagram related to fuel control dial

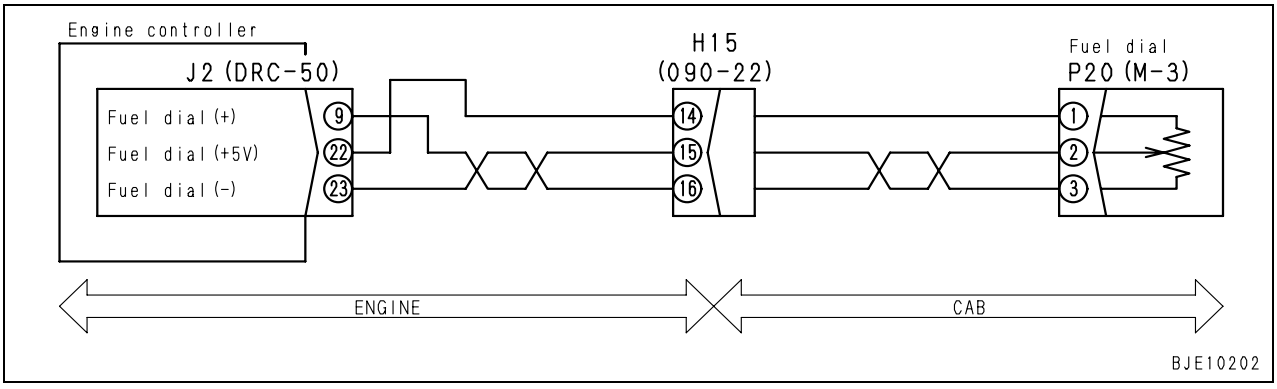


E-49 Code [2186/CA2186] Abnormally low level in throttle sensor power supply

Error code	Failure code	Trouble	Abnormally low level in throttle sensor power supply
2186	**CA2186**		
Contents of trouble	• Low voltage occurred in throttle sensor power supply circuit.		
Action of controller	• Depends on machine model.		
Problem that appears on machine	• Depends on machine model.		
Related information	• Method of reproducing failure code: Starting switch ON.		

		Causes	Standard value in normal state/Remarks on troubleshooting		
Possible causes and standard value in normal state	1	Ground fault in wiring harness (Short circuit with GND circuit)	★ Troubleshoot with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			Wiring harness between J2 (female) (22) – P20 (female) (1)	Resistance	Min. 100 kΩ
	2	Short circuit in wiring harness (With another wiring harness)	★ Troubleshoot with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			Wiring harness among all pins between J2 (female) (9) – J2 (female) (With also P20 disconnected)	Resistance	Min. 100 kΩ
	3	Defective wiring harness connector	Connecting parts among fuel control dial, engine unit wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation		
	4	Defective engine controller	If causes 1 – 3 are not detected, engine controller may be defective. (Since trouble is in system, troubleshooting cannot be carried out.)		

Circuit diagram related to fuel control dial



E-50 Code [2249/CA2249] No-pressure feed by supply pump 2

Error code	Failure code	Trouble	No-pressure feed by supply pump 2
2249	**CA2249**		
Contents of trouble	• No-pressure feed (2) occurred in common rail circuit.		
Action of controller	• Limits output and continues operation.		
Problem that appears on machine	• Starting performance is poor • Exhaust smoke is black • Output drops.		
Related information	• Method of reproducing failure code: Start engine.		

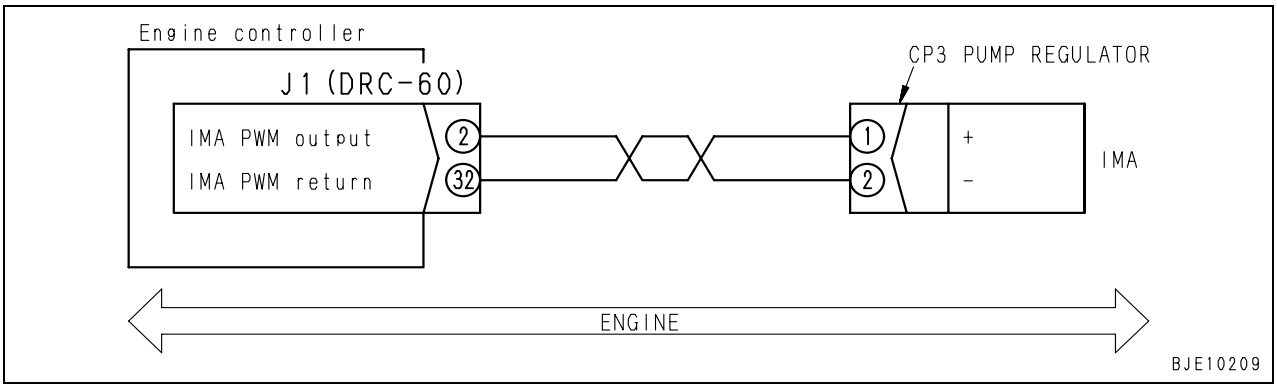
Possible causes and standard value in normal state	Causes		Standard value in normal state/Remarks on troubleshooting
	1	Carry out troubleshooting for failure code [**CA559**].	

E-51 Code [2311/CA2311] Abnormality in IMV solenoid

Error code	Failure code	Trouble	Abnormality in IMV solenoid
2311	**CA2311**		
Contents of trouble	• Resistance in supply pump actuator is abnormally high or low.		
Action of controller	• None in particular.		
Problem that appears on machine	• Engine output drops.		
Related information	• Method of reproducing failure code: Start engine.		

	Causes		Standard value in normal state/Remarks on troubleshooting			
Possible causes and standard value in normal state	1	Defective related system	If another failure code is indicated, carry out troubleshooting for it.			
	2	Defective supply pump actuator	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			CP3 PUMP REGULATOR (Male)	Resistance		
			Between (1) – (2)	Max. 5 Ω		
			Between (1) – chassis ground	Min. 100 kΩ		
	3	Disconnection in wiring harness (Disconnection in wiring or defective contact in connector)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness between J1 (female) (2) – CP3 PUMP REGULATOR (female) (1)	Resistance	Max. 5 Ω	
			Wiring harness between J1 (female) (32) – CP3 PUMP REGULATOR (female) (2)	Resistance	Max. 5 Ω	
	4	Ground fault in wiring harness (Short circuit with GND circuit)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			Wiring harness between J1 (female) (2) – CP3 PUMP REGULATOR (female) (1)	Resistance	Min. 100 kΩ	
	5	Defective wiring harness connector	Connecting parts among supply pump actuator, engine wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation			
	6	Defective engine controller	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.			
			J1 (Female)	Resistance		
			Between (2) – (32)	Max. 5 Ω		
			Between (2) – chassis ground	Min. 100 kΩ		

Circuit diagram related to supply pump actuator (metering unit)

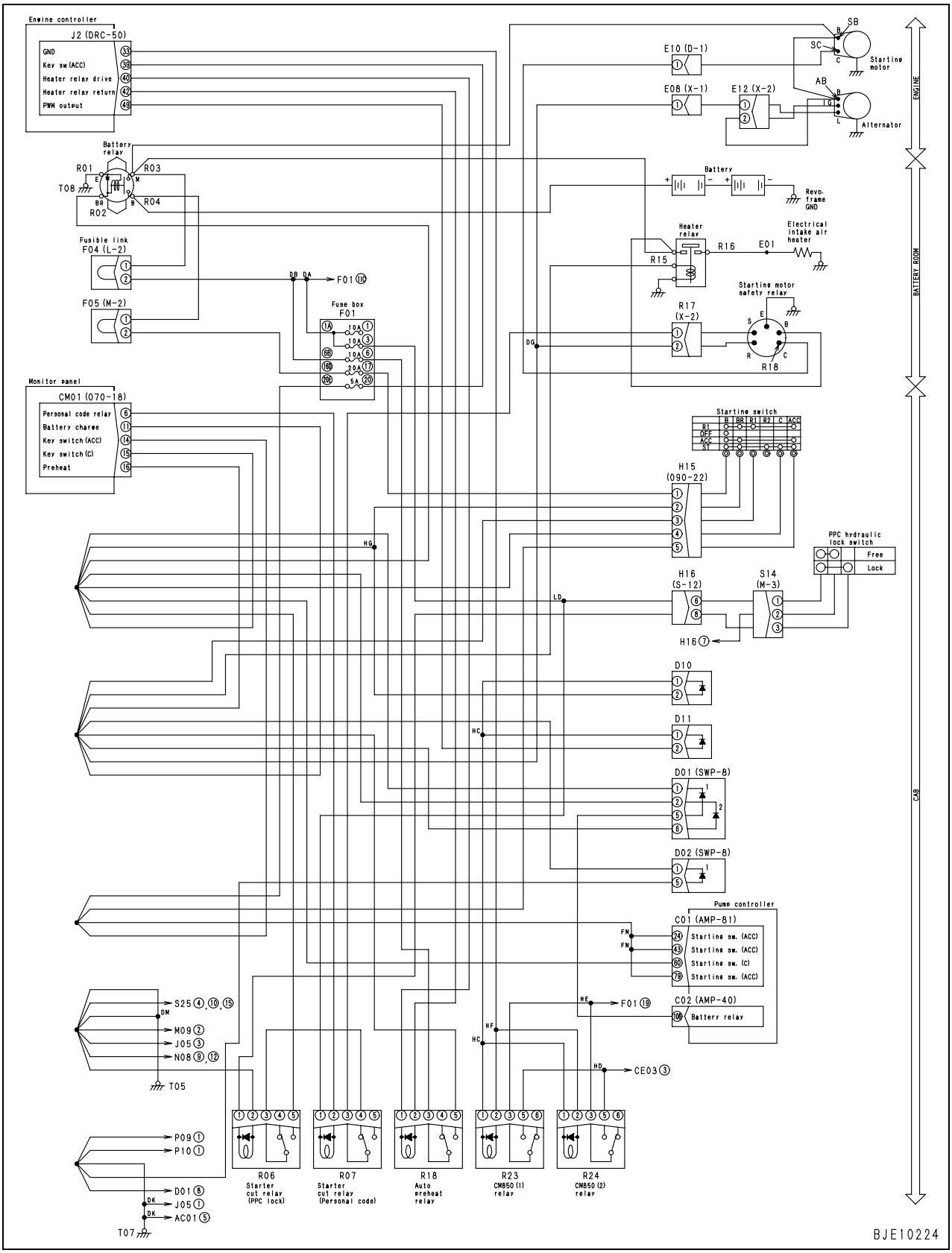


E-52 Code [2555/CA2555] Disconnection in intake air heater relay

Error code	Failure code	Trouble	Disconnection in intake air heater relay
2555	**CA2555**		
Contents of trouble	• There is disconnection in intake air heater relay.		
Action of controller	• None in particular.		
Problem that appears on machine	• Intake air heater does not function (Poor starting performance, white exhaust smoke at low temperature)		
Related information	• Method of reproducing failure code: Starting switch ON when coolant temperature is –4°C or less.		

	Causes		Standard value in normal state/Remarks on troubleshooting		
Possible causes and standard value in normal state	1	Defective auto preheat relay (Internal disconnection)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON. (Test relay unit).		
			R18 (Male)	Resistance	
			Between (1) – (2)	300 – 600 Ω	
			★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting. (Test by replacing relay unit.)		
			Replace auto preheat relay (R18) with other relay. If mark E of failure code disappears after reproduction operation is conducted, that relay has a defect.		
	2	Disconnection in wiring harness (Disconnection in wiring or defective contact in connector)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			Wiring harness between J2 (female) (40) – R18 (female) (1)	Resistance	Max. 10 Ω
			Wiring harness between J2 (female) (42) – R18 (female) (2)	Resistance	Max. 10 Ω
	3	Short circuit in wiring harness (With another wiring harness)	★ Troubleshoot with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			Wiring harness among all pins between J2 (female) (40) – J2 (female) (With also R18 disconnected)	Resistance	Min. 100 kΩ
	4	Defective wiring harness connector	Connecting parts among auto preheat relay, engine unit wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation		
	5	Defective engine controller	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			J2 (Female)	Resistance	
			Between (40) – (42)	300 – 600 Ω	

Circuit diagram related to preheating, starting and charging engine

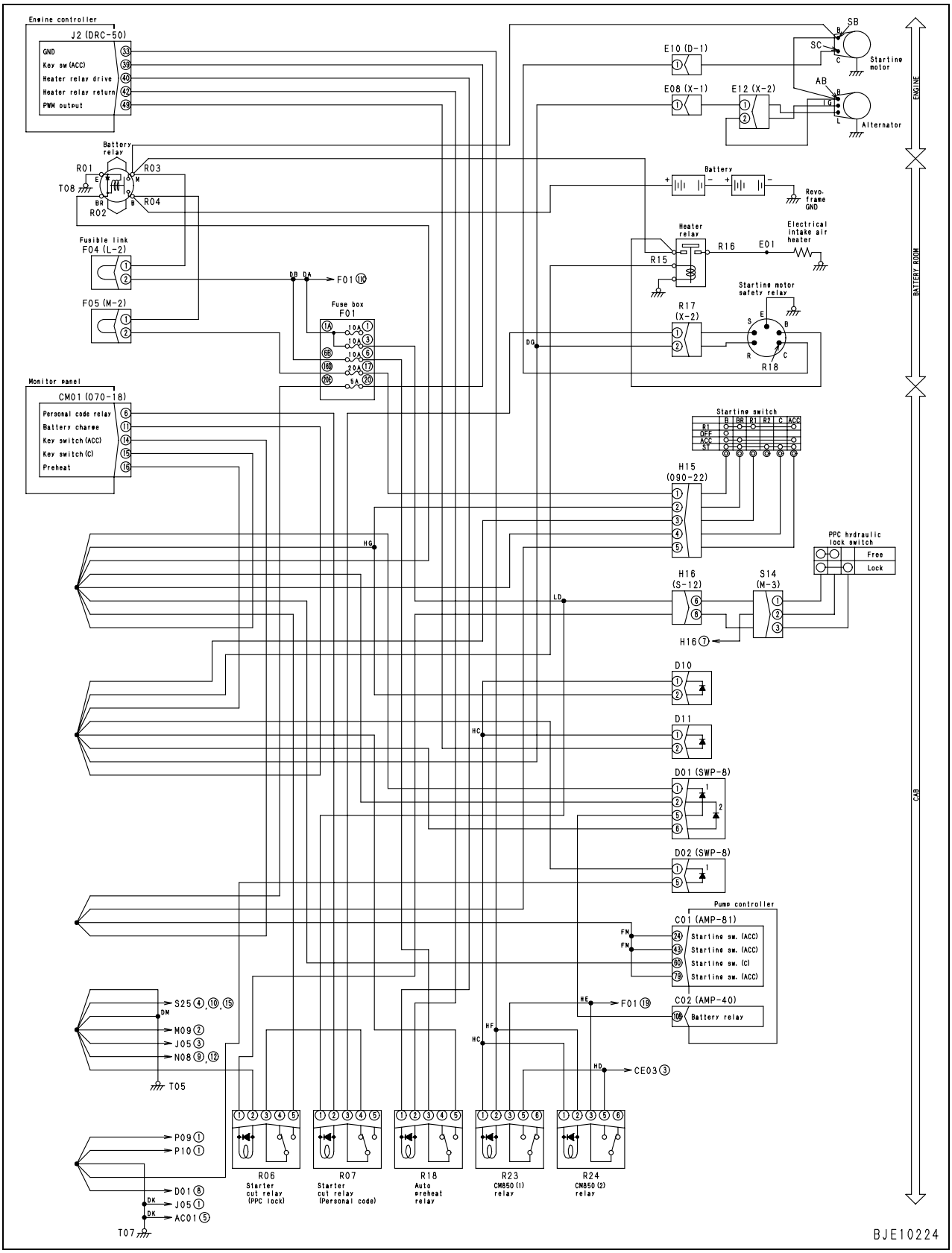


E-53 Code [2556/CA2556] Short circuit in intake air heater relay

Error code	Failure code	Trouble	Short circuit in intake air heater relay
2556	**CA2556**		
Contents of trouble	• There is short circuit in intake air heater relay.		
Action of controller	• None in particular.		
Problem that appears on machine	• Intake air heater does not function. (Poor starting performance, white exhaust smoke at low temperature)		
Related information	• Method of reproducing failure code: Starting switch ON when coolant temperature is –4°C or less.		

	Causes		Standard value in normal state/Remarks on troubleshooting		
Possible causes and standard value in normal state	1	Defective auto preheat relay (Internal disconnection)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON. (Test relay unit).		
			R18 (Male)	Resistance	
			Between (1) – (2)	300 – 600 Ω	
			★ Prepare with starting switch OFF, then turn starting switch ON and carry out troubleshooting. (Test by replacing relay unit.)		
			Replace auto preheat relay (R18) with other relay. If mark E of failure code disappears after reproduction operation is conducted, that relay has a defect.		
	2	Ground fault in wiring harness (Short circuit with GND circuit)	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			Wiring harness between J2 (female) (40) – R18 (female) (1)	Resistance	Min. 100 kΩ
	3	Short circuit in wiring harness (With another wiring harness)	★ Troubleshoot with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			Wiring harness among all pins between J2 (female) (40) – J2 (female) (With also R18 disconnected)	Resistance	Min. 100 kΩ
	4	Defective wiring harness connector	Connecting parts among auto preheat relay, engine unit wiring harness, and engine controller are suspected. Inspect them directly. • Loose connector, broken lock, broken seal • Corrosive, bent, broken, forced-in, or extended pin • Humidity in connector, entry of dirt or dust, poor insulation		
	5	Defective engine controller	★ Prepare with starting switch OFF, then carry out troubleshooting without turning starting switch ON.		
			J2 (Female)	Resistance	
			Between (40) – (42)	300 – 600 Ω	

Circuit diagram related to preheating, starting and charging engine



E-54 Code [---/B@BAZG] Derating of speed by engine oil pressure reduction

Error code	Failure code	Trouble	Derating of speed by engine oil pressure reduction
---	**B@BAZG**		
Contents of trouble	• Engine oil pressure is below operating range.		
Action of controller	• Limits output and continues operation (Limits fuel injection rate and engine speed).		
Problem that appears on machine	• Output drops.		
Related information			

Possible causes and standard value in normal state	Causes		Standard value in normal state/Remarks on troubleshooting
	1	Carry out troubleshooting on machine side	This fault is indicated on only machine side when detected. Accordingly, contents of troubleshooting depend on machine model. See Shop manual for machine.

E-55 Code [---/B@BAZG] Derating of torque by engine oil pressure reduction

Error code	Failure code	Trouble	Derating of torque by engine oil pressure reduction
---	**B@BAZG**		
Contents of trouble	• Engine oil pressure is below operating range.		
Action of controller	• Limits output and continues operation (Limits fuel injection rate and engine speed).		
Problem that appears on machine	• Output drops.		
Related information			

Possible causes and standard value in normal state	Causes		Standard value in normal state/Remarks on troubleshooting
	1	Carry out troubleshooting on machine side	This fault is indicated on only machine side when detected. Accordingly, contents of troubleshooting depend on machine model. See Shop manual for machine.

E-56 Code [---/B@BCNS] Engine overheat

Error code	Failure code	Trouble	Engine overheat
---	**B@BCNS**		
Contents of trouble	• Engine coolant temperature is above operating range.		
Action of controller	• Limits output and continues operation.		
Problem that appears on machine	• Output drops.		
Related information			

Possible causes and standard value in normal state	Causes		Standard value in normal state/Remarks on troubleshooting
	1	Carry out troubleshooting on machine side	This fault is indicated on only machine side when detected. Accordingly, contents of troubleshooting depend on machine model. See Shop manual for machine.

Engine

107E-1 Series

50 Disassembly and assembly

General information on disassembly and assembly

General information on disassembly and assembly 2
How to read this manual 2
Coating materials list 3
Special tools list 6

General information on disassembly and assembly

How to read this manual

Disassembly

- In Disassembly section, the work procedures, precautions and know-how for carrying out those procedures, and quantity of the oil and coolant drained are described.
- The meanings of the symbols used in Disassembly section are as follows.

⚠ : **This mark indicates safety-related precautions that must be followed when implementing the work.**

★ : Know-how or precautions for work

: Quantity of oil or coolant drained

: Weight of part or component

Assembly

- In Assembly section, the work procedures, precautions and know-how for carrying out those procedures, and quantity of the oil and coolant added are described.
- The meanings of the symbols used in Assembly section are as follows.

⚠ : **Precautions related to safety in execution of work.**

★ : Know-how or precautions for work

: Type of coating material

: Tightening torque

: Quantity of oil or coolant to be added

: Weight of part or component

Coating materials list

★ The recommended coating materials such as adhesives, gasket sealants, and greases used for disassembly and assembly are listed below.
★ For coating materials not listed below, use the equivalent of products shown in this manual.

Category	Komatsu code	Part No.	Q'ty	Container	Main features and applications
Adhesive	LT-1A	790-129-9030	150 g	Tube	• Used to prevent rubber gaskets, rubber cushions, and cork plugs from coming out.
	LT-1B	790-129-9050	20 g (2 pcs.)	Polyethylene container	• Used for plastic (except polyethylene, polypropylene, tetrafluoroethylene and vinyl chloride), rubber, metal, and non-metal parts which require immediate and strong adhesion.
	LT-2	09940-00030	50 g	Polyethylene container	• Features: Resistant to heat and chemicals. • Used to fix and seal bolts and plugs.
	LT-3	790-129-9060 Set of adhesive and hardener	Adhesive: 1 kg Hardener: 500 g	Can	• Used to stick and seal metal, glass, and plastics.
	LT-4	790-129-9040	250 g	Polyethylene container	• Used to seal plugs for drilled holes.
	Holtz MT 705	790-129-9120	75 g	Tube	• Heat-resistant sealant used to repair engines.
	Three Bond 1735	790-129-9140	50 g	Polyethylene container	• Quick-setting adhesive. • Setting time: Within 5 sec. to 3 min. • Used mainly to stick metals, rubbers, plastics, and woods.
	Aron-alpha 201	790-129-9130	2 g	Polyethylene container	• Quick-setting adhesive. • Quick-hardening type. (max. strength is obtained after 30 minutes of application) • Used mainly to stick rubbers, plastics, and metals.
	Loctite 648-50	79A-129-9110	50 cc	Polyethylene container	• Features: Resistant to heat and chemicals. • Used for fitted portions used at high temperatures.

Category	Komatsu code	Part No.	Q'ty	Container	Main features and applications
Gasket sealant	LG-1	790-129-9010	200 g	Tube	• Used to stick or seal gaskets and packings of power train case, etc.
	LG-5	790-129-9080	1 kg	Polyethylene container	• Used to seal various threaded portions, pipe joints, and flanges. • Used to seal tapered plugs, elbows, and nipples of hydraulic piping.
	LG-6	790-129-9160	250 g	Tube	• Features: Silicon-based heat and cold-resistant sealant. • Used to seal flanged surfaces and threaded portions. • Used to seal oil pan, final drive case, etc.
	LG-7	790-129-9170	150 g	Tube	• Features: Silicon-based quick-drying sealant. • Used to seal flywheel housing, intake manifold, oil pan, thermostat housing, etc.
	Three Bond 1211	790-129-9090	100 g	Tube	• Gasket sealant used to repair engine.
	Three Bond 1207B	419-15-18131	100 g	Tube	• Features: Silicon-based, heat-resistant, vibration-resistant and impact-resistant sealant. • Used to seal transfer case, etc.
Molybdenum disulfide lubricant	LM-G	09940-00051	60 g	Can	• Used to lubricate sliding portions (to prevent squeaking)
	LM-P	09940-00040	200 g	Tube	• Used to prevent scuffing and seizure of press-fitted portions, shrink-fitted portions, and threaded portions • Used to lubricate linkages, bearings, etc.
Grease	G2-LI	SYG2-400LI SYG2-350LI SYG2-400LI-A SYG2-160LI SYGA-160CNLI	Various	Various	• General purpose type
	G2-CA	SYG2-400CA SYG2-350CA SYG2-400CA-A SYG2-160CA SYGA-160CNCA	Various	Various	• Used for bearings used at normal temperature under light load in contact with water or steam
	Molybdenum disulfide lubricant LM-G (G2-M)	SYG2-400M SYG2-400M-A SYGA-16CNM	400 g×10 400 g×20 16 kg	Bellows-type container Can	• Used for parts under heavy load
	Hyper White Grease G2-T, G0-T (*) *: For cold district	SYG2-400T-A SYG2-16CNT SYG0-400T-A (*) SYG0-16CNT (*)	400 g 16 kg	Bellows-type container Can	• Seizure resistance and heat resistance higher than molybdenum disulfide • Not conspicuous on machine since color is white
	Biogrease G2-B, G2-BT (*) *: For use at high temperature and under high load	SYG2-400B SYGA-16CNB SYG2-400BT (*) SYGA-16CNBT (*)	400 g 16 kg	Bellows-type container Can	• Since this grease is decomposed by natural bacteria in short period, it has less effects on microorganisms, animals, and plants

Category	Komatsu code	Part No.	Q'ty	Container	Main features and applications	
Primer	SUNSTAR PAINT PRIMER 580 SUPER	417-926-3910	20 ml	Glass container	Adhesive for cab glass	• Used as primer for cab side (Using limit: 4 months after date of manufacture)
	SUNSTAR GLASS PRIMER 580 SUPER		20 ml	Glass container		• Used as primer for glass side (Using limit: 4 months after date of manufacture)
	SUNSTAR PAINT PRIMER 435-95	22M-54-27230	20 ml	Glass container		• Used as primer for painted surface on cab side (Using limit: 4 months after date of manufacture)
	SUNSTAR GLASS PRIMER 435-41	22M-54-27240	150 ml	Can		• Used as primer for black ceramic-coated surface on glass side and for hard polycarbonate-coated surface (Using limit: 4 months after date of manufacture)
	SUNSTAR SASH PRIMER GP-402	22M-54-27250	20 ml	Glass container		• Used as primer for sash (anodized aluminum) (Using limit: 4 months after date of manufacture)
Adhesive	SUNSTAR PENGUINE SEAL 580 SUPER “S” or “W”	417-926-3910	320 ml	Polyethylene container	Adhesive for cab glass	• “S” is used for high-temperature season (April – October) and “W” for low-temperature season (October – April) as adhesive for glass (Using limit: 4 months after date of manufacture)
	Sika Japan, Sikaflex 256HV	20Y-54-39850	310 ml	Polyethylene container		• Used as adhesive for glass (Using limit: 6 months after date of manufacture)
	SUNSTAR PENGUINE SUPER 560	22M-54-27210	320 ml	Ecocart (Special container)		• Used as adhesive for glass (Using limit: 6 months after date of manufacture)
Caulking material	SUNSTAR PENGUINE SEAL No. 2505	417-926-3920	320 ml	Polyethylene container		• Used to seal joints of glass parts (Using limit: 4 months after date of manufacture)
	SEKISUI SILICONE SEALANT	20Y-54-55130	333 ml	Polyethylene container		• Used to seal front window (Using limit: 6 months after date of manufacture)
	GE TOSHIBA SILICONES TOSSEAL 381	22M-54-27220	333 ml	Cartridge		• Used to seal joint of glasses Translucent white seal (Using limit: 12 months after date of manufacture)

Special tools list

★ Tools with part number 79○T - ○○○ - ○○○○ cannot be supplied (they must be locally manufactured)
★ Necessity : ■ ------ Cannot be substituted, should always be installed (used)
: ● ------ Extremely useful if available, can be substituted with commercially available part
★ New / Remodel : N---------- Tools newly developed for this model and given new part Nos.
: R---------- Tools remodeled from tools already available for other models and given upgraded part Nos.
: Blank------- Tools already available for other models, usable without any modification
★ Tools marked with ○ in the sketch column are tools introduced in the sketches of the special tools (See Sketches of special tools)
★ Nos. in the upper line of the Part No. column are KOMATSU part Nos. and those in the lower line are CUMMINS part Nos.

Work item	Sym-bol	Part No.	Part name	Necessity	Q'ty	New/Remodel	Sketches	Remarks
Disassembly and assembly of engine assembly	**A**	790-501-2001 795-799-1150	Repair stand	■	1			
	B	— 795-799-7210	Adapter	■	1	N N		Specialized for Komatsu repair stand Specialized for CUMMINS repair stand
Removal and installation of air intake and exhaust valves	**C**	795-102-2103 795-799-8800	Spring pusher	●	1	 N		
Removal and installation of piston ring	**D**	795-100-2800	Piston ring tool	■	1			
Installation of piston	**E**	795-921-1100	Piston holder	■	1			ϕ75 – ϕ175 mm
Removal of injector	**F**	795-799-6700	Puller	■	1	N		
Removal and installation of rear seal	**G**	795-799-6500	Seal driver	●	1	N		
Removal and installation of front seal	**H**	795-799-6400	Seal driver	●	1	N		
Angle tightening of bolt	**J**	790-331-1120 795-799-2240	Wrench (angle) Angle gauge	●	1			
Adjusting valve clearance	**K**	Commercially available	Clearance gauge	●	1			
Adjusting gear backlash and end play	**L**	Commercially available	Dial gauge and stand	●	1			
Removal of fuel inlet connector	**M**	795-799-8150	Remover	●	1	N		
Removal and installation of main bearing	**U**	795-799-2270	Remover and installer	●	1			
Rotation of crankshaft	**V**	795-799-1131	Barring tool	●	1			
Tightening of injector harness nut	**W**	795-799-7110	Torque wrench	●	1	N		
Measurement of cylinder head bolt length	**X**	795-790-4510	Gauge	●	1	N		

Engine

107E-1 Series

50 Disassembly and assembly

Disassembly and assembly, Part 1

Disassembly and assembly, Part 1 2
General disassembly of engine 2

Disassembly and assembly, Part 1

General disassembly of engine

★ This document describes the general disassembly procedure of SAA6D107E-1. The shapes, quantity, locations, etc. of the parts depend on each applicable machine. Take care.

1. Preparation work

Before disassembling the engine, check its parts for cracking, damage, etc. and clean it generally and carefully for accurate inspection of its parts and quick disassembly and assembly.

★ Before cleaning the engine, carefully seal or remove the openings, electric parts, and wiring connectors so that water will not enter them.

● Prepare a stable engine stand (Blocks [1]) on both sides and place the engine assembly on the stands so that it may not tip over.

Remove oil gauge assembly (2).

2. Installation on engine repair stand

1) Lift engine assembly (1) and then install it on repair stand **A**.

Engine assembly: **Approx. 520 kg** (The weight depends on the applicable machines.)

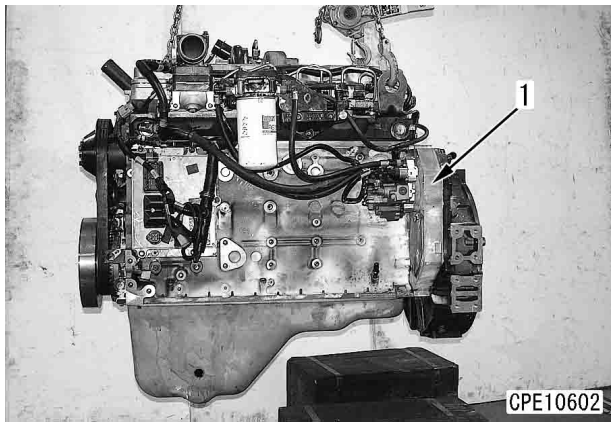
2) Install tool **B**.

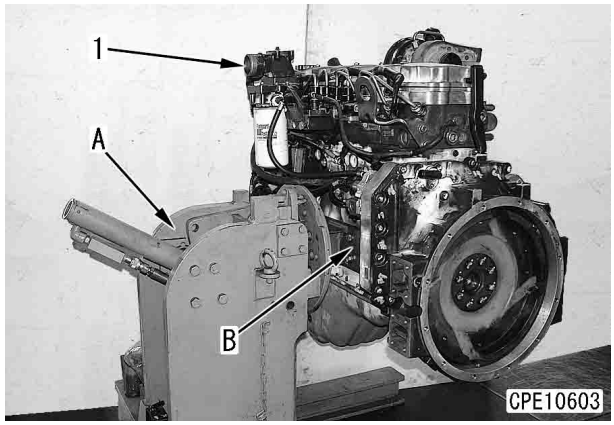
● Engine side mounting bolt: **M12 x 1.75**

Mounting bolt: **77 ± 12 Nm {7.9 ± 1.2 kgm}**

3) Drain the engine coolant and engine oil.

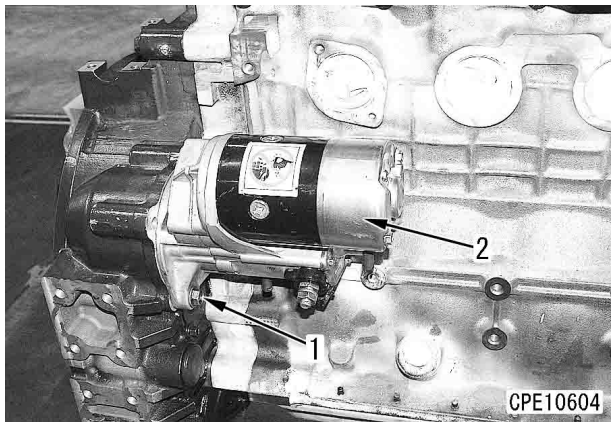
Engine oil: **Approx. 23 ℓ**



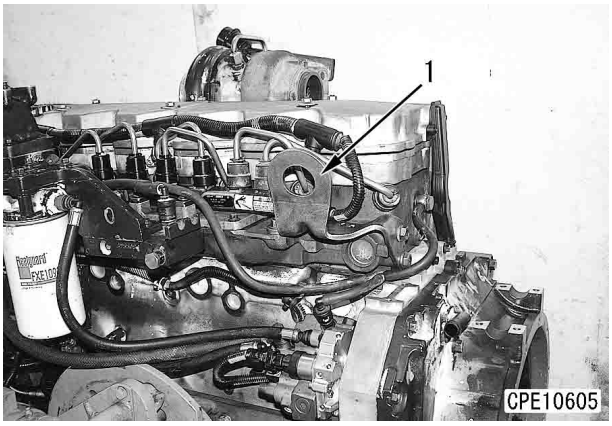


3. Starting motor assembly

Remove mounting bolt (1) and then remove starting motor assembly (2).

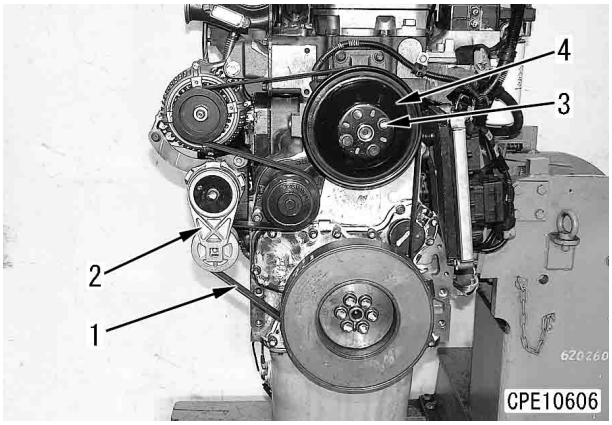


4. Engine sling bracket on rear side
Remove rear engine sling bracket (1) from the cylinder head.

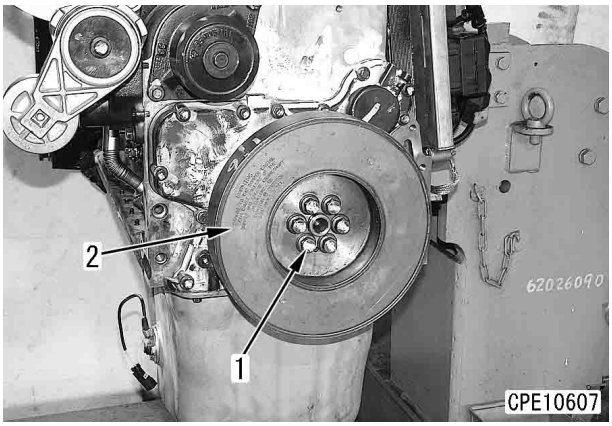


5. Fan pulley
1) Remove fan belt (1).
- Turn tensioner (2) in the opposite direction to the winding direction to reduce the belt tension.

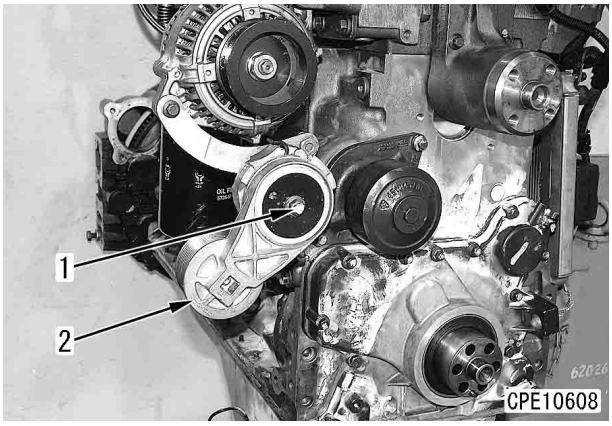
2) Remove bolt (3) and then remove fan pulley (4).



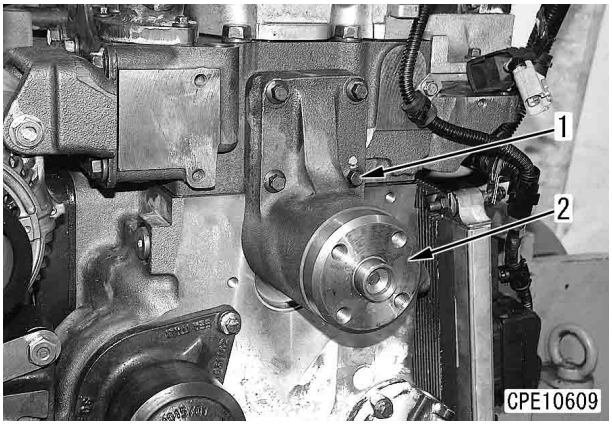
6. Vibration damper and crankshaft pulley
Remove bolt (1) and then remove vibration damper (2).



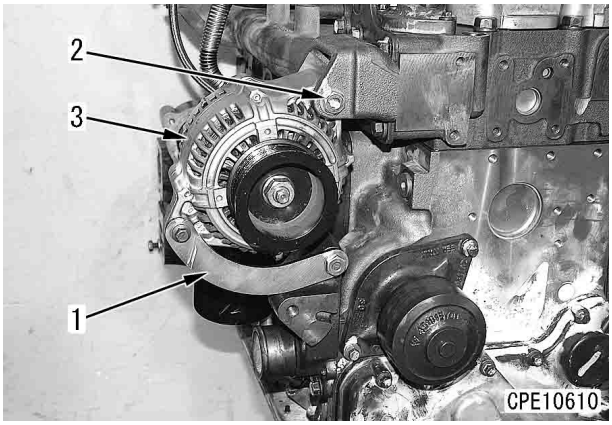
7. Belt tensioner
Remove bolt (1) and then remove tensioner (2).



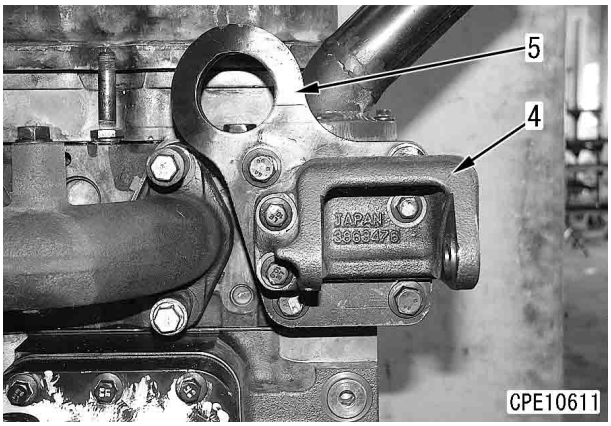
8. Fan hub
Remove 4 bolts (1) and then remove fan hub (2).



9. Alternator
1) Remove plate (1).
2) Remove bolt (2) and then remove alternator (3).



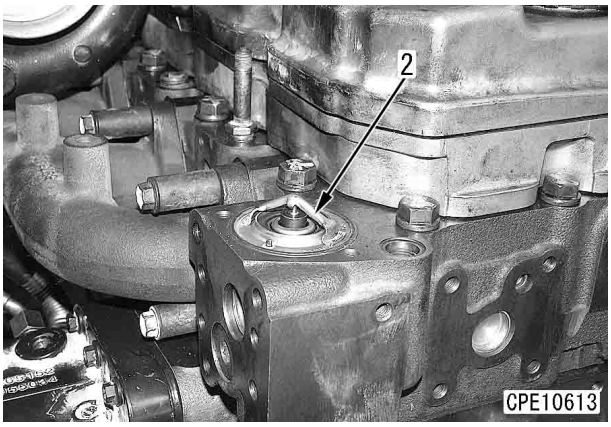
3) Remove alternator bracket (4) and front side sling bracket (5).



10. Thermostat

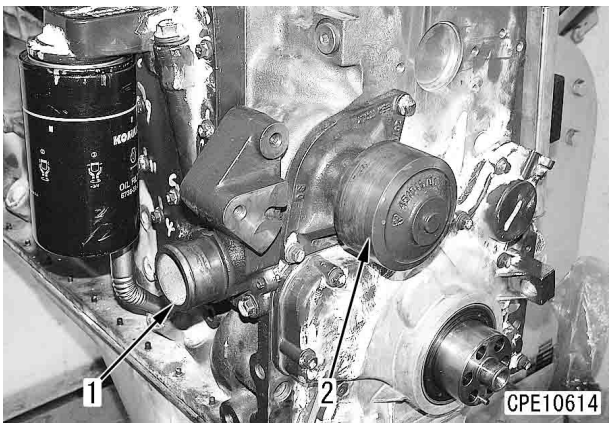
1) Remove coolant outlet connector (1).

2) Remove thermostat (2).



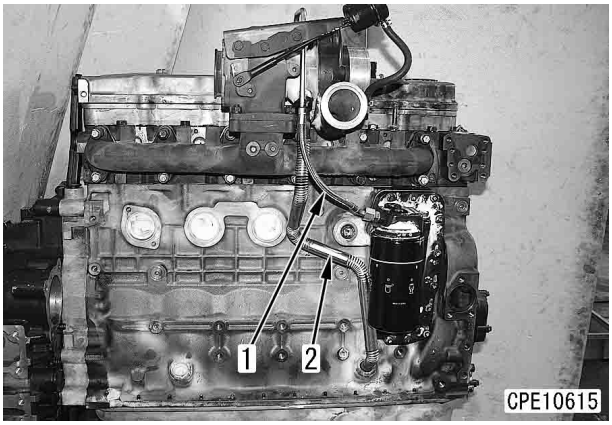
11. Water pump

1) Remove coolant inlet connector (1).
2) Remove water pump (2).

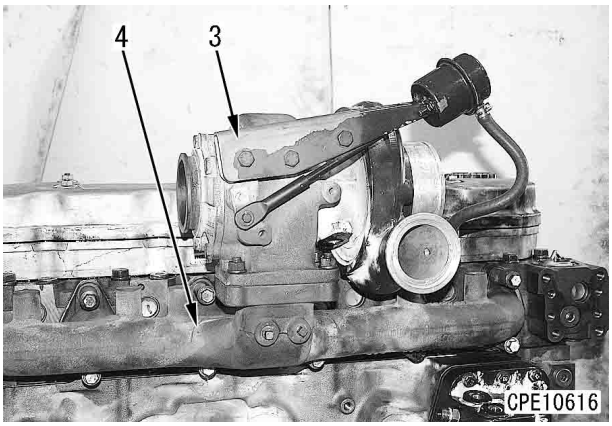


12. Turbocharger and exhaust manifold

1) Remove turbocharger lubrication inlet tube (1) and drain tube (2).

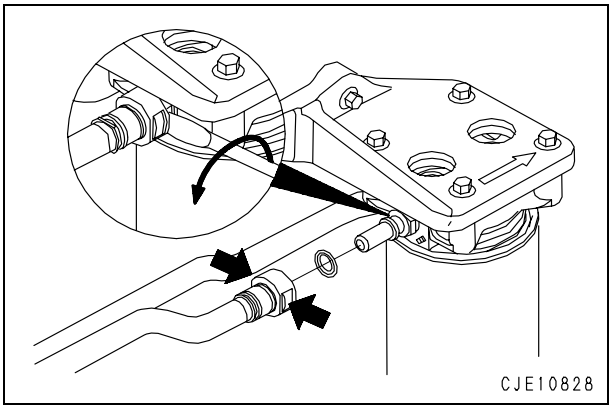


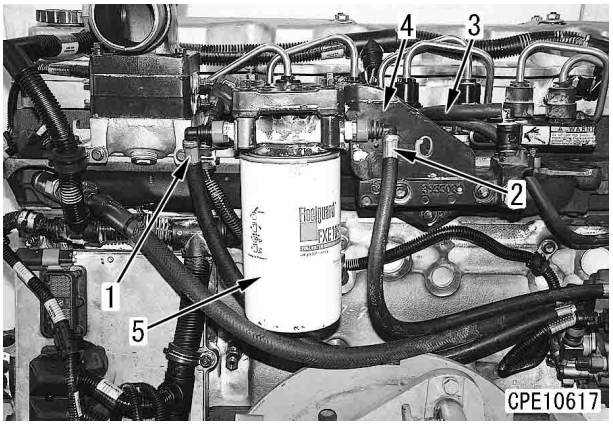
2) Remove turbocharger assembly (3) and exhaust manifold (4).



13. Fuel filter assembly

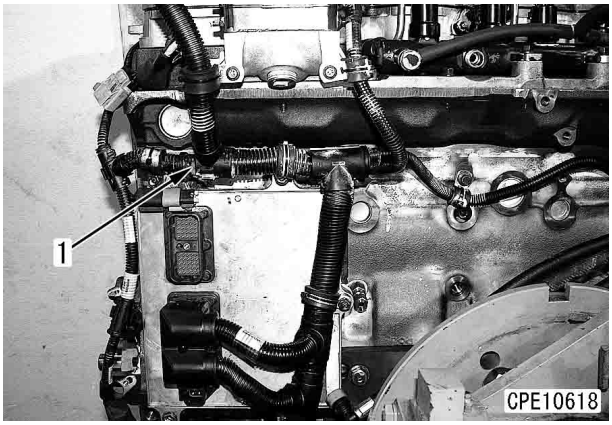
1) Disconnect hoses (1) and (2) from the filter.
2) Disconnect drain hose (3) from filter bracket (4).
3) Remove fuel filter assembly (5).
   - Removal of hose
     Remove the hose by prying the hose end face with screwdriver while pushing the mouthpiece (on both sides) of the hose termination.





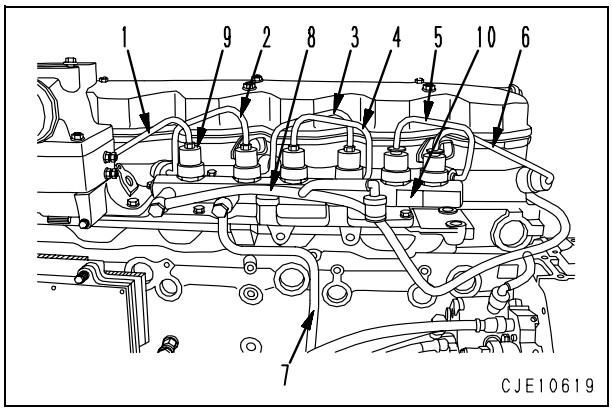
14. Harness assembly

Disconnect the harness from the sensor and controller, and then remove the clamp to remove harness assembly (1).



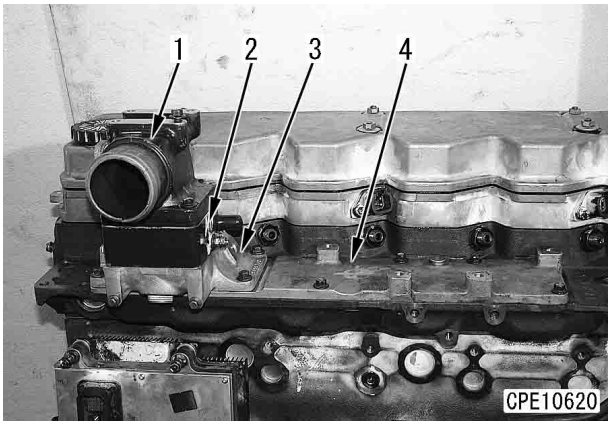
15. Fuel piping and common rail

1) Remove boots (9) of high-pressure pipings (1) to (7) and then remove the high-pressure pipings.
2) Remove fuel drain hose (8).
3) Remove common rail (10).
   - Removal of hose
     Remove the hose by prying the hose end face with screwdriver while pushing the mouthpiece (on both sides) of the hose termination.



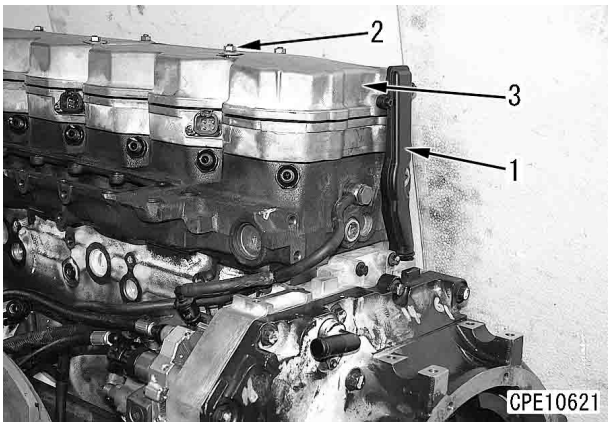
16. Air intake manifold cover

1) Remove tube (1) and electrical intake air heater (2).
2) Remove manifold covers (3) and (4).



17. Head cover housing

1) Remove duct (1).
2) Remove nut (2) and then remove head cover (3).

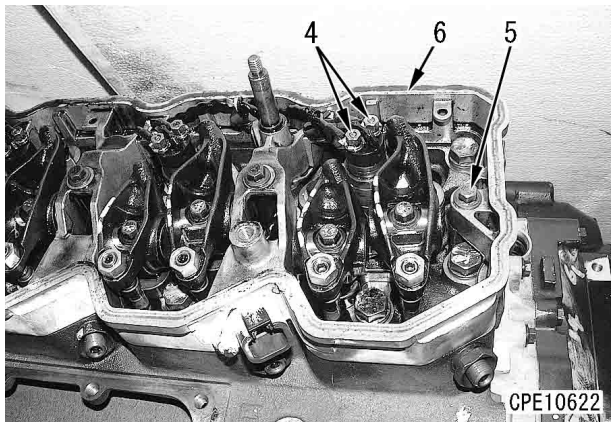


3) Remove nuts (4) of the harness from injector.

★ Harness installation position

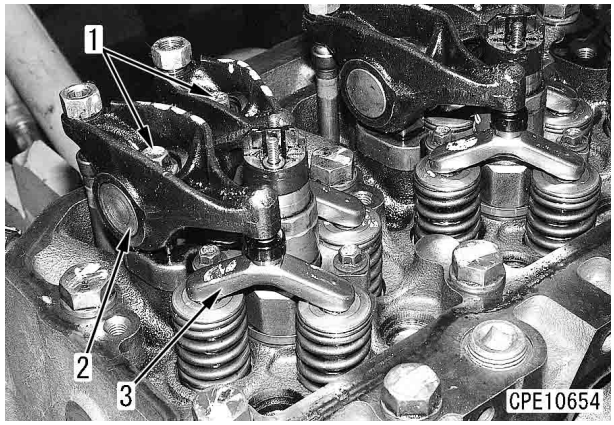
Color of cable	Cylinder No.
White	1, 3, 5
Black	2, 4, 6

4) Remove 7 bolts (5) and then remove rocker housing assembly (6).

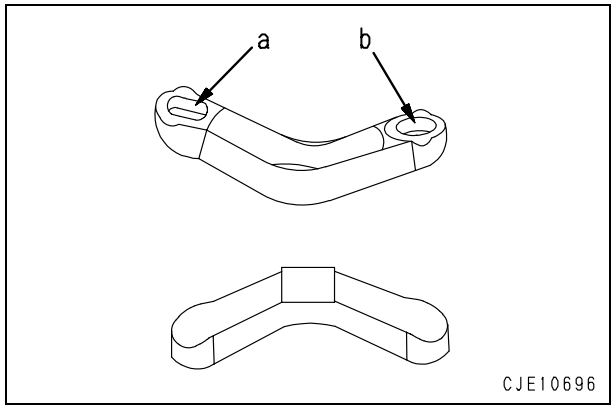


18. Rocker arm assembly and crosshead

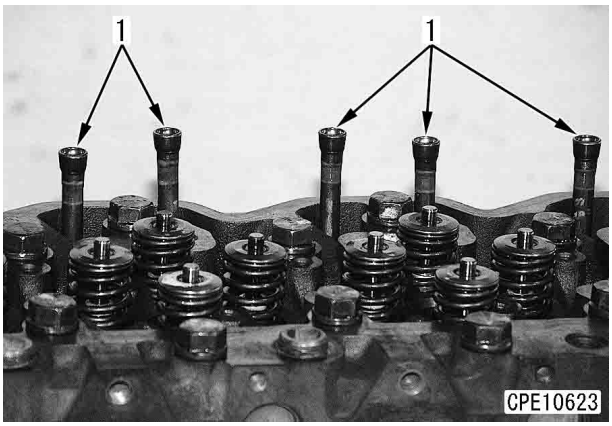
1) Remove 12 bolts (1) and then remove rocker arm assembly (2).
   ★ Loosen locknut to loosen adjustment screw by 2 – 3 turns so that excessive force may not be applied to push rod when the rocker arm is installed.
2) Remove crosshead (3).



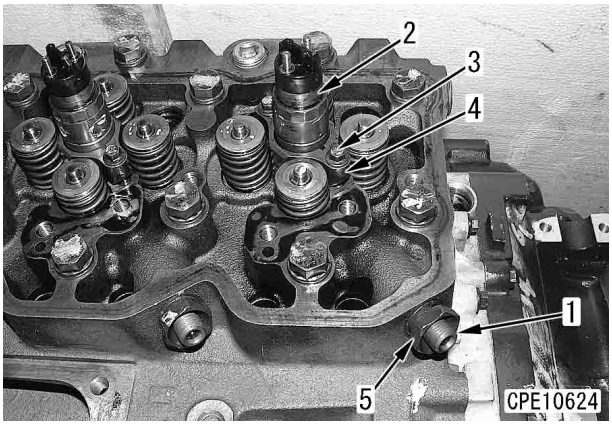
★ Record the installation position and shape of holes in parts (a) and (b). (So that it may be installed in the same direction in re-assembly.)



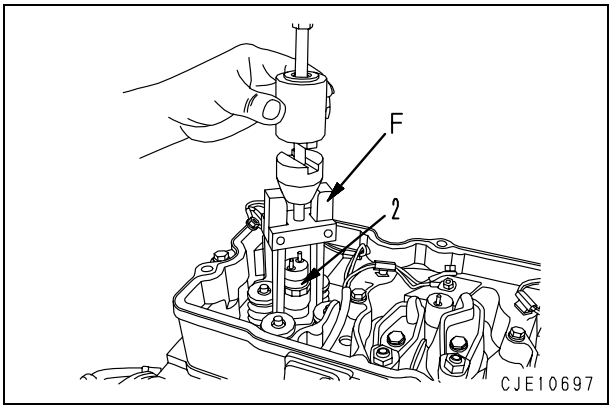
19. Push rod
Remove 12 push rods (1).



20. Fuel injector assembly
1) Remove retainer (5) and then remove 6 inlet connectors (1).
   - Use tool **M** when removing the inlet connector.
2) Remove mounting bolts (3) of injector (2) and then remove holder (4).
3) Remove injector (2).

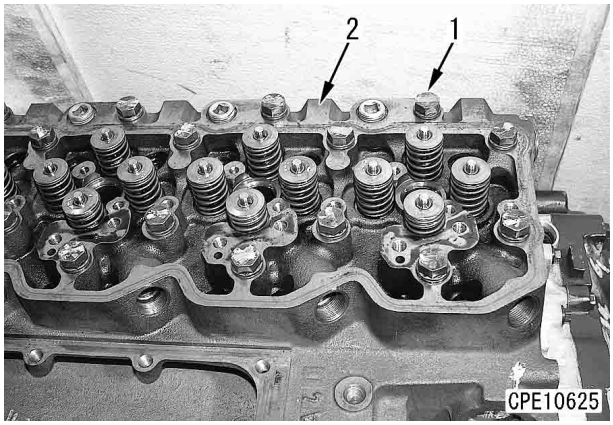


★ Lift and remove the injector using tool **F**. (Don't try to pry the top of the injector.)

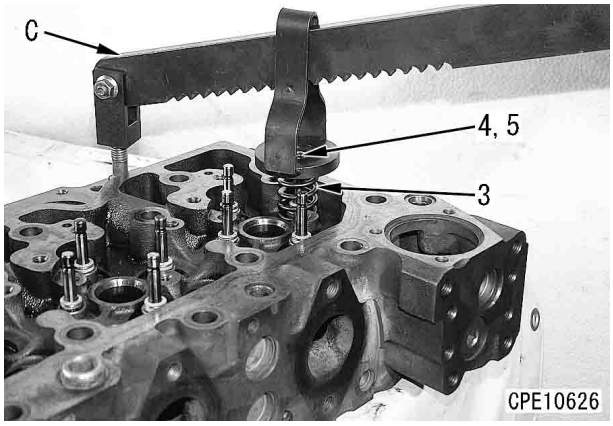


21. Cylinder head assembly
Remove 26 bolts (1) and cylinder head assembly (2).

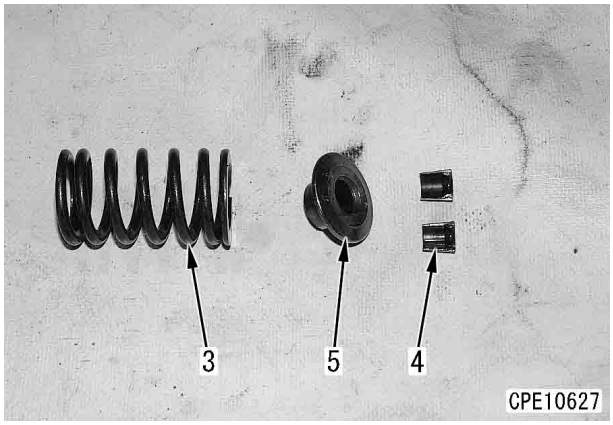
Cylinder head assembly: **70 kg**



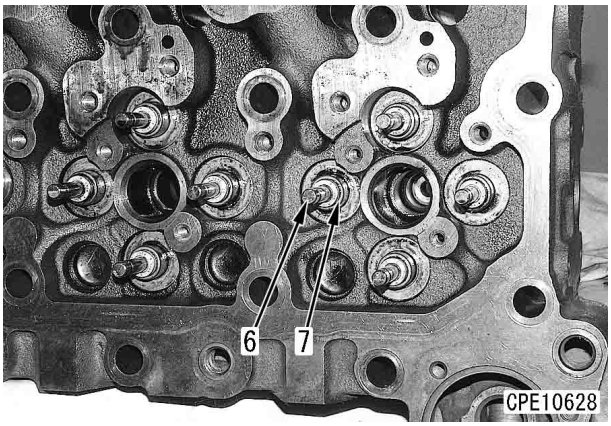
- Disassemble the cylinder head assembly according to the following procedure.
  1) Using tool **C**, compress valve spring (3) to remove valve cotter (4).



  2) Remove upper seat (5) and valve spring (3).

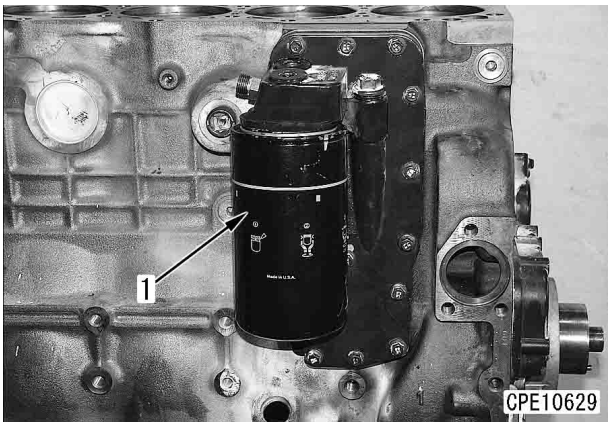


3) Raise the cylinder head assembly, pull and remove valve (6) and then remove valve seal (7).
★ Apply a marking to valve (6) to make it identifiable to which cylinder head it should be combined.

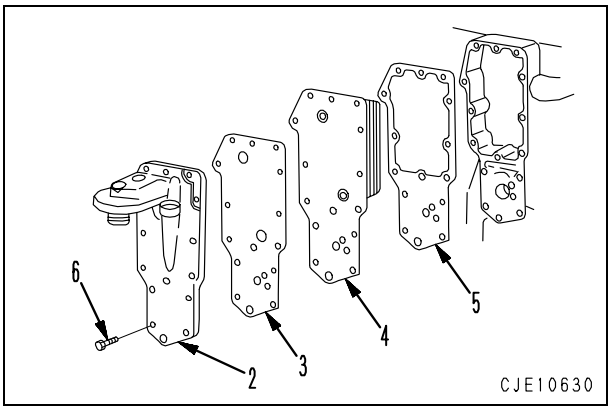


22. Oil cooler

1) Remove oil filter (1).

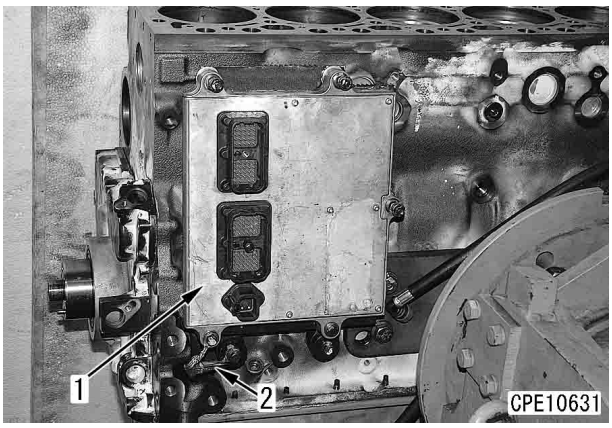


2) Remove cover (2) and cover gasket (3).
3) Remove oil cooler (4) and cooler gasket (5).

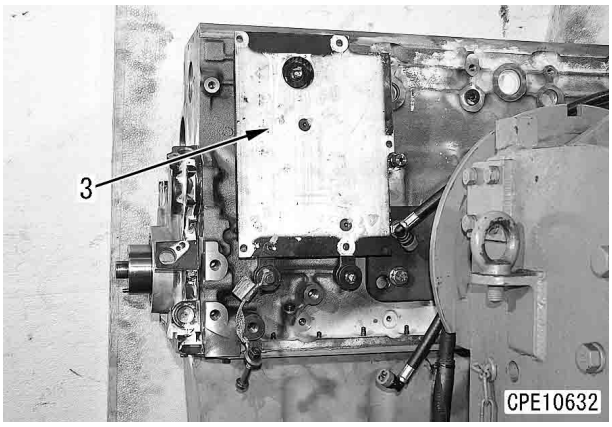


23. Engine controller

1) Remove engine controller (1).
2) Disconnect ground cable (2).

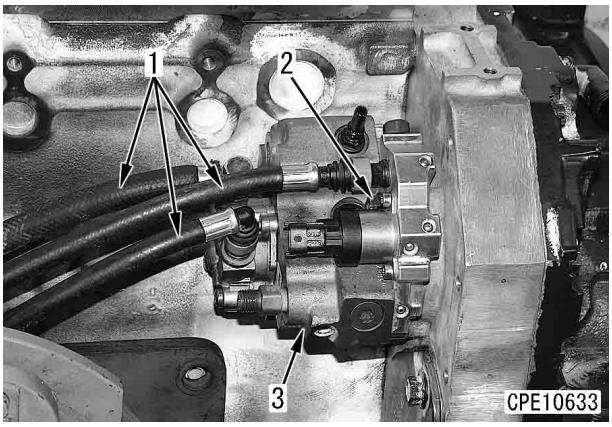


3) Remove cooling plate (3).



24. Fuel supply pump

1) Remove hose (1).
2) Remove 3 nuts (2) and then remove supply pump (3).

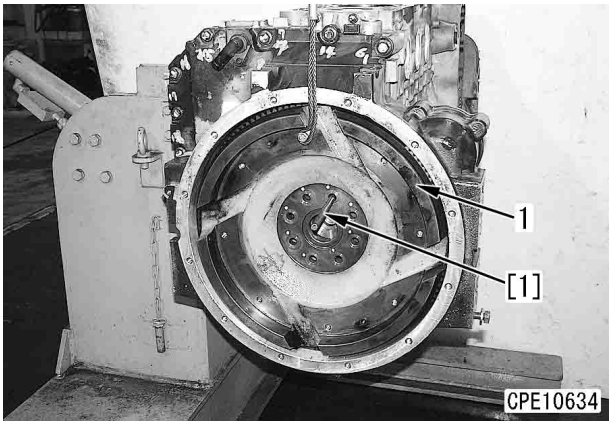


25. Flywheel

Install guide bolt [1] and then remove flywheel (1).

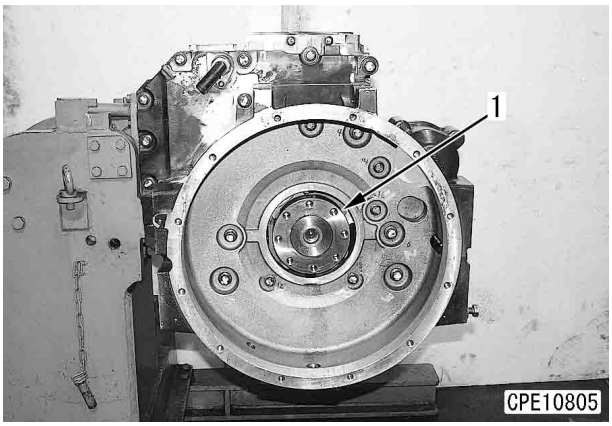
Flywheel: **35 kg**

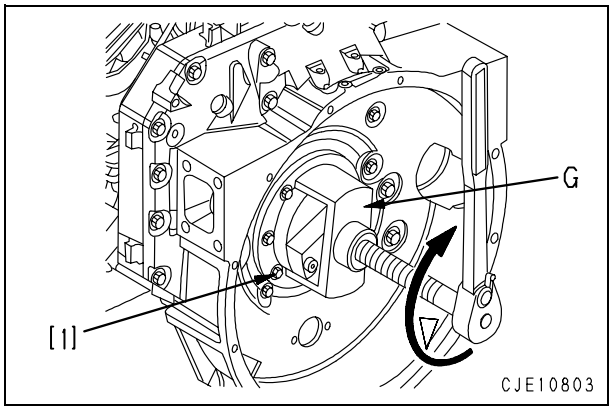
⚠ Take care in this operation so that your fingers may not be damaged by the flywheel that can fall off from the shallow spigot joint.



26. Rear oil seal

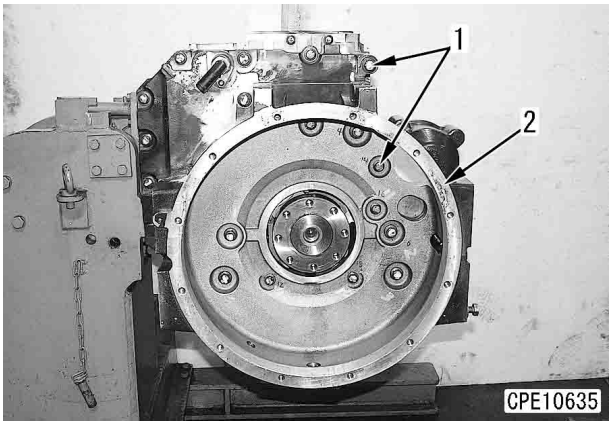
1) Install tool **G** to the crankshaft.
2) Screw in tapping screw [1] of tool **G** to the seal carrier section of rear oil seal (1).
3) Turn the handle counterclockwise to remove rear oil seal (1).





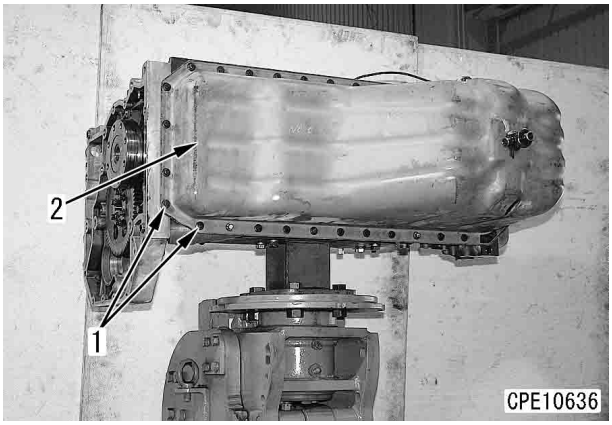
27. Flywheel housing

Remove 20 bolts (1) and then remove flywheel housing (2).

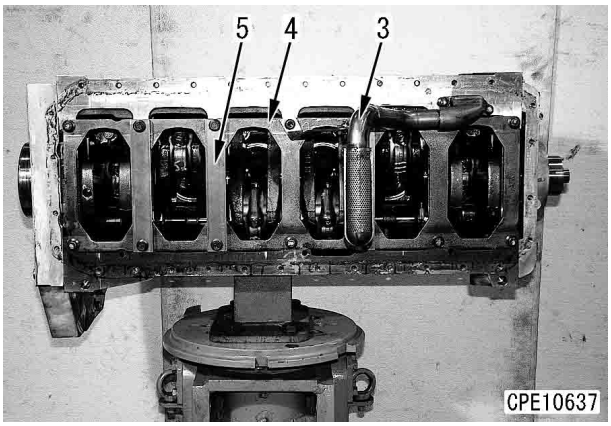


28. Oil pan

1) Remove 32 bolts (1) and then remove oil pan (2).

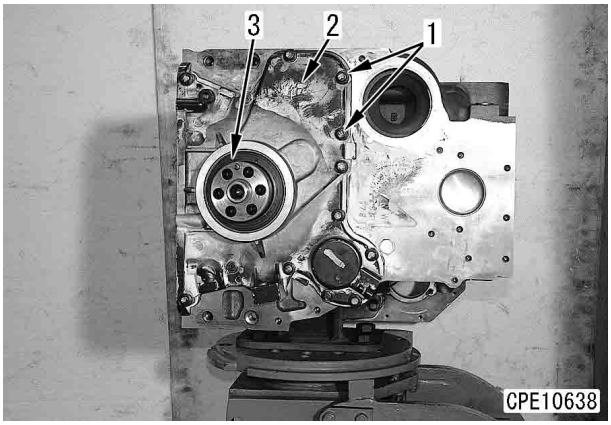


2) Remove suction tube (3).
3) Remove plate (4) and 3 plates (5).



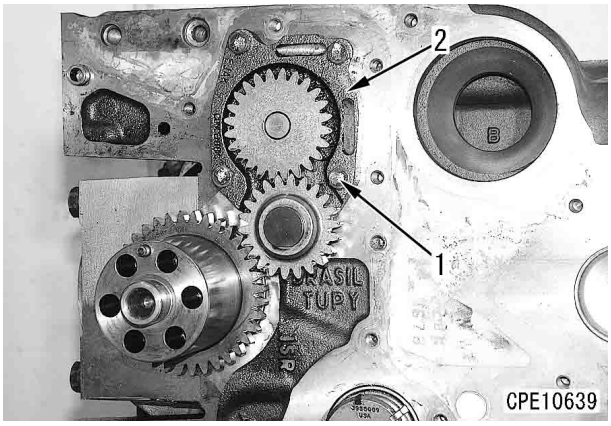
29. Front cover

1) Remove 13 bolts (1) and then remove front cover (2).
2) Remove front seal (3).



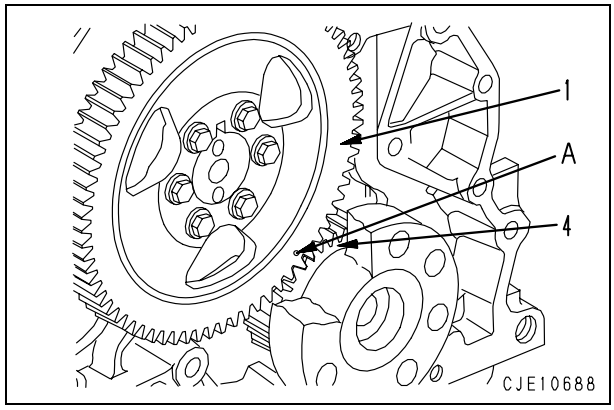
30. Oil pump assembly

Remove 4 bolts (1) and then remove oil pump assembly (2).

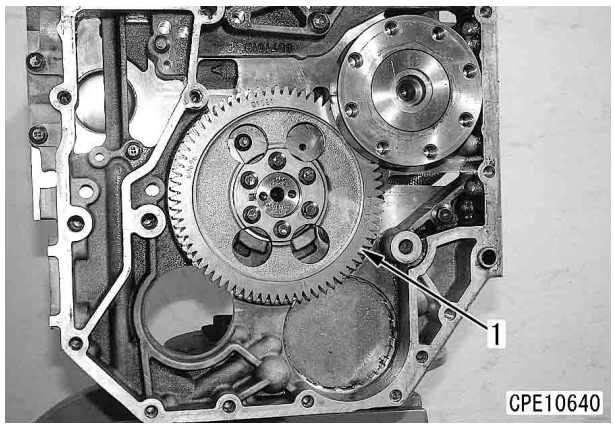


31. Camshaft

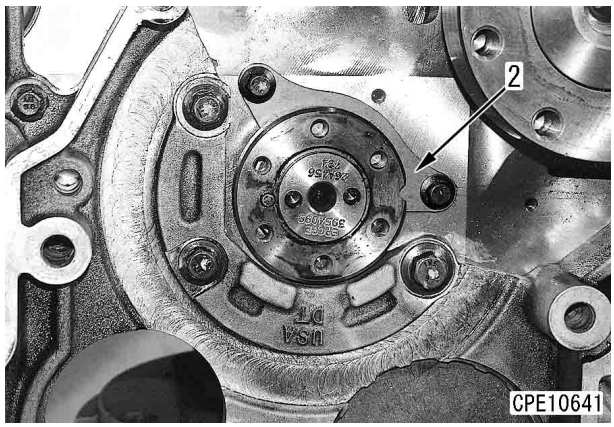
★ Rotate the crankshaft to align punch mark **A** on camshaft gear (1) to the chamfered tooth of crankshaft gear (4) (preventing of interference with the connecting rod when removing the camshaft).



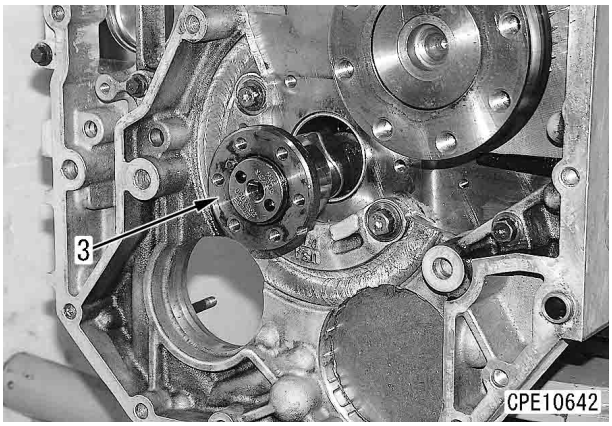
1) Remove gear (1).



2) Remove thrust plate (2).

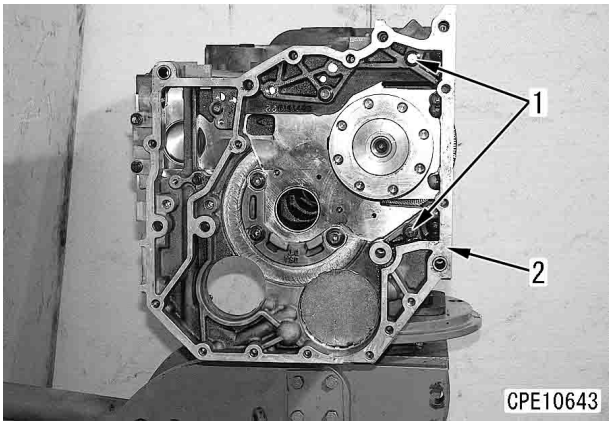


3) Rotate camshaft (3) when removing the plate.



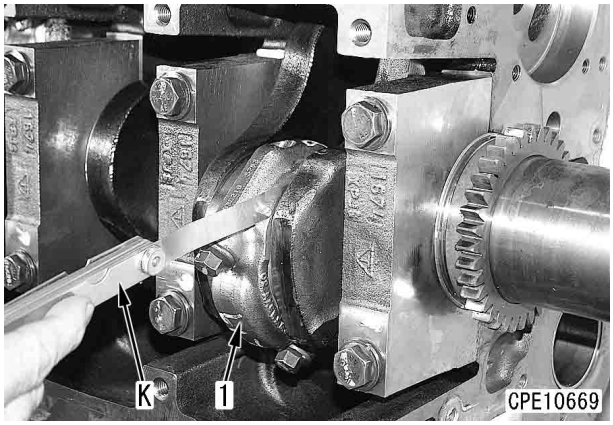
32. Gear housing

Remove 10 bolts (1) and then remove gear housing (2).

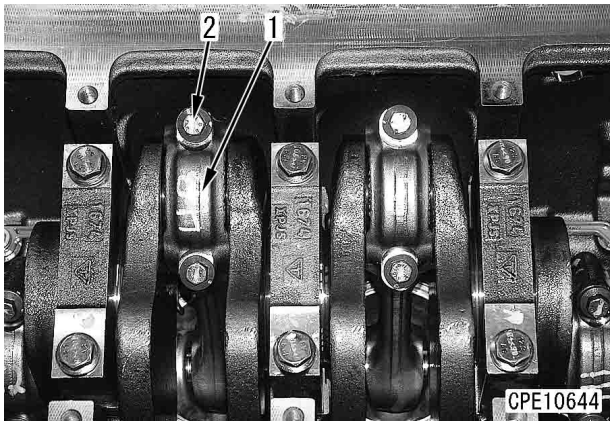


33. Piston and connecting rod assembly

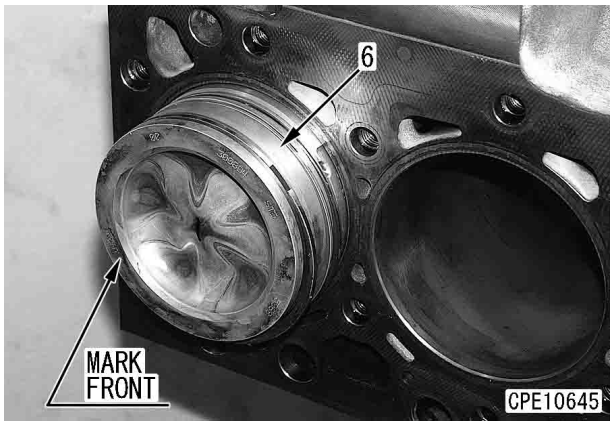
1) Rotate the crankshaft and bring the piston to be removed to the bottom dead center.
2) Scrub the carbon off the top wall of the cylinder liner with fine sandpaper.
3) Apply a marking to each of connecting rod cap (1) assembly position on respective cylinders.
   ★ Before removing the piston and connecting rod assembly, measure the side clearance between the connecting rod and crankshaft using tool **K**.



4) Remove bolt (2).
5) Hit connecting rod cap (1) with a plastic hammer lightly, and remove connecting rod cap (1) and connecting rod bearing as a unit.

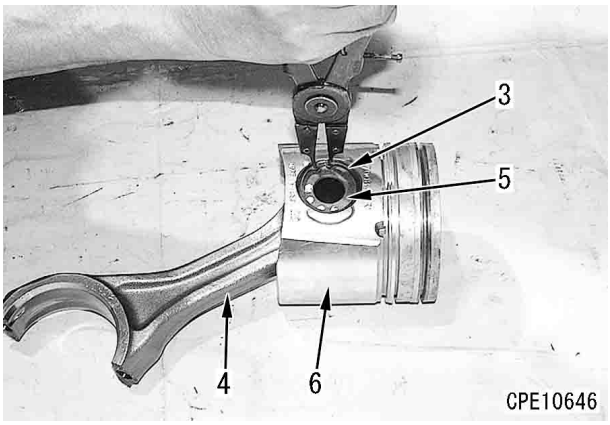


6) Push in and remove the piston and connecting rod assembly with a wood bar, etc. from the oil pan side, while supporting piston (6) on the cylinder head side.
   ★ Take care in the removal so that inner surface of the cylinder liner may not be damaged by corners of the connecting rod.
   ★ Make a cylinder No. mark on the piston.

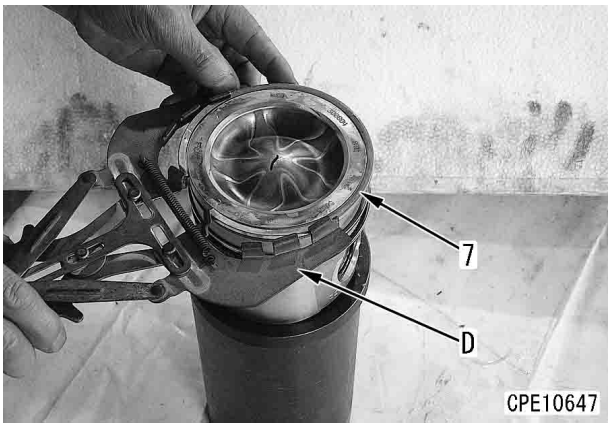


7) Remove the other piston and connecting rod assemblies according to the above procedure.
   ★ Take care not to damage the sliding portions of the pistons and bearings.

- Disassemble the piston and connecting rod assembly according to the following procedure.
  1] Remove snap ring (3).
  2] Separate piston (6) from connecting rod by pulling out piston pin (5) while supporting connecting rod (4) with hand.
  3] Remove the snap ring on the opposite side.

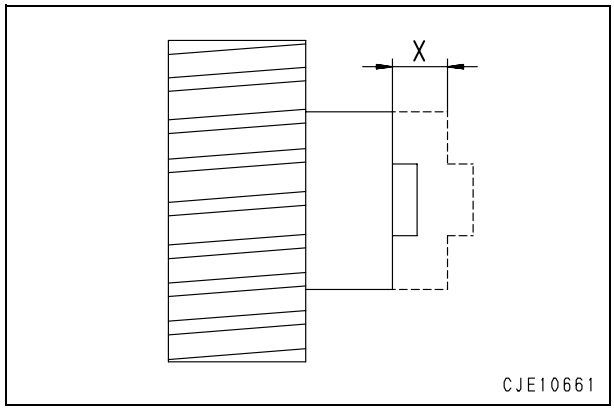


  4] Using tool **D**, remove piston ring (7).
     ★ Keep the piston, connecting rod, connecting rod bearing, piston rings, and piston pin by cylinder.

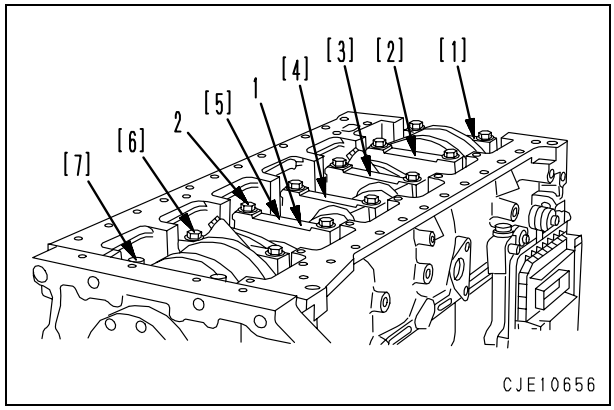


34. Crankshaft

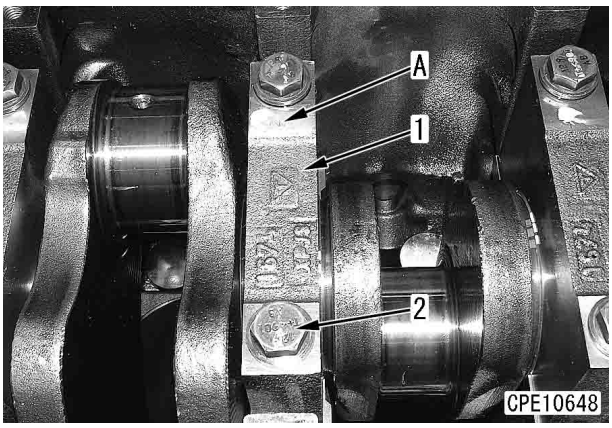
★ Before removing the crankshaft, measure its end play **X** with dial gauge.



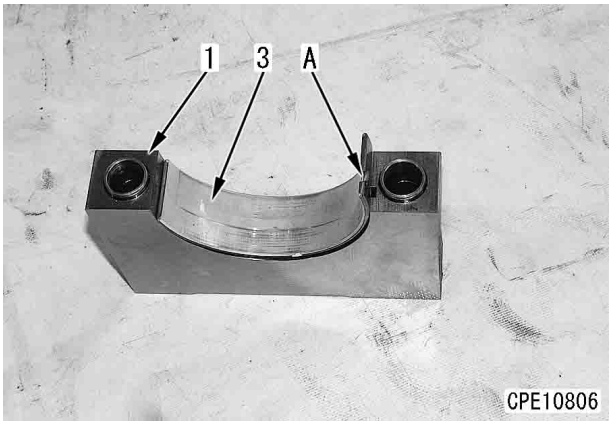
1) Remove mounting bolts (2) of main cap (1).
2) Insert the bolts in the bolt holes of main cap (1) and remove the cap, while shaking it.
   ★ Installation position numbers are marked on the main cap. No. 1 starts from the front side of the cylinder block.



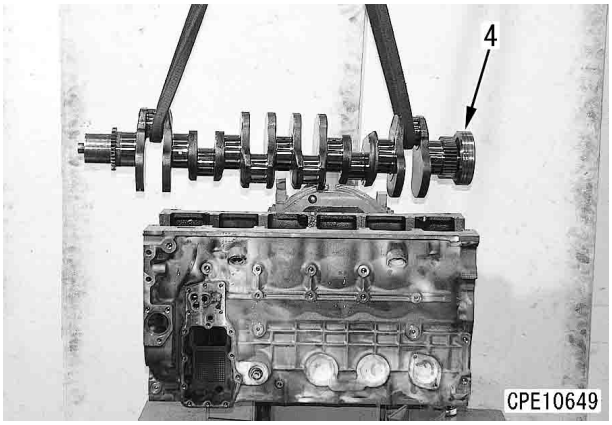
★ Make sure that markings (numbers are marked in section **A**) on the main cap are faced to the oil cooler.



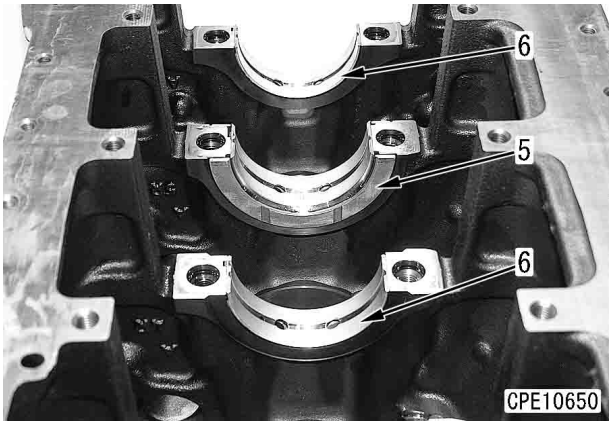
3) Remove bottom lower bearing (3) from main cap (1).
★ Mark the installation position number on claw (A) of the removed bearing. (Don't apply the number on the sliding surface of the bearing.)



4) Lift off crankshaft (4) using the nylon sling.
★ When lifting and removing the crankshaft, take care not to damage it by bumping against the cylinder block.
★ Keep the crankshaft in a safe place so that its sliding portions will not be damaged.
Crankshaft: **60 kg**

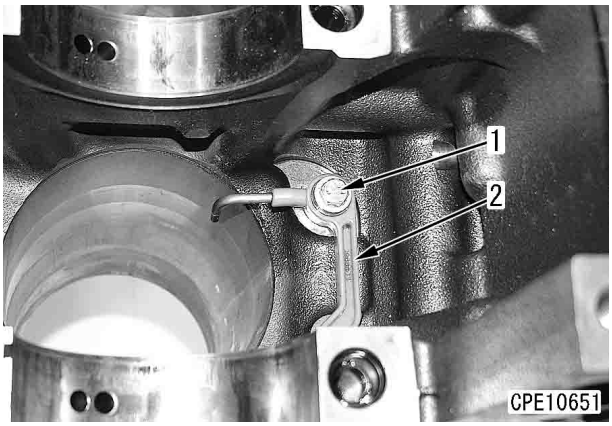


5) Remove upper thrust bearing (5) and 6 upper bearings (6).



Mark the assembly position on the main cap, upper bearing and upper thrust bearing (in the claw of the bearing). It is advised to maintain the bearings of a main cap separately from that for another main cap. And take care not to damage these parts.

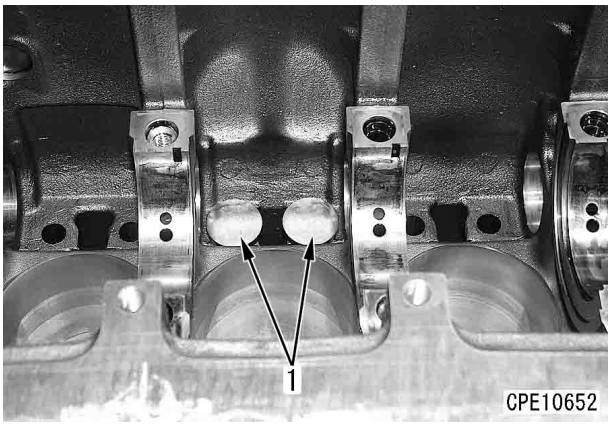
35. Piston cooling nozzle
Remove bolt (1) and then remove 6 piston cooling nozzles (2).



36. Tappet

Remove 12 tappets (1) and then mark the installation position on them.

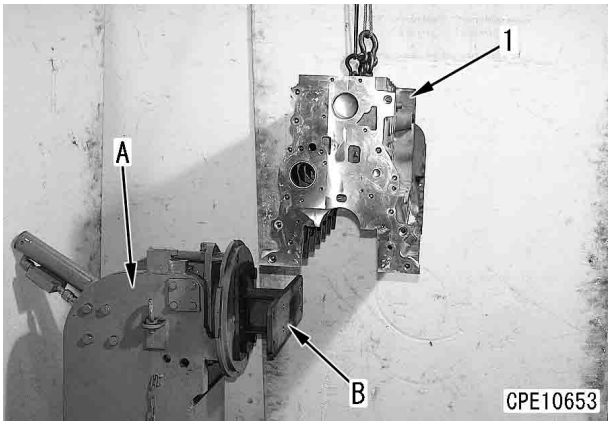
★ When reusing the camshafts and tappets, be sure to install them in the original position.



37. Cylinder block

Remove cylinder block (1) from repair stand **A**.

Cylinder block: **140 kg**



Engine

107E-1 Series

50 Disassembly and assembly

Disassembly and assembly, Part 2

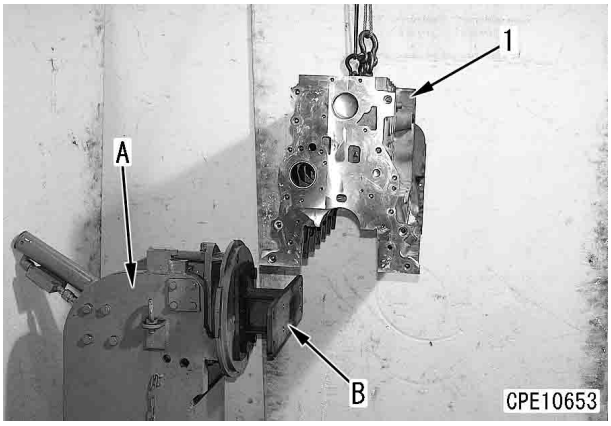
Disassembly and assembly, Part 2 2
General assembly of engine 2

Disassembly and assembly, Part 2

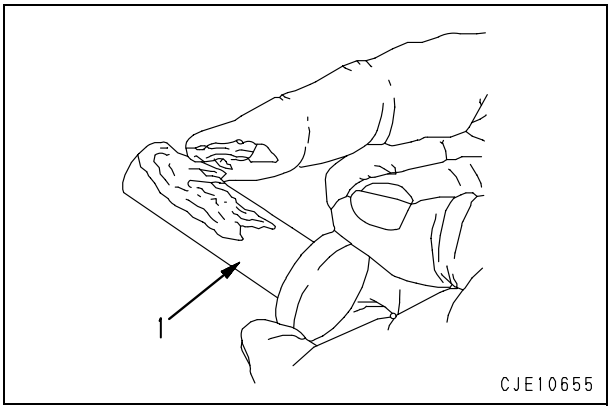
General assembly of engine

★ This document describes the general assembly procedure of SAA6D107E-1. The shapes, quantity, locations, etc. of the parts depend on each applicable machine. Take care.
★ When reusing a part, be sure to install it to its original position.

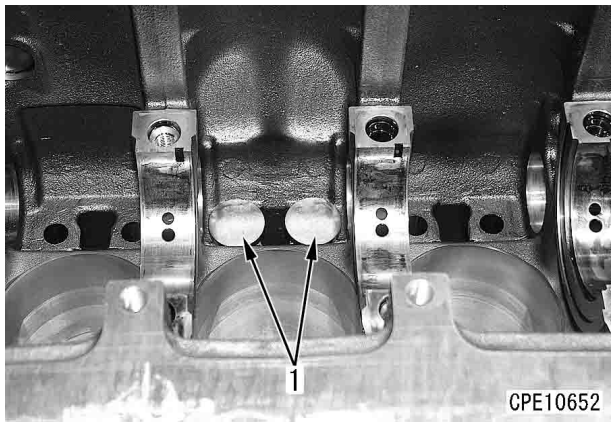
1. Preparation work
1) Install tool **B**.
2) Install cylinder block (1) to repair stand **A**.
Mounting bolt:
77 ± 12 Nm {7.9 ± 1.2 kgm}
Cylinder block: **140 kg**



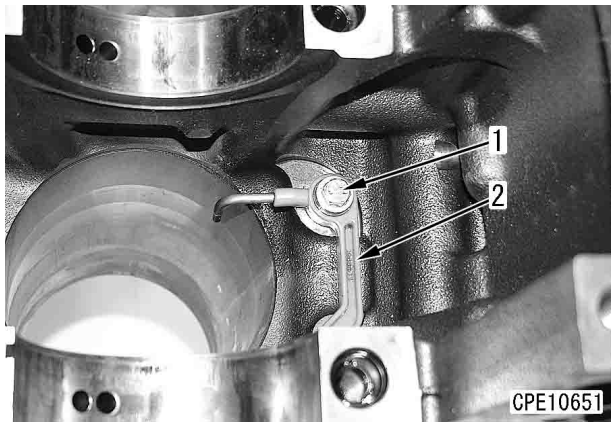
2. Tappet
1) Apply engine oil (EO15W-40) to tappets (1).



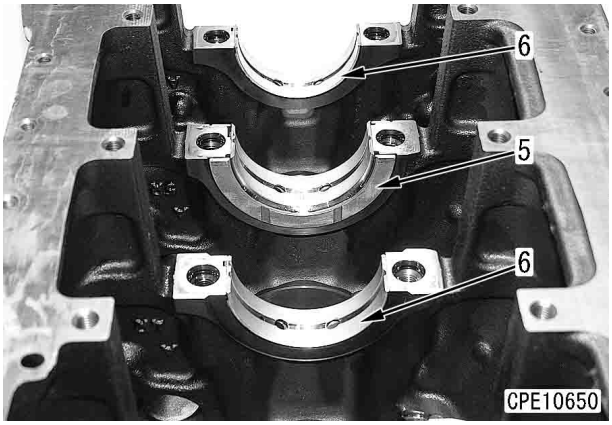
2) Install 12 tappets (1).
★ When reusing the camshafts and tappets, be sure to install them in the original position.
★ Don't invert the cylinder block by 90° or more. Otherwise, the tappet may fall off from the mounting hole during assembly.



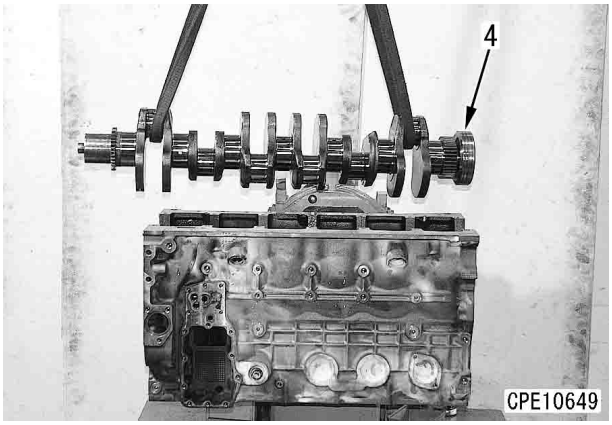
3. Piston cooling nozzle
Set 6 piston cooling nozzles (2) and then tighten them using mounting bolt (1).
Mounting bolt:
15 ± 2 Nm {1.5 ± 0.2 kgm}



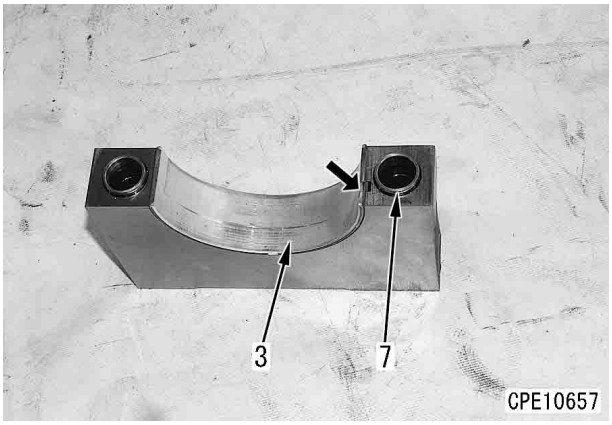
4. Crankshaft
1) Install 6 upper bearings (6) and an upper thrust bearing (5).
★ Set the projection of each bearing to the notch of the cylinder block.
★ When installing the bearings, check that their back sides are free from foreign matter.
★ Apply engine oil (EO15W-40) to inner surface of metal. Don't apply the oil to the backside.
★ When reusing the bearings, check the marks made on them when removed and install them to their original positions.



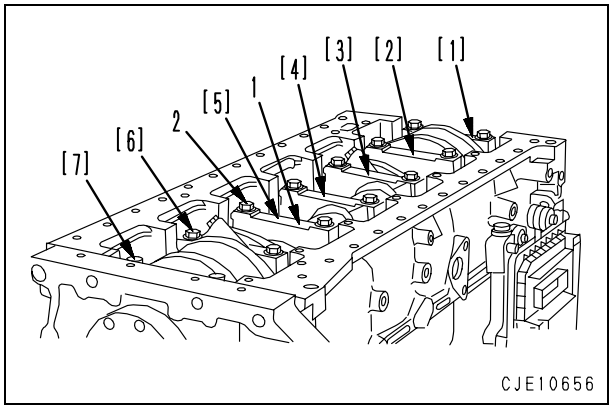
2) Using a nylon sling, lift up and install crankshaft (4).
   ★ When installing, take care so that the upper thrust bearing may not be damaged.
   Crankshaft: **60 kg**



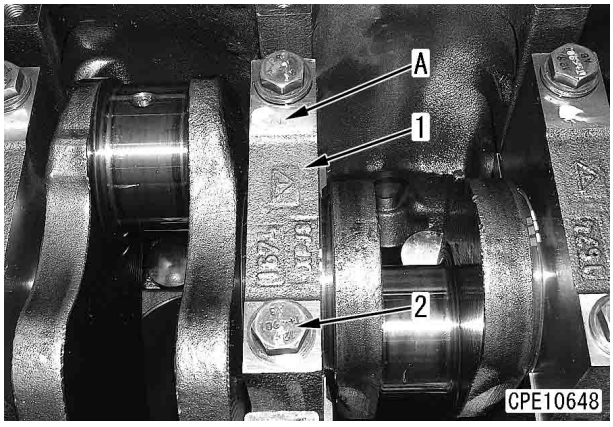
3) Fit lower bearing (3) to the main cap, matching the projection of the former to the notch of the latter.
   ★ Make sure ring dowel (7) is installed on the main cap.



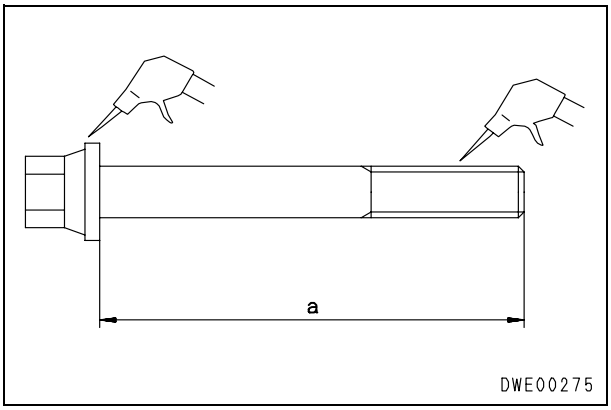
4) Apply engine oil (EO15W-40) to the journal surfaces of the crankshaft.
5) Before installation, check the No. stamped on main cap (1).
   - Installation shall start from the front side of the cylinder block in a numerical order.

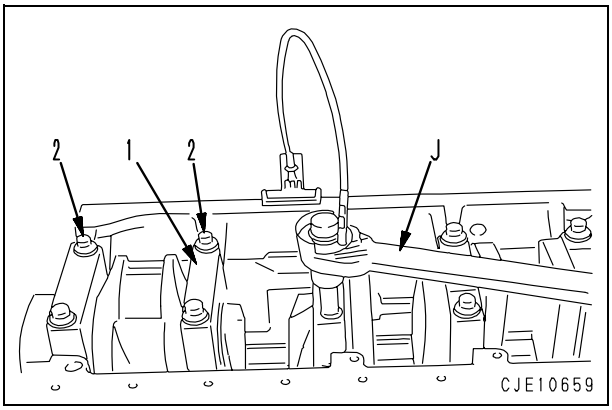


   - In the installation, make sure the stamped No. (section **A**) is faced to the oil cooler.
   - Push in each main cap by hitting it lightly with a plastic hammer or rubber hammer.

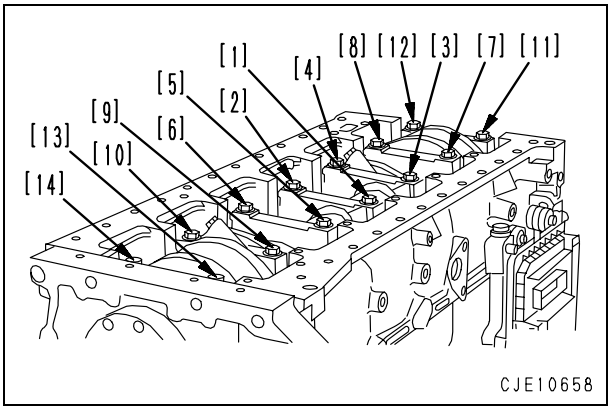


6) Before tightening the main cap mounting bolt (2), check the following.
   1] Measure stem length (a) of every bolt to make sure that they are within the using limit.
      - Using limit length of bolt stem:
        Less than 120 mm
      ★ If a bolt is longer than the using limit, do not reuse it but replace it.





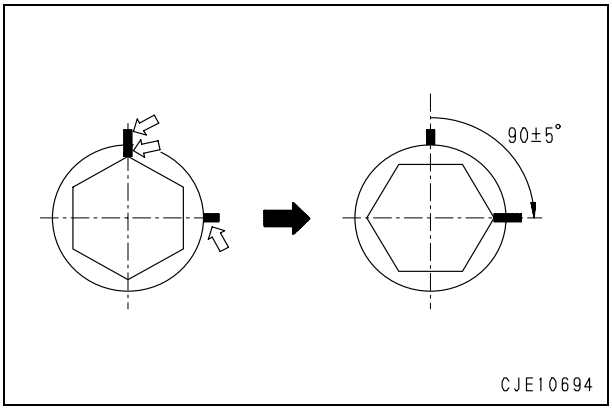
7) Tightening procedure of main cap mounting bolts (2).
   ★ Apply engine oil (EO15W-40) to the bolts.
   ★ Tightening shall be done according to the sequence [1] → [14] shown in the figure.



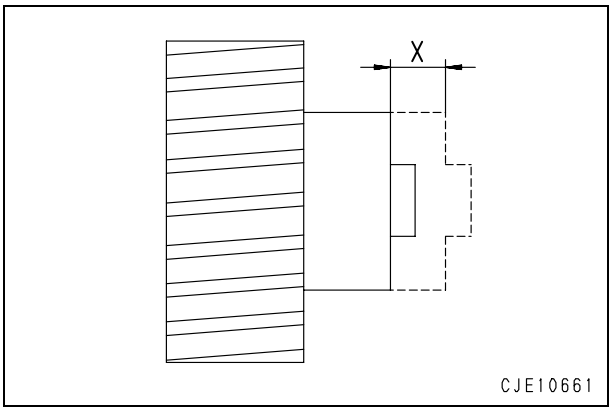
Main cap mounting bolt
When current bolts are used again:
1] 60 ± 5 Nm {6.1 ± 0.5 kgm}
2] 80 ± 5 Nm {8.2 ± 0.5 kgm}
3] 90° ± 5° (When angle tightening tool J is used)

When new bolts are used:
1] 120 ± 5 Nm {12.2 ± 0.5 kgm}
2] Loosen by 360°
3] 60 ± 5 Nm {6.1 ± 0.5 kgm}
4] 85 ± 5 Nm {8.7 ± 0.5 kgm}
5] 120° ± 5° (When angle tightening tool J is used)

- When tool **J** is not used:
  After applying a paint marking to the main cap and bolt, tighten the bolt by **90° ± 5° or 120° ± 5°**.

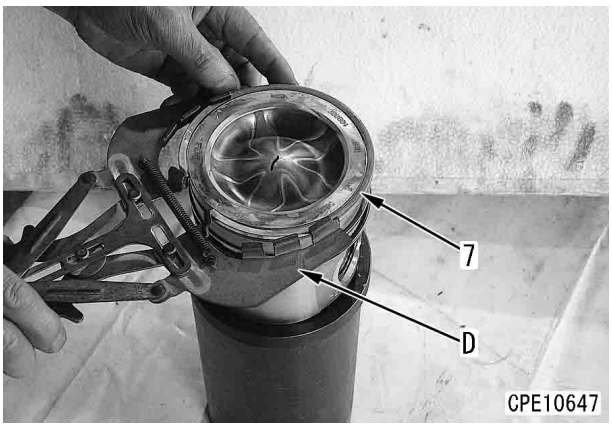


8) After tightening the bolts, make sure that the crankshaft turns smoothly.
9) Measure the end play of the crankshaft with a dial gauge.
   ★ End play **X**: **0.24 ± 0.175 mm**

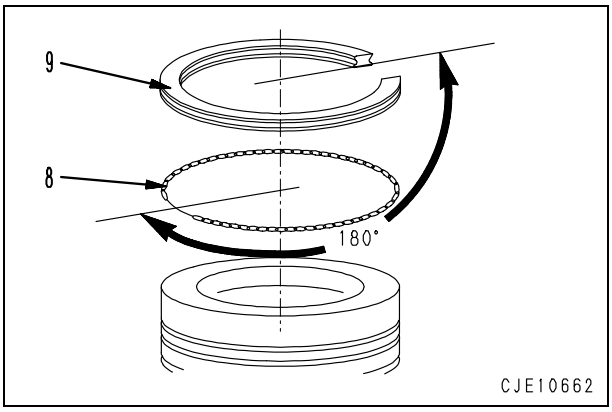


5. Piston and connecting rod assembly

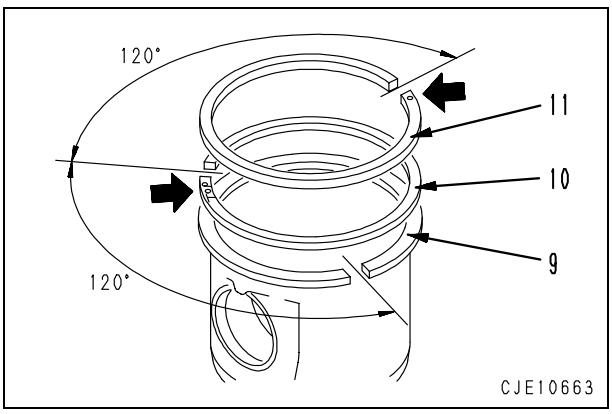
1) Using tool **D**, fit piston rings (7) to the piston.



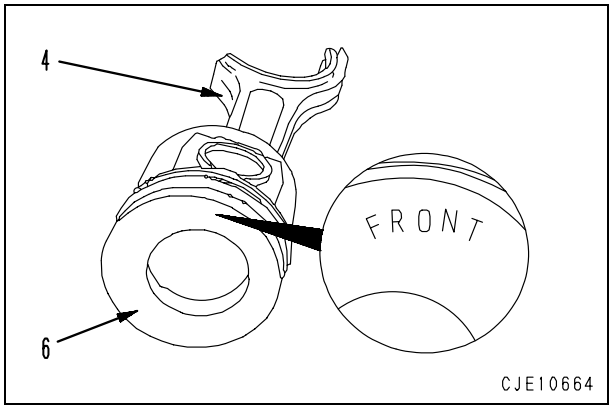
2) Installation of oil ring
Set the oil ring so that its abutment joint comes in a position at 180° to the end of expander (8).



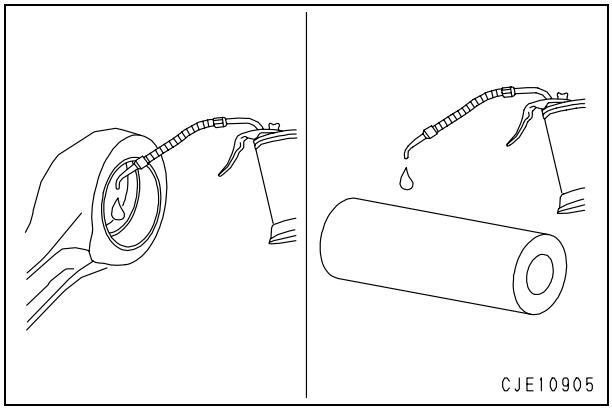
3) Fit second ring (10) with the stamp of "TOP" near the abutment joint up.
4) Fit top ring (11) with the "dot" mark near the abutment joint up.
5) When fitting each ring, set the abutment joint off the thrust, anti-thrust, and piston axis directions.



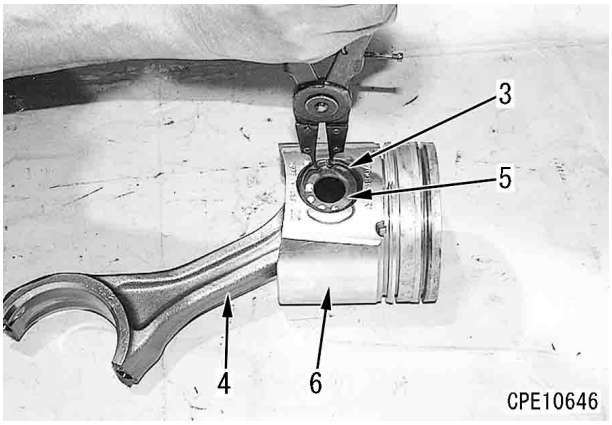
6) Apply engine oil (EO15W-40) to piston ring and piston skirt.
7) Set the piston and connecting rod.
   - When "FRONT" mark of piston (6) head is set at the top, connecting rod (4) must be facing to the direction shown in the figure.



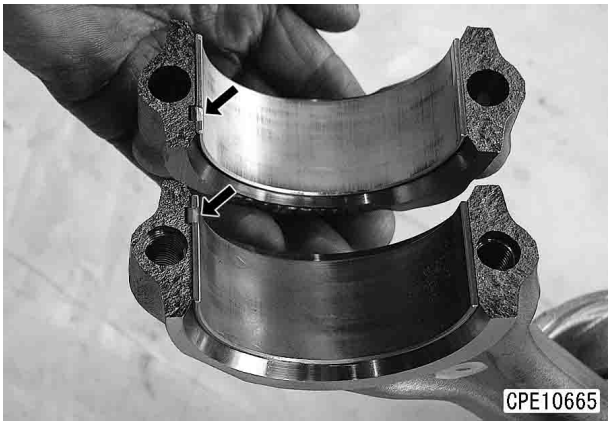
1] Apply engine oil (EO15W-40) to the piston pin hole and piston pin.



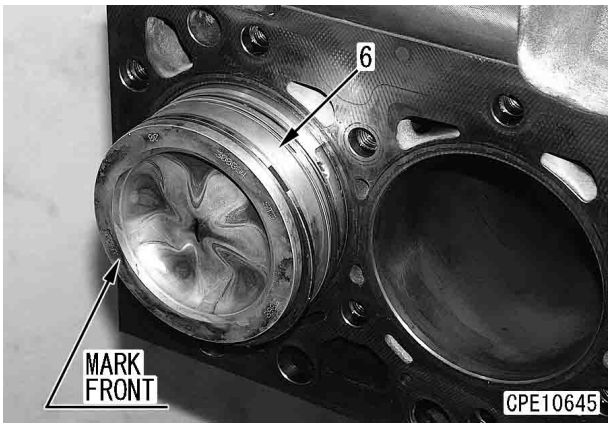
2] Insert piston pin (5) and then assemble piston (6) to connecting rod (4).
3] Install snap ring (3) to both sides.



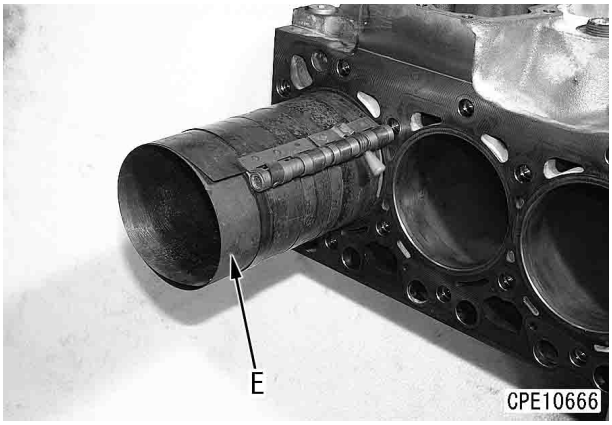
8) Install the bearing to the connecting rod and connecting rod cap.
   - Check that the back side of the bearing is free from foreign matter.
   - Fit the bearing to the connecting rod and cap, matching the projection of the former to the notch of the connecting rod and cap.



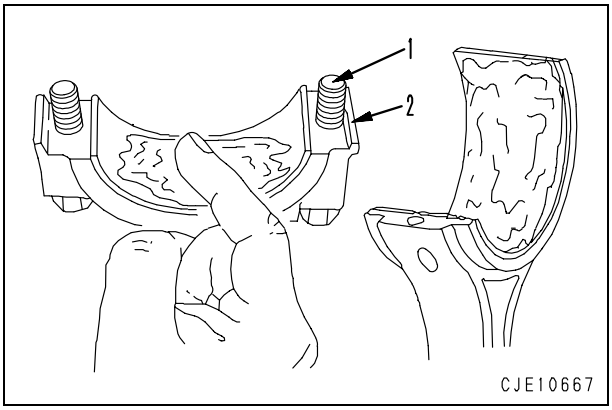
9) Set the crankshaft to the bottom dead center of the cylinder to install the piston and apply engine oil (EO15W-40) to the inside of the cylinder.
10) Insert the piston and connecting rod assembly setting “FRONT” mark of piston (6) to the front side of the cylinder.
    ★ Check the directions of the abutment joints of the piston rings again.
    ★ When inserting the connecting rod, take care not to damage the wall of the cylinder.
    ★ When installing the assembly, take care so that it may hit the cooling nozzle.



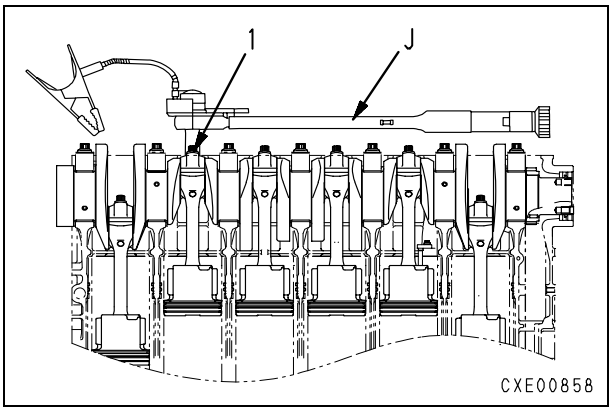
   - Using tool **E**, reduce the piston rings and push in the piston head with a wood bar, etc.



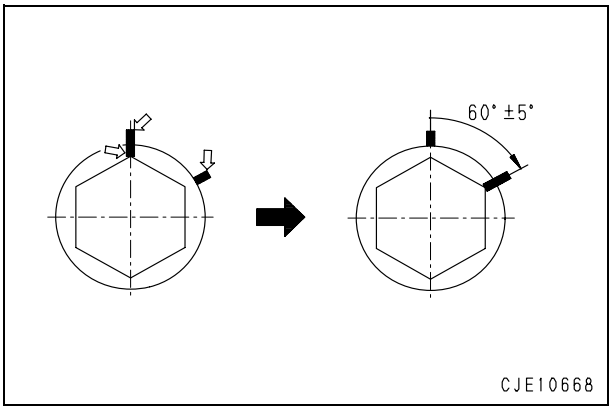
11) Install connecting rod cap (2) by tightening bolt (1) in the following procedure.
    ★ Apply engine oil (EO15W-40) to the threaded portions of the bolts and inside of the connecting rod bearings.



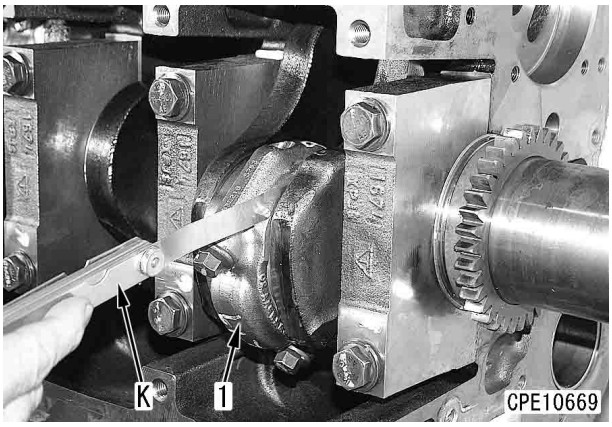
Mounting bolts (Tighten both bolts alternately):
1] 30 ± 3 Nm {3.06 ± 0.31 kgm}
2] 60 ± 5 Nm {6.12 ± 0.51 kgm}
3] 60° ± 5° (When angle tightening tool J is used)



- When not using tool **J**
  Make marks on the main cap and bolt with paint and then tighten the bolt by **60° ± 5°**.

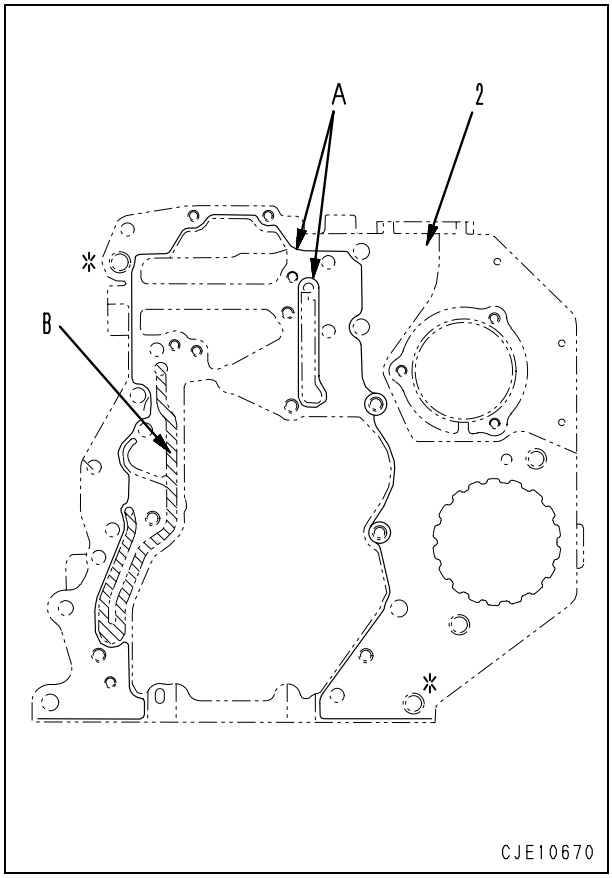


12) After installing the piston and connecting rod assembly, rotate the crankshaft to make sure that it turns smoothly without getting hung up.
13) Measure the side clearance between the connecting rod and crankshaft with tool **K**.
    ★ Side clearance: **0.10 – 0.33 mm**

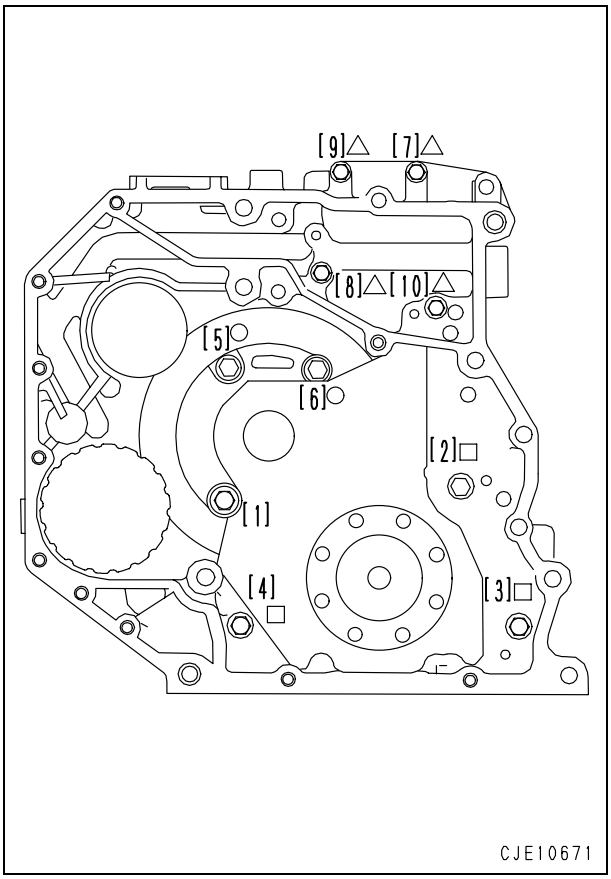


6. Gear housing

1) Apply gasket sealant to installation surface (A) of housing (2).
   Housing: **Gasket sealant (LG-7)**
   - Diameter of gasket sealant line shall be 1 – 2 mm.
   ★ Excessive gasket sealant can fill up blow-by oil groove (B). Use care to avoid excessive use of the sealant.
2) Make sure that each of the dowel ring of the gear housing is correctly inserted to the dowel hole at both the top and bottom (indicated with ∗ mark).
   ★ Otherwise, they can be damaged when bolt-tightened.
3) Press the gear housing against the cylinder block so that its entire mating surface with the cylinder block is closely contacted.
   ★ Don't apply torque to the mounting bolts until the entire mating surface is closely contacted.



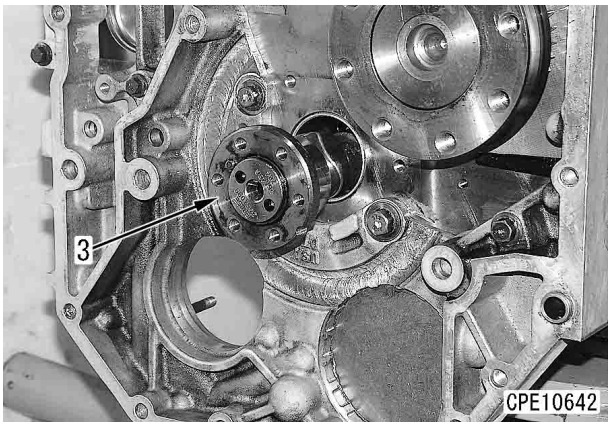
4) Tighten the mounting bolts.
   Bolts tightening procedure
   Tighten the bolts according to the numbers [1] → [10] shown in the figure.
   ★ 4 types of bolts are used. Make sure when tightening that specified torque is applied.
   Mounting bolt:
   1] M12:
   77 ± 7 Nm {7.9 ± 0.7 kgm}
   (No marking)
   2] M10:
   43 ± 5 Nm {4.4 ± 0.5 kgm}
   (○ mark ℓ = 30, □ mark ℓ = 50)
   3] M8:
   24 ± 4 Nm {2.5 ± 0.4 kgm}
   (△ mark)



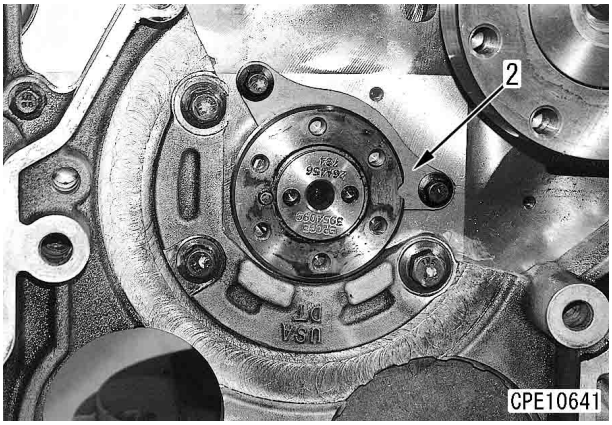
7. Camshaft

★ Rotate the crankshaft to set the No. 1 cylinder to the top dead center.

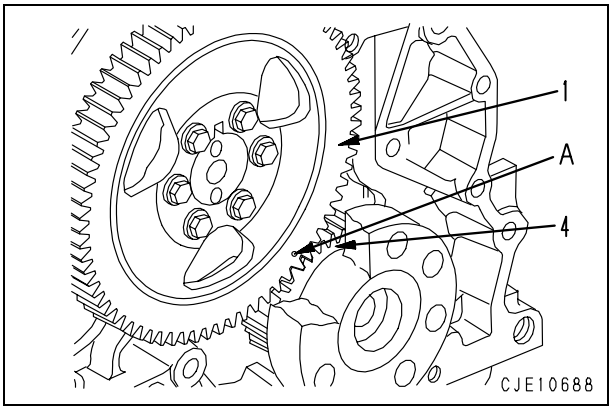
1) Apply engine oil (EO15W-40) to the camshaft bore, camshaft journal and lobed surface. Then lightly push in and rotate camshaft (3) to install.



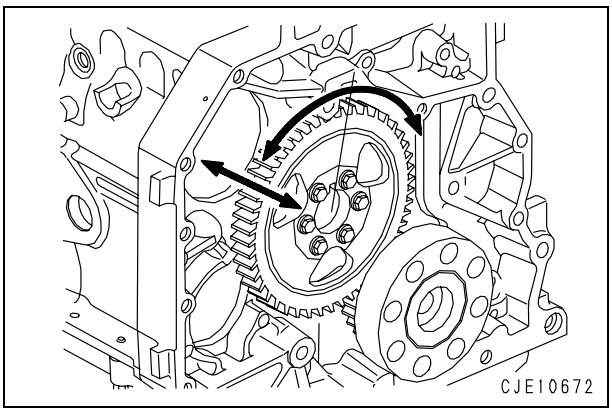
2) Install thrust plate (2).
Mounting bolt:
24 ± 4 Nm {2.4 ± 0.4 kgm}



3) When installing, fit in the thrust plate to timing mark **A** on camshaft gear (1) as well as to the chamfered tooth (a single location) of crankshaft gear (4).
Mounting bolt:
36 ± 4 Nm {3.7 ± 0.4 kgm}

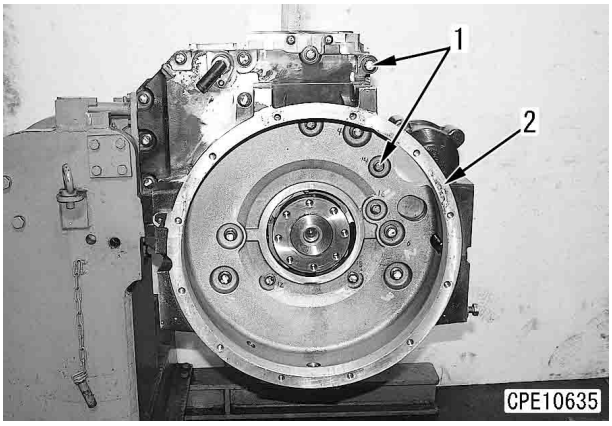


4) Measure the end play and backlash of the camshaft with a dial gauge.
★ End play: **0.10 – 0.36 mm**
● The end play is determined by thickness of the thrust plate and groove of the camshaft.
★ Backlash: **0.076 – 0.280 mm**

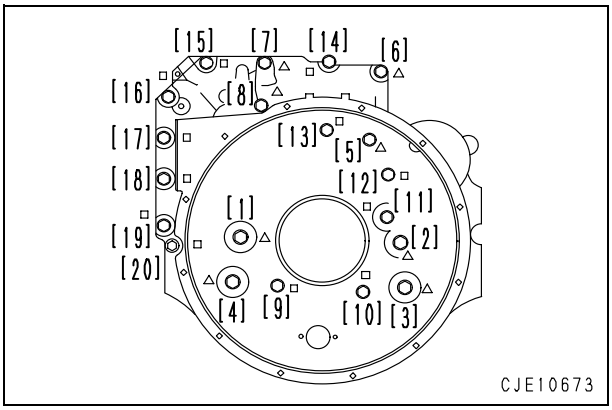


8. Flywheel housing

1) Install flywheel housing (2) using bolt (1).
   ★ Apply gasket sealant (LG-7) to the flanged housing installation surface as well as circumference of the mounting bolt hole. Diameter of gasket sealant line shall be 1 – 3 mm.

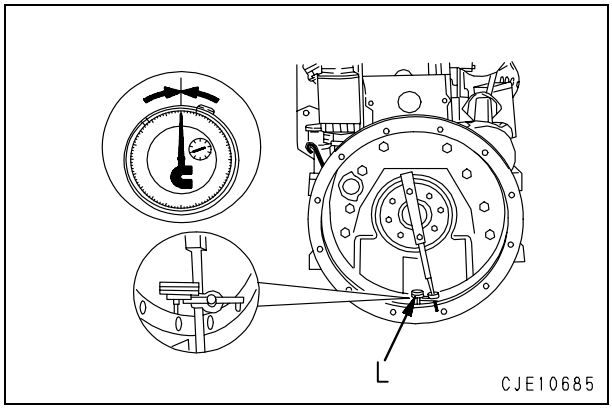


2) Bolts tightening procedure
   Tighten the bolts according to the numbers [1] → [20] shown in the figure.
   ★ 2 types of bolts are used. Make sure when tightening that the specified torque is used.
      1] M12:
      85 ± 10 Nm {8.7 ± 1.0 kgm}
      (△ mark)
      2] M10:
      49 ± 5 Nm {5.0 ± 0.5 kgm}
      (□ mark)

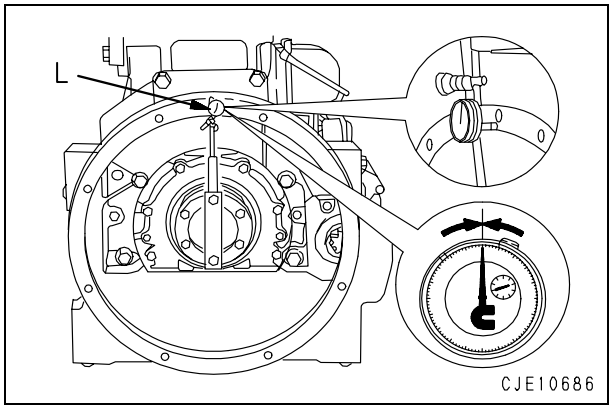


3) After installing the flywheel housing, measure the radial runout and facial runout with tool **L**.
   ★ Radial runout: **Max. 0.20 mm**
   ★ Facial runout: **Max. 0.20 mm**

- Measurement of radial runout
  1] Install tool **L** to the crankshaft end face.
  2] Contact the probe of the dial gauge at right angle against part of the spigot joint of the flywheel housing.
  3] Set graduations of the dial gauge to “0” and then rotate the crankshaft by a turn to measure differences in the maximum deflections of the gauge pointer.
     ★ Make sure after rotating the crankshaft by a turn that the gauge pointer is returned to the position where it had situated when the crankshaft was started to turn.

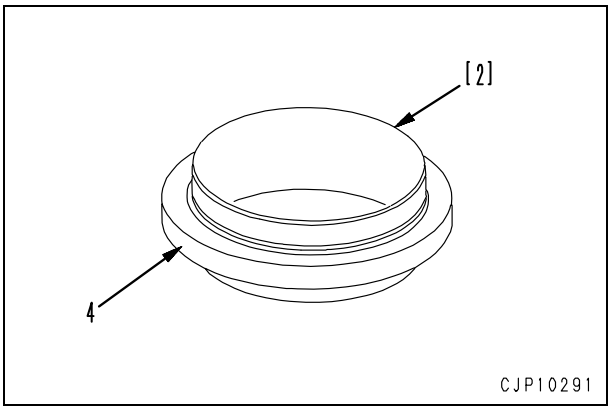


- Measurement of facial runout
  1] Like in measurement of the radial runout, contact the probe of the dial gauge at right angle against the end face of the flywheel housing.
     ★ Before conducting the measurement, move the crankshaft either in the front or rear side in order to prevent errors due to the end play.
  2] Set graduations of the dial gauge to “0” and then rotate the crankshaft by a turn to measure the difference in the maximum deflections of the gauge pointer.

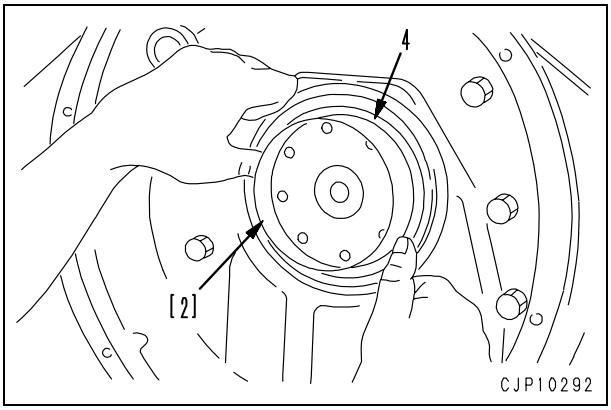


9. Rear oil seal

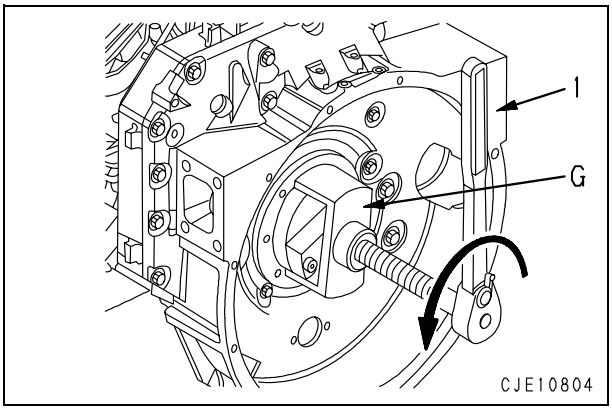
1) Insert pilot [2] to rear oil seal (4).
   ★ Before inserting the pilot, clean, degrease and dry the oil seal lip surface.



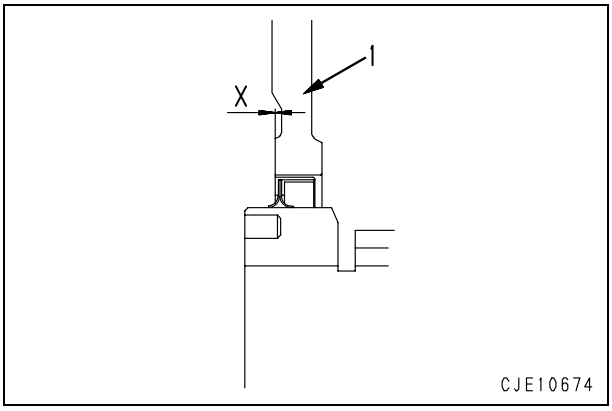
2) Insert oil seal (4), together with pilot [2], to the crankshaft and then push in the seal carrier of oil seal (4) to the flywheel housing (1).
   ★ Before inserting the oil seal, clean, degrease and dry the surface where the seal is contacted against the crankshaft.
3) Pull out pilot [2] and then further push in oil seal (4).



4) Using tool **G**, insert the oil seal to an appropriate depth.

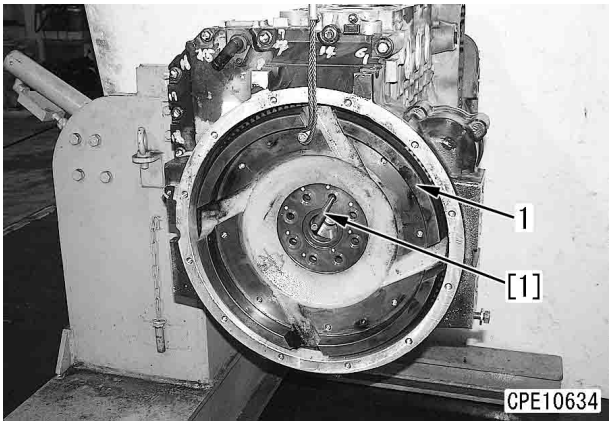


   ★ When pushing in the oil seal, make sure that the seal carrier is free from bending.
   ★ Installed dimension
   X = Max. 0.38 mm

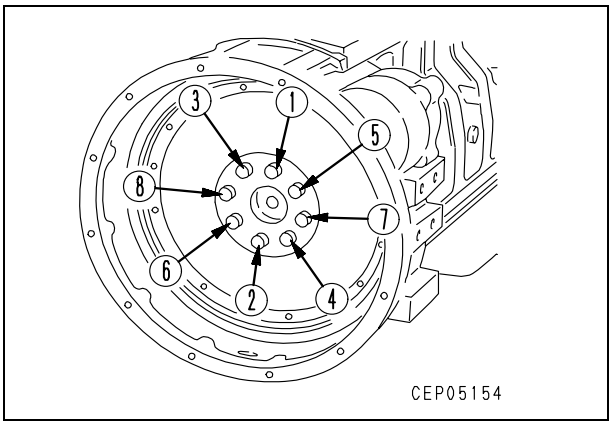


10. Flywheel

1) Install guide bolt [1] and flywheel (1).
   ● Apply gasket sealant to the circumference of the bolt hole (eight locations) on the mounting surface of the crankshaft. Diameter of gasket sealant line shall be 1 – 3 mm.
   Flywheel mounting surface: **Gasket sealant (LG-7)**
   Flywheel assembly: **35 kg**

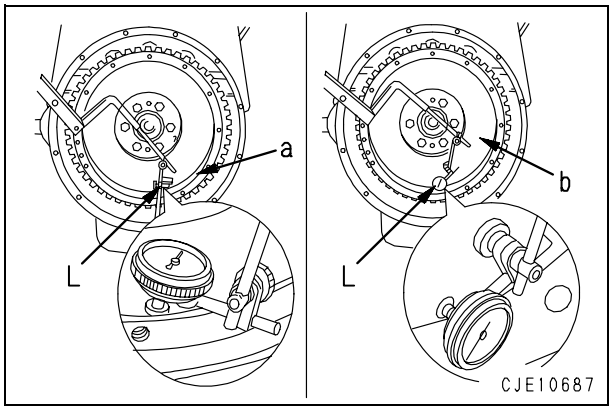


2) Tighten the mounting bolts in the order indicated in the figure.
   Mounting bolt:
   137 ± 7 Nm {14.0 ± 0.7 kgm}



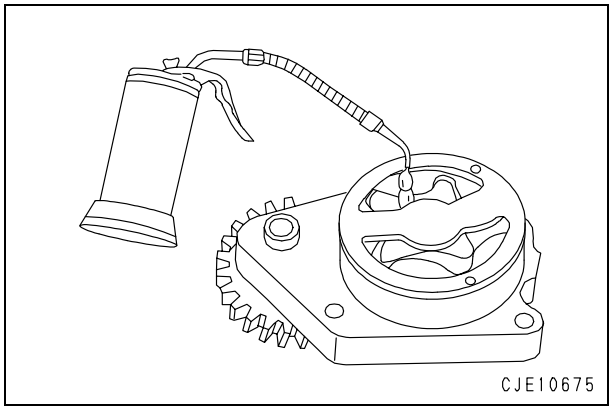
3) After installing the flywheel, measure the radial runout and facial runout with tool **L**.
   ★ Radial runout: **Max. 0.13 mm**
   ★ Facial runout: **Max. 0.20 mm**

- Measurement of radial runout
  1] Install tool **L** to the flywheel housing.
  2] Contact the probe of the dial gauge at right angle against section (a) of the spigot joint or periphery of the flywheel housing.
  3] Rotate the flywheel by a turn to measure the differences in the maximum deflections of the dial gauge pointer.
     ★ Make sure after rotating the flywheel by a turn that the dial gauge pointer is returned to the position where it had situated when the flywheel was started to turn.
- Measurement of facial runout
  1] Like in measurement of the radial runout, contact the probe of the dial gauge at right angle against end face (b) situated near the periphery of the flywheel housing.
     ★ Before conducting the measurement, move the crankshaft either in the front or rear side in order to prevent errors due to the end play.
  2] Rotate the flywheel by a turn to measure the differences in the maximum deflections of the dial gauge pointer.

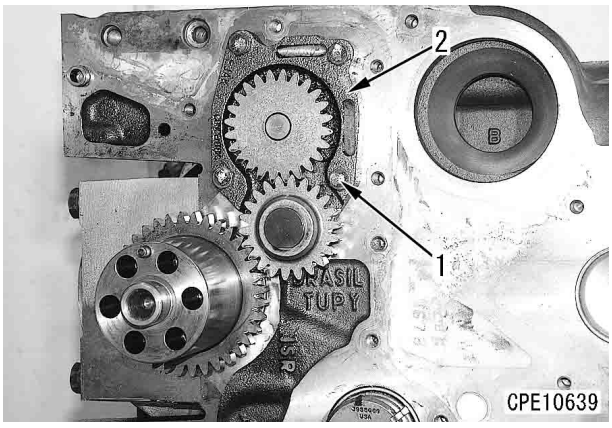


11. Oil pump assembly

★ Fill up the oil pump rotor space and the idler shaft bore with engine oil (EO15W-40) and then rotate the rotor by two turns.

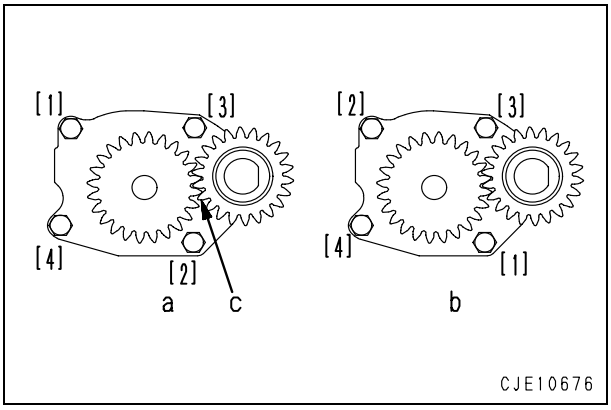


1) Install oil pump assembly (2) with bolts (1).

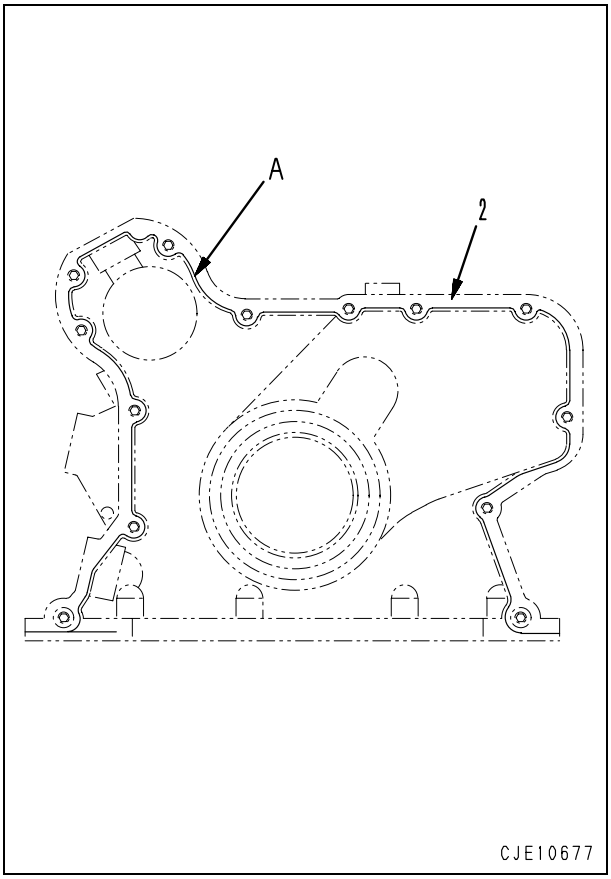


2) Tighten the mounting bolts in the order indicated in the figure.
   Mounting bolt:
   1st time:
   8 ± 1 Nm {0.8 ± 0.1 kgm} (Fig.**a**)
   2nd time:
   24 ± 4 Nm {2.4 ± 0.4 kgm} (Fig.**b**)

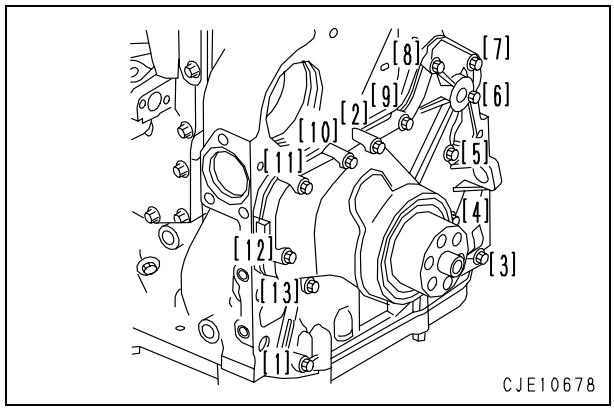
3) Measurement of backlash
Stop move of a gear of either side and then measure backlash (C) of the pump and idler gear.
★ Magnitude of backlash (C):
0.2 ± 0.05 mm



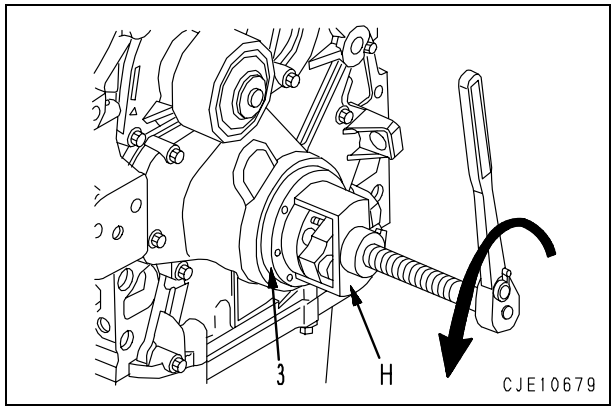
12. Front cover
1) Apply gasket sealant to installation surface (A) of cover (2).
Front cover: **Gasket sealant (LG-7)**
● Diameter of gasket sealant line shall be 1 – 2 mm.



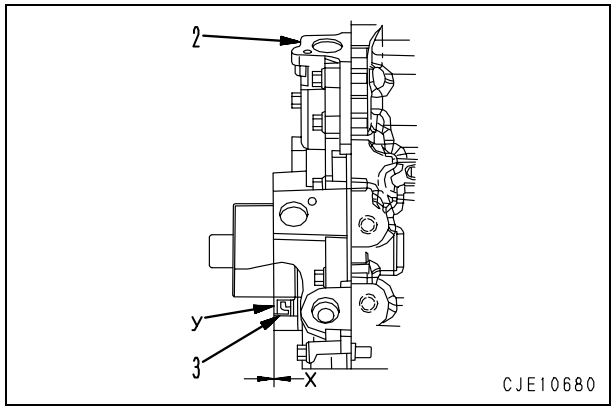
2) Bolts tightening procedure
Tighten the bolts according to the numbers [1] → [13] shown in the figure.
Mounting bolt:
24 ± 4 Nm {2.4 ± 0.4 kgm}



3) Using tool **H**, install front seal (3).
★ Before installing the seal, check that the end corners and lip sliding surfaces of the crankshaft are free from flaw, burr, and rust of the housing.
★ When installing the seal, do not apply oil or grease to the shaft and seal lip. Wipe off the oil from the shaft.

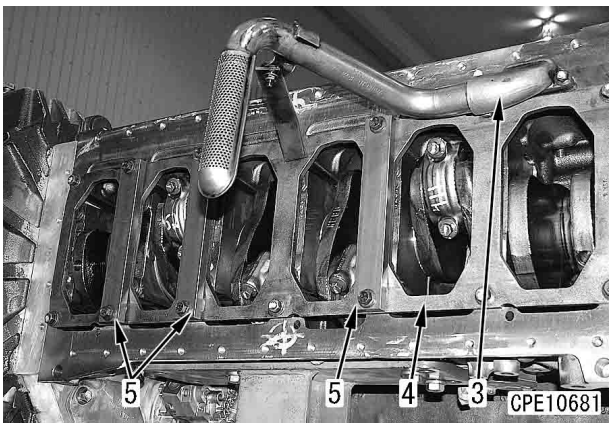


★ When measured from cover (2) of seal (3):
● Extrusion (x): Max. 0.38 mm
● Facial runout (TIR) (y): Max. 0.25 mm

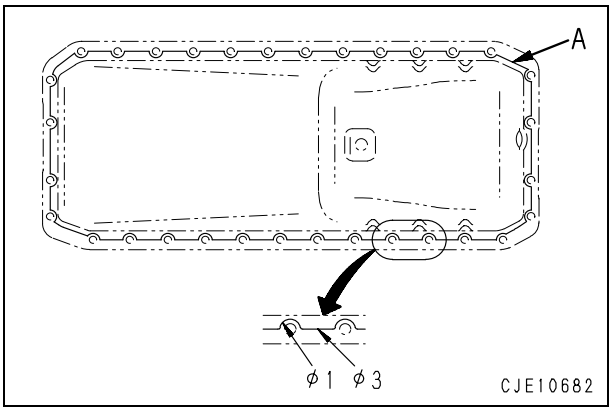


13. Oil pan

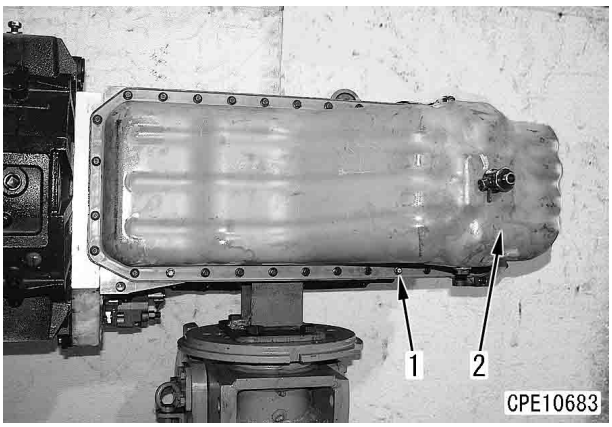
1) Install plate (4) and 3 plates (5).
   Mounting bolt:
   43 ± 5 Nm {4.4 ± 0.5 kgm}
2) Install suction tube (3).
   Flange mounting bolt:
   24 ± 4 Nm {2.4 ± 0.4 kgm}



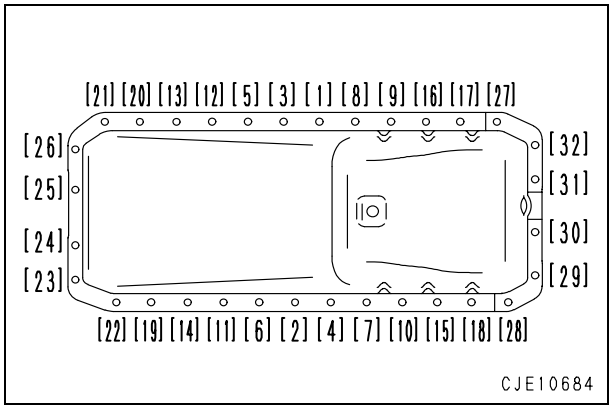
3) Apply gasket sealant to installation surface (A).
   Oil pan: **Gasket sealant (LG-7)**
   - Diameter of gasket sealant line:
     Straight line: 3 mm, curved line: 1 mm



4) Install oil pan (2) with bolts (1).

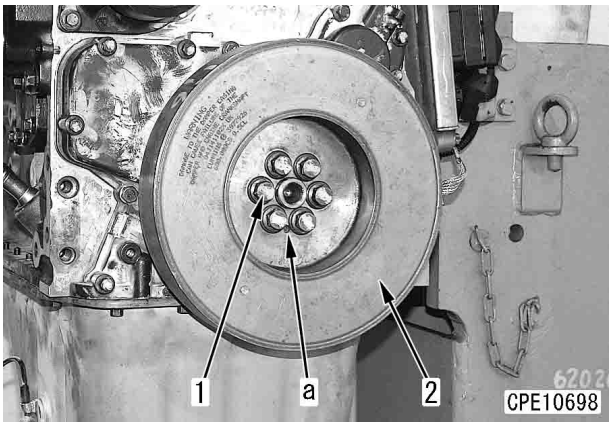


5) Bolts tightening procedure
   Tighten the bolts according to the numbers [1] → [32] shown in the figure.
   Mounting bolt:
   28 ± 2 Nm {2.9 ± 0.2 kgm}

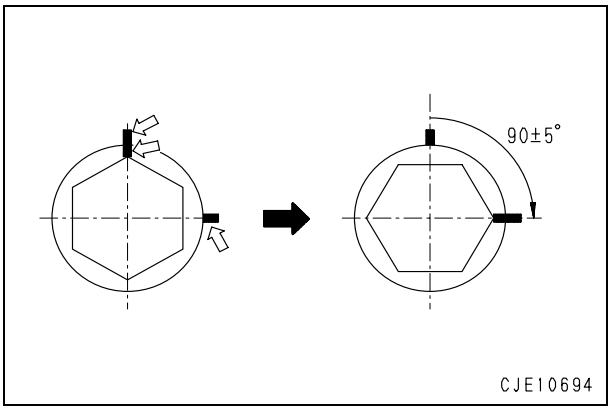


14. Vibration damper

1) Install damper (2) by fitting in its knock hole to dowel pin (a) of the crankshaft.
2) Bolt (1) tightening procedure
   1] Tighten the bolt's opposing corner sequentially with 55 ± 5 Nm {5.6 ± 0.5 kgm}.
   2] Loosen the bolt by 180°.
   3] Tighten the bolt's opposing corner sequentially with 55 ± 5 Nm {5.6 ± 0.5 kgm}.
   4] 90° ± 5° (When angle tightening tool J is used)

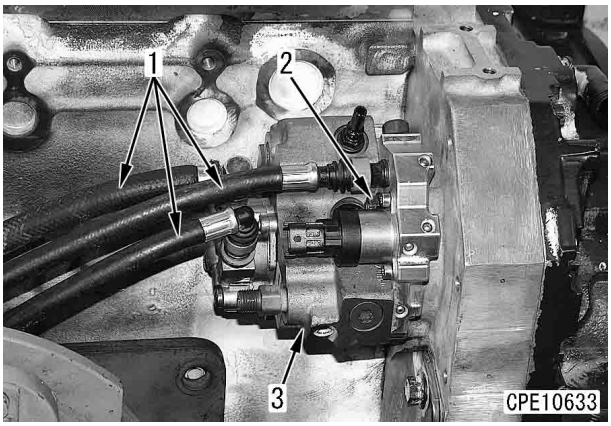


- When the angle tightening tool is not used:
  After applying a paint marking to the vibration damper and bolt, tighten the bolt by **90° ± 5°**.



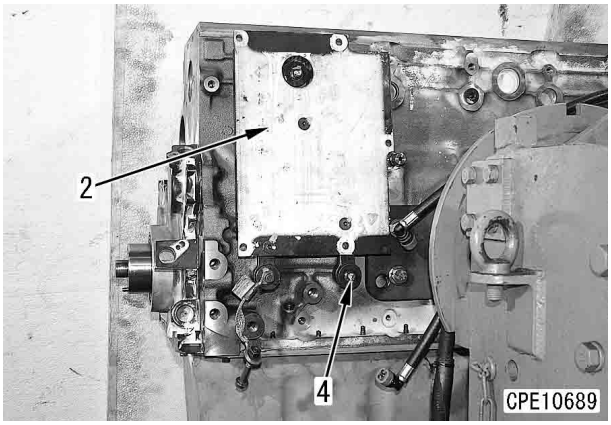
15. Fuel supply pump

1) Install supply pump (3) with 3 nuts (2).
   Mounting nut:
   24 ± 4 Nm {2.4 ± 0.4 kgm}
2) Install hose (1).

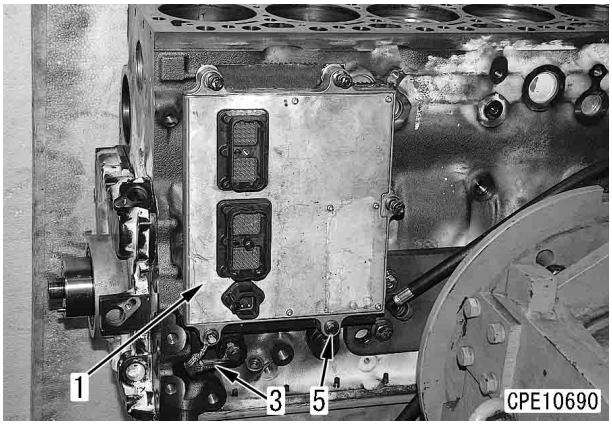


16. Engine controller

1) Install cooling plate assembly (2) with 3 bolts (4).
   Mounting bolt:
   24 ± 4 Nm {2.4 ± 0.4 kgm}

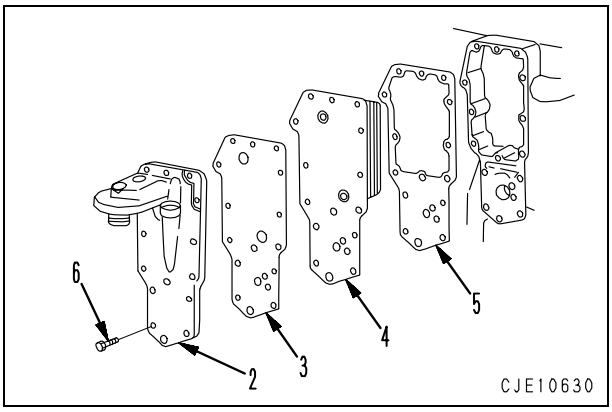


2) Install engine controller (1) with 5 bolts (5).
   Mounting bolt:
   24 ± 4 Nm {2.4 ± 0.4 kgm}
3) Connect ground cable (3).

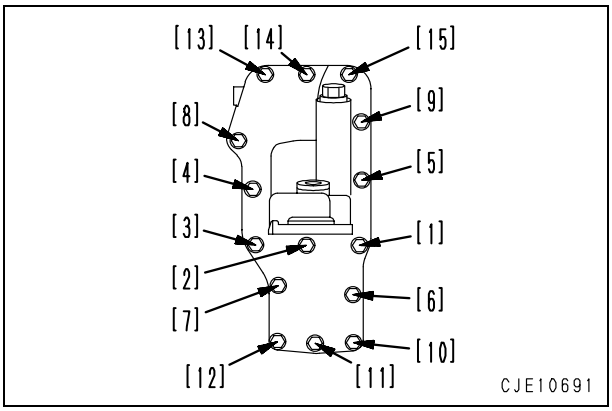


17. Oil cooler

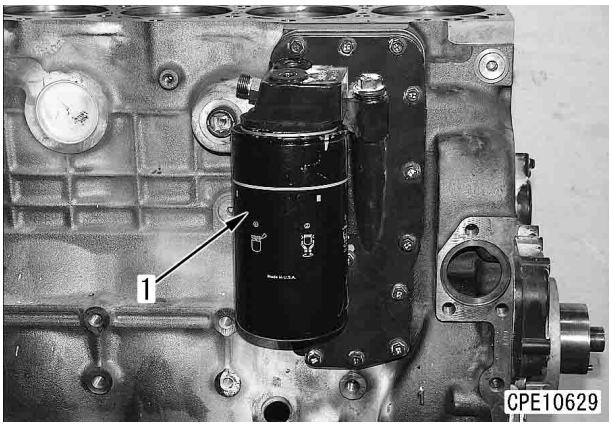
1) Using bolts (6), install gaskets (5) and (3), oil cooler (4) and cover (2).



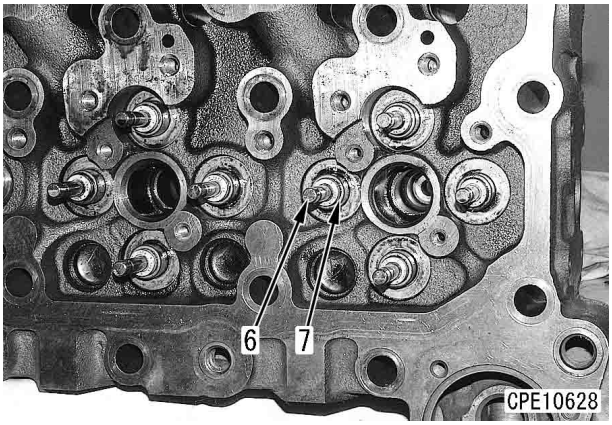
★ Tighten the bolts according to the numbers shown in the figure.
Mounting bolt:
24 ± 4 Nm {2.4 ± 0.4 kgm}



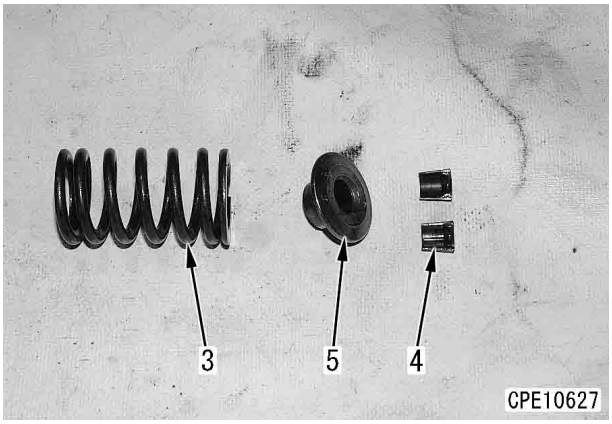
2) Install oil filter (1).
★ Apply engine oil (EO15W-40) on the sealed portion.
Filter: **13 ± 2 Nm {1.3 ± 0.2 kgm}**
Or tighten the bolt by 3/4 turns after contact is made.



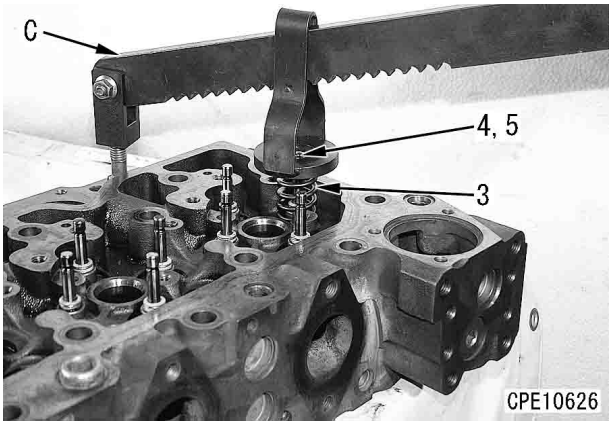
18. Cylinder head assembly
1) Assemble the cylinder head assembly according to the following procedure.
1] Install valve seal (7).
2] Apply engine oil (EO15W-40) to the valve stem and inside of valve guide and install valve (6).



3] Raise the cylinder head and install valve spring (3) and upper seat (5).

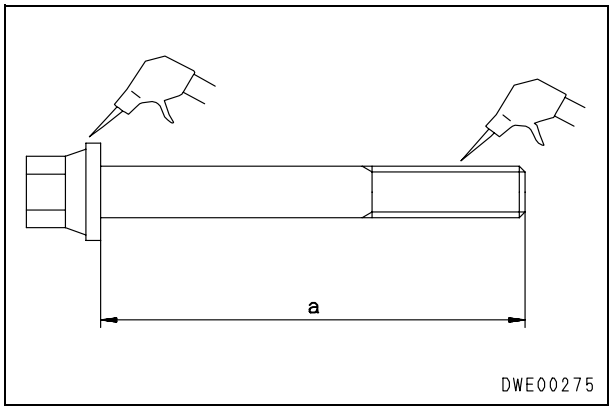


4] Using tool **C**, compress valve spring (3) to install valve cotter (4).

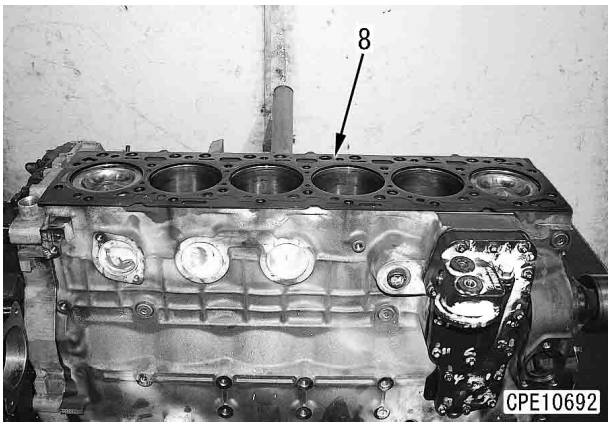


★ Hit the valve stem lightly with a plastic hammer to make sure that the valve collet is positively fitted to the groove of the valve stem.

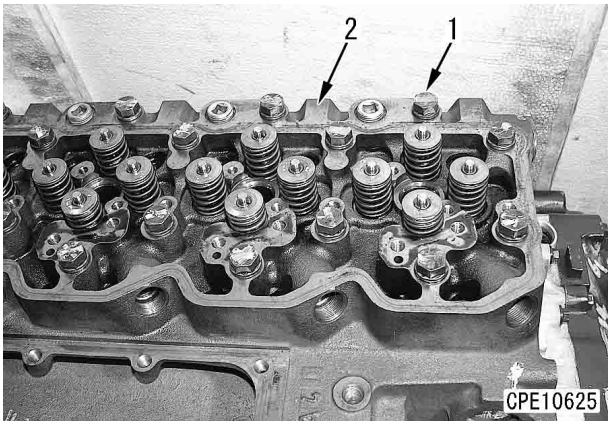
2) Before tightening the cylinder head mounting bolt, check the following.
1] Measure stem length (a) of every bolt to make sure that they are within the using limit.
● Using limit length of bolt stem (a):
Max. 132.1 mm
★ If a bolt is longer than the using limit, do not reuse it but replace it.



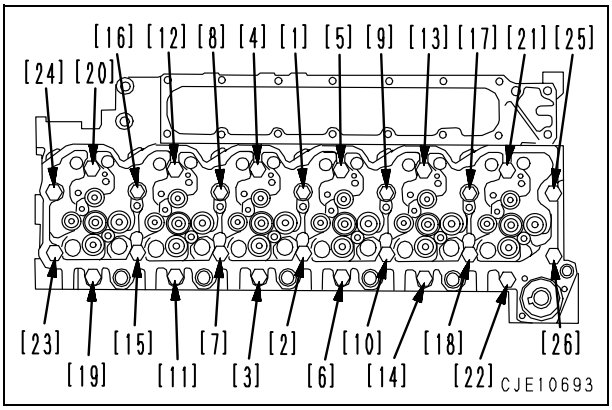
3) Check that the cylinder head mounting face and inside of the cylinder are free from foreign matter and then set cylinder head gasket (8).
   ★ Make sure that the gasket is fitted in the hole on the cylinder head.



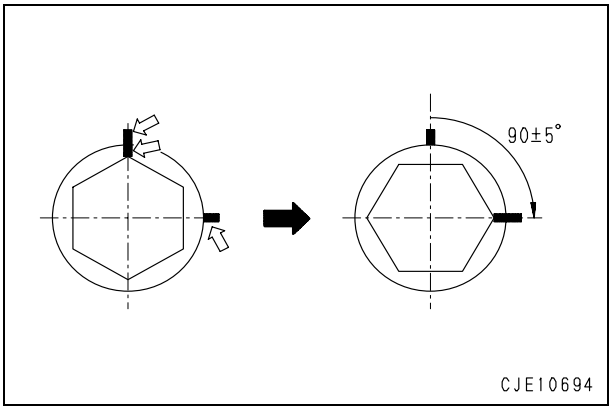
4) Set cylinder head assembly (2) on the cylinder block and then tighten it using mounting bolts (1).



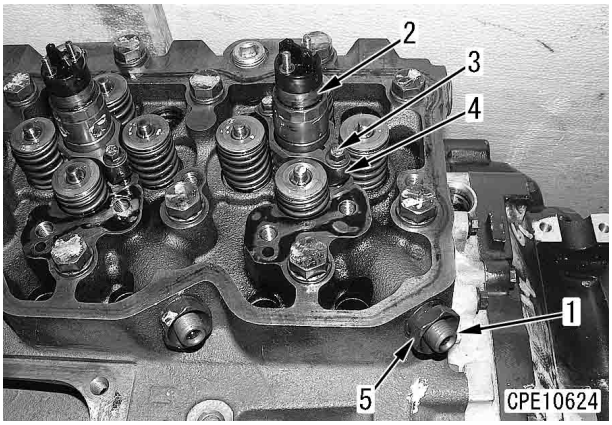
★ First, tighten the mounting bolts by 2 – 3 turns or more by hand, and then tighten them according to the following procedure.
   Apply engine oil (EO15W-40) to the threaded portion and head seat of each bolt.
★ Tighten the bolts in the order shown below.
1] 90 ± 3 Nm {9.2 ± 0.3 kgm}
2] 90 ± 3 Nm {9.2 ± 0.3 kgm}
3] 90° ± 5° (When tightened with angle tightening tool)



- When the angle tightening tool is not used:
  After applying a paint marking to the cylinder head and bolt, tighten the bolt by **90° ± 5°**.

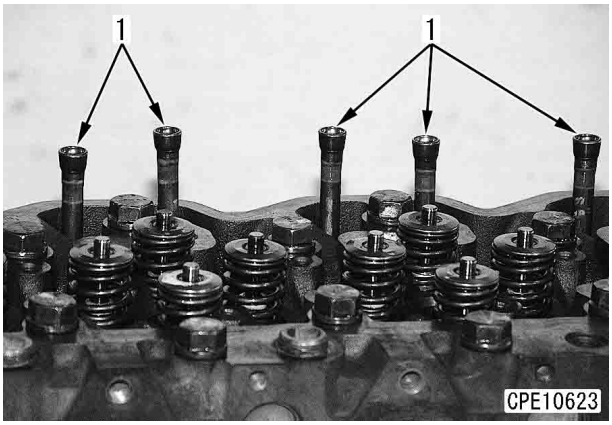


19. Fuel injector assembly
★ Make sure that the injector sleeve is free from damages or dirt.
1) Attach the gasket and O-ring to injector (2).
2) Apply engine oil (EO15W-40) to the O-ring of injector (2) as well as the head side.
3) Insert injector (2) to the cylinder head while directing its fuel inlet hole to the manifold.
4) Install holder (4) and then tighten bolt (3) 3 to 4 threads.
5) Apply engine oil (EO15W-40) to the O-ring of inlet connector (1) as well as the head side.
6) Install inlet connector (1) temporarily (push it into the injector hole).
7) Tighten bolts (3) alternately.
   Bolt: **8 ± 0.8 Nm {0.8 ± 0.08 kgm}**
8) Tighten retainer (5).
   Retainer:
   50 ± 5 Nm {5.1 ± 0.5 kgm}



20. Push rod

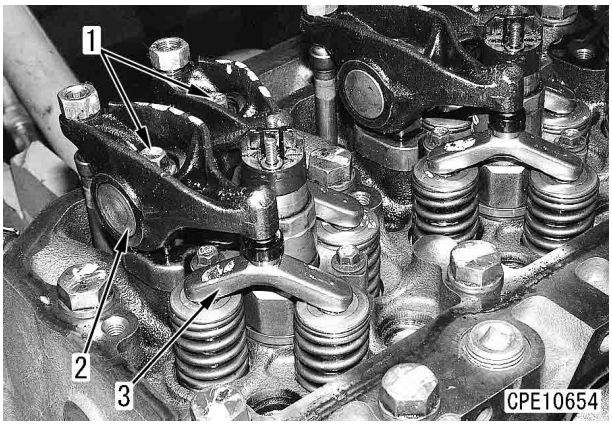
1) Install push rod (1).
   - ★ Check that the push rod is fitted in the tappet.
   - ★ Refill the push rod socket with engine oil (EO15W-40).



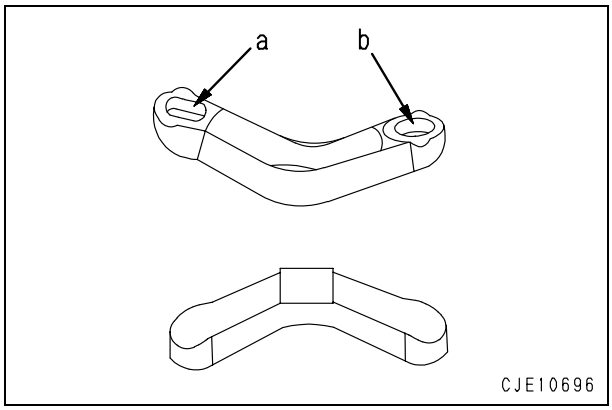
21. Rocker arm assembly

1) Install crosshead (3).
2) Install rocker arm assembly (2) and tighten bolts (1).
   - ★ Before tightening the mounting bolts, check that the ball of the adjustment screw is fitted to the socket of the push rod.

   Bolt: **36 ± 5 Nm {3.7 ± 0.5 kgm}**



- ★ Shape of the holes (a) and (b) on the crosshead is not identical. Thus when using this part again, the same air intake and exhaust valves must be assembled in the same direction.



22. Adjusting valve clearance

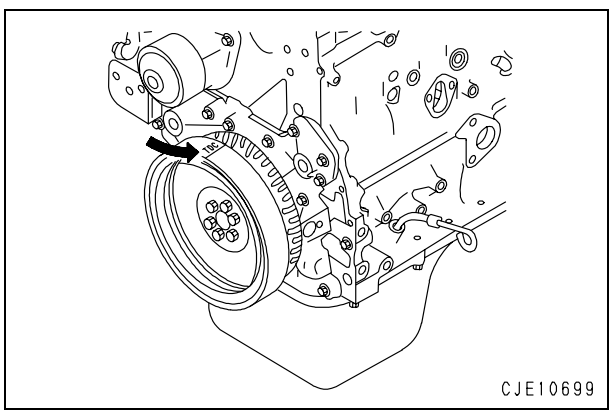
Adjust the valve clearance according to the following procedure.

- ★ As the valve clearance, adjust the clearance between the crosshead and rocker arm to the following value.
- ★ Valve clearance (when cold)

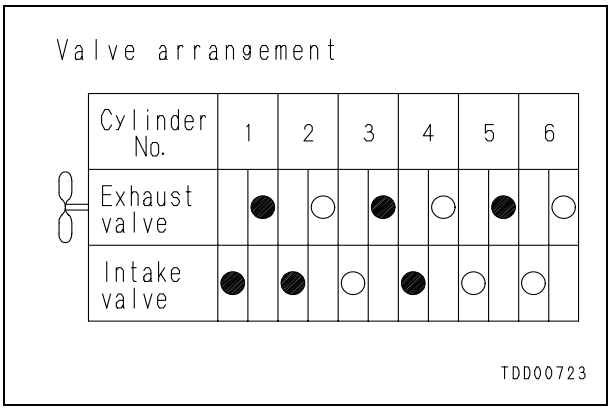
Unit: mm

Air intake valve	Exhaust valve
0.25 ± 0.05	0.51 ± 0.05

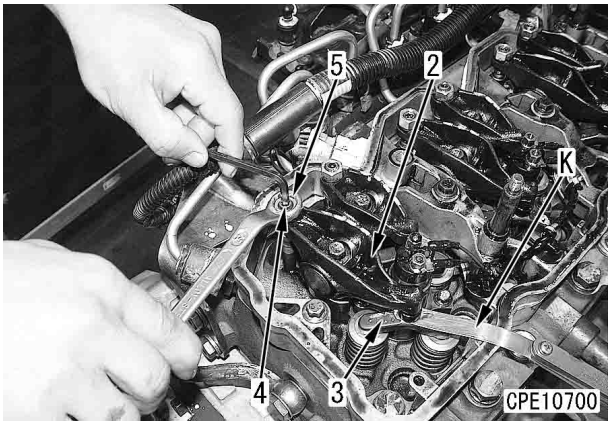
1) While watching move of the rocker arm, rotate the crankshaft forward to set No. 1 cylinder to compression top dead center (T.D.C). Above operation shifts the stamped (T.D.C) of the vibration damper to the directly above position.
   - ★ When the No. 1 cylinder is at the compression top dead center, the rocker arm of the No. 1 cylinder can be moved by the valve clearance with the hand. If the rocker arm cannot be moved, rotate the crankshaft one more turn.



2) While the No. 1 cylinder is at the compression top dead center, adjust the valve clearances marked with ● in the valve arrangement drawing.



3) Insert clearance gauge **K** between rocker arm (3) and crosshead (4) and tighten adjustment screw (5) to the degree that the clearance gauge can be moved lightly.
4) Tighten locknut (5) under this condition.
Locknut:
24 ± 4 Nm {2.4 ± 0.4 kgm}
★ After tightening the locknut, check the valve clearance again.



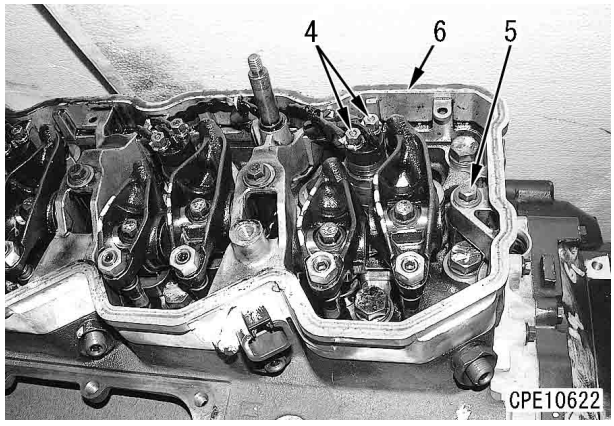
- After adjusting every ● marked valve, rotate the crankshaft forward one turn to set No. 6 cylinder to the compression top dead center with the stamped (T.D.C) of the vibration damper being located at the directly above position.

5) While the No. 6 cylinder is at the compression top dead center, adjust the valve clearances marked with ○ in the valve arrangement drawing.
★ Adjust the valve clearance according to steps 3 and 4 above.

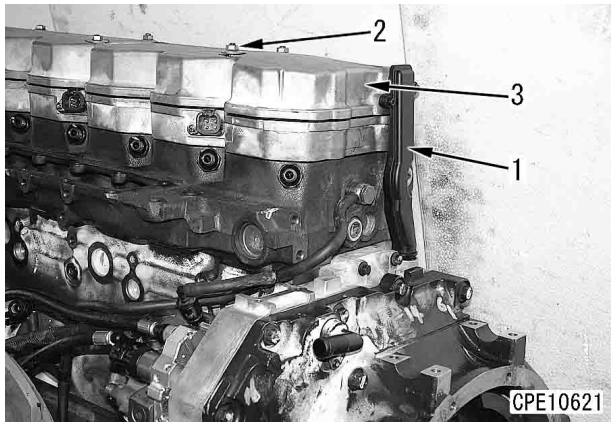
23. Head cover housing
1) Install housing assembly (6) with bolt (5).
Mounting bolt:
24 ± 4 Nm {2.4 ± 0.4 kgm}
2) Install nuts (4) of the harness.
★ Harness installation position

Color of cable	Cylinder No.
White	1, 3, 5
Black	2, 4, 6

Mounting nut:
1.5 ± 0.25 Nm {0.15 ± 0.03 kgm}

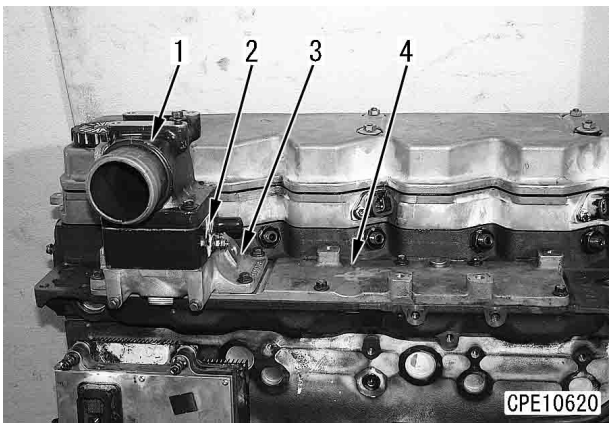


3) Fit the gasket and then install head cover (3) using nut (2).
Mounting bolt:
24 ± 4 Nm {2.4 ± 0.4 kgm}
4) Install duct (1).



24. Air intake manifold cover

1) Fit the gasket and then install manifold covers (3) and (4).
2) Fit the gasket and then install tube (1) and electrical intake air heater (2).
   Tube mounting bolt:
   24 ± 4 Nm {2.4 ± 0.4 kgm}



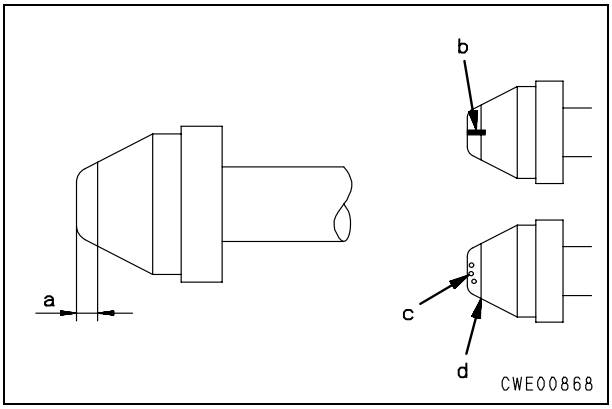
25. Fuel piping and common rail

⚠ **It is prohibited to bend high-pressure pipings used.**

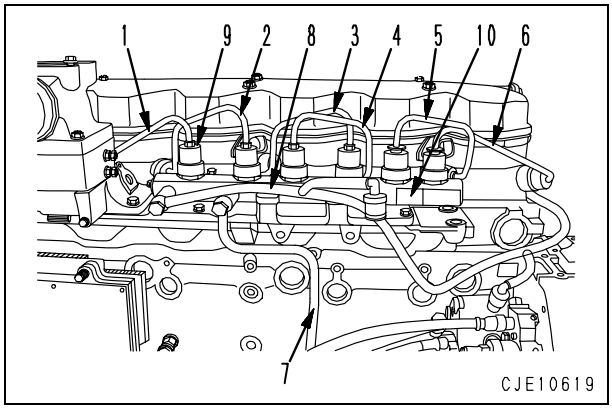
⚠ **The clamps used for fixing high-pressure pipings must be the genuine part. And the specified tightening torque must be strictly observed.**

⚠ **A clearance of 10 mm minimum must be provided between the installed high-pressure piping and harness.**

★ Fuel leakage can occur on a high-pressure piping that has visible lengthwise slits (b) or dents such as patchy damages (c) on its taper seal of the piping connection (part (a): 2 mm space from the tip). The same holds true for a high-pressure piping that catches claw because of fatigued steps in part (d) (end of the taper seal: a 2 mm space from the tip). Such high-pressure piping must be replaced.



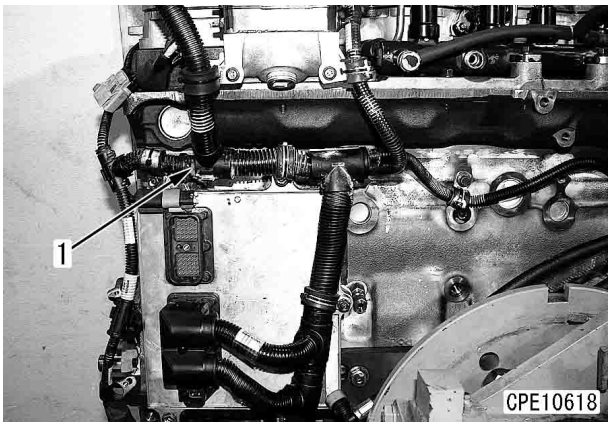
1) Temporarily install common rail (10) and high-pressure pipings (1) to (7).
   Mounting nut and bolt:
   0.2 – 0.8 Nm {0.02 – 0.08 kgm}
2) Tighten high-pressure pipings (1) to (7) according to the following procedure.
   Mounting nut:
   35 ± 3.5 Nm {3.57 ± 0.36 kgm}
   1] Head side of high-pressure pipings (6) and (1)
   2] Rail side of high-pressure pipings (6) and (1)
   3] Head side of high-pressure pipings (2) → (3) → (4) → (5)
   4] Rail side of high-pressure pipings (2) → (3) → (4) → (5)
   5] Pump side of high-pressure piping (7) and then its rail side
3) Tighten the clamp of high-pressure piping (7).
   Mounting bolt:
   24 ± 4 Nm {2.4 ± 0.4 kgm}
4) Tighten common rail (10) with 4 bolts.
   ★ Apply sealant (LG-7) to the threaded part of 2 bolts inside the engine.
   Mounting bolt:
   24 ± 4 Nm {2.4 ± 0.4 kgm}
5) Install boots (9). (14 pieces)
   ★ Set the slits of each boots out and down.
   ★ The boots are installed so that fuel will not spout over the hot parts of the engine and catch fire when it leaks for some reason.
6) Install fuel drain hose (8).
   Joint bolt:
   24 ± 4 Nm {2.4 ± 0.4 kgm}



26. Harness assembly

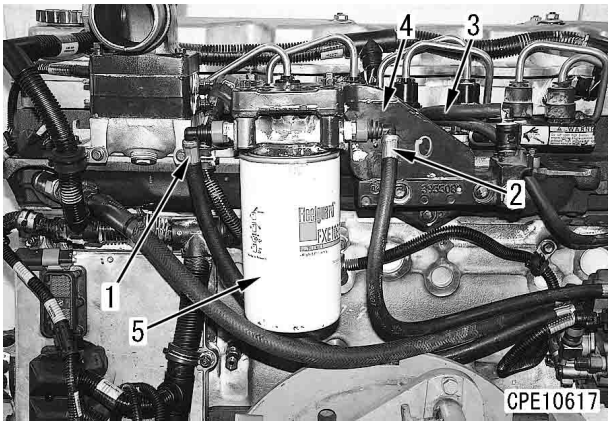
Connect harness assembly (1) to the controller and sensor and then fix it to the engine.

⚠ **Before connecting the harness connector, clean sands, dusts or water in the controller side connector.**



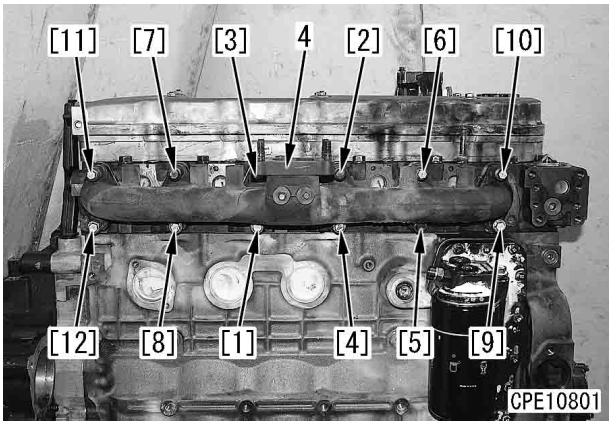
27. Fuel filter assembly

1) Install filter bracket assembly (4).
2) Install drain hose (3) to the filter bracket.
3) Install filter inlet hose (1) and outlet hose (2).
4) Install fuel filter (5).
   ★ As the gasket touched the filter head, tighten the bolt 1/2 turn to install it.

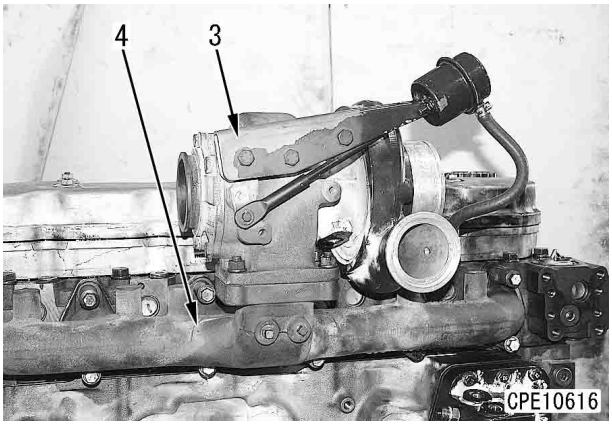


28. Exhaust manifold and turbocharger

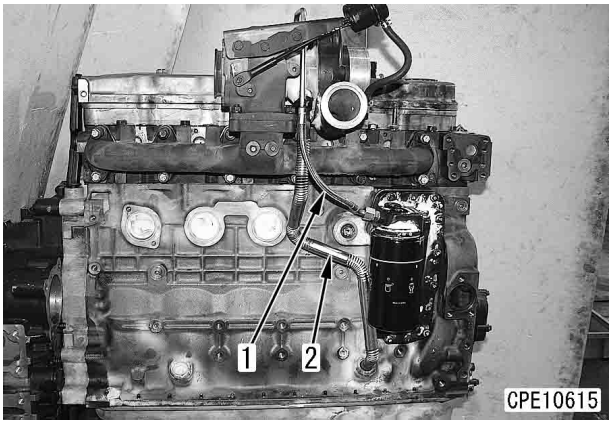
1) Fit the gasket and then install exhaust manifold (4) to the cylinder head. Tighten the mounting bolts according to the numbers shown in the figure.
   Mounting bolts tightening procedure:
   1] Tighten every bolt with **24 ± 4 Nm {2.4 ± 0.4 kgm}**.
   2] Tighten every bolt with **53 ± 6 Nm {5.4 ± 0.6 kgm}**.
   3] Tighten [1] to [4] bolts alone in the figure in this order with **53 ± 6 Nm {5.4 ± 0.6 kgm}**.



2) Fit the gasket and install turbocharger assembly (3).
   Mounting nut:
   43 ± 6 Nm {4.4 ± 0.6 kgm}
   ★ Tighten the mounting nut in the diagonal position alternately.

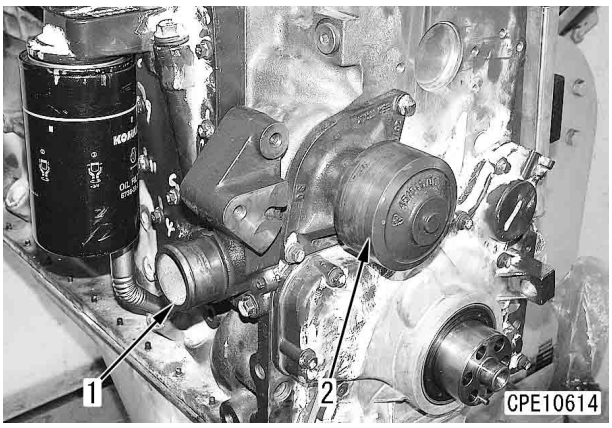


3) Install turbocharger lubrication inlet tube (1).
   Mounting nut:
   35 ± 5 Nm {3.6 ± 0.5 kgm}
4) Fit the gasket to the turbocharger side and then install drain tube (2).
   Mounting bolt:
   24 ± 4 Nm {2.4 ± 0.4 kgm}



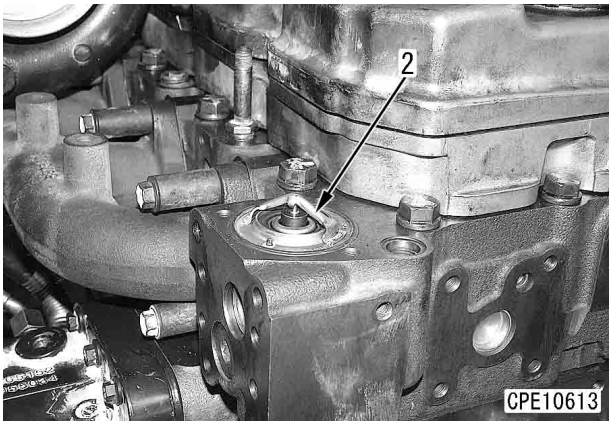
29. Water pump

1) Attach the seal ring and then install water pump (2).
2) Attach the seal ring and then install coolant inlet connector (1).
   Mounting bolt:
   43 ± 6 Nm {4.4 ± 0.6 kgm}



30. Thermostat

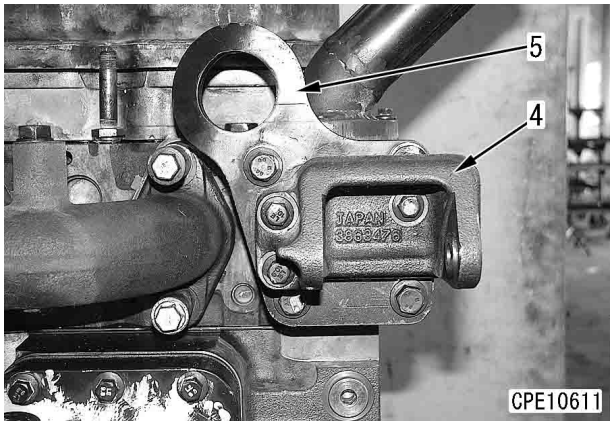
1) Install thermostat (2).



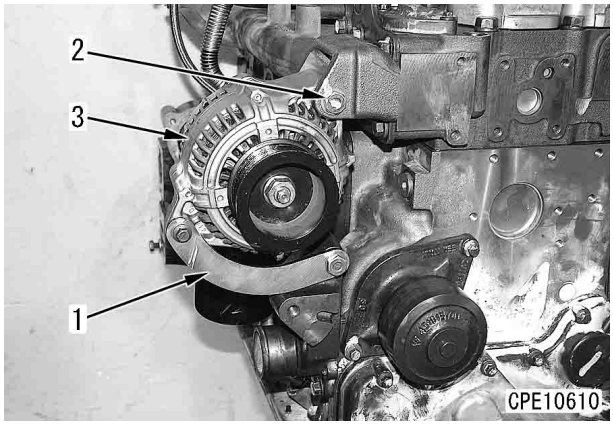
2) Install coolant outlet connector (1).

31. Alternator

1) Install front side sling bracket (5) and alternator bracket (4).
   Mounting bolt:
   24 ± 4 Nm {2.4 ± 0.4 kgm}

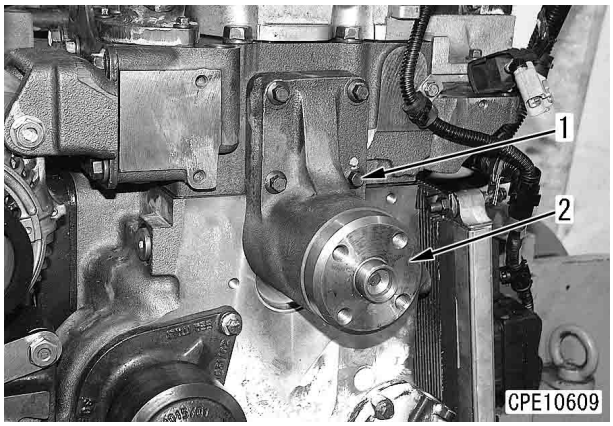


2) Install alternator (3) temporarily using mounting bolt (2).
3) Install plate (1).
4) Tightening alternator mounting bolt (2).



32. Fan hub

Install fan hub (2) with mounting bolt (1).
Mounting bolt:
32 ± 5 Nm {3.3 ± 0.5 kgm}



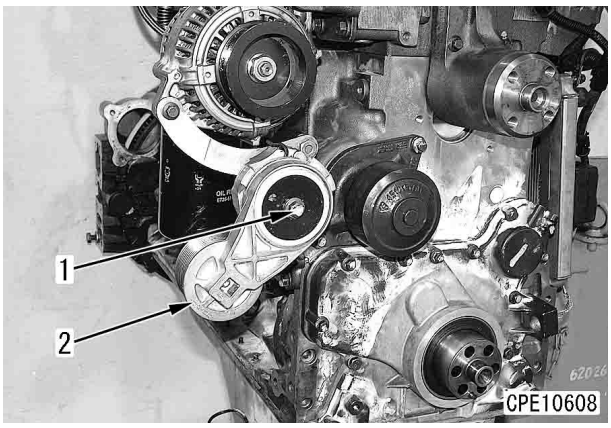
33. Belt tensioner

Install tensioner (2) with mounting bolt (1).

Mounting bolt:

43 ± 6 Nm {4.4 ± 0.6 kgm}

★ Check visually or using a ruler, that the belt mounts of the tensioner pulley and alternator pulley are parallel.



34. Fan pulley

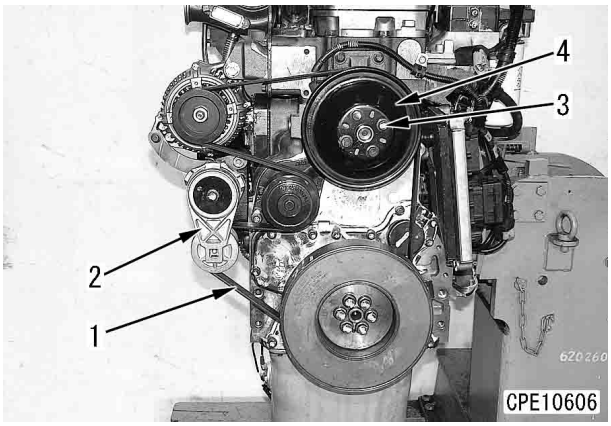
1) Install fan pulley (4) with mounting bolt (3).

Mounting bolt:

43 ± 6 Nm {4.4 ± 0.6 kgm}

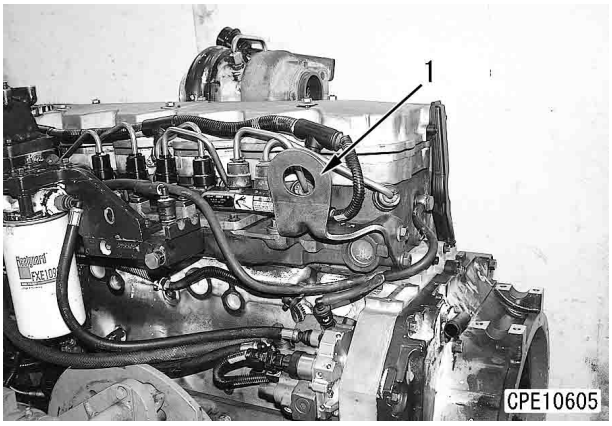
2) Install fan belt (1).

- Install tensioner (2) rotating it in the direction opposite to the hoisting direction.



35. Engine sling bracket on rear side

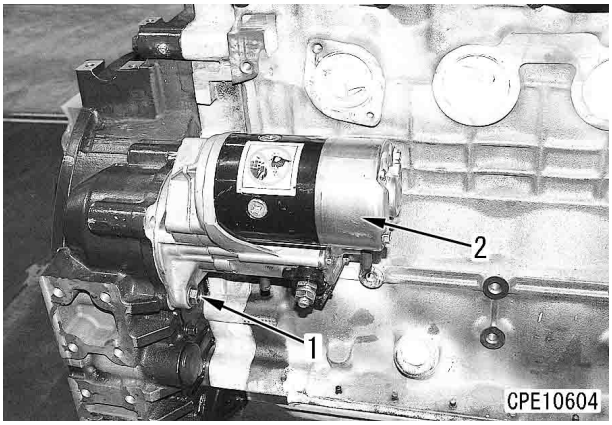
Install engine sling bracket (1) to the cylinder bracket.



36. Starting motor assembly

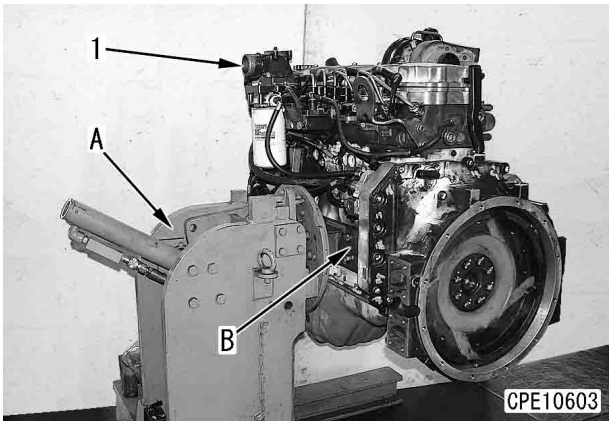
Fit the gasket and then install starting motor assembly (2) with bolt (1).

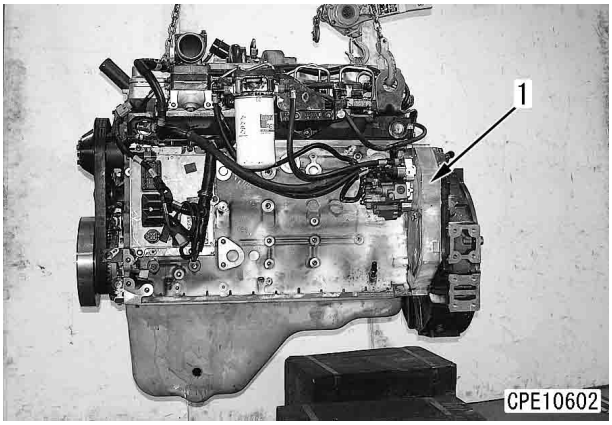
Apply 1 mm wide gasket sealant (LG-7) to the entire circumference of the front and back sides of the gasket.



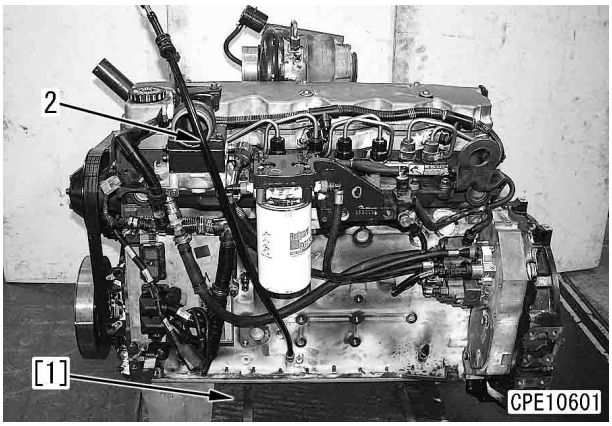
37. Removal from engine repair stand

1) Lift engine assembly (1) temporarily and then separate it from repair stand **A**.
   Engine assembly: **Approx. 520 kg** (The weight depends on the applicable machines.)
2) Remove tool **B**.





3) Set engine assembly (1) on stable engine stand [1] provided on both sides.
4) Install oil gauge assembly (2).



38. Refilling with oil

1) Check that the engine oil drain plug is tightened.
2) Add oil through the oil filler to the specified level.
   Engine oil pan: **Approx. 23ℓ**